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UNITED STATES DISTRICT COURT
DISTRICT OF NEW JERSEY

TAKEDA PHARMACEUTICAL COMPANY LIMITED, *Plaintiff*, v. ANNORA PHARMA PRIVATE LIMITED, *Defendant*.	Civil Action No. ________________ **COMPLAINT FOR PATENT INFRINGEMENT** (Filed Electronically)

Plaintiff Takeda Pharmaceutical Company Limited ("Plaintiff" or "Takeda"), by its attorneys, for its complaint against Annora Pharma Private Limited ("Defendant" or "Annora") alleges as follows:

NATURE OF THE ACTION

1. This is a civil action for infringement of United States Patent Nos. 11,684,632 ("the '632 patent"), 12,213,989 ("the '989 patent"), 12,295,940 ("the '940 patent"), 12,433,907

("the '907 patent"), 12,447,169 ("the '169 patent"), and 12,447,170 ("the '170 patent") (collectively, the "Patents-in-Suit"), all owned by Takeda, under the patent laws of the United States, 35 U.S.C. §100, *et seq.* This action arises from Annora's submission of Abbreviated New Drug Application ("ANDA") No. 221113 ("Annora's ANDA") to the United States Food and Drug Administration ("FDA") seeking approval to manufacture, use, import, distribute, offer to sell, and/or sell a generic version of LIVTENCITY® prior to the expiration of the Patents-in-Suit.

THE PARTIES

2. Takeda is a corporation organized and existing under the laws of Japan, having an office and place of business at 1-1, Doshomachi 4-chome, Chuo-ku, Osaka, Japan.

3. On information and belief, Annora is a corporation organized and existing under the laws of India, having a principal place of business at Sy. No. 261, Plot Nos. 13 and 14, Gummadidala Mandal, Sangareddy Dist., Telangana State, 502313, India.

THE PATENTS-IN-SUIT

4. On June 27, 2023, the United States Patent and Trademark Office ("USPTO") duly and lawfully issued the '632 patent, entitled "Maribavir Isomers, Compositions, Methods of Making and Methods of Using." A copy of the '632 patent is attached as Exhibit A.

5. On February 4, 2025, the USPTO duly and lawfully issued the '989 patent, entitled "Use of Maribavir in Treatment Regimens." A copy of the '989 patent is attached as Exhibit B.

6. On May 13, 2025, the USPTO duly and lawfully issued the '940 patent, entitled "Viral Inhibitors, the Synthesis Thereof, and Intermediates Thereto." A copy of the '940 patent is attached as Exhibit C.

7. On October 7, 2025, the USPTO duly and lawfully issued the ’907 patent, entitled “Use of Maribavir in Treatment Regimens.” A copy of the ’907 patent is attached as Exhibit D.

8. On October 21, 2025, the USPTO duly and lawfully issued the ’169 patent, entitled “Maribavir Isomers, Compositions, Methods of Making and Methods of Using.” A copy of the ’169 patent is attached as Exhibit E.

9. On October 21, 2025, the USPTO duly and lawfully issued the ’170 patent, entitled “Use of Maribavir in Treatment Regimens.” A copy of the ’170 patent is attached as Exhibit F.

THE LIVTENCITY® DRUG PRODUCT

10. LIVTENCITY® is a cytomegalovirus (“CMV”) pUL97 kinase inhibitor indicated for the treatment of adults and pediatric patients (12 years of age and older and weighing at least 35 kg) with post-transplant CMV infection/disease that is refractory to treatment (with or without genotypic resistance) with ganciclovir, valganciclovir, cidofovir, or foscarnet.

11. Pursuant to 21 U.S.C. § 355(b)(1) and attendant FDA regulations, the Patents-in-Suit are listed in the FDA publication titled “Approved Drug Products with Therapeutic Equivalence Evaluations” (the “Orange Book”) with respect to LIVTENCITY®.

12. The claims of the Patents-in-Suit cover, *inter alia*, compositions comprising maribavir, pharmaceutical compositions comprising maribavir, and methods of using maribavir.

13. The FDA-approved labeling for LIVTENCITY® instructs and encourages physicians, pharmacists, other healthcare workers, and patients to orally administer LIVTENCITY® according to one or more methods claimed in the Patents-in-Suit.

JURISDICTION AND VENUE

14. This action arises under the patent laws of the United States, 35 U.S.C. §§ 100, *et seq*., and this Court has jurisdiction over the subject matter of this action under 28 U.S.C. §§ 1331, 1338(a), 2201, and 2202.

15. On information and belief, Annora is in the business of, among other things, developing, manufacturing, marketing, importing, offering for sale, and selling pharmaceutical products, including generic drug products, throughout the United States, including in this Judicial District.

16. On information and belief, Annora prepares and submits ANDAs to the FDA, including Annora's ANDA. The Orange Book identifies "ANNORA PHARMA PRIVATE LTD" as the holder of an ANDA for more than one hundred generic prescription products. On information and belief, Annora is the ANDA holder for generic pharmaceutical products that Annora sells and/or distributes throughout the United States.

17. On information and belief, Annora derives substantial revenue, directly or indirectly, from selling generic pharmaceutical products throughout the United States, including in this Judicial District.

18. On information and belief, Annora will work towards the regulatory approval, manufacturing, use, importation, marketing, offer for sale, sale, and distribution of generic pharmaceutical products, including those described in Annora's ANDA, throughout the United States, including in New Jersey and in this Judicial District, prior to the expiration of the Patents-in-Suit.

19. On information and belief, Annora intends to benefit directly if Annora's ANDA is approved, including by participating in the manufacture, importation, distribution, and/or sale of the generic drug product that is the subject of Annora's ANDA.

20. On information and belief, this Judicial District is a likely destination for the generic drug product described in Annora's ANDA.

21. This Court has personal jurisdiction over Annora because, *inter alia*, it: (1) has purposefully availed itself of the privilege of doing business in the State of New Jersey; and (2) maintains extensive and systematic contacts with the State of New Jersey, including through the marketing, distribution, and/or sale of generic pharmaceutical drugs in New Jersey (*see, e.g.*, *supra* ¶¶ 15–17).

22. In the alternative, this Court has personal jurisdiction over Annora because the requirements of Federal Rule of Civil Procedure 4(k)(2)(A) are met, as (a) Plaintiff's claims arise under federal law; (b) Annora is a foreign defendant not subject to general personal jurisdiction in the courts of any state; and (c) Annora has sufficient contacts with the United States as a whole, including, but not limited to, preparing and submitting an ANDA to the FDA and/or manufacturing and/or selling pharmaceutical products distributed throughout the United States, such that this Court's exercise of jurisdiction over Annora satisfies due process.

23. This Court also has personal jurisdiction over Annora because, *inter alia*, it has committed an act of patent infringement under 35 U.S.C. § 271(e)(2), and on information and belief, Annora intends a future course of conduct that includes acts of patent infringement in New Jersey. These acts have led and will continue to lead to foreseeable harm and injury to Plaintiff in New Jersey and in this Judicial District.

24. On information and belief, Annora has previously invoked, stipulated, and/or consented to personal jurisdiction in this Judicial District in numerous prior patent cases.

25. Annora has previously been sued in this Judicial District, has availed itself of New Jersey courts (including this Court) through the assertion of counterclaims in suits brought

in New Jersey (including this Court), and has not challenged personal jurisdiction in New Jersey. *See, e.g.*, *Azurity Pharm., Inc. v. Annora Pharma Private Ltd.*, Civil Action No. 24-8809 (D.I. 12) (D.N.J. Feb. 3, 2025); *Azurity Pharm., Inc. v. Annora Pharma Private Ltd.*, Civil Action No. 23-18420 (D.I. 18) (D.N.J. Nov. 6, 2023); *Rigel Pharm., Inc. v. Annora Pharma Private Ltd., et al.*, Civil Action No. 22-4732 (D.I. 7) (D.N.J. Sept. 21, 2022); *Celgene Corp. v. Annora Pharma Private Ltd., et al.*, Civil Action No. 18-11220 (D.I. 11) (D.N.J. Sept. 10, 2018).

26. Venue is proper in this Judicial District pursuant to 28 U.S.C. §§ 1391 and/or 1400(b).

ACTS GIVING RISE TO THIS SUIT

27. Pursuant to Section 505 of the Federal Food, Drug, and Cosmetics Act ("FD&C Act"), Annora submitted Annora's ANDA seeking approval to engage in the commercial manufacture, use, sale, or offer for sale in, or importation into, the United States of maribavir tablets, 200 mg ("Annora's ANDA Product") before expiration of the Patents-in-Suit.

28. On information and belief, following FDA approval of Annora's ANDA, Annora, unless enjoined by this Court, will make, use, offer for sale, or sell Annora's ANDA Product throughout the United States, or import such generic product into the United States.

29. On information and belief, in connection with the submission of Annora's ANDA as described above, Annora provided written certification to the FDA pursuant to Section 505 of the FD&C Act and 21 U.S.C. § 355(j)(2)(A)(vii)(IV) ("Paragraph IV Certification").

30. By letter dated January 12, 2026, Annora purported to provide notice pursuant to Section 505(j)(2)(B)(iv) of the FD&C Act and 21 C.F.R. § 314.95 to Takeda ("Annora's Notice Letter") that Annora had submitted Annora's ANDA to the FDA with a Paragraph IV Certification, seeking approval to commercially manufacture, use, sell, offer for sale, or import Annora's ANDA Product before expiration of the Patents-in-Suit.

31. Annora's ANDA Product is intended to be a generic version of LIVTENCITY®.

<u>COUNT I</u>
Infringement of U.S. Patent No. 11,684,632

32. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

33. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '632 patent constitutes infringement of one or more claims of that patent under 35 U.S.C. § 271(e)(2)(A).

34. There is a justiciable controversy between the parties hereto as to infringement of the '632 patent.

35. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will infringe one or more claims of the '632 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States.

36. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will induce infringement of one or more claims of the '632 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, upon FDA approval of Annora's ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the '632 patent and knowledge that its acts are encouraging infringement.

37. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will contributorily infringe one or more claims of the '632 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that

Annora's ANDA Product is especially adapted for a use that infringes one or more claims of the '632 patent and that there is no substantial non-infringing use for Annora's ANDA Product.

38. Annora has had knowledge of the '632 patent since at least the date of Annora's ANDA submission.

39. Plaintiff will be substantially and irreparably harmed if Annora's infringement of the '632 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

40. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT II
Infringement of U.S. Patent No. 12,213,989

41. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

42. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '989 patent constitutes infringement under 35 U.S.C. § 271(e)(2)(A).

43. There is a justiciable controversy between the parties hereto as to infringement of the '989 patent.

44. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will infringe one or more claims of the '989 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States.

45. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will induce infringement of one or more claims of the '989 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the

United States. On information and belief, upon FDA approval of Annora's ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the '989 patent and knowledge that its acts are encouraging infringement.

46. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will contributorily infringe one or more claims of the '989 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that Annora's ANDA Product is especially adapted for a use that infringes one or more claims of the '989 patent and that there is no substantial non-infringing use for Annora's ANDA Product.

47. Annora has had knowledge of the '989 patent since at least the date of Annora's ANDA submission.

48. Plaintiff will be substantially and irreparably harmed if Annora's infringement of the '989 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

49. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT III
Infringement of U.S. Patent No. 12,295,940

50. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

51. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '940 patent constitutes infringement under 35 U.S.C. § 271(e)(2)(A).

52. There is a justiciable controversy between the parties hereto as to infringement of the ’940 patent.

53. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will infringe one or more claims of the ’940 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States.

54. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will induce infringement of one or more claims of the ’940 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States. On information and belief, upon FDA approval of Annora’s ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the ’940 patent and knowledge that its acts are encouraging infringement.

55. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will contributorily infringe one or more claims of the ’940 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that Annora’s ANDA Product is especially adapted for a use that infringes one or more claims of the ’940 patent and that there is no substantial non-infringing use for Annora’s ANDA Product.

56. Annora has had knowledge of the ’940 patent since at least the date of Annora’s ANDA submission.

57. Plaintiff will be substantially and irreparably harmed if Annora’s infringement of the ’940 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

58. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys’ fees under 35 U.S.C. § 285.

<u>COUNT IV</u>
Infringement of U.S. Patent No. 12,433,907

59. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

60. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '907 patent constitutes infringement under 35 U.S.C. § 271(e)(2)(A).

61. There is a justiciable controversy between the parties hereto as to infringement of the '907 patent.

62. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will infringe one or more claims of the '907 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States.

63. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will induce infringement of one or more claims of the '907 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, upon FDA approval of Annora's ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the '907 patent and knowledge that its acts are encouraging infringement.

64. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will contributorily infringe one or more claims of the '907 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that

Annora's ANDA Product is especially adapted for a use that infringes one or more claims of the '907 patent and that there is no substantial non-infringing use for Annora's ANDA Product.

65. Annora has had knowledge of the '907 patent since at least the date of Annora's ANDA submission.

66. Plaintiff will be substantially and irreparably harmed if Annora's infringement of the '907 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

67. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT V
Infringement of U.S. Patent No. 12,447,169

68. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

69. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '169 patent constitutes infringement under 35 U.S.C. § 271(e)(2)(A).

70. There is a justiciable controversy between the parties hereto as to infringement of the '169 patent.

71. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will infringe one or more claims of the '169 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States.

72. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will induce infringement of one or more claims of the '169 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the

United States. On information and belief, upon FDA approval of Annora's ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the '169 patent and knowledge that its acts are encouraging infringement.

73. Unless enjoined by this Court, upon FDA approval of Annora's ANDA, Annora will contributorily infringe one or more claims of the '169 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora's ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that Annora's ANDA Product is especially adapted for a use that infringes one or more claims of the '169 patent and that there is no substantial non-infringing use for Annora's ANDA Product.

74. Annora has had knowledge of the '169 patent since at least the date of Annora's ANDA submission.

75. Plaintiff will be substantially and irreparably harmed if Annora's infringement of the '169 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

76. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT VI
Infringement of U.S. Patent No. 12,447,170

77. Plaintiff repeats and realleges the preceding paragraphs above as if fully set forth herein.

78. Annora's submission of ANDA No. 221113 for the purpose of obtaining approval to engage in the commercial manufacture, use, sale, offer for sale, or importation into the United States of Annora's ANDA Product before the expiration of the '170 patent constitutes infringement under 35 U.S.C. § 271(e)(2)(A).

79. There is a justiciable controversy between the parties hereto as to infringement of the ’170 patent.

80. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will infringe one or more claims of the ’170 patent under 35 U.S.C. § 271(a) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States.

81. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will induce infringement of one or more claims of the ’170 patent under 35 U.S.C. § 271(b) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States. On information and belief, upon FDA approval of Annora’s ANDA, Annora will intentionally encourage acts of direct infringement with knowledge of the ’170 patent and knowledge that its acts are encouraging infringement.

82. Unless enjoined by this Court, upon FDA approval of Annora’s ANDA, Annora will contributorily infringe one or more claims of the ’170 patent under 35 U.S.C. § 271(c) by making, using, offering to sell, selling, and/or importing Annora’s ANDA Product in or for the United States. On information and belief, Annora has had and continues to have knowledge that Annora’s ANDA Product is especially adapted for a use that infringes one or more claims of the ’170 patent and that there is no substantial non-infringing use for Annora’s ANDA Product.

83. Annora has had knowledge of the ’170 patent since at least the date of Annora’s ANDA submission.

84. Plaintiff will be substantially and irreparably harmed if Annora’s infringement of the ’170 patent is not enjoined. Plaintiff does not have an adequate remedy at law.

85. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys’ fees under 35 U.S.C. § 285.

PRAYER FOR RELIEF

WHEREFORE, Plaintiff respectfully requests the following relief:

A. A Judgment that Annora has infringed one or more claims of the Patents-in-Suit by submitting ANDA No. 221113;

B. A Judgment that Annora's making, using, selling, offering to sell, or importing Annora's ANDA Product will infringe one or more claims of each of the Patents-in-Suit;

C. An Order, pursuant to 35 U.S.C. § 271(e)(4)(A), that the effective date of FDA approval of ANDA No. 221113 be a date that is no earlier than the later of the expiration of each of the Patents-in-Suit, or any later expiration of exclusivity to which Plaintiff is or becomes entitled;

D. Preliminary and permanent injunctions restraining and enjoining Annora and its officers, agents, attorneys, and employees, and those acting in privity and/or concert with it and/or them, from making, using, offering to sell, selling, or importing Annora's ANDA Product until after the expiration of each of the Patents-in-Suit, or any later expiration of exclusivity to which Plaintiff is or becomes entitled;

E. A permanent injunction, pursuant to 35 U.S.C. § 271(e)(4)(B), restraining and enjoining Annora and its officers, agents, attorneys, and employees, and those acting in privity and/or concert with it and/or them, from practicing any of the subject matter claimed in the Patents-in-Suit, or actively inducing or contributing to the infringement of any claim of the Patents-in-Suit, until after the expiration of each of the Patents-in-Suit, or any later expiration of exclusivity to which Plaintiff is or becomes entitled;

F. A Judgment that the commercial manufacture, use, importation into the United States, offer for sale, and/or sale of Annora's ANDA Product will directly infringe, induce, and/or contribute to infringement of one or more claims of each of the Patents-in-Suit;

G. To the extent that Annora committed any acts with respect to the subject matter claimed in the Patents-in-Suit, other than those acts expressly exempted by 35 U.S.C. § 271(e)(1), a Judgment awarding Plaintiff damages for such acts;

H. If Annora, its officers, agents, attorneys, and employees, and/or those acting in privity and/or concert with it and/or them, engages in the commercial manufacture, use, sale, offer for sale, and/or importation into the United States of Annora's ANDA Product prior to the expiration of the Patents-in-Suit, a Judgment awarding damages to Plaintiff resulting from such infringement, together with interest;

I. A Judgment declaring that each of the Patents-in-Suit remain valid;

J. A Judgment that this is an exceptional case pursuant to 35 U.S.C. § 285 and awarding Plaintiff its attorneys' fees, costs, and expenses incurred in this action; and

K. Any such other and further relief as this Court may deem just and proper.

Dated: February 23, 2026

By: s/ Charles M. Lizza

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CERTIFICATION PURSUANT TO LOCAL CIVIL RULES 11.2 & 40.1

Pursuant to Local Civil Rules 11.2 and 40.1, I hereby certify that the matter captioned *Takeda Pharmaceutical Company Limited v. Qilu Pharmaceutical (Hainan) Co., Ltd.* Civil Action 26-1633 (ES)(MAH) is related to the matter in controversy involves the same plaintiff and the same patents, and because Annora is seeking FDA approval to market a generic version of the same pharmaceutical product.

I further certify that, to the best of my knowledge, the matter in controversy is not the subject of any other action pending in any court or of any pending arbitration or administrative proceeding.

Dated: February 23, 2026

By: s/ Charles M. Lizza
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EXHIBIT A

US011684632B2

(12) **United States Patent**
Peabody, III

(10) **Patent No.: US 11,684,632 B2**
(45) **Date of Patent: *Jun. 27, 2023**

(54) **MARIBAVIR ISOMERS, COMPOSITIONS, METHODS OF MAKING AND METHODS OF USING**

(71) Applicant: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

(72) Inventor: **John D. Peabody, III**, West Chester, PA (US)

(73) Assignee: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 69 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **16/983,310**

(22) Filed: **Aug. 3, 2020**

(65) **Prior Publication Data**

US 2020/0360414 A1 Nov. 19, 2020

Related U.S. Application Data

(60) Division of application No. 16/663,743, filed on Oct. 25, 2019, now Pat. No. 10,765,692, which is a division of application No. 15/291,639, filed on Oct. 12, 2016, now Pat. No. 10,485,813, which is a division of application No. 15/055,043, filed on Feb. 26, 2016, now abandoned, which is a continuation of application No. 14/595,548, filed on Jan. 13, 2015, now abandoned, which is a continuation of application No. 13/282,501, filed on Oct. 27, 2011, now Pat. No. 8,940,707.

(60) Provisional application No. 61/407,637, filed on Oct. 28, 2010.

(51) **Int. Cl.**

A61K 31/7056	(2006.01)
G09B 19/00	(2006.01)
A61K 45/06	(2006.01)
G01N 33/94	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/7056*** (2013.01); ***A61K 45/06*** (2013.01); ***G01N 33/94*** (2013.01); ***G09B 19/00*** (2013.01); *G01N 2430/00* (2013.01); *G01N 2800/52* (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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Primary Examiner — Traviss C McIntosh, III

(74) *Attorney, Agent, or Firm* — Choate, Hall & Stewart LLP; John P. Rearick; Nicholas J. Pace

(57) **ABSTRACT**

The invention relates to novel compositions and methods of using maribavir which enhance its effectiveness in medical therapy, as well as to maribavir isomers and methods of use thereof for counteracting the potentially adverse effects of maribavir isomerization in vivo in the event it occurs.

16 Claims, 4 Drawing Sheets

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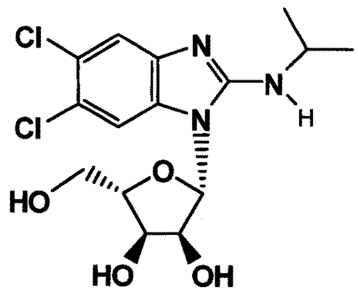
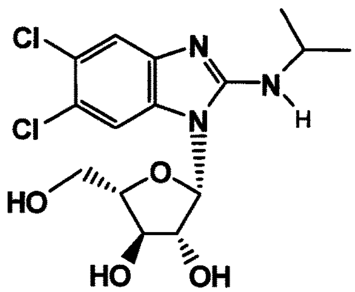


Maribavir



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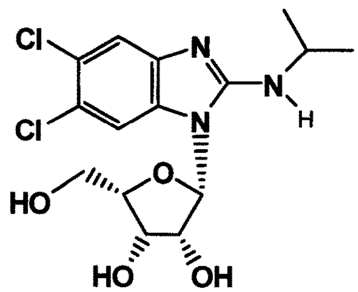

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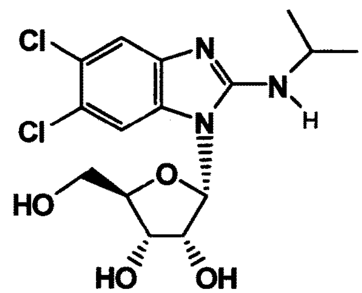

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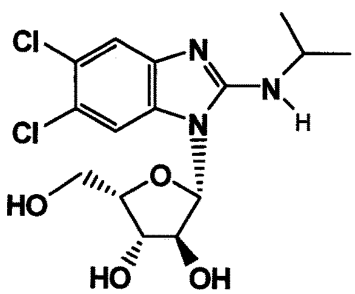

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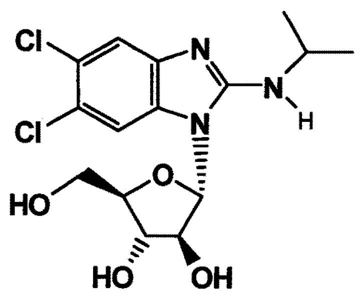

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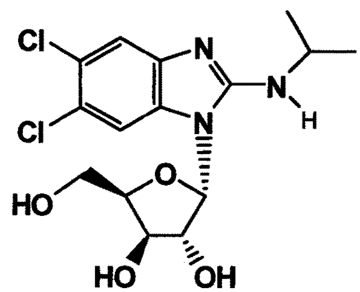

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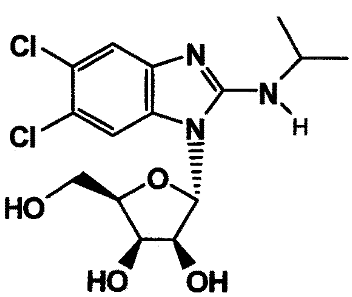

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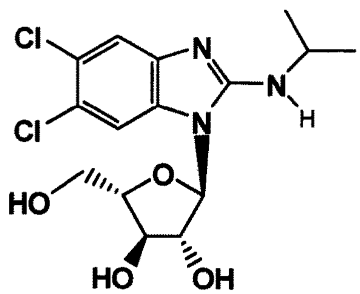
FIG. 1



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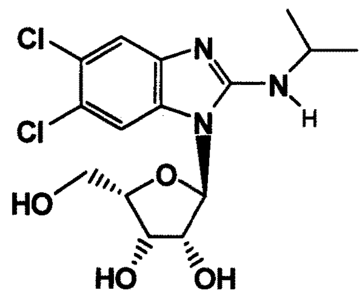

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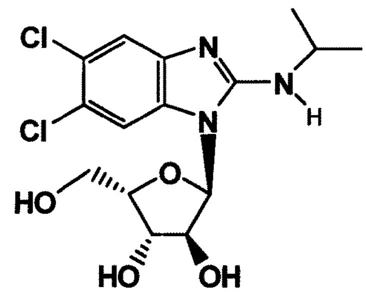

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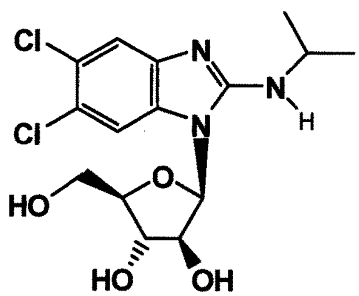

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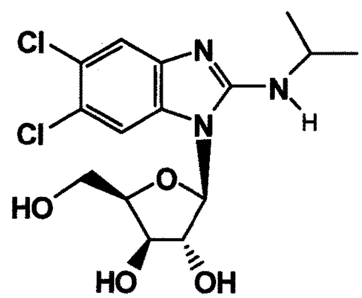

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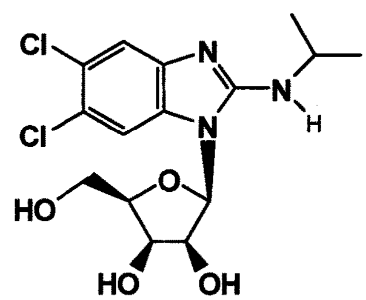

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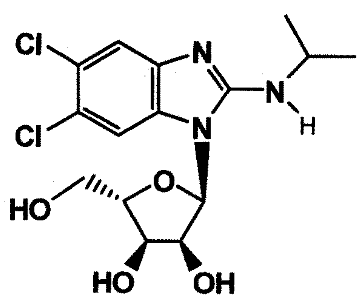

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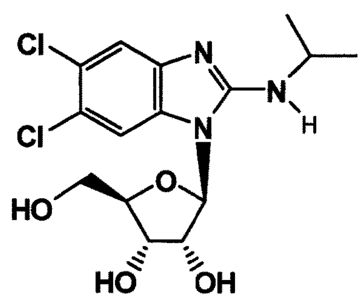

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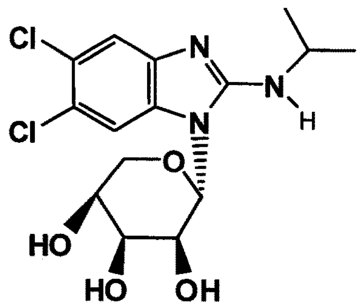
FIG. 2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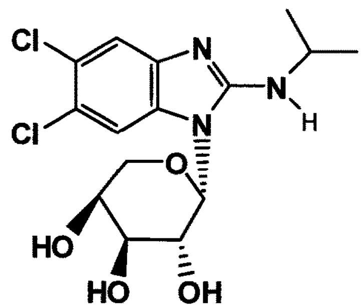
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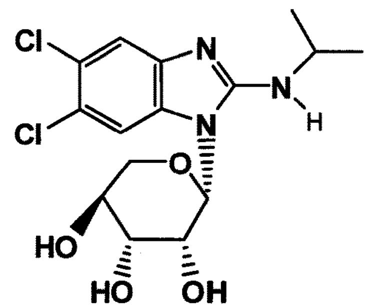
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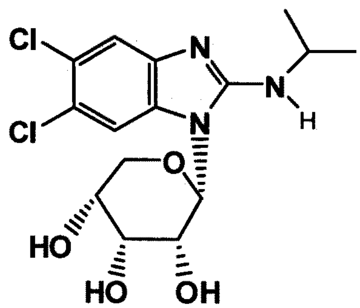
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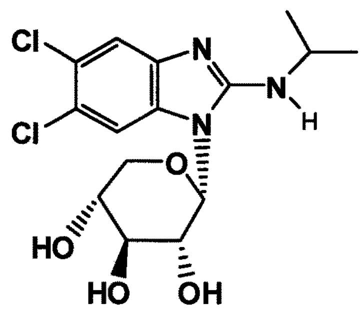
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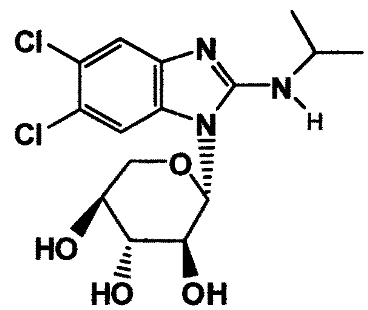
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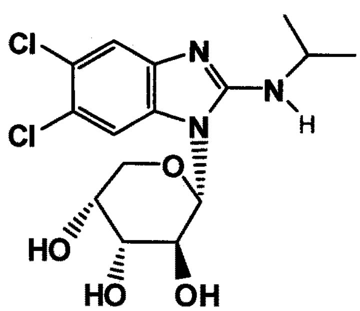
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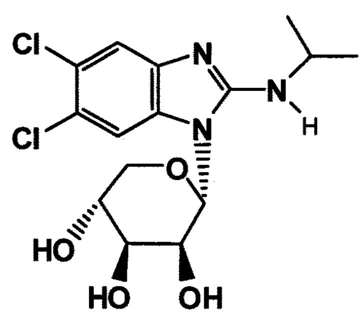
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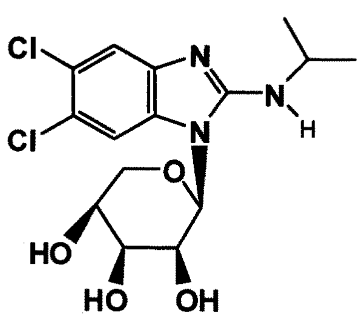
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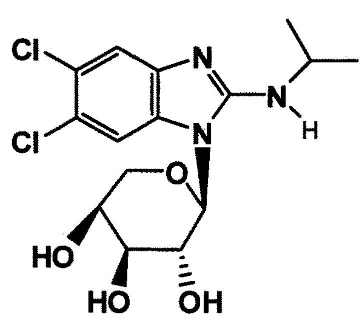
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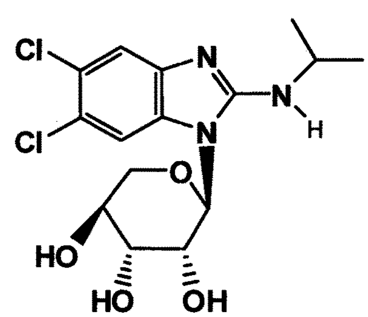
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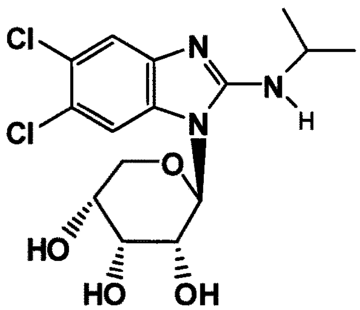
FIG. 3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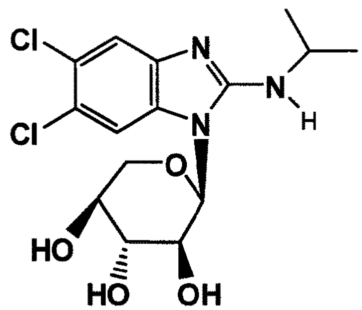
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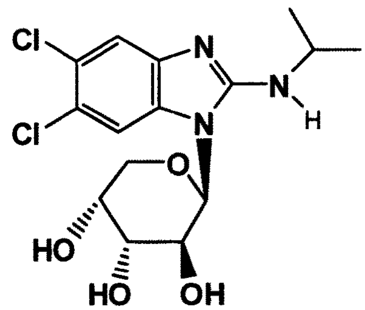
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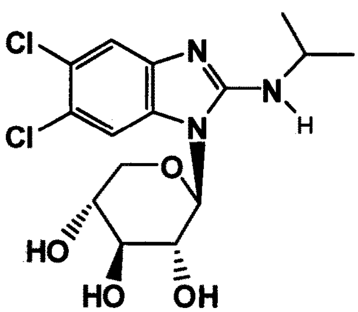
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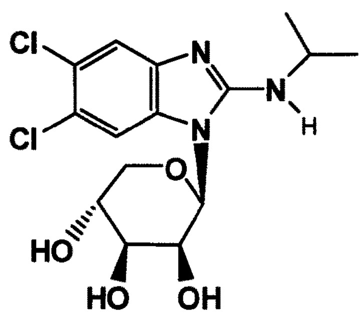
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FIG. 3 (CONT'D)

MARIBAVIR ISOMERS, COMPOSITIONS, METHODS OF MAKING AND METHODS OF USING

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a divisional of U.S. Pat. No. 10,765,692, filed on Oct. 25, 2019, which is a divisional of U.S. Pat. No. 10,485,813, filed Oct. 12, 2016, which is a divisional of U.S. patent application Ser. No. 15/055,043, filed Feb. 26, 2016, which is a continuation of U.S. patent application Ser. No. 14/595,548, filed Jan. 13, 2015, which is a continuation of U.S. Pat. No. 8,940,707, filed Oct. 27, 2011, which claims the benefit of U.S. Provisional Patent Application No. 61/407,637, filed Oct. 28, 2010, the entire disclosure of each of which is incorporated by reference herein.

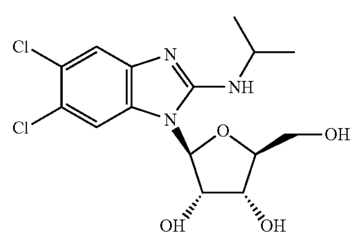
BACKGROUND OF THE INVENTION

The present invention relates to compositions and methods for enhancing the therapeutic efficacy of the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole (also known as 1263W94 and maribavir), as well as to maribavir isomers and a method of making such isomers.

5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole is a benzimidazole derivative useful in medical therapy. U.S. Pat. No. 6,077,832 discloses 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole and its use for the treatment or prophylaxis of viral infections such as those caused by herpes viruses. The compound as disclosed in U.S. Pat. No. 6,077,832 is an amorphous, non-crystalline material.

The structure of 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole is:

(I)



The preparation of certain unique crystalline forms and solvate forms of maribavir, as well as pharmaceutical formulations thereof and their use in therapy are described in U.S. Pat. Nos. 6,469,160 and 6,482,939.

The present invention has arisen out of the unexpected discovery that maribavir may isomerize under in vivo conditions to one or more configurational stereoisomers or constitutional isomers. The maribavir compound contains 4 (four) chiral carbon centers in the ribofuranosyl moiety and therefore maribavir is just one of 16 (sixteen) potential stereoisomers that may be formed under various in vivo conditions.

Under in vivo conditions, maribavir can isomerize to other compounds that may (or may not) have the same or similar chemical, physical and biological properties. The in vivo isomerization of maribavir results in conversion of maribavir to other isomers that have the same molecular formula but a different molecular structure. The different molecular structures can be grouped into isomers that have different connectivity of the constituent atoms (constitutional isomers) or grouped into isomers that have the same "connectivity" but differ in the way the atoms and groups of atoms are oriented in space (configurational stereoisomers). Such molecular conversion in vivo is believed to result in the dilution of the effective maribavir concentration in the host that was treated with maribavir. The in vivo isomerization transforms and distributes dosed maribavir material into other molecular entities that do not necessarily have the same or similar biological activity. If the isomerization results in the formation of isomers that have a lower degree of corresponding biological activity relative to the activity of maribavir, then the isomerization will decrease the effective biological activity of maribavir dose administered to a host.

A recent maribavir Phase 3 clinical trial conducted by ViroPharma Incorporated (the 300 Study) that evaluated maribavir for cytomegalovirus (CMV) prophylaxis in allogeneic stem cell, or bone marrow, transplant (SCT) patients did not achieve its primary endpoint. In the primary analysis, there was no statistically significant difference between maribavir and placebo in reducing the rate of CMV disease. In addition, the study failed to meet its key secondary endpoints (ViroPharma Press Release dated Feb. 9, 2009). The 300 Study result appeared at first blush to be inconsistent with an earlier proof-of-concept (POC) maribavir Phase 2 clinical trial (the 200 Study) wherein ViroPharma reported positive preliminary results that showed that maribavir inhibited CMV reactivation in SCT patients. The data from this study demonstrate that prophylaxis with maribavir displays strong antiviral activity, as measured by significant reduction in the rate of reactivation of CMV in recipients of allogeneic stem cell (bone marrow) transplants, and that administration of maribavir for up to 12 weeks has a favorable tolerability profile in this very sick patient population (ViroPharma Press Release date Mar. 29, 2006).

However, the 300 Study result can be explained in terms of the instant maribavir isomerization theory/discovery. An unrecognized key difference between the 200 Study and the 300 Study was that the former provided for a fasted dosing protocol of maribavir, whereas the latter allowed the dosing protocol to be either under fasted or fed conditions (at the discretion of the clinician). The nature of the patient population in the 300 Study suggests that probably very few patients were dosed under the strict fasted dosing protocol that was previously used in the 200 Study. The change in dosing protocol in the 300 Study changed not only the in vivo dosing conditions for maribavir, but also the nature and/or degree of isomerization of maribavir that occurs in vivo, so that more maribavir was isomerized to other less effective compounds, thereby reducing the effective bioavailable concentration of maribavir drug below levels necessary to adequately prevent CMV infection and/or CMV re-activation in the host.

The degree and nature of the isomerization of maribavir depends on the particular in vivo conditions to which the drug is exposed, which are variable. The potential mechanisms for isomerizing maribavir in vivo are by chemical isomerization (acid, base and/or metal catalyzed isomerization), microbially-mediated isomerization, and/or host metabolism induced isomerization. See, for example, Okano, Kazuya, Tetrahedron, 65: 1937-1949 (2009); Kelly, James A. et al., J. Med. Chem., 29: 2351-2358 (1986); and Ahmed, Zakaria et al., Bangladesh J. Sci. Ind. Res., 25(1-4): 90-104 (2000).

Thus, maribavir should be formulated, administered, packaged and promoted in ways that will prevent or at least reduce the unwanted occurrence of maribavir isomerization in vivo, thereby enhancing the drug's bioavailablity and efficacy, and/or counteract the potential adverse effect(s) of maribavir isomerization in vivo, if it occurs.

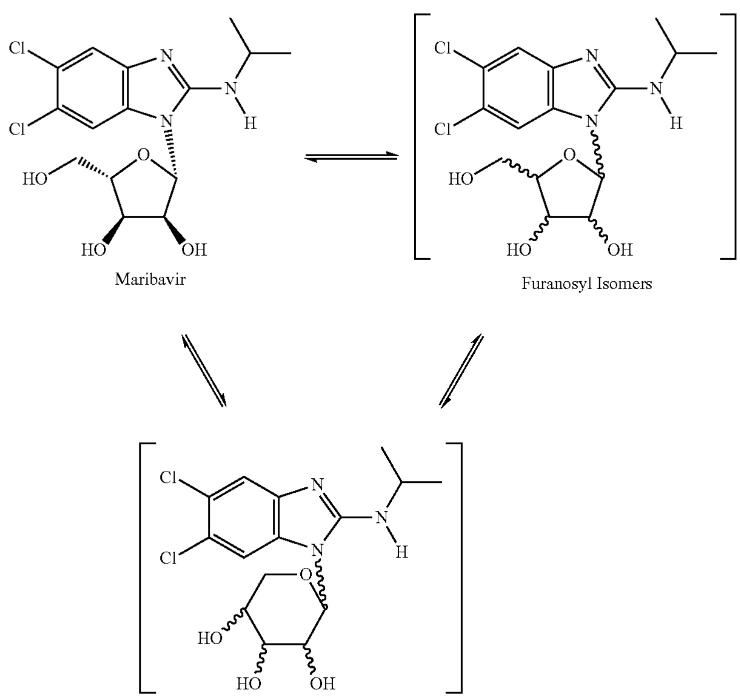
SUMMARY OF THE INVENTION

The invention generally relates to the unexpected discovery that maribavir may isomerize under in vivo conditions to one or more configurational stereoisomers and/or constitutional isomers. Aspect of the invention is illustrated below.

(5) The invention includes methods of using one or more maribavir isomers as reference standards in analytical methods for monitoring the blood plasma concentrations of maribavir and related isomers.

(6) The invention includes methods of monitoring maribavir and maribavir isomers, and using the information to adjust treatment protocols (increase/decrease dose, change dosing regimen fed/fasted, discontinue maribavir therapy, start other therapy). The analytical methods for monitoring in vivo maribavir concentrations must be able discriminate between maribavir and maribavir

Maribavir In Vivo Isomerization Scheme



Pyranosyl Isomers

The practical applications of the instant discovery and related inventions are as follows:

(1) The invention includes methods of making maribavir isomers under in vivo conditions (method of administering maribavir as a prodrug to make other maribavir isomers).

(2) The invention includes methods dosing maribavir so as to mitigate the impact of in vivo maribavir isomerization (such as dosing under fasted conditions, or increasing the dose of maribavir).

(3) The invention includes maribavir formulations that effectively mitigate the adverse effects of in vivo maribavir isomerization (quick release formulations, delayed/controlled release formulations, combination with antacids, intravenous (IV) formulations, combination formulations with antibiotics to prevent microbial isomerization).

(4) The invention includes methods of using one or more maribavir isomers to prevent or treat disease in host (as antivirals, for example for treating/preventing CMV, EBV, HCV).

isomers (for example chiral chromatography, and in particular LC-MS-MS using a chiral sorbent material in the LC column).

(7) The invention includes a method of more safely and effectively using maribavir to treat or prevent disease in humans by including information and guidance in the product label and/or promotional materials to inform the public as to how to use the maribavir product so as to avoid or to mitigate the adverse impact of in vivo maribavir isomerization, and thus optimize therapeutic efficacy and safety.

(8) The invention includes the corresponding inventions related to the maribavir isomers that are pyranosyl constitutional isomers of maribavir.

The present invention also relates to a package or kit comprising therapeutically effective dosage forms of maribavir, prescribing information and a container for the dosage form. The prescribing information includes advice to a patient receiving maribavir therapy regarding the administration of maribavir without food to improve bioavailability.

DETAILED DESCRIPTION OF THE INVENTION

The following table contains list of useful dosing protocols that may be used to improve the efficacy and safety for treating a patient with maribavir.

Maribavir dosing Protocol	Dosing amount	Fasting conditions (before/after dosing)	Route of administration and dosage form.
01	3200 mg/2 × day	None	Oral - tablet - immediate release
02	3200 mg/2 × day	None	IV
03	1600 mg/2 × day	None	Oral - tablet - w/antacids
04	1600 mg/1 × day	None	Oral - tablet - w/antibiotics
05	800 mg/3 × day	None	Oral - tablet - delayed release
06	800 mg/2 × day	None	Oral - tablet - immediate release
07	800 mg/1 × day	None	IV
08	400 mg/4 × day	None	Oral - tablet - w/antacids
09	400 mg/3 × day	None	Oral - tablet - w/antibiotics
10	400 mg/2 × day	None	Oral - tablet - delayed release
11	400 mg/1 × day	None	Oral - tablet - immediate release
12	3200 mg/2 × day	12 hrs/3 hrs	IV
13	3200 mg/2 × day	12 hrs/3 hrs	Oral - tablet - w/antacids
14	1600 mg/2 × day	12 hrs/3 hrs	Oral - tablet - w/antibiotics
15	1600 mg/1 × day	12 hrs/3 hrs	Oral - tablet - delayed release
16	800 mg/3 × day	12 hrs/3 hrs	Oral - tablet - immediate release
17	800 mg/2 × day	12 hrs/3 hrs	IV
18	800 mg/1 × day	12 hrs/3 hrs	Oral - tablet - w/antacids
19	400 mg/4 × day	12 hrs/3 hrs	Oral - tablet - w/antibiotics
20	400 mg/3 × day	12 hrs/3 hrs	Oral - tablet - delayed release
21	400 mg/2 × day	12 hrs/3 hrs	Oral - tablet - immediate release
22	400 mg/1 × day	12 hrs/3 hrs	Oral - tablet - immediate release
23	3200 mg/2 × day	6 hrs/2 hrs	Oral - tablet - immediate release
24	3200 mg/2 × day	6 hrs/2 hrs	IV
25	1600 mg/2 × day	6 hrs/2 hrs	Oral - tablet - w/antacids
26	1600 mg/1 × day	6 hrs/2 hrs	Oral - tablet - w/antibiotics
27	800 mg/3 × day	6 hrs/2 hrs	Oral - tablet - delayed release
28	800 mg/2 × day	6 hrs/2 hrs	Oral - tablet - immediate release
29	800 mg/1 × day	6 hrs/2 hrs	IV
30	400 mg/4 × day	6 hrs/2 hrs	Oral - tablet - w/antacids
31	400 mg/3 × day	6 hrs/2 hrs	Oral - tablet - w/antibiotics
32	400 mg/2 × day	6 hrs/2 hrs	Oral - tablet - delayed release
33	400 mg/1 × day	6 hrs/2 hrs	Oral - tablet - immediate release
34	3200 mg/2 × day	3 hrs/1 hr	IV
35	3200 mg/2 × day	3 hrs/1 hr	Oral - tablet - w/antacids
36	1600 mg/2 × day	3 hrs/1 hr	Oral - tablet - w/antibiotics
37	1600 mg/1 × day	3 hrs/1 hr	Oral - tablet - delayed release
38	800 mg/3 × day	3 hrs/1 hr	Oral - tablet - immediate release
39	800 mg/2 × day	3 hrs/1 hr	IV
40	800 mg/1 × day	3 hrs/1 hr	Oral - tablet - w/antacids
41	400 mg/4 × day	3 hrs/1 hr	Oral - tablet - w/antibiotics
42	400 mg/3 × day	3 hrs/1 hr	Oral - tablet - delayed release
43	400 mg/2 × day	3 hrs/1 hr	Oral - tablet - immediate release
44	400 mg/1 × day	3 hrs/1 hr	Oral - tablet - immediate release

In carrying out the method of the invention, it is preferably to determine the presence and/or concentration of maribavir isomers, especially isomers of diminished therapeutic efficacy in patient blood plasma samples as part of the method.

As used herein, the terms "fasted conditions", "fasting conditions" and "without food" are defined to mean, in general, the condition of not having consumed food during the period between from at least about 3 to 12 hours prior to the administration of maribavir to at least about 1 to 3 hours after the administration of maribavir. Other narrower "fasted conditions" are also contemplated herein and described below.

The term "with food" is defined to mean, in general, the condition of having consumed food prior to, during and/or after the administration of maribavir that is consistent with the relevant intended definition of "fasted conditions" (which may be narrow or broad depending on the circumstances). Preferably, the food is a solid food sufficient bulk and fat content that it is not rapidly dissolved and absorbed in the stomach. More preferably, the food is a meal, such as breakfast, lunch or dinner.

The term "isomers" means compounds that have the same molecular formula but a different molecular structure.

The term "constitutional isomers" is defined to mean isomers that have the same molecular formula but a different molecular structure wherein the molecular structures of the isomers have different connectivity of the constituent atoms.

The term "configurational stereoisomers" is defined to mean isomers that have the same "connectivity" but differ in the molecular structure in the way the atoms and groups of atoms are oriented in space.

The term "immediate release" is defined to mean release of drug from drug formulation by dissolution is less than 60 minutes or is otherwise release from the drug formulation in less than 60 minutes.

The term "IV" is defined to mean intravenous.

The chemical structure of maribavir and some maribavir isomers are shown below. The instant invention contemplates novel formulations, dosage levels and methods of use of maribavir, the maribavir isomers MFI-01 to MFI-015 (configurational stereoisomers), as well as the maribavir isomers MPI-01 to MPI-016 (constitutional isomers). The invention also contemplates the corresponding acyclic constitutional isomers wherein the sugar moiety is an open chain and attached to the benzimidazole.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows chemical structures of maribavir and maribavir configurational stereoisomers that have the same configuration at the furanoysl anomer carbon.

FIG. **2** shows chemical structures of maribavir configurational stereoisomers that have the opposite configuration at the furanosyl anomer carbon.

FIG. **3** shows chemical structures of maribavir "pyranosyl" constitutional isomers)

A number of patent and non-patent documents are cited in the foregoing specification in order to describe the state of the art to which this invention pertains. The entire disclosure of each of these citations is incorporated by reference herein.

While certain of the preferred embodiments of the present invention have been described and specifically exemplified above, it is not intended that the invention be limited to such embodiments. Various modifications may be made thereto without departing from the scope and spirit of the present invention, as set forth in the following claims. Furthermore, the transitional terms "comprising", "consisting essentially of" and "consisting of" define the scope of the appended claims, in original and amended form, with respect to what unrecited additional claim elements or steps, if any, are excluded from the scope of the claims. The term "comprising" is intended to be inclusive or open-ended and does not exclude additional, unrecited elements, methods step or materials. The phrase "consisting of" excludes any element, step or material other than those specified in the claim, and, in the latter instance, impurities ordinarily associated with the specified materials. The phrase "consisting essentially of" limits the scope of a claim to the specified elements, steps or materials and those that do not materially affect the basic and novel characteristic(s) of the claimed invention. All compositions or formulations identified herein can, in alternate embodiments, be more specifically defined by any of the transitional phases "comprising", "consisting essentially of" and "consisting of".

What is claimed is:

1. A method for treatment of a herpes viral infection in a patient in need thereof comprising orally administering to said patient the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole, or an isomer of said compound, in an amount of 400 mg twice a day, wherein said patient is a stem cell transplant recipient.

2. The method according to claim **1**, wherein said herpes viral infection is cytomegalovirus.

3. The method according to claim **2**, wherein said compound is administered to said patient under fasted conditions.

4. The method according to claim **2**, wherein said compound is administered as a composition comprising a therapeutically acceptable adjuvant, excipient, or carrier medium.

5. The method according to claim **4**, wherein said composition is an immediate release formulation, a delayed release formulation, or a controlled release formulation.

6. A method for treatment of a herpes viral infection in a patient in need thereof comprising orally administering to said patient the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole, or an isomer of said compound, in an amount of 400 mg twice a day, wherein said patient is a kidney transplant recipient.

7. The method according to claim **6**, wherein said herpes viral infection is cytomegalovirus.

8. The method according to claim **7**, wherein said compound is administered to said patient under fasted conditions.

9. The method according to claim **7**, wherein said compound is administered as a composition comprising a therapeutically acceptable adjuvant, excipient, or carrier medium.

10. The method according to claim **9**, wherein said composition is an immediate release formulation, a delayed release formulation, or a controlled release formulation.

11. A method for treatment of a herpes viral infection in a patient in need thereof comprising orally administering to said patient the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole, or an isomer of said compound, in an amount of 400 mg twice a day, wherein said patient is a liver transplant recipient.

12. The method according to claim **11**, wherein said herpes viral infection is cytomegalovirus.

13. The method according to claim **12**, wherein said compound is administered to said patient under fasted conditions.

14. The method according to claim **12**, wherein said compound is administered as a composition comprising a therapeutically acceptable adjuvant, excipient, or carrier medium.

15. The method according to claim **14**, wherein said composition is an immediate release formulation, a delayed release formulation, or a controlled release formulation.

16. The method according to any one of claim **1**, **2**, **3**, **5**, or **6-15**, wherein the compound administered is 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 11,684,632 B2
APPLICATION NO. : 16/983310
DATED : June 27, 2023
INVENTOR(S) : John D. Peabody, III

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 8, Lines 48-50 Claim 16: Please replace the text:
“The method according to any one of claims 1, 2, 3, 5, or 6-15”
With:
--The method according to any one of claims 1-15--.

Signed and Sealed this
Twelfth Day of September, 2023

Katherine Kelly Vidal

Katherine Kelly Vidal
Director of the United States Patent and Trademark Office

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 11,684,632 B2
APPLICATION NO. : 16/983310
DATED : June 27, 2023
INVENTOR(S) : Peabody

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

On the Title Page:

The first or sole Notice should read --

Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 73 days.

Signed and Sealed this
Twelfth Day of November, 2024

Katherine Kelly Vidal

Katherine Kelly Vidal
Director of the United States Patent and Trademark Office

EXHIBIT B

US012213989B2

(12) **United States Patent**
Song et al.

(10) **Patent No.: US 12,213,989 B2**
(45) **Date of Patent: Feb. 4, 2025**

(54) **USE OF MARIBAVIR IN TREATMENT REGIMENS**

(71) Applicant: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

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(73) Assignee: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/646,035**

(22) Filed: **Apr. 25, 2024**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 18/400,304, filed on Dec. 29, 2023, now abandoned, which is a continuation of application No. PCT/US2022/050341, filed on Nov. 18, 2022.

(60) Provisional application No. 63/281,206, filed on Nov. 19, 2021.

(51) **Int. Cl.**

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A61K 31/4166	(2006.01)
A61K 31/513	(2006.01)
A61K 31/55	(2006.01)

(52) **U.S. Cl.**
CPC ***A61K 31/7056*** (2013.01); ***A61K 31/4166*** (2013.01); ***A61K 31/513*** (2013.01); ***A61K 31/55*** (2013.01)

(58) **Field of Classification Search**
CPC A61K 31/7056; A61K 31/4166; A61K 31/513; A61K 31/55
USPC 514/43
See application file for complete search history.

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Primary Examiner — Yih-Horng Shiao
(74) *Attorney, Agent, or Firm* — Honigman LLP; Andrew S. Chipouras; Jonathan P. O'Brien

(57) **ABSTRACT**

Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

7 Claims, 4 Drawing Sheets

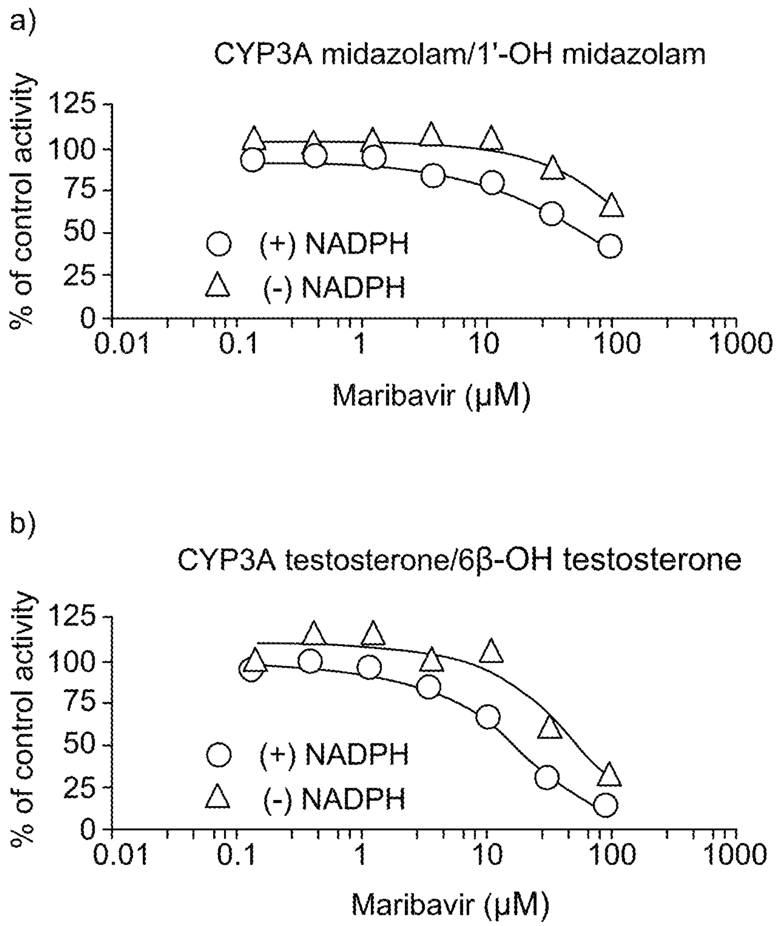

a)
CYP3A midazolam/1'-OH midazolam
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (μM)
b)
CYP3A testosterone/6β-OH testosterone
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (μM)

Figure 1

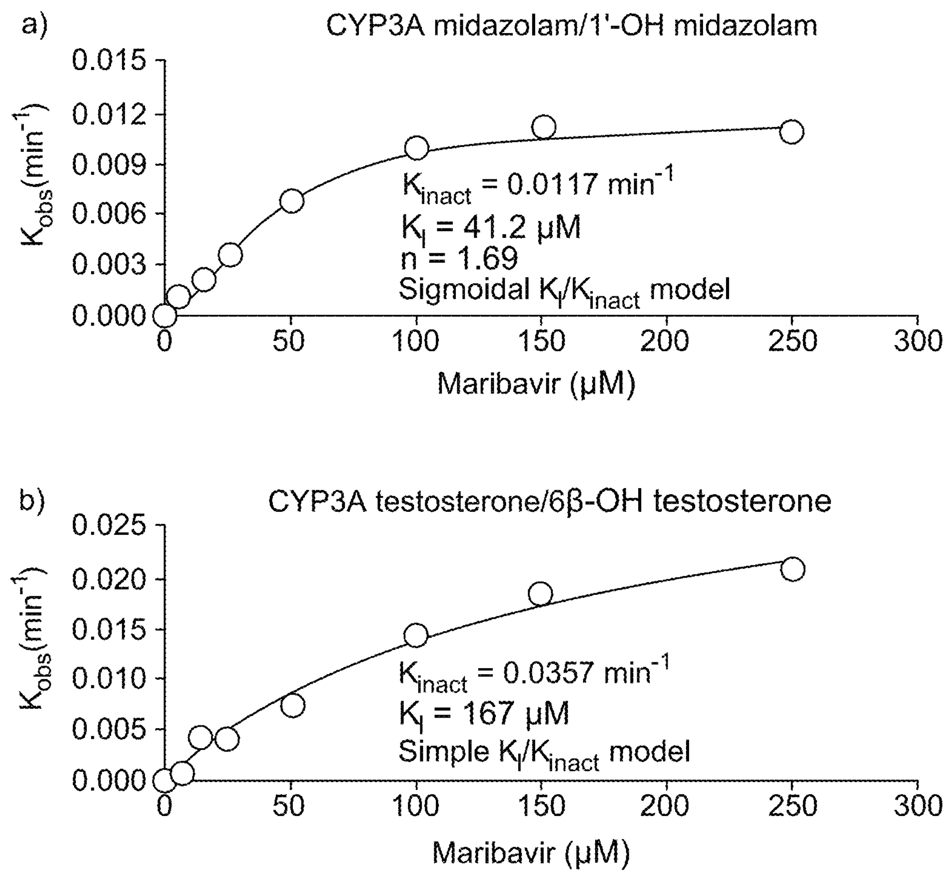

a)
CYP3A midazolam/1'-OH midazolam
0.015
0.012
0.009
0.006
0.003
0.000
Kobs(min-1)
Kinact = 0.0117 min-1
KI = 41.2 µM
n = 1.69
Sigmoidal KI/Kinact model
0
50
100
150
200
250
300
Maribavir (µM)
b)
CYP3A testosterone/6β-OH testosterone
0.025
0.020
0.015
0.010
0.005
0.000
Kobs(min-1)
Kinact = 0.0357 min-1
KI = 167 µM
Simple KI/Kinact model
0
50
100
150
200
250
300
Maribavir (µM)

Figure 2

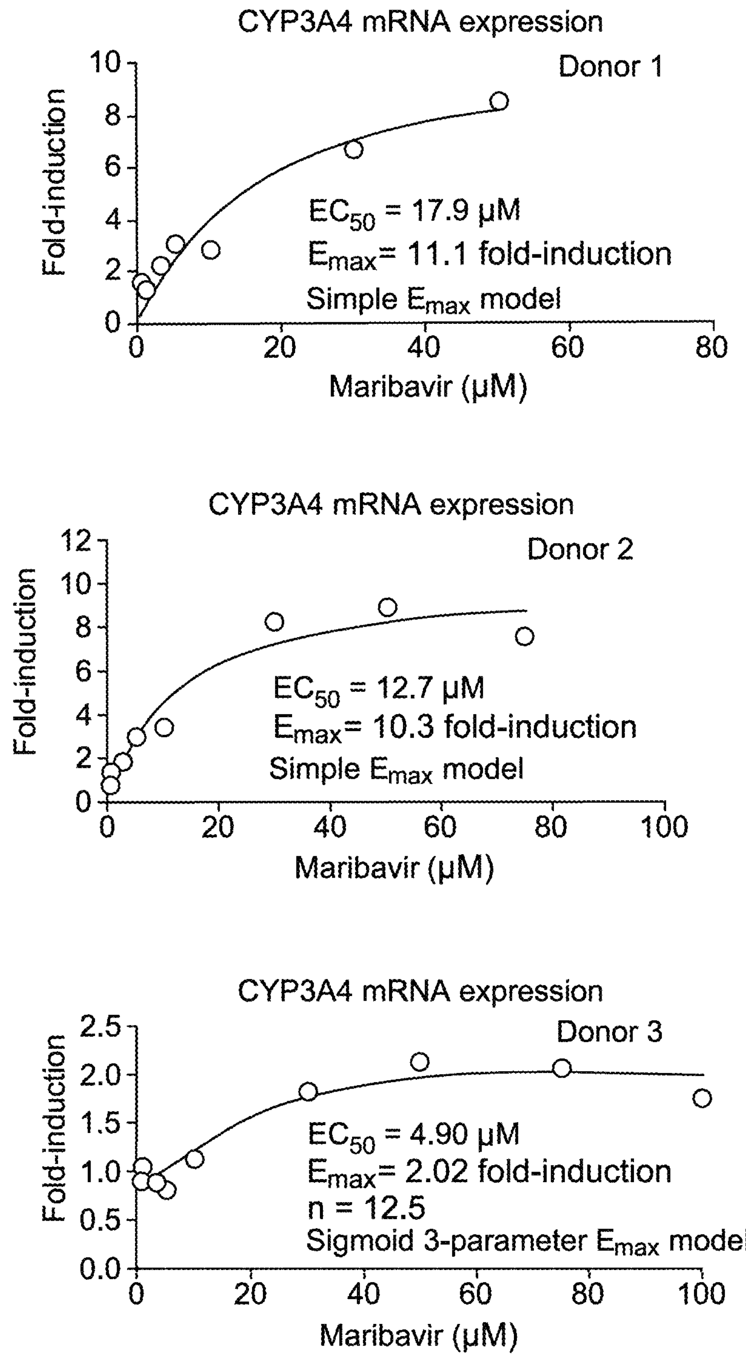

CYP3A4 mRNA expression
Donor 1
Fold-induction
EC50 = 17.9 μM
Emax= 11.1 fold-induction
Simple Emax model
Maribavir (μM)
CYP3A4 mRNA expression
Donor 2
Fold-induction
EC50 = 12.7 μM
Emax= 10.3 fold-induction
Simple Emax model
Maribavir (μM)
CYP3A4 mRNA expression
Donor 3
Fold-induction
EC50 = 4.90 μM
Emax= 2.02 fold-induction
n = 12.5
Sigmoid 3-parameter Emax model
Maribavir (μM)

Figure 3

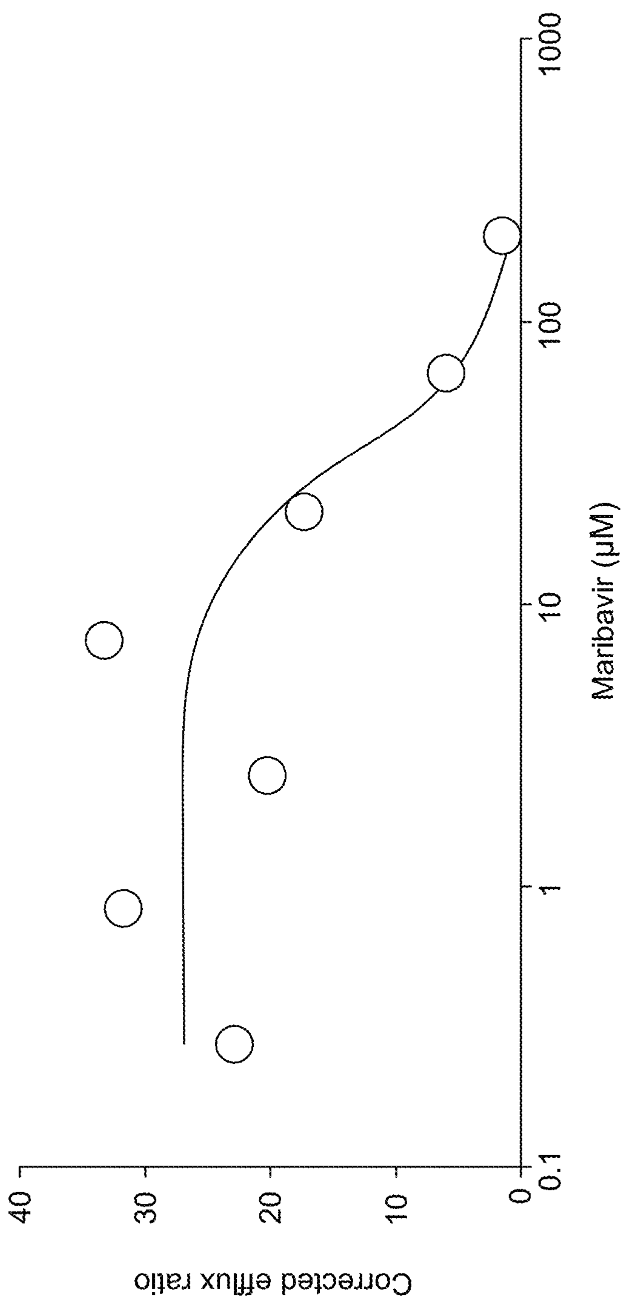

Corrected efflux ratio
40
30
20
10
0
0.1
1
10
100
1000
Maribavir (μM)

Figure 4

USE OF MARIBAVIR IN TREATMENT REGIMENS

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation application of U.S. application Ser. No. 18/400,304, filed Dec. 29, 2023, which is a continuation application of PCT Application No. PCT/US22/50341, filed Nov. 11, 2022, which claims priority to U.S. Provisional Application No. 63/281,206, filed Nov. 19, 2021, each of which is incorporated herein by reference in its entirety.

BACKGROUND

Cytomegalovirus (CMV) infection and disease are significant post-transplant complications and are associated with substantial morbidity and reduced long-term survival among transplant recipients. There are currently no approved therapies for the treatment of CMV infection in transplant recipients.

Transplant patients often receive numerous concomitant medications to manage their comorbidities. Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

SUMMARY

Maribavir is a benzimidazole riboside, and is an orally available antiviral medication against cytomegalovirus (CMV). Maribavir is also known under its trade name LIVTENCITY™. Typically, patients (e.g., adults and/or children over 12 years of age and weighing at least 35 kg) are administered 400 mg if maribavir orally, twice daily. However, in some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a cytochrome P450 3A4 (CYP3A4) inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or a CYP3A4 inducer is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir; and

discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a CYP3A4 inducer; and

increasing the amount of maribavir administered to the patient.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments a CYP3A4 inducer decreases maribavir exposure. In some embodiments, an initial therapeutically effective amount of maribavir is administered to the patient, and the amount of maribavir is increased when co-administering a CYP3A4 inducer. In some embodiments, an initial therapeutically effective amount of maribavir is about 400 mg, orally twice daily. In some embodiments, an amount of maribavir is increased to an amount between about 800 mg and about 1200 mg, orally twice daily. In some embodiments, the provided methods further comprising a step of monitoring patient blood (e.g., whole blood or plasma) concentration of maribavir and/or the CYP3A4 inducer.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an immunosuppressant is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, provided methods further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL.

In some embodiments, tacrolimus is administered at an initial dose of between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax.

In some embodiments, a dose of tacrolimus is adjusted when co-administered with maribavir. In some embodiments, maribavir is co-administered at a concentration of 400 mg, 800 mg, or 1200 mg, administered twice daily. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an antiarrhythmic (e.g., digoxin) is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an antiarrhythmic (e.g., digoxin). In some embodiments, provided methods further comprise monitoring the levels of digoxin in patient serum. In some embodiments, provided methods further comprise the step of reducing the amount of digoxin administered to the patient (e.g., to the amount of antiarrhythmic that was administered prior to commencing administration of maribavir).

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or ganciclovir or valganciclovir is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and

discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.

In some embodiments, concentration of maribavir and/or the other drug being co-administered (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration.

In some embodiments, the patient is refractory to treatment with one or more other drugs to treat CMV infection (e.g., ganciclovir, valganciclovir, cidofovir, or foscarnet). In some embodiments, maribavir is administered with or without food. In some embodiments, the patient is a transplant recipient (e.g., a hematopoietic stem cell transplant recipient or a solid organ transplant recipient). In some embodiments, the patient is an adult or a child over 12 years old and greater than 35 kg.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**. IC_{50} shift of maribavir in a concentration-dependent assay with HLM for CYP3A with midazolam (A) or testosterone (B) as a probe substrate. CYP, cytochrome P450; HLM, human liver microtomes; IC_{50}, half maximal inhibitory concentration; NADPH, nicotinamide-adenine dinucleotide phosphate.

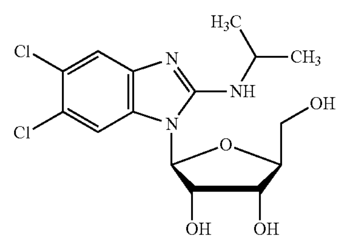
FIG. **2**. Observed enzyme inactivation rate constant vs. inhibitor concentration for TDI of CYP3A by maribavir with (A) midazolam or (B) testosterone as probe substrate. CYP, cytochrome P450; k_1, inhibitor concentration at half maximal enzyme inactivation; k_{inact}, maximum enzyme inactivation rate constant; k_{obs}, observed enzyme inactivation rate constant; TDI, time-dependent inhibition.

FIG. **3**. Estimation of induction E_{max} and EC_{50} of CYP3A4 mRNA expression by maribavir through non-linear regression of data obtained from three donors of human hepatocytes. For donor 3, n is the sigmoidicity coefficient that accommodates the shape of the curve; EC_{50}, half maximal effective concentration; E_{max}, maximum effect.

FIG. **4**. Estimation of IC_{50} of maribavir on P-gp efflux of digoxin (as determined by the corrected efflux ratio) in cultured Caco-2 cells. IC_{50}, half maximal inhibitory concentration; P-gp, P-glycoprotein.

DETAILED DESCRIPTION

Maribavir ((2S,3S,4R,5S)-2-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)-5-(hydroxymethyl)tetrahydrofuran-3,4-diol), a compound having the chemical structure:



is a potent and orally bioavailable antiviral for the treatment of cytomegalovirus (CMV) infection and disease in transplant recipients. Transplant recipients are at significant risk of CMV infection and also often receive numerous concomitant medications to manage their comorbidities. Thus, evaluating the drug-drug interaction potential between maribavir and prospective therapies is useful. Further, understanding the clinical pharmacology of maribavir is necessary to define the optimal dosing strategy in transplant recipients who often have multiple comorbidities and require complex concurrent medication regimens.

1. Definitions

As used herein, the term "about," when used in reference to a numerical value, means±10% of that value. For example, a dose that comprises "about 100 mg" of maribavir encompasses any amount of maribavir within a range of 90 mg to 110 mg.

As used herein, the term "reference" describes a standard or control relative to which a comparison is performed. For example, in some embodiments, an agent, animal, individual, population, sample, sequence or value of interest is compared with a reference or control agent, animal, individual, population, sample, sequence or value. In some embodiments, a reference or control is tested and/or determined substantially simultaneously with the testing or determination of interest. In some embodiments, a reference or control is a historical reference or control, optionally embodied in a tangible medium. Typically, as would be understood by those skilled in the art, a reference or control is determined or characterized under comparable conditions or circumstances to those under assessment. Those skilled in the art will appreciate when sufficient similarities are present to justify reliance on and/or comparison to a particular possible reference or control.

The terms "treat" or "treating," as used herein, refers to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition. As used herein, the terms "treatment," "treat," and "treating" refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed. In some embodiments, the term "treating" includes halting the progression of a disease or disorder. Treatment may also be continued after symptoms have resolved, for example to prevent or delay their recurrence. Thus, in some embodiments, the term "treating" includes preventing relapse or recurrence of a disease or disorder.

2. Dose Regimens

In some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some embodiments, maribavir is administered to patients at about 400 mg orally, twice daily ("standard dose"). In some embodiments, a patient has received an initial therapeutically effective amount of maribavir, e.g., prior to receiving the additional co-administered drug. In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, maribavir is administered as a tablet. In some embodiments, maribavir is administered as a tablet comprising 200 mg of maribavir. In some embodiments, provided methods comprise administering maribavir to a patient with or without food.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir is altered (e.g., increased or decreased) from the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of the co-administered drug is altered (e.g., increased or decreased) from its recommended amount provided by its label. In some embodiments, the dose of the co-administered drug is decreased as compared to its standard dosing regimen. In some embodiments, provided methods further comprising the step of monitoring the levels of a coadministered drug after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of a coadministered drug administered to the patient (e.g., to the amount of a coadministered drug that was administered prior to commencing administration of maribavir).

a. Maribavir and CYP3A4 Inducers

In some embodiments, the present disclosure provides the recognition that maribavir is primarily metabolized by CYP3A4. Additionally or alternatively, the present disclosure provides the recognition that drugs that are CYP3A4 inducers decrease maribavir plasma concentrations and may result in a reduced virologic response. In some embodiments, coadministration of maribavir with certain CYP3A4 inducers is not recommended, and/or dosing regimens of one or both drugs should be altered.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir. In some embodiments, methods further comprise discontinuing administration of the CYP3A4 inducer prior to administration of maribavir. In some embodiments, method further comprise increasing the amount of maribavir administered to the patient.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital. In some embodiments, a CYP3A4 inducer is a strong CYP3A4 inducer. Strong CYP3A4 inducer include, e.g., rifampin, rifabutin, and St. John's wort. In some embodiments, CYP3A4 inducers may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for a given indication).

In some embodiments, a CYP3A4 inducer is also a p-Gp inducer. In some embodiments, a CYP3A4 inducer also decreases maribavir exposure.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

- administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and
- discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

- administering an initial therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and
- increasing the amount of maribavir administered to the patient.

In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is carbamazepine, the amount of maribavir is increased to about 800 mg of maribavir orally twice daily. In some embodiments, wherein the CYP3 A4 inducer is phenytoin or phenobarbital, the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient (e.g., 800 mg or about 1200 mg orally, twice daily), wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.

In some embodiments, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir, a therapeutically effective amount of maribavir is about 800 mg or about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3 A4 inducer is carbamazepine, the amount of maribavir administered is about 800 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is phenytoin or phenobarbital, the amount of maribavir administered is about 1200 mg orally, twice daily.

Drug-drug interaction potential of maribavir with cytochrome P450 enzymes and P-glycoprotein (P-gp) was thoroughly characterized. The reversible inhibition, time-dependent inhibition, and induction of CYPs were evaluated in the presence of maribavir using human cells or human-derived cell lines. The inhibition of P-gp was also assessed.

Maribavir was not a reversible inhibitor of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, or CYP3A4 but was a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 (the half maximal inhibitory concentration, or IC_{50}, was 40, 18, and 35 μM, respectively).

Maribavir was not a time-dependent inhibitor of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. However, it was a time-dependent inhibitor of CYP3 A4, evident from IC_{50} shift values presented in FIG. **1**. With midazolam and testosterone as probe substrates, IC_{50} values for maribavir were greater than 1.9 and 3.2 μM, respectively. Maribavir's concentration at half-maximal enzyme inactivation, or k_1, of CYP3A was 41.2 μM and 167 μM with midazolam and testosterone, respectively, as shown in FIG. **2**.

Maribavir was not an inducer of CYP1A2 or CYP2B6 mRNA, but was a weak in vitro inducer for CYP3A4 mRNA, exhibiting a greater than 14-fold increase in mRNA induction and a greater than 30% increase in positive control levels. However, effects varied across donors, as FIG. **3** shows. The half-maximal effective concentration, or EC_{50}, of maribavir was greater than 5 μM for all donors.

As shown in FIG. **4**, maribavir demonstrated concentration-dependent inhibition of P-gp-mediated digoxin efflux. The IC_{50} of maribavir in this case was 33.8 μM.

The data presented herein, specifically in Example 1, demonstrate the concentrations of maribavir required for the observed levels of in vitro inhibition or induction of P-gp and/or some CYPs.

b. Maribavir and Immunosuppressants

In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of certain immunosuppressant drugs that are CYP3A4 and/or P-gp substrates, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, immunosuppressant drug levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of maribavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received an immunosuppressant. In some embodiments, methods further comprise monitoring the levels of immunosuppressant (e.g., in whole blood or plasma). In some embodiments, methods further comprise reducing the amount of immunosuppressant administered to a patient. In some embodiments, maribavir also increases immunosuppressant exposure. In some embodiments, provided methods further comprise the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprise the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is everolimus. In some embodiments, an immunosuppressant is cyclosporine. In some embodiments, an immunosuppressant is sirolimus.

In some embodiments, immunosuppressants may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for tacrolimus). In some embodiments, tacrolimus is administered as a stable dose, twice daily, with a total daily dose between about 0.5 mg and about 16 mg.

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from the initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL. In some embodiments, an initial dose of tacrolimus administered is between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some embodiments, the present disclosure provides methods of administering to a patient in need thereof a therapeutically effective amount of an immunosuppressant for the prophylaxis or treatment of organ rejection and/or graft versus host disease, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of an immunosuppressant, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of maribavir, the improvement comprising administering a therapeutically effective amount of an immunosuppressant to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

c. Maribavir and Digoxin

Digoxin is an antiarrhythmic drug most frequently used for atrial fibrillation, atrial flutter, and heart failure. In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of digoxin, which is a P-gp substrate, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, digoxin levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of mariabavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin. In some embodiments, provided methods further comprise monitoring the levels of digoxin (e.g., in whole blood or plasma). In some embodiments, provided methods further comprise reducing the amount of digoxin administered to a patient, e.g., as compared to the amount prior to co-administering maribavir, such as the US FDA approved dosage. In some embodiments, digoxin is administered as 0.5 mg, once daily.

d. Maribavir and Ganciclovir or Valganciclovir

In some embodiments, the present disclosure recognizes that maribavir may antagonize the antiviral activity of ganciclovir and/or valganciclovir by inhibiting human CMV pUL97 kinase, which is required for activation/phosphorylation of ganciclovir and valganciclovir.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir and/or valganciclovir. In some embodiments, provided methods further comprise discontinuing administration of ganciclovir and/or valganciclovir prior to administration of maribavir. In some embodiments, provided methods further comprising the step of monitoring the levels of ganciclovir and/or valganciclovir (e.g., in whole blood or plasma) after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of ganciclovir and/or valganciclovir administered to the patient (e.g., to the amount of ganciclovir and/or valganciclovir that was administered prior to commencing administration of maribavir).

3. Patients

As described above and herein, the present methods provide administering maribavir and a co-administered drug (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) to a patient in need thereof. In some embodiments, a patient suffers from CMV infection. In some embodiments, a patient suffers from post-transplant CMV infection. In some embodiments, a patient is a transplant recipient. In some embodiments, a patient is a hematopoietic stem cell transplant recipient. In some embodiments, a patient is a solid organ transplant recipient (e.g., liver, kidney, lung, heart, pancreas, intestine).

In some embodiments, a patient is refractory to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory with genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory without genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet. In some embodiments, a patient is refractory to treatment with ganciclovir or valganciclovir. In some embodiments, a patient is refractory to treatment with ganciclovir. In some embodiments, a patient is refractory to treatment with valganciclovir.

In some embodiments, a patient is an adult or a child over 12 years old and greater than 35 kg. In some embodiments, a patient is an adult. In some embodiments, a patient is a child. In some embodiments, a patient is a child over 12 years old. In some embodiments, a patient is a child over 35 kg. In some embodiments, a patient is a child over 12 years old and over 35 kg.

4. Evaluating Maribavir Drug-Drug Interaction Potential Using Data from Phase 1 Clinical Studies and Nonclinical In Vitro Studies

Maribavir's drug-drug interaction potential are described in Example 3.

Accumulated data have shown that maribavir is metabolized primarily in the liver, and renal clearance is a minor route (representing less than 5 percent). VP 44469 (N-dealkylated Maribavir) was the principal metabolite of maribavir in both urine and feces. In plasma, unchanged maribavir and VP 44469 represented approximately 69% and 9.8% of total radioactivity, respectively. The hepatic metabolism of maribavir was primarily driven by cytochrome P450s. CYP3A4 and CYP1A2 were responsible for 70 to 85% and 15 to 30% of the CYP-driven pathways, respectively. Glucuronidation accounted for less than 20% of metabolism.

Clinically significant drug interactions are summarized in Table 1-A of Example 3. Potent inducers of CYP34A and P-gp decrease maribavir exposure, necessitating maribavir dose increases. Inhibitors of CYP3 A4 and/or P-gp may increase maribavir exposure. However, previous safety and tolerability data indicate that dose reduction is not necessary with CYP3A4 and/or P-gp inhibitors. Maribavir may increase the exposure of immunosuppressants, so monitoring of concomitant immunosuppressants should be considered.

5. Clinical Pharmacology of Maribavir

Following oral administration, maribavir was rapidly and well absorbed, with peak concentrations generally achieved within 1 to 3 hours; exposure was unaffected by food; and bioavailability wasn't affected by crushing the tablet or when it was taken with antacids.

Maribavir was metabolized in the liver through the cytochrome P450 3A4 and 1A2 pathways. Less than 5 percent of maribavir was eliminated through the kidneys. Maribavir demonstrated a half-life of approximately 5 to 7 hours.

Maribavir was found to present a low risk for drug-to-drug interactions. Co-administration of potent CYP3A4 inducers was found to decrease the exposure of maribavir, necessitating a maribavir dose increase. Some immunosuppressants, such as tacrolimus, may be affected by maribavir, which increased tacrolimus exposure by 51%.

In conclusion, maribavir is a suitable treatment for CMV infections in a broad range of transplant recipients. It can be taken with or without food, and no dose adjustment is needed in patients with mild-to-moderate hepatic impairment or renal impairment. Maribavir has a low drug-interaction potential and there are minimal dose adjustments required (for example, only when taken concurrently with CYP3 A4 inducers or certain immunosuppressants) and there are no effects on QT interval.

Exemplary Embodiments

The following numbered embodiments, while non-limiting, are exemplary of certain aspects of the present disclosure:

1 A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and

discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

2. The method of embodiment 1, wherein the CYP3 A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

3. The method of embodiments 1 or 2, wherein the CYP3A4 inducer is a strong CYP3A4 inducer.

4. The method of any one of embodiments 1-3, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort.

5. The method of any one of embodiments 1-4, comprising administering about 400 mg of maribavir orally to the patient twice daily.

6. The method of any one of embodiments 1-5, comprising administering maribavir to the patient with or without food.

7. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and

increasing the amount of maribavir administered to the patient.

8. The method of embodiment 7, wherein the CYP3 A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

9. The method of embodiment 7 or 8, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

10. The method of any one of embodiments 7-9, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.

11. The method of any one of embodiments 7-10, wherein the patient had received a therapeutically effective amount of maribavir prior to receiving a CYP3A4 inducer ("an initial therapeutically effective amount") of about 400 mg of maribavir orally twice daily.

12. The method of any one of embodiments 7-11, wherein the amount of maribavir is increased to about 800 or about 1200 mg of maribavir orally twice daily.

13. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir is increased to about 800 mg of maribavir orally twice daily.

14. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.

15. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.

16. The method of embodiment 15, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

17. The method of embodiment 15 or 16, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

18. The method of any one of embodiments 15-17, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.

19. The method of any one of embodiments 15-18, wherein the amount of maribavir administered is about 800 or about 1200 mg orally twice daily.

20. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir administered is about 800 mg orally twice daily.

21. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir administered is about 1200 mg orally twice daily.

22. The method of any one of embodiments 1-21, wherein the CYP3A4 inducer decreases maribavir exposure.

23. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant.

24. The method of embodiment 23, further comprising the step of monitoring the levels of immunosuppressant after commencing administration of Maribavir (e.g., with comparison to a reference or standard level).

25. The method of embodiment 23 or 24, further comprising the step of reducing the amount of the immunosuppressant administered to the patient.

26. The method of any one of embodiments 23-25, further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level).

27. The method of embodiment 26, further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

28. The method of any one of embodiments 23-27, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate.

29. The method of any one of embodiments 23-28, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus.

30. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin.

31. The method of embodiment 30, further comprising the step of monitoring the levels of digoxin.

32. The method of embodiment 30 or 31, further comprising the step of reducing the amount of the digoxin administered to the patient.

33. The method of one of embodiments 1-32, wherein the patient is refractory to treatment with one or more other drugs to treat CMV infection.

34. The method of one of embodiments 1-33, wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.

35. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment with genotypic resistance.

36. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment without genotypic resistance.

37. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and

discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.

38. The method of any one of embodiments 7-37, comprising administering maribavir to the patient with or without food.

39. The method of any one of embodiments 1-38, wherein the patient is a transplant recipient.

40. The method of any one of embodiments 1-39, wherein the patient is a hematopoietic stem cell transplant recipient.

41. The method of any one of embodiments 1-39, wherein the patient is a solid organ transplant recipient.

42. The method of any one of embodiments 1-41, wherein the patient is an adult or a child over 12 years old and greater than 35 kg.

EXAMPLES

Example 1—In Vitro Profiling for Potential Cytochrome P450 Drug-Drug Interaction by Maribavir

Inhibition or induction of the cytochrome P450 enzymes (CYPs) are among the most commonly observed mechanisms of drug-drug interactions.

In vitro systems such as human liver microsomes (HLM), recombinant enzymes, human hepatocytes, and other human-derived cells lines have long been used as validation systems for characterization of potential drug-drug interactions prior to clinical studies.

Reversible CYP Inhibition

HLM were used to evaluate maribavir's half maximal inhibitory concentration (IC_{50}) on inhibition of activities of nine CYP isoforms. Probe substrates were phenacetin (CYP1A2), coumarin (CYP2A6), bupropion (CYP2B6), paclitaxel (CYP2C8), tolbutamide (CYP2C9), (S)-mephenytoin (CYP2C19), dextromethorphan (CYP2D6), clorzoxazone (CYP2E1), and midazolam and testosterone (CYP3A4).

CYP activities with 0, 0.1, 0.3, 1, 3, 10, 30, and 100 μM of maribavir were evaluated.

Incubation of maribavir in HLM at up to 100 M resulted in no significant inhibition of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, and CYP3A4. Maribavir is a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 with ICso of 40, 18, and 35 μM, respectively.

Time-Dependent CYP Inhibition

Time-dependent inhibition (TDI) potentials for various CYP enzymes were evaluated by a 30 minute pre-incubation of maribavir with HLM in the presence and absence of nicotinamide-adenine dinucleotide diphosphate (NADPH) followed by a CYP enzyme activity assay.

Enzyme inactivation kinetics of maribavir for the TDI of CYP3A, with midazolam and testosterone as probe substrates, was evaluated by pre-incubation of HLM with maribavir at various concentrations with NADPH for six different pre-incubation time frames. CYP3A activity was measured by determining the formation of CYP3A probe metabolite.

Inhibitor concentration at half maximal enzyme inactivation (k_1), and maximum enzyme inactivation rate constant (k_{inact}), were estimated using non-linear least-square regression.

The IC_{50} shift values of maribavir were <1 μM for CYP1A2, CYP2C8, CYP2C9, CYP2C19, and CYP2D6 in HLM. The IC_{50} shift values of maribavir for CYP3A were greater than 1.9 and 3.2 μM, with midazolam and testosterone as probe substrate, respectively (FIG. 1). Therefore, maribavir, in concentrations of up to 100 μM, is unlikely a

TDI of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6; but it is likely a TDI of CYP3A.

k_1 and k_{intact} of maribavir for TDI of CYP3A with midazolam as substrate (FIG. **2**A) were 41.2 μM and 0.0117 min-1, respectively, k_1 and k_{intact} of maribavir for TDI of CYP3A with testosterone as a substrate (FIG. **2**B) were 167 μM and 0.0357 min-1, respectively.

CYP Induction

Fresh human hepatocytes were treated with culture medium with maribavir concentrations of 36 μM, 144 μM, and 480 μM.

Positive controls were 50 μM omeprazole (CYP1A2), 1 mM phenobarbital (CYP2B6), and 50 μM rifampicin (CYP3A).

Incubation was conducted at 37° C. for 72 hours, with daily replacement of medium with compound. Total RNA was isolated, and cDNA was synthesized from up to 1 μg of total isolated RNA. Analysis of CYP expression was performed using qPCR with CYP-specific probes.

Half maximal effective concentration (EC_{50}) and maximum effect (E_{max}) values for CYP3A4 mRNA induction were further determined by a concentration-response curve in three donors with maribavir at a concentration between 0-100 μM.

Maribavir displayed varying degrees of CYP1A2 and CYP2B6 mRNA induction in human hepatocytes. The fold-increase in mRNA was not concentration-dependent, it was variable among donors, and it did not exceed 11% of the levels of positive controls. Therefore, maribavir is likely not an inducer of CYP1A2 or of CYP2B6 mRNA at clinically relevant concentrations.

At 36 μM, maribavir displayed a greater than 14-fold increase in CYP3A4 mRNA induction and greater than 30% of positive control (rifampicin) levels. The EC_{50} and E_{max} values for CYP3A4 induction were then further determined by a concentration-response curve in three donors.

Maribavir also displayed donor-dependent induction of CYP3A4 mRNA upon determination of its induction E_{max} and EC_{50} properties (FIG. **3**).

Example 2—Quantitative Prediction of Exposure of Maribavir Using Prior In Vitro and In Vivo Data

A Physiologically-based Pharmacokinetic ("PBPK") model based on prior in vitro and in vivo information on the metabolism and kinetics of maribavir was constructed with the aim of predicting plasma concentration-time profiles of maribavir and to evaluate the likely impact of coadministration of CYP3 A4 inhibitors and inducers on the kinetics of maribavir in healthy subjects.

Model Development

A combination of in vitro data and clinical pharmacokinetic data obtained following a single dose of 400 mg maribavir were used to develop the PBPK model. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 2A. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data. The C_{max} was marginally under predicted, which could be expected because the PBPK model was optimized for better prediction of C_{12h}.

TABLE 2A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (400 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	97.98	11.98
Median	91.97	11.48
Geometric Mean	90.40	11.56
90% CI around geometric mean (lower limit)	84.45	11.06
90% CI around geometric mean (upper limit)	96.76	12.08
5^{th} percentile	46.52	7.61
95^{th} percentile	165.11	18.08
SD	39.33	3.26
% CV	40	27
OBSERVED		
Mean	97.8	16.7
SD	28.6	5.72
% CV	29.2	34.3
RATIO MEAN PREDICTED/OBSERVED	1.00	0.72

A renal clearance of 0.051 L/H was obtained following oral administration of 50 mg to 1600 mg single doses of maribavir. In vitro human liver microsomal and recombinant CYP data combined with oral clearance values reported from two studies (n=46 individuals; Ma et al., 2006) were used to assign the relative contribution of CYP3A4 to the clearance of maribavir. Distribution models evaluated included a full PBPK model and a minimal PBPK model, both of which consider liver and intestinal metabolism. Mass balance data from the [^{14}C] ADME study indicated a fraction absorbed of 0.83 or greater following oral administration of 400 mg dose of maribavir. Absorption models evaluated included a simple first-order absorption model and a more mechanistic "Advanced Dissolution Absorption and Metabolism" (ADAM) model. Although maribavir has been shown to be a P-gp substrate in vitro, relatively linear pharmacokinetics across the dose range of interest, (400 mg to 1600 mg) informed the selection of the simpler first order absorption model for the use in this PBPK study.

Model Verification

A comparison of observed and predicted plasma concentrations of maribavir following single oral doses of 800 mg and 1600 mg to healthy patients were. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Tables 3A and 4A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 3A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (800 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	195.71	24.05
Median	179.48	23.22
Geometric Mean	180.00	23.23
90% CI around geometric mean (lower limit)	167.95	22.23
90% CI around geometric mean (upper limit)	192.91	24.27
5^{th} percentile	93.03	15.22
95^{th} percentile	330.22	36.17

TABLE 3A-continued

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (800 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
SD	80.04	6.47
% CV	41	27
OBSERVED		
Mean	183	26.4
SD	69.1	6.85
% CV	38.0	26.0
RATIO MEAN PREDICTED/OBSERVED	1.07	0.91

TABLE 4A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (1600 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	391.43	48.10
Median	358.95	46.45
Geometric Mean	359.99	46.45
90% CI around geometric mean (lower limit)	335.89	44.46
90% CI around geometric mean (upper limit)	385.82	48.54
5^{th} percentile	186.06	30.44
95^{th} percentile	660.44	72.34
SD	106.09	12.94
% CV	41	27
OBSERVED		
Mean	437	48.8
SD	163	7.88
% CV	37.4	16.1
RATIO MEAN PREDICTED/OBSERVED	0.90	0.99

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral 400 mg BID doses of maribavir was performed. Mean concentrations for the total virtual population (n=150) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 5A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 5A

Predicted and Observed $C_{12\ h}$, C_{max} and $AUC_{(0-\infty)}$ values for maribavir after multiple oral dose of maribavir (400 mg BBID, 5 doses).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L
PREDICTED			
Mean	96.49	14.31	2.99
Median	90.13	14.05	2.44
Geometric Mean	89.31	13.78	2.13
90% CI around geometric mean (lower limit)	85.01	13.31	1.89
90% CI around geometric mean (upper limit)	93.82	14.26	2.15
5^{th} percentile	46.35	8.80	0.36
95^{th} percentile	162.88	20.59	7.76
SD	37.93	3.96	2.2
% CV	39	28	73
OBSERVED			
Mean	89.9	16.9	2.54
SD	24.5	5.22	1.37
% CV	27.2	30.8	53.8
RATIO MEAN PREDICTED/OBSERVED	1.07	0.85	1.18

A comparison of observed and predicted plasma concentrations of maribavir following a single oral dose of 400 mg administered in the absence of ketoconazole (a CYP3A4 inhibitor) and 1 hour after a single 400 mg dose of ketoconazole to healthy subjects was performed. Mean simulated and observed plasma maribavir concentrations are compared, and associated geometric mean C_{max} and $AUC_{(0-\infty)}$ values for maribavir in the presence and absence of ketoconazole are shown in Table 6A and are within 1.25-fold of the observed data.

TABLE 6A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$AUC_{(0-\infty)}$	C_{max}
PREDICTED						
Mean	98.96	12.27	150.60	14.31	1.54	1.17
Median	93.51	11.91	140.38	13.75	1.49	1.15
Geometric Mean	92.29	11.81	141.16	13.77	1.53	1.17
90% CI around geometric mean (lower limit)	88.30	11.43	135.28	13.32	1.51	1.16
90% CI around geometric mean (upper limit)	96.45	12.20	147.30	14.23	1.55	1.17
5^{th} percentile	48.52	7.50	77.98	8.85	1.27	1.09
95^{th} percentile	169.37	17.86	246.30	21.20	1.97	1.29
SD	37.19	3.4	54.7	4.0	0.21	0.06
% CV	38	28	36	28	14	5

TABLE 6A-continued

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$	C_{max}	$AUC_{(0-\infty)}$	C_{max}		
	mg/L*h	mg/L	mg/L*h	mg/L	$AUC_{(0-\infty)}$	C_{max}
OBSERVED						
Mean	126	19.8	194	21.8	1.55	1.12
SD	41.6	3.9	64.3	4.4	0.23	0.22
% CV	33	20	33	20	15	20
Geometric mean	119	19.4	183	21.3	1.53	1.10
90% CI around geometric mean (lower limit)					1.44	1.01
90% CI around geometric mean (upper limit)					1.63	1.19
MEAN RATIO PREDICTED/OBSERVED	0.79	0.62	0.78	0.66	0.99	1.05
GEOMEAN RATIO PREDICTED/OBSERVED	0.78	0.61	0.77	0.65	1.00	1.06

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral doses of 400 mg administered in the absence of rifampicin and during concurrent treatment with rifampicin was performed. Subjects took maribavir 400 mg BID on Days 1-3 (5 doses), then rifampicin 600 mg QD on Days 4 through 12, followed by both maribavir 400 mg (BID) and rifampicin 600 mg QD on Days 13 and 14, with final doses of maribavir and rifampicin on the morning of Day 15. Associated geometric mean C_{12h}, C_{max}, and $AUC_{(0-12h)}$ values for maribavir in the presence or absence of rifampicin are shown in Table 7A. Predicted values for C_{max} and AUC are within 1.25-fold of the observed parameter values.

TABLE 7A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-12\ h)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of rifampicin with associated variability in healthy adults.

	Maribavir 400 mg BID			400 mg BID Maribavir + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
PREDICTED									
Mean	95.28	14.26	2.88	35.13	7.97	0.340	0.37	0.56	0.09
Median	90.02	13.98	2.61	30.52	7.57	0.145	0.35	0.55	0.06
Geometric Mean	88.77	13.78	2.07	30.53	7.44	0.088	0.34	0.54	0.04
90% CI around geometric mean (lower limit)	84.25	13.30	2.03	28.36	7.07	0.083	0.33	0.52	0.04
90% CI around geometric mean (upper limit)	93.54	14.29	2.10	32.86	7.83	0.093	0.36	0.56	0.04
5^{th} percentile	41.15	8.76	0.39	12.44	3.95	0.002	0.17	0.33	0.01
95^{th} percentile	151.23	20.23	6.49	67.37	13.43	1.507	0.62	0.79	0.33
SD	35.28	3.7	2.1	18.64	2.93	0.53	0.14	0.14	0.11
% CV	37	26	73	53	37	151	37	25	112
OBSERVED									
Mean	89.9	16.9	2.54	35.1	10.2	0.41	0.41	0.65	0.17
SD	24.5	5.22	1.37	7.07	2.3	0.21	0.08	0.24	0.07
% CV	86.6	16.2	2.24	34.5	9.88	0.41	0.40	0.61	0.18
RATIO MEAN PREDICTED/OBSERVED	1.06	0.84	1.14	1.00	0.78	0.83	0.91	0.87	0.56
RATIO GEOMETRIC MEAN PREDICTED/OBSERVED	1.03	0.85	0.92	0.88	0.75	0.22	0.85	0.89	0.24

Model Verification

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 8A. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for this interaction is compared with the PK parameters for a 400 mg BID dose of maribavir in Table 9A.

Maribavir C_{max} from 800 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone, however, $AUC_{(0\text{-}12)}$ was 23% lower and C_{12h} was 75% lower. This indicates that increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of rifampicin on the reduced therapeutic exposure to maribavir.

TABLE 8A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L* h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	194.99	28.96	5.95	73.25	16.40	0.73	0.37	0.56	0.10
Median	189.24	28.37	5.34	68.57	15.57	0.31	0.37	0.56	0.06
Geometric Mean	182.49	27.95	4.39	63.52	15.22	0.20	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	174.67	27.09	4.36	59.54	14.54	0.18	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	190.66	28.85	4.43	67.77	15.94	0.21	0.36	0.56	0.05
5^{th} percentile	94.23	17.97	0.93	24.65	7.93	0.00	0.18	0.35	0.00
95^{th} percentile	311.05	43.06	13.85	136.31	27.80	2.82	0.62	0.79	0.34
SD	68.94	7.68	4.00	38.65	6.25	1.09	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 9A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	95.28	14.26	2.88	73.25	16.40	0.73	0.77	1.15	0.25
Median	90.02	13.98	2.61	68.57	15.57	0.31	0.76	1.11	0.12
Geometric Mean	88.77	13.78	2.07	63.52	15.22	0.20	0.72	1.10	0.10
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	59.54	14.54	0.18	0.71	1.09	0.09
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	67.77	15.94	0.21	0.72	1.12	0.10
5^{th} percentile	41.15	8.76	0.39	24.65	7.93	0.00	0.60	0.91	0.00
95^{th} percentile	151.23	20.34	6.49	136.31	27.80	2.82	0.90	1.37	0.43
SD	35.28	3.7	2.1	38.65	6.25	1.09	1.10	1.69	0.52
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 10A. Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 11A.

Maribavir AUC from 1200 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone while the C_{max} was slightly higher. However, the geometric mean C_{12h} was 86% lower, indicating that increasing maribavir dose from 400 mg BID to 1200 mg BID cannot counteract the effect of rifampicin from an efficacy perspective (mainly driven by C_{12h}). Increasing the maribavir dose to 1600 mg BID had a minimal effect on the geometric mean C_{12h}, which was 81% lower than that with a 400 mg BID dose of maribavir.

TABLE 10A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	292.48	43.44	8.93	109.87	24.59	1.10	0.37	0.56	0.10
Median	283.87	42.55	8.00	102.85	23.35	0.47	0.37	0.56	0.06
Geometric Mean	273.73	41.93	6.59	95.28	22.84	0.30	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	262.00	40.63	6.55	89.31	21.81	0.28	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	285.99	43.27	6.62	101.65	23.91	0.31	0.36	0.56	0.05
5^{th} percentile	141.35	26.96	1.40	36.97	11.90	0.00	0.18	0.35	0.00
95^{th} percentile	466.58	64.59	20.77	204.46	41.71	4.23	0.62	0.79	0.34
SD	108.42	11.45	6.00	57.57	9.37	1.63	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 11A

Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	95.28	14.26	2.89	109.87	24.59	1.10	1.15	1.72	0.38
Median	90.02	13.98	2.61	102.85	23.35	0.47	1.14	1.67	0.18
Geometric Mean	88.77	13.78	2.07	95.28	22.84	0.30	1.07	1.66	0.14
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	89.31	21.81	0.28	1.06	1.64	0.14
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	101.65	23.91	0.31	1.09	1.67	0.15
5^{th} percentile	41.15	8.76	0.39	36.97	11.90	0.00	0.90	1.36	0.00
95^{th} percentile	151.23	20.34	6.49	204.46	41.71	4.23	1.35	2.05	0.65
SD	35.28	3.7	2.1	57.57	9.37	1.63	1.63	2.53	0.78
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 12A. Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 39%, 27% and 63% in $AUC_{(0-12h)}$, C_{max} and C_{12h}, respectively.

TABLE 12A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12\,h)}$ mg/L*h	C_{max} mg/L	$C_{12\,h}$ mg/L	$AUC_{(0-12\,h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\,h}$ mg/L	$AUC_{(0-12\,h)}$	C_{max}	$C_{12\,h}$
Mean	99.72	14.66	3.12	61.20	10.66	1.33	0.61	0.73	0.37
Median	93.26	14.23	2.40	55.41	10.44	0.85	0.63	0.75	0.37
Geometric Mean	92.65	14.09	2.29	55.66	10.21	0.72	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	52.88	9.86	0.70	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	58.60	10.57	0.73	0.62	0.74	0.31
5^{th} percentile	47.52	9.12	0.50	27.75	6.18	0.07	0.39	0.56	0.07
95^{th} percentile	169.70	21.94	7.30	112.27	16.47	4.17	0.78	0.86	0.65
SD	38.6	4.1	2.26	26.96	3.12	1.25	0.12	0.09	0.18
% CV	38	28	72	44	29	101	20	13	47

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h} for the population are shown in Table 13A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 14A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 100 mg phenobarbital QD were marginally lower than those from maribavir 800 mg BID alone. However, C_{min} was 63% lower than with maribavir 800 mg alone. Increasing the maribavir dose given with phenobarbital from 400 mg BID to 800 mg BID results in a mean C_{min} that is 15% lower than that with 400 mg BID maribavir alone. Thus, by increasing the dose of maribavir to 800 mg BID, the interaction is counteracted significantly.

TABLE 13A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12\,h)}$ mg/L*h	C_{max} mg/L	$C_{12\,h}$ mg/L	$AUC_{(0-12\,h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\,h}$ mg/L	$AUC_{(0-12\,h)}$	C_{max}	$C_{12\,h}$
Mean	199.43	29.32	6.24	122.40	21.32	2.66	0.61	0.73	0.37
Median	186.51	28.46	4.81	110.83	20.88	1.70	0.63	0.75	0.37
Geometric Mean	185.31	28.19	4.58	111.33	20.42	1.43	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	105.75	19.72	1.41	0.59	0.71	0.31

TABLE 13A-continued

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
				PREDICTED					
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	117.20	21.14	1.45	0.62	0.74	0.31
5^{th} percentile	95.04	18.24	1.00	55.49	12.35	0.14	0.39	0.56	0.07
95^{th} percentile	339.40	43.89	14.60	224.54	32.93	8.33	0.78	0.86	0.65
SD	36.3	8.2	4.53	53.9	6.24	2.7	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	48

TABLE 14A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
				PREDICTED					
Mean	99.72	14.66	3.12	122.40	21.32	2.66	1.23	1.45	0.85
Median	93.26	14.23	2.40	110.83	20.88	1.70	1.19	1.47	0.71
Geometric Mean	92.65	14.09	2.29	111.33	20.42	1.43	1.20	1.45	0.62
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	105.75	19.72	1.41	1.19	1.45	0.62
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	117.20	21.14	1.45	1.21	1.45	0.63
5^{th} percentile	47.52	9.12	0.50	55.49	12.35	0.14	1.17	1.35	0.28
95^{th} percentile	169.70	21.94	7.30	224.54	32.93	8.33	1.32	1.50	1.14
SD	38.6	4.1	2.26	53.9	6.24	2.7	1.40	1.52	1.19
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 15A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 16A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 100 mg phenobarbital QD were about 2 times higher those from maribavir 400 mg BID alone while C_{min} is about 28% higher. Increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenobarbital. If the higher exposure (as seen with AUC and Cmax) is not detrimental to treatment, it can be concluded that the maribavir dose should be adjusted from 400 mg BID to 1200 mg BID when co-administration with phenobarbital is needed.

TABLE 15A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L			
Mean	299.15	43.98	9.36	183.60	31.98	3.99	0.61	0.73	0.37
Median	279.77	42.69	7.21	166.24	31.31	2.55	0.63	0.75	0.37
Geometric Mean	277.96	42.28	6.88	166.99	30.63	2.15	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	158.62	29.58	2.11	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	175.80	31.71	2.18	0.62	0.74	0.31
5^{th} percentile	142.56	27.35	1.50	83.24	18.53	0.21	0.39	0.56	0.07
95^{th} percentile	509.10	65.83	21.89	336.81	49.40	12.50	0.78	0.86	0.65
SD	114.5	12.3	6.8	80.9	9.36	4.04	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	47.97

TABLE 16A

Predicted PK parameters for marvibavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L			
Mean	99.72	14.66	3.12	183.60	31.98	3.99	1.84	2.18	1.28
Median	93.26	14.23	2.40	166.24	31.31	2.55	1.78	2.20	1.06
Geometric Mean	92.65	14.09	2.29	166.99	30.63	2.15	1.80	2.17	0.94
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	158.62	29.58	2.11	1.79	2.17	0.93
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	175.80	31.71	2.18	1.81	2.18	0.94
5^{th} percentile	47.52	9.12	0.50	83.24	18.53	0.21	1.75	2.03	0.42
95^{th} percentile	169.70	21.94	7.30	336.81	49.40	12.50	1.98	2.25	1.71
SD	38.6	4.1	2.26	80.9	9.36	4.04	2.10	2.28	1.79
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 17A.

Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 42%, 31% and 64% in $AUC_{(0\text{-}12)}$, C_{max} and C_{12h}, respectively.

TABLE 17A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	56.81	10.04	1.16	0.58	0.69	0.36
Median	93.26	14.23	2.40	56.31	9.95	0.81	0.58	0.69	0.35
Geometric Mean	92.65	14.09	2.29	52.42	9.63	0.68	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	49.99	9.30	0.67	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	54.98	9.96	0.69	0.58	0.70	0.30
5^{th} percentile	47.52	9.12	0.50	26.75	5.59	0.09	0.35	0.50	0.09
95^{th} percentile	169.70	21.94	7.30	99.79	14.78	3.33	0.82	0.88	0.68
SD	38.6	4.1	2.26	23.0	2.91	1.1	0.14	0.11	0.18
% CV	38	28	72	40	29	95	24	17	51.07

Simulation of Plasma Concertation-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 18A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 19A.

Maribavir AUC and Cmax from 800 mg BID in the presence of 300 mg phenytoin QD was marginally higher than those from maribavir 400 mg BID alone, however, C_{min} was 26% lower. Increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}).

TABLE 18A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	199.43	29.32	6.24	113.63	20.08	2.32	0.58	0.69	0.36
Median	186.51	28.46	4.81	112.62	19.90	1.62	0.58	0.69	0.35
Geometric Mean	185.31	28.19	4.58	104.85	19.26	1.36	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	99.97	18.61	1.34	0.55	0.67	0.30

TABLE 18A-continued

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
	PREDICTED								
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	109.96	19.93	1.38	0.58	0.70	0.30
5^{th} percentile	95.04	18.24	1.00	53.51	11.18	0.18	0.35	0.50	0.09
95^{th} percentile	339.40	43.89	14.60	199.59	29.57	6.65	0.82	0.88	0.68
SD	76.33	8.2	4.53	46.1	5.83	2.2	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	51

TABLE 19A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
	PREDICTED								
Mean	99.72	14.66	3.12	113.63	20.08	2.32	1.14	1.37	0.74
Median	93.26	14.23	2.40	112.62	19.90	1.62	1.21	1.40	0.68
Geometric Mean	92.65	14.09	2.29	104.85	19.26	1.36	1.13	1.37	0.59
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	99.97	18.61	1.34	1.13	1.37	0.59
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	109.96	19.93	1.38	1.13	1.37	0.60
5^{th} percentile	47.52	9.12	0.50	53.51	11.18	0.18	1.13	1.23	0.36
95^{th} percentile	169.70	21.94	7.30	199.59	29.57	6.65	1.18	1.35	0.91
SD	38.6	4.1	2.26	46.1	5.83	2.2	1.21	1.42	0.97
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-24h)}$ and C_{12h}) for the population are shown in Table 20A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 21A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 300 mg phenytoin QD were about 2-fold higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}). The impact of the higher exposure (AUC and C_{max}) will have to be evaluated.

TABLE 20A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	299.15	43.98	9.36	170.44	30.13	3.48	0.58	0.69	0.36
Median	279.77	42.69	7.21	168.93	29.85	2.42	0.58	0.69	0.35
Geometric Mean	277.96	42.28	6.88	157.27	28.88	2.05	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	149.96	27.91	2.02	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	164.94	29.89	2.08	0.58	0.70	0.30
5^{th} percentile	142.56	27.35	1.50	80.26	16.76	0.27	0.35	0.50	0.09
95^{th} percentile	509.10	65.83	21.89	299.38	44.35	9.98	0.82	0.88	0.68
SD	114.5	12.3	6.8	69.01	8.7	3.29	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	52

TABLE 21A

Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	170.44	30.13	3.48	1.71	2.06	1.12
Median	93.26	14.23	2.40	168.93	29.85	2.42	1.81	2.10	1.01
Geometric Mean	92.65	14.09	2.29	157.27	28.88	2.05	1.70	2.05	0.90
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	149.96	27.91	2.02	1.69	2.05	0.89
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	164.94	29.89	2.08	1.70	2.05	0.90
5^{th} percentile	47.52	9.12	0.50	80.26	16.76	0.27	1.69	1.84	0.54
95^{th} percentile	169.70	21.94	7.30	299.38	44.35	9.98	1.76	2.02	1.37
SD	38.2	4.1	2.26	69.01	8.7	3.29	1.81	2.12	1.46
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles for Maribavir 400 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 400 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 22A.

Due to CYP3A4 induction by carbamazepine, a mean decrease of 46% in maribavir C_{12h} (efficacy marker) and a 29% and 23% decrease in AUC and C_{max} respectively, is predicted when 400 mg BID maribavir is combined with carbamazepine.

TABLE 22A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of carbamazepine.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	67.85	11.12	1.65	0.71	0.77	0.54
Median	88.34	13.91	2.34	63.37	10.80	1.29	0.71	0.79	0.56
Geometric Mean	91.25	14.01	2.18	63.82	10.74	1.15	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	61.25	10.41	1.14	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	66.49	11.08	1.16	0.71	0.78	0.53
5^{th} percentile	48.05	8.74	0.52	36.23	7.02	0.25	0.54	0.61	0.36
95^{th} percentile	176.30	21.99	7.77	108.25	16.29	4.07	0.84	0.90	0.70
SD	40.96	4.23	2.53	24.40	2.99	1.34	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.20
3^{rd} quartile	123.18	17.26	4.22	82.72	12.92	2.34	0.77	0.09	0.61
1^{st} quartile	69.63	11.43	1.20	51.60	8.95	0.63	0.65	0.84	0.46
10^{th} percentile	55.45	9.76	0.75	40.08	7.78	0.42	0.58	0.72	0.39
90^{th} percentile	154.24	20.66	6.58	100.38	15.22	3.34	0.82	0.65	0.67

Simulation of Plasma Concentration-Time Profile Maribavir 800 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 800 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 23A. Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 24A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 400 mg carbamazepine QD were marginally higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID can counteract the effect of carbamazepine from an efficacy perspective (mainly driven by C_{min}).

TABLE 23A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of carbamazepine.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	197.86	29.19	6.21	135.70	22.25	3.30	0.71	0.77	0.54
Median	176.68	27.81	4.68	126.73	21.61	2.58	0.71	0.79	0.56
Geometric Mean	182.50	28.01	4.37	127.63	21.48	2.30	0.70	0.77	0.53

TABLE 23A-continued

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of carbamazepine.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
90% CI around geometric mean (lower limit)	174.10	27.08	4.35	122.50	20.82	2.29	0.69	0.76	0.52
90% CI around geometric mean (upper limit)	191.31	28.97	4.39	132.99	22.16	2.31	0.71	0.78	0.53
5^{th} percentile	96.09	17.47	1.03	72.46	14.03	0.50	0.54	0.61	0.36
95^{th} percentile	352.60	43.98	15.54	216.50	32.57	8.13	0.84	0.90	0.70
SD	81.92	8.47	5.05	48.79	5.97	2.67	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3^{rd} quartile	246.37	34.52	8.44	165.43	25.84	4.67	0.77	0.84	0.61
1^{st} quartile	139.25	22.86	2.41	103.20	17.89	1.27	0.65	0.72	0.46
10^{th} percentile	110.89	19.53	1.50	80.17	15.56	0.85	0.58	55.45	0.39
90^{th} percentile	308.48	41.32	13.15	200.76	30.43	6.68	0.82	154.24	0.67

TABLE 24A

Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	135.70	22.25	3.30	1.37	1.52	1.06
Median	88.34	13.91	2.34	126.73	21.61	2.58	1.43	1.55	1.10
Geometric Mean	91.25	14.01	2.18	127.63	21.48	2.30	1.40	1.53	1.06
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	122.50	20.82	2.29	1.41	1.54	1.06
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	132.99	22.16	2.31	1.39	1.53	1.05
5^{th} percentile	48.05	8.74	0.52	72.46	14.03	0.50	1.51	1.61	0.96
95^{th} percentile	176.30	21.99	7.77	216.50	32.57	8.13	1.23	1.48	1.05
SD	40.96	4.23	2.53	48.79	5.97	2.67	1.19	1.41	1.06
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
3^{rd} quartile	123.18	17.26	4.22	165.43	25.84	4.67	1.34	1.50	1.11
1^{st} quartile	69.63	11.43	1.20	103.20	17.89	1.27	1.48	1.57	1.06
10^{th} percentile	55.45	9.76	0.75	80.17	15.56	0.85	1.45	1.59	1.13
90^{th} percentile	154.24	20.66	6.58	200.76	30.43	6.68	1.30	1.47	1.02

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 1200 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h} for the population are shown in Table 25A. Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 26A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 400 mg carbamazepine QD were significantly higher (about 2-fold) than those from maribavir 400 mg BID alone. The C_{min} was about 50% higher than that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID might be preferred to a 1200 mg BID maribavir dose to counteract the effect of carbamazepine from an efficacy perspective.

TABLE 25A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of carbamazepine.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	296.79	43.79	9.32	203.54	33.37	4.95	0.71	0.77	0.54
Median	265.02	41.72	7.02	190.10	32.41	3.88	0.71	0.79	0.56
Geometric Mean	273.75	42.02	6.55	191.45	32.22	3.45	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	261.14	40.62	6.48	183.75	31.23	3.42	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	286.97	43.46	6.62	199.48	33.24	3.49	0.71	0.78	0.53
5^{th} percentile	144.14	26.21	1.55	108.68	21.05	0.76	0.54	0.61	0.36
95^{th} percentile	528.90	65.97	23.31	324.74	48.86	12.20	0.84	0.90	0.70
SD	122.87	12.70	7.58	73.19	8.96	4.01	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3^{rd} quartile	369.55	51.78	12.65	248.14	38.76	7.01	0.77	0.84	0.61
1^{st} quartile	208.88	34.29	3.61	154.80	26.84	1.90	0.65	0.72	0.46
10^{th} percentile	166.34	29.29	2.26	120.25	23.34	1.27	0.58	55.45	0.39
90^{th} percentile	462.72	61.98	19.73	301.14	45.65	10.02	0.82	154.24	0.67

TABLE 26A

Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	203.54	33.37	4.95	2.06	2.29	1.59
Median	88.34	13.91	2.34	190.10	32.41	3.88	2.15	2.33	1.66
Geometric Mean	91.25	14.01	2.18	191.45	32.22	3.45	2.10	2.30	1.58
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	183.75	31.23	3.42	2.11	2.31	1.58
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	199.48	33.24	3.49	2.09	2.29	1.58
5^{th} percentile	48.05	8.74	0.52	108.68	21.05	0.76	2.26	2.41	1.46
95^{th} percentile	176.30	21.99	7.77	324.74	48.86	12.20	1.84	2.22	1.57
SD	40.96	4.23	2.53	73.19	8.96	4.01	1.79	2.12	1.58
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
	123.18	17.26	4.22	248.14	38.76	7.01	2.01	2.25	1.66
	69.63	11.43	1.20	154.80	26.84	1.90	2.22	2.35	1.58
	55.45	9.76	0.75	120.25	23.34	1.27	2.17	2.39	1.69
	154.24	20.66	6.58	301.14	45.65	10.02	1.95	2.21	1.52

Discussion

Prospective use of the model to predict the likely outcomes of interaction with rifampicin (800 mg and 1200 mg BID maribavir), phenobarbital (400 mg, 800 mg and 1200 mg BID maribavir), phenytoin (400 mg, 800 mg and 1200 mg BID maribavir), and carbamazepine (400 mg, 800 mg and 1200 mg BID maribavir) indicated geometric mean (arithmetic mean) C_{max} ratios of 0.54 (0.56), 0.72 (0.73), 0.68 (0.69), and 0.77 (0.77), respectively, for maribavir. Corresponding predicted geometric mean (arithmetic mean) $AUC_{0\text{-}12h}$ ratios were 0.35 (0.37), 0.60 (0.61), 0.57 (0.58), and 0.70 (0.71), respectively. Corresponding predicted geometric mean (arithmetic mean) C_{12h} ratios were 0.04 (0.10), 0.31 (0.37), 0.30 (0.36), and 0.53 (0.54) respectively. As there was no non-linearity in the PBPK model for maribavir, the changes in exposure as a consequence of CYP3A4-mediated induction by the perpetrators were not dependent on the dose of maribavir.

Efficacy of maribavir is correlated to the trough concentrations (C_{12h}) of the drug. Significant reductions in C_{12h} were predicted when the standard dose of 400 mg BID maribavir was combined with CYP3 A4 inducers such as rifampicin, phenobarbital, phenytoin, and carbamazepine. Simulations with higher doses of maribavir (i.e. 800 mg BID, 1200 mg BID and 1600 mg BID) were performed to determine whether the reduction in C_{12h}, (and hence reduction in maribavir efficacy) could be overcome by dose increase. The reduction in C_{12h} due to the rifampicin induction of CYP3 A4 could not be corrected by the use of up to 1600 mg BID maribavir. Use of a dose of 800 mg BID maribavir with phenobarbital resulted in a C_{12h} that was 15% lower than that with 400 mg BID maribavir alone while 1200 mg BID resulted a 1.28-fold increase in C_{12h}. A 1200 mg BID dose of maribavir with phenytoin was also predicted to overcome the reduction in C_{12h}, due to CYP3A4 induction. Increasing the maribavir dose to 800 or 1200 mg BID can be used to overcome the reduction in C_{12h} due to carbamazepine. Maribavir demonstrated acceptable safety/tolerability profiles with no limiting toxicity at doses up to 1200 mg BID in Phase 2 studies for the treatment of CMV infection/diseases and the predicted mean $AUC_{(0-12h)}$ and C_{max} values are 292 mg/L*h and 43.4 mg/L for 1200 mg BID maribavir. The predicted $AUC_{(0-12h)}$ and C_{max} values seen with maribavir dose increases, as discussed above, in the presence of inducers are below the exposure predicted for 1200 mg BID maribavir alone, therefore there is no safety concern with increasing maribavir dose from 400 mg to 800 mg, or 1200 mg in the presence of inducer. Overall, CYP3A4 inducers significantly reduce systemic exposure of maribavir; therefore maribavir dose increase is necessary when co-administration with CYP3 A4 inducers is necessary. When co-administration with carbamazepine or phenobarbital is needed, maribavir dose increase to 800 or 1200 mg BID is recommended. When co-administration with phenytoin is needed, maribavir dose increase to 1200 mg BID recommendation. Rifampicin caused significant reduction in maribavir exposure which cannot be overcome by maribavir dose increase up to 1600 mg BID, therefore coadministration with rifampicin should be prohibited or alternative antibacterial therapy of lower CYP3A4 induction potential should be considered.

General note: The dose recommendations for inducers were based on arithmetic mean ratios due to the bias in the geometric mean estimates with coadministration. However, geometric mean ratios were used for dose recommendation for inhibitors.

Example 3—Co-Administration of Maribavir with Other Drugs

Drug interaction studies were summarized based on Example 2 and other available data (e.g., in vitro and clinical data). Considerations for each coadministered drug are provided in Table 1-A, and the effects of co-administration of other drugs on the pharmacokinetics of maribavir are summarized in Tables 1-B and 1-C.

TABLE 1-A

Summary of established and other potentially significant drug interactions (observed and predicted).

Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Antiarrhythmics		
digoxin	↑digoxin	Use caution when maribavir and digoxin are co-administered. Monitor serum digoxin concentrations. The dose of digoxin may need to be reduced with co-administered with maribavir
Anticonvulsants		
Carbamazepine	↓Maribavir	A dose adjustment of maribavir to 800 mg, twice daily when co-administered with carbamazepine
Phenobarbital	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenobarbital
Phenytoin	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenytoin
Antimycobacterials		
Rifabutin	↓Maribavir	Co-administration of maribavir and rifamutin may decrease efficacy of maribavir
Rifampin	↓Maribavir	Co-administration of maribavir and rifampin may decrease efficacy of maribavir
Herbal Products		
St. John's wort	↓Maribavir	Co-administration of maribavir and St. John's wort may decrease efficacy of maribavir
Immunosuppressants		
Cyclosporine	↑Cyclosporine	Frequently monitor cyclosporine levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Everolimus	↑Everolimus	Frequently monitor everolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Sirolimus	↑Sirolimus	Frequently monitor sirolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed

TABLE 1-A-continued

Summary of established and other potentially significant drug interactions (observed and predicted).		
Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Tacrolimus	↑Tacrolimus	Frequently monitor tacrolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed

TABLE 1-B

Summary of Changes in Pharmacokinetics of Maribavir in the Presence of Co-administered Drugs.

Co-administered Drug and Regimen		Maribavir Regimen	N	Geometric Mean Ratio (90% CI] of Maribavir PK with/without Co-administered Drug (No Effect = 1.00) AUC	C_{max}	C_{tau}[C]
Anticonvulsants						
Carbamazepine[a]	400 mg once daily	800 mg twice daily/ 400 mg twice daily	200	1.40 (1.09, 1.67)	1.53 (1.22, 1.79)	1.05 (0.71, 1.40)
Phenobarbital[a]	100 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.80 (1.18, 2.35)	2.17 (1.69, 2.57)	0.94 (0.22, 1.97)
Phenytoin[a]	300 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.70 (1.06, 2.46)	2.05 (1.49, 2.63)	0.89 (0.26, 2.04)
Antimycobacterials						
Rifampin	600 mg once daily	400 mg twice daily	14	0.40 (0.36, 0.44)	0.61 (0.52, 0.72)	0.18 (0.14, 0.25)
Antifungals						
Ketoconazole	400 mg single dose	400 mg single dose	19	1.53 (1.44, 1.63)	1.10 (1.01, 1.19)	—
Aluminum hydroxide and magnesisum hydroxide antacid	20 mL[b] single dose	100 mg single dose	15	0.89 (0.83, 0.96)	0.84 (0.75, 0.94)	

[a]Based on physiologically based pharmacokinetic modelings results from 10 trials of 20 subjects each. The maribavir dosing regimen and geometric mean ratios (5th percentile, 95 percentile) correspond to dose-adjusted maribavir with inducer vs. 400 mg twice daily without inducer.
[b]Containing 800 mg aluminum hydroxide and 800 mg magnesium hydroxide.
[C]tau is maribavir dosing interval: 12 hours.

TABLE 1-C

Changes of Pharmacokinetics for Co-administered Drug in the Presence of 400 mg Twice Daily Maribavir.

Co-administered Drug and Regimen		N	Geometric Mean Ratio [90% CI] of Maribavir PK with/without Co-maribavir (No Effect = 1.00) AUC	C_{max}	C_{trough}
Immunosuppressants					
Tacrolimus	stable dose, twice daily (total daily dose: 0.5-16 mg)	20	1.51 (1.39, 1.65)	1.38 (1.20, 1.57)	1.57 (1.41, 1.74)
P-gp substrate					
digoxin	0.5 mg single dose	18	1.21 (1.10, 1.32)	1.25 (1.13, 1.38)	—

While we have described a number of embodiments of this invention, it is apparent that our basic examples may be altered to provide other embodiments that utilize the compounds and methods of this invention. Therefore, it will be appreciated that the scope of this invention is to be defined by the appended claims rather than by the specific embodiments that have been represented by way of example.

The invention claimed is:

1. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom, the method comprising

administering maribavir in an amount of 1200 mg orally twice daily, wherein the patient is a transplant recipient concomitantly exposed to or receiving an anticonvulsant selected from phenytoin or phenobarbital,

wherein maribavir is administered prior to, concurrently with, or subsequently from the administration of the anticonvulsant.

2. The method of claim **1**, wherein the anticonvulsant is phenytoin.

3. The method of claim **1**, wherein the anticonvulsant is phenobarbital.

4. The method of claim **1**, wherein the patient is an adult or a child of 12 years of age or older, and wherein the patient weighs at least 35 kg.

5. The method of claim **1**, wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.

6. The method of claim **1**, wherein the patient is a hematopoietic stem cell transplant recipient.

7. The method of claim **1**, wherein the patient is a solid organ transplant recipient.

* * * * *

EXHIBIT C

US012295940B2

(12) **United States Patent**
Depue et al.

(10) **Patent No.: US 12,295,940 B2**
(45) **Date of Patent: May 13, 2025**

(54) **VIRAL INHIBITORS, THE SYNTHESIS THEREOF, AND INTERMEDIATES THERETO**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/777,473**

(22) Filed: **Jul. 18, 2024**

(65) **Prior Publication Data**

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(63) Continuation of application No. 18/379,047, filed on Oct. 11, 2023.

(60) Provisional application No. 63/415,438, filed on Oct. 12, 2022.

(51) **Int. Cl.**

A61K 31/4184	(2006.01)
A61K 9/00	(2006.01)
A61K 31/7056	(2006.01)
C07H 19/052	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/4184*** (2013.01); ***A61K 9/0053*** (2013.01); ***A61K 31/7056*** (2013.01); ***C07H 19/052*** (2013.01)

(58) **Field of Classification Search**

CPC A61K 31/7056; A61K 31/4184; C07H 19/052

See application file for complete search history.

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Primary Examiner — Jonathan S Lau

(74) *Attorney, Agent, or Firm* — Honigman LLP; Andrew S. Chipouras; Jonathan P. O'Brien

(57) **ABSTRACT**

The present disclosure discloses compositions comprising maribavir, methods of providing the same, and compositions providing intermediates useful in providing maribavir. Maribavir (((2S,3S,4R,5S)-2-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)-5-(hydroxymethyl)tetrahydrofuran-3,4-diol) is a benzimidazole riboside and is an orally available antiviral medication against cytomegalovirus (CMV).

20 Claims, 4 Drawing Sheets

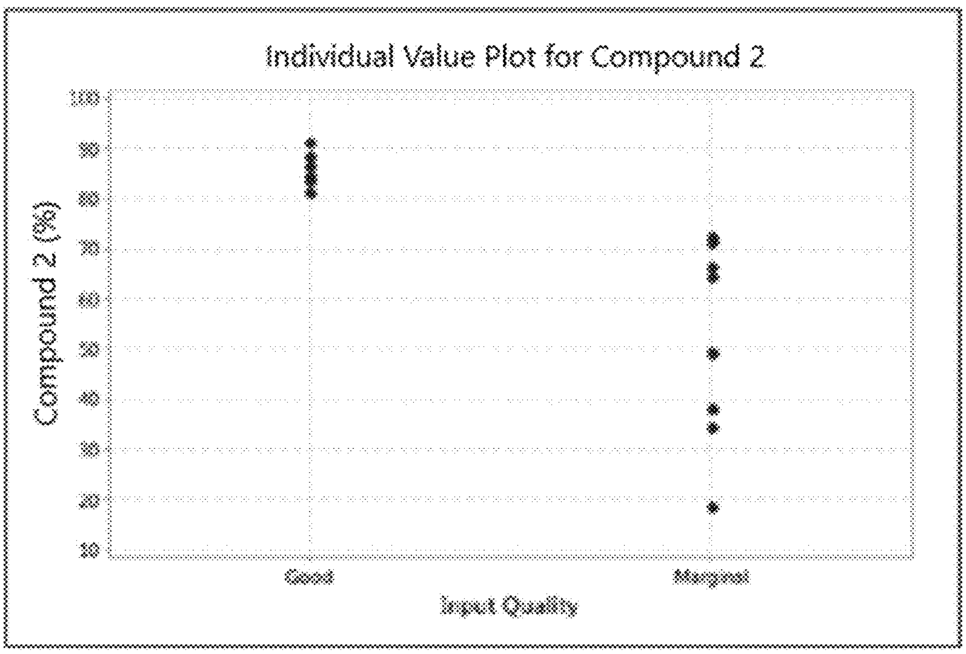
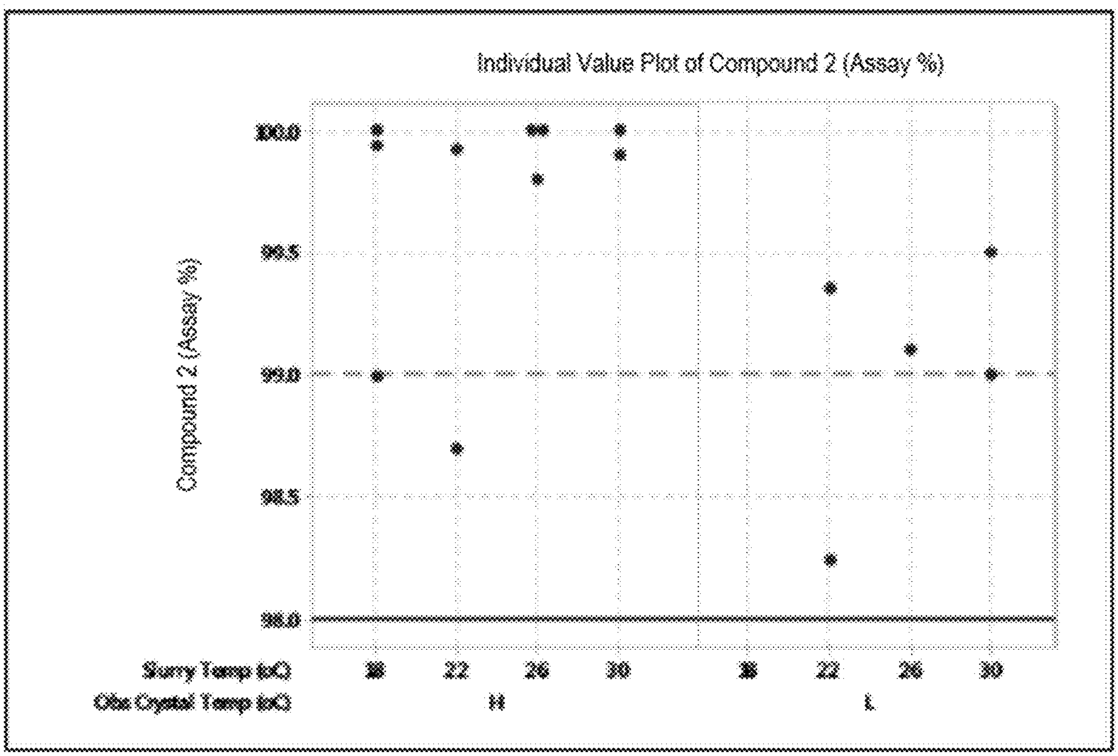

Individual Value Plot for Compound 2
Compound 2 (%)
100
90
80
70
60
50
40
30
20
10
Good
Marginal
Input Quality

Figure 1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Compound 2 (Assay %)
22
26
30
22
26
30
H
L

Figure 2

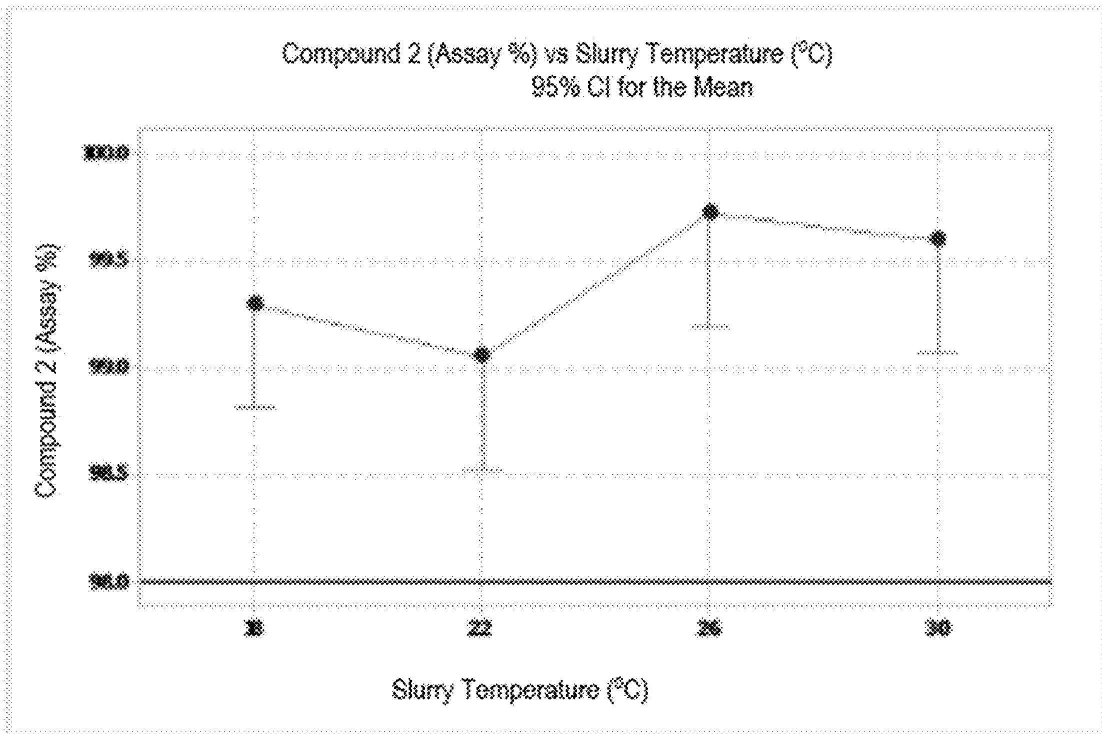

Compound 2 (Assay %) vs Slurry Temperature (°C)
95% CI for the Mean
Compound 2 (Assay %)
22
26
30
Slurry Temperature (°C)

Figure 3

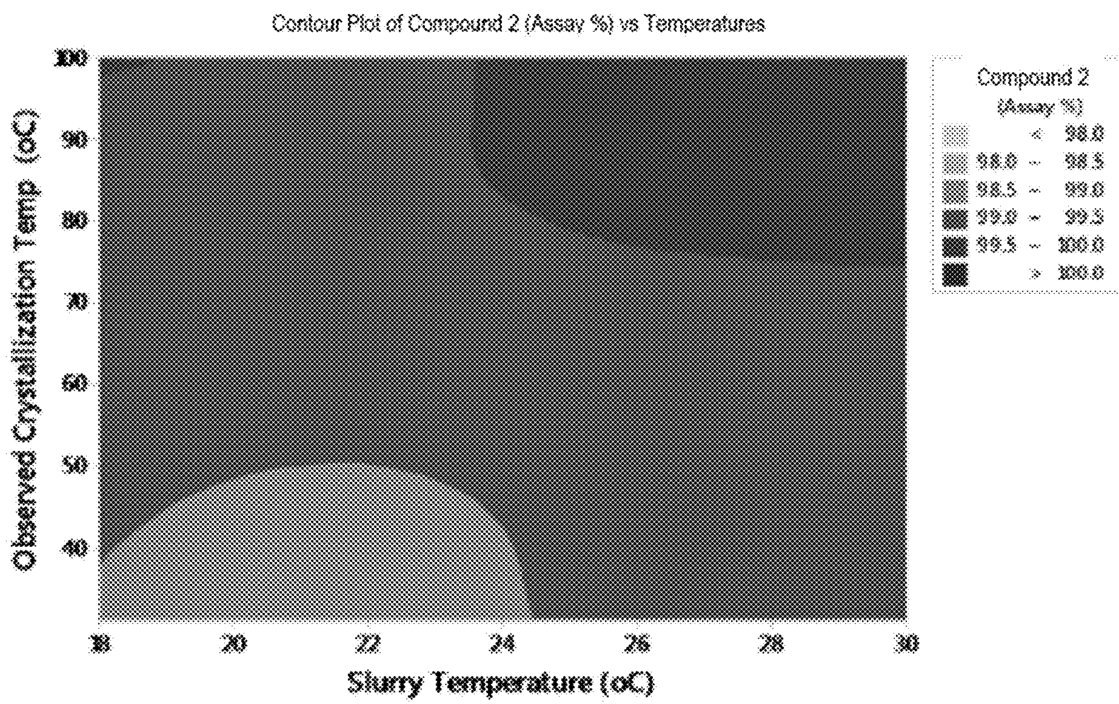

Contour Plot of Compound 2 (Assay %) vs Temperatures
Observed Crystallization Temp (oC)
100
90
80
70
60
50
40
18
20
22
24
26
28
30
Slurry Temperature (oC)
Compound 2
(Assay %)
< 98.0
98.0 – 98.5
98.5 – 99.0
99.0 – 99.5
99.5 – 100.0
> 100.0

Figure 4

VIRAL INHIBITORS, THE SYNTHESIS THEREOF, AND INTERMEDIATES THERETO

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 18/379,047, filed Oct. 11, 2023, which claims the benefit of U.S. Provisional Patent Application No. 63/415,438, filed Oct. 12, 2022, the entire disclosures of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD OF INVENTION

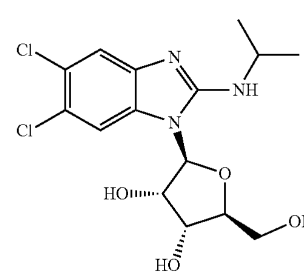
The present invention relates to compounds useful as antivirals, pharmaceutical compositions thereof, and methods of making said compounds and compositions.

BACKGROUND

Cytomegalovirus (CMV) is a member of the herpes virus family. Human cytomegalovirus (HCMV) infection is common, with serologic evidence of prior infection in 40% to 100% of various adult populations. However, serious HCMV disease occurs almost exclusively in individuals with compromised or immature immune systems. Cytomegalovirus remains a significant problem for patients undergoing various types of transplants that are associated with the use of potent immunosuppressive chemotherapy, including hematopoietic stem cell transplants (HSCTs) and solid organ transplants (SOTs).

SUMMARY

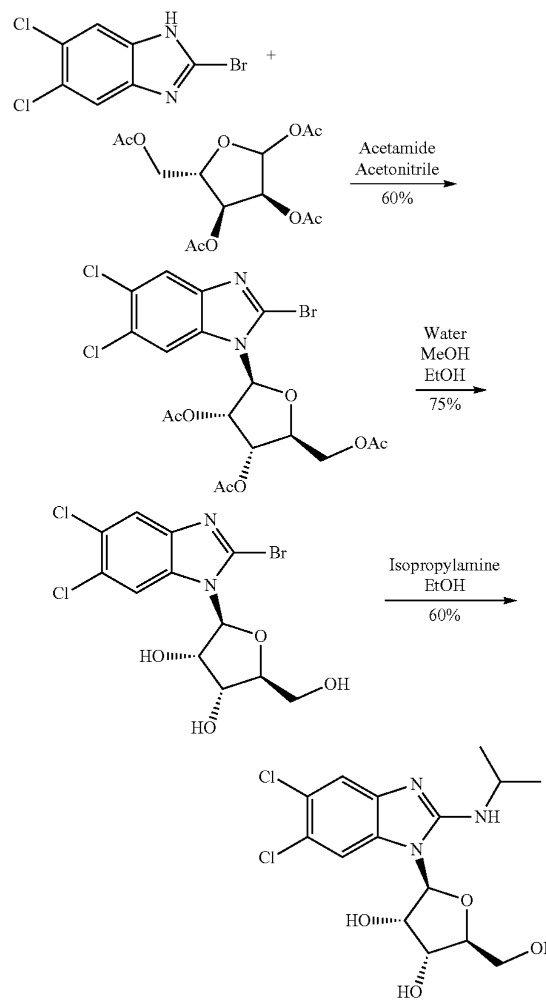
Maribavir is a benzimidazole riboside, and is an orally available antiviral medication against CMV. Maribavir is also known under its trade name LIVTENCITY™. Maribavir ((2S,3S,4R,5S)-2-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)-5-(hydroxymethyl)tetrahydrofuran-3,4-diol), a compound having the chemical structure:



is a potent and orally bioavailable antiviral for the treatment of CMV infection and disease in transplant recipients. Transplant recipients are at significant risk of CMV infection.

The synthesis of maribavir is described in Examples 1, 2, and 5 of U.S. Pat. No. 6,077,832 (the '832 patent). This synthesis, depicted in Scheme 1 below, consists of three chemical transformation steps with a combined yield of about 27%.

Scheme 1

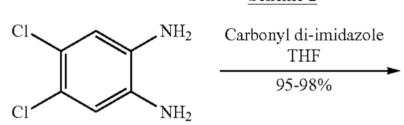


Maribavir

The synthesis depicted in Scheme 1 first couples 2-bromo-5,6-dichlorobenzimidazole with 1,2,3,5-tetra-O-acetyl-L-ribofuranose. Next, the acetyl groups are removed on the ribofuranose moiety followed by installation of the isopropylamine to provide maribavir. Notably, the first step also produced the alpha anomer in about 6% yield (see the '832 patent at Example 1).

The synthesis of maribavir is also described in WO 2001/077083 (the '083 publication) at Examples 1-4 and 7. This synthesis, depicted in Scheme 2 below, consists of five chemical transformations with a combined yield of between about 18-20%.

Scheme 2



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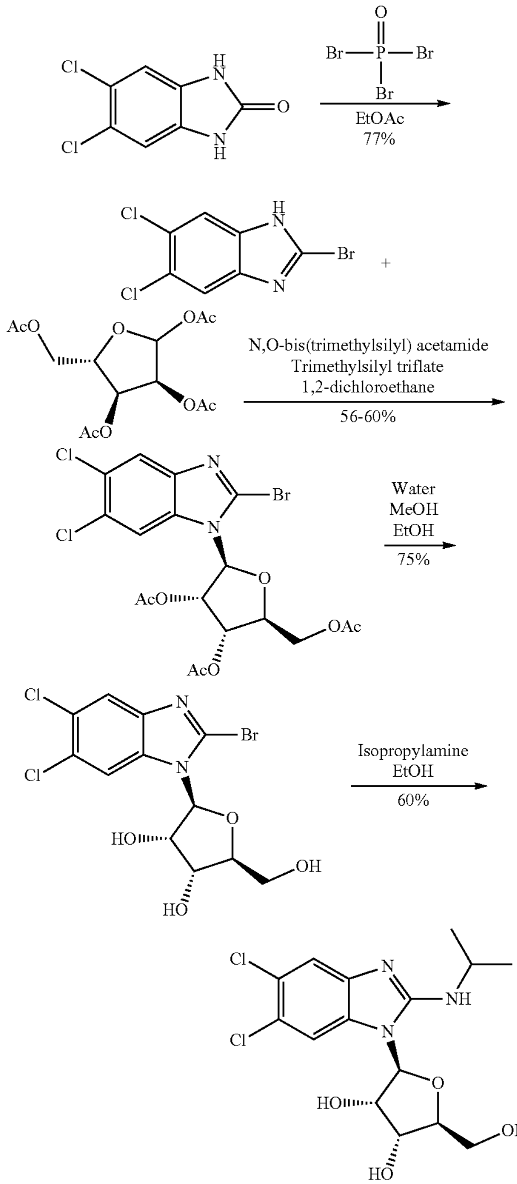
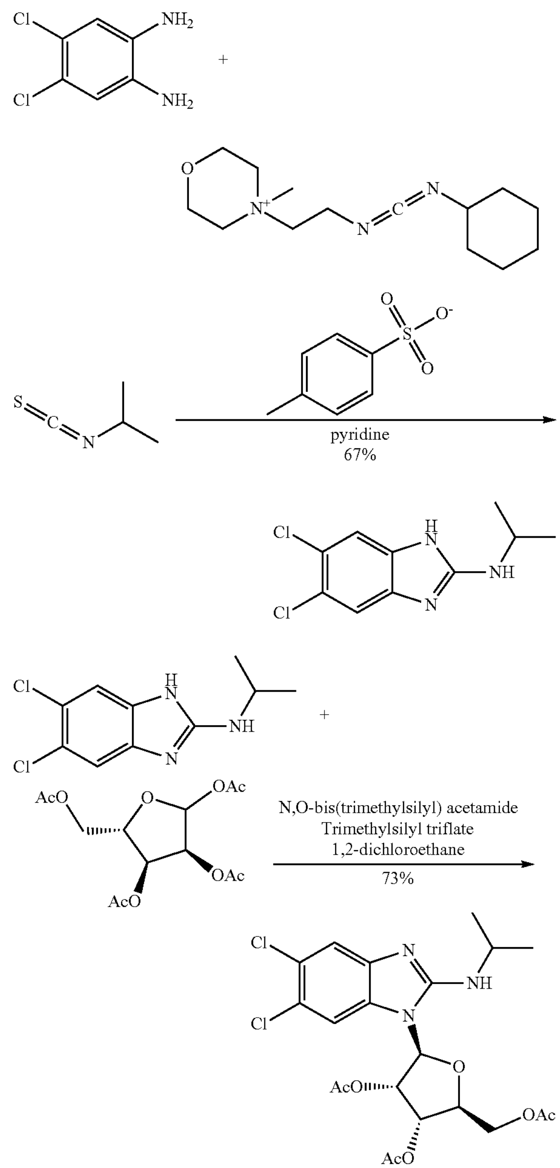


Scheme 3



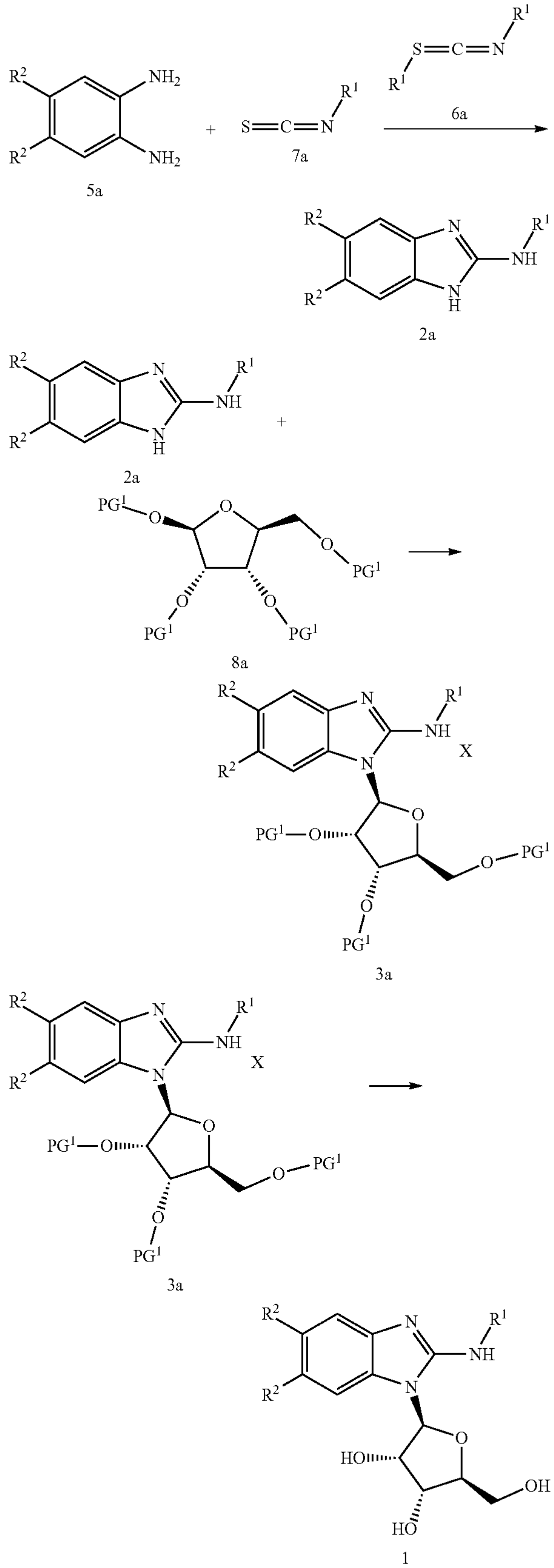
The synthesis in Scheme 2 follows a similar route as that provided in Scheme 1 as disclosed within the '832 patent, but replaces some reagents in the first step. Additionally, the '083 publication also discloses a two-step process for providing 2-bromo-5,6-dichlorobenzimidazole.

U.S. Pat. No. 6,617,315 (the '315 patent) discloses an alternative route: synthesis of 2-(alkylamino)-1H-benzimidazoles using 1-cyclohexyl-3-(2-morpholinoethyl)carbodiimde metho-p-toluenesulfonate as a desulfurizing agent; coupling 2-(alkylamino)-1H-benzimidazoles with 1,2,3,5-tri-O-acetyl-ribofuranose; and deprotection of 2-(alkylamino)-1-(2,3,5-tri-O-acetyl-beta-L-ribofuranosyl-1H-benzimidazoles. See General Procedures I, II, and III at columns 27-28 of the '315 patent. Specifically, the '315 patent discloses the synthesis of the acetyl-protected intermediate in Scheme 3 at Examples 24 and 25:

These two-steps have a combined yield of about 49%. Notably, the '315 patent does not exemplify deacetylation of 5,6-dichloro-2-(isopropylamino)-1-(2,3,5-tri-O-acetyl-beta1-L-ribofuranosyl)-1H-benzimidazole to afford maribavir. While the synthesis disclosed in the '315 patent appears to be a marked improvement over the syntheses described in the '832 patent and the '083 publication, the yield of maribavir is still expected to be below 49% (e.g., 37% when assuming 75% yield for deacetylation as exemplified in the '832 patent and the '083 publication).

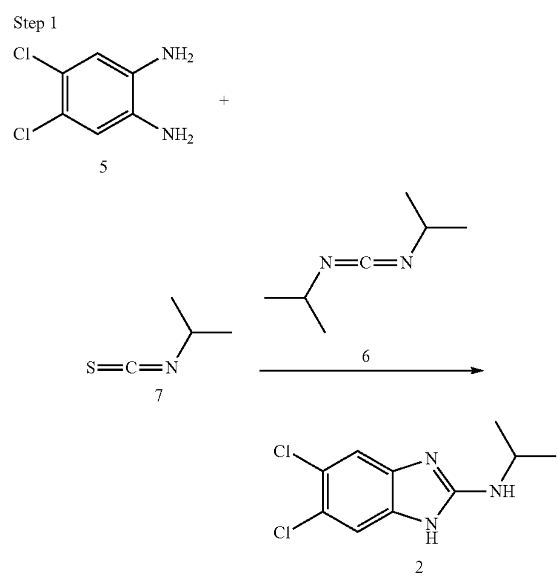
Accordingly, in some embodiments, the present invention encompasses the recognition that the synthesis of maribavir can be modified to increase the overall yield. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, is prepared according to Scheme 4 or Scheme 5, set forth below:

Scheme 4

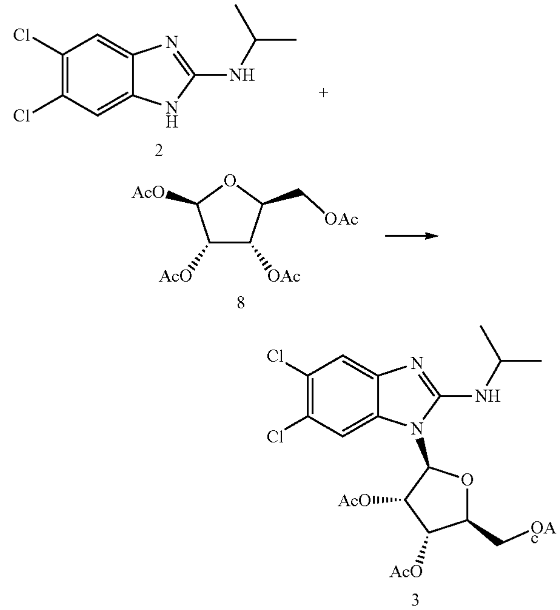


Scheme 5

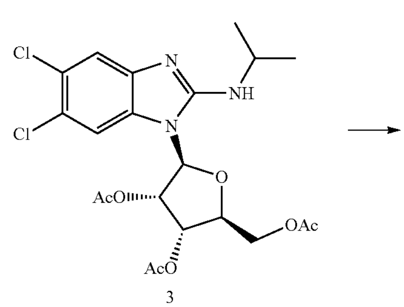
Step 1



Step 2



Step 3



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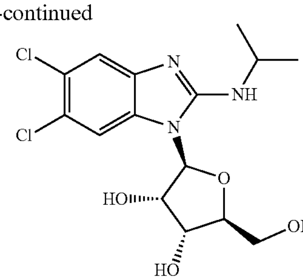


Maribavir

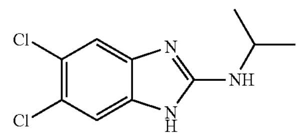
In some embodiments, the present disclosure provides an improved synthesis of maribavir in Scheme 5 with overall yields of at least 45%. It will be appreciated that the physical and/or chemical properties of certain intermediate compounds (e.g., compounds 2-3 or 5-8), solvents and/or reagents, as well as reaction conditions may contribute to the overall yield of maribavir and/or help control impurities, particularly when scaling-up the synthesis.

Additionally or alternatively, the present disclosure also provides the recognition that maribavir of acceptable quality is important to be properly milled and formulated into a tablet. For example, in some embodiments, a particular polymorphic form of maribavir (e.g., Form VI as disclosed in U.S. Pat. No. 6,482,939) is desired, free of other crystal forms, solvates, or hydrates. In some embodiments, the present disclosure provides methods of preparing maribavir in a particular polymorphic form (e.g., Form VI as disclosed in U.S. Pat. No. 6,482,939), free of other crystal forms, solvates, or hydrates. Additionally or alternatively, in some embodiments, the particular size distribution of maribavir is important for tablet manufacture. In some embodiments, the present disclosure provides methods of preparing maribavir with particular particle size distributions that are amenable for milling and formulating into a tablet. Additionally or alternatively, in some embodiments, the particular size distribution of maribavir is important for manufacturing an oral solid formulation. In some embodiments, the present disclosure provides methods of preparing maribavir with particular particle size distributions that are amenable for formulating into an oral solid formulation. Additionally or alternatively, in some embodiments, the particular size distribution of maribavir may affect bioavailability of maribavir. In some embodiments, the present disclosure provides a solid oral formulation comprising maribavir polymorphic Form VI having particular particle size distributions (e.g., d(50) is between about 50 and about 400 μm). In some embodiments, the present disclosure provides methods of crystallization of maribavir polymorphic Form VI having particular particle size distributions (e.g., d(50) is between about 50 and about 400 μm).

Additionally or alternatively, the present disclosure provides the recognition that minimizing certain impurities is important when manufacturing a drug product. For example, impurity profiles should be consistent to maintain consistency of efficacy and minimize potential adverse effects. In some embodiments, where synthesis of a drug product consists of multiple steps, it will be appreciated that impurities formed in an early step may be carried through subsequent steps to form additional impurities. Therefore, reducing impurities at each step in the synthesis of maribavir (e.g., Steps 1, 2, or 3) is important to maintain consistency in the manufactured drug product.

Accordingly, in some embodiments, the present disclosure provides compositions comprising maribavir, or a pharmaceutically acceptable salt thereof, and methods of preparing the same. In some embodiments, provided compositions comprise maribavir and one or more compounds selected from:

2



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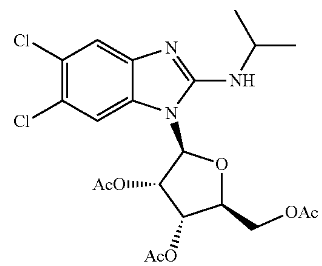
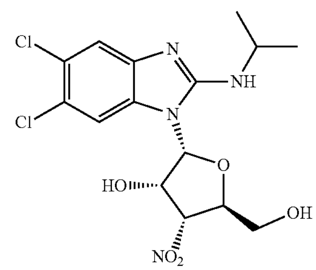


or

4



or pharmaceutically acceptable salts thereof.

In some embodiments, provided compositions are prepared according to methods disclosed herein (e.g., Steps 1-3). In some embodiments, provided composition comprise at least 90%, 95%, 99%, 99.5%, or 99.9% by weight of maribavir. In some embodiments, provided composition comprise maribavir substantially free of impurities. As used herein, the term "substantially free of impurities" means that the composition or compound contains no significant amount of extraneous matter. Such extraneous matter may include starting materials, residual solvents, or other impurities that may result from the preparation of, and/or isolation of maribavir. In some embodiments, provided compositions comprise maribavir, or a pharmaceutically acceptable salt thereof, and compound 2, 3, and/or 4 in an amount of less than 0.10% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise maribavir, or a pharmaceutically acceptable salt thereof, and compound 2, 3, and/or 4 in an amount of less than 0.10% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise maribavir, or a pharmaceutically acceptable salt thereof, and compound 2, 3, and/or 4 in an amount of less than 0.10% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise maribavir, or a pharmaceutically acceptable salt thereof, and compound 2, 3, and/or 4 in an amount not detectable by HPLC. In some embodiments, provided compositions comprise maribavir as Form VI, as described in U.S. Pat. No. 6,482,939. In some embodiments, provided compositions comprise at least 90%, 95%, 99%, 99.5%, or 99.9% by weight of Form VI maribavir. In some embodiments, provided compositions comprise maribavir substantially free of other polymorphic forms, e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777. U.S. Pat. No. 8,546,344, or U.S. Pat. No. 11,130,777. In some embodiments, provided compositions comprise maribavir having a particular size distribution as defined and described herein. Impurities (e.g., intermediates, enantiomeric impurities) would potentially impact drug substance and drug product quality and may represent a risk to the drug product safety. There is a possible impact to patient safety if certain impurities are present in quantities above qualified limits. There are unknown risks to drug product safety for unknown impurities.

BRIEF DESCRIPTION OF THE DRAWING

FIG. **1** displays a plot of Compound 2 purity as a function of input quality, after filtration of the recrystallized Compound 2 (i.e., before the wash steps).

FIG. **2** depicts a plot of Compound 2 (HPLC purity) versus Recrystallization Temperature and Observed Crystallization Temperature (H=high temperature wherein T>90° C. & L=low temperature wherein T<40° C.).

FIG. **3** depicts an Interval Plot of Compound 2 versus Recrystallization Temp. The pooled standard deviation is used to calculate the intervals.

FIG. **4** depicts a Contour Plot of Compound 2 purity examining the effects of Recrystallization Temperature and Observed Crystallization Temperature

DETAILED DESCRIPTION

Definitions

Compounds of this invention include those described generally above, and are further illustrated by the classes, subclasses, and species disclosed herein. As used herein, the following definitions shall apply unless otherwise indicated. For purposes of this invention, the chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, Handbook of Chemistry and Physics, 75th Ed. Additionally, general principles of organic chemistry are described in "Organic Chemistry", Thomas Sorrell, University Science Books, Sausalito: 1999, and "March's Advanced Organic Chemistry", 5th Ed., Ed.: Smith, M. B. and March, J., John Wiley & Sons, New York: 2001, the entire contents of which are hereby incorporated by reference.

The term "aliphatic" or "aliphatic group", as used herein, means a straight-chain (i.e., unbranched) or branched, substituted or unsubstituted hydrocarbon chain that is completely saturated or that contains one or more units of unsaturation, or a monocyclic hydrocarbon or bicyclic hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic (also referred to herein as "carbocycle," "carbocyclic", "cycloaliphatic" or "cycloalkyl"), that has a single point of attachment to the rest of the molecule. Unless otherwise specified, aliphatic groups contain 1-6 aliphatic carbon atoms. In some embodiments, aliphatic groups contain 1-5 aliphatic carbon atoms. In other embodiments, aliphatic groups contain 1-4 aliphatic carbon atoms. In still other embodiments, aliphatic groups contain 1-3 aliphatic carbon atoms, and in yet other embodiments, aliphatic groups contain 1-2 aliphatic carbon atoms. In some embodiments, "carbocyclic" (or "cycloaliphatic" or "carbocycle" or "cycloalkyl") refers to a monocyclic C_3-C_8 hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic, that has a single point of attachment to the rest of the molecule. Suitable aliphatic groups include, but are not limited to, linear or branched, substituted or unsubstituted alkyl, alkenyl, alkynyl groups and hybrids thereof such as (cycloalkyl)alkyl, (cycloalkenyl)alkyl or (cycloalkyl)alkenyl.

The term "halogen" means F, Cl, Br, or I.

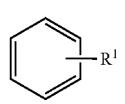
The term "aryl" used alone or as part of a larger moiety as in "aralkyl," "aralkoxy," or "aryloxyalkyl," refers to monocyclic or bicyclic ring systems having a total of five to fourteen ring members, wherein at least one ring in the system is aromatic and wherein each ring in the system contains 3 to 7 ring members. The term "aryl" may be used interchangeably with the term "aryl ring." In certain embodiments of the present invention, "aryl" refers to an aromatic ring system and exemplary groups include phenyl, biphenyl, naphthyl, anthracyl and the like, which may bear one or more substituents. Also included within the scope of the term "aryl," as it is used herein, is a group in which an aromatic ring is fused to one or more non-aromatic rings, such as indanyl, phthalimidyl, naphthimidyl, phenanthridinyl, or tetrahydronaphthyl, and the like.

The terms "heteroaryl" and "heteroar-," used alone or as part of a larger moiety, e.g., "heteroaralkyl," or "heteroaralkoxy," refer to groups having 5 to 10 ring atoms, preferably 5, 6, or 9 ring atoms; having 6, 10, or 14 p electrons shared in a cyclic array; and having, in addition to carbon atoms, from one to five heteroatoms. The term "heteroatom" refers to nitrogen, oxygen, or sulfur, and includes any oxidized form of nitrogen or sulfur, and any quaternized form of a basic nitrogen. Exemplary heteroaryl groups include thienyl, furanyl, pyrrolyl, imidazolyl, pyrazolyl, triazolyl, tetrazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiazolyl, isothiazolyl, thiadiazolyl, pyridyl, pyridazinyl, pyrimidinyl, pyrazinyl, indolizinyl, purinyl, naphthyridinyl, and pteridinyl. The terms "heteroaryl" and "heteroar-", as used herein, also include groups in which a heteroaromatic ring is fused to one or more aryl, cycloaliphatic, or heterocyclyl rings, where the radical or point of attachment is on the heteroaromatic ring. Exemplary groups include indolyl, isoindolyl, benzothienyl, benzofuranyl, dibenzofuranyl, indazolyl, benzimidazolyl, benzthiazolyl, quinolyl, isoquinolyl, cinnolinyl, phthalazinyl, quinazolinyl, quinoxalinyl, 4H-quinolizinyl, carbazolyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl, tetrahydroquinolinyl, tetrahydroisoquinolinyl, and pyrido[2,3-b]-1,4-oxazin-3(4H)-one. A heteroaryl group may be mono- or bicyclic. The term "heteroaryl" may be used interchangeably with the terms "heteroaryl ring," "heteroaryl group," or "heteroaromatic," any of which terms include rings that are optionally substituted. The term "heteroaralkyl" refers to an alkyl group substituted by a heteroaryl, wherein the alkyl and heteroaryl portions independently are optionally substituted.

As used herein, the terms "heterocycle," "heterocyclyl," "heterocyclic radical," and "heterocyclic ring" are used interchangeably and refer to a stable 5- to 7-membered monocyclic or 7-10-membered bicyclic heterocyclic moiety that is either saturated or partially unsaturated, and having, in addition to carbon atoms, one or more, preferably one to four, heteroatoms, as defined above. When used in reference to a ring atom of a heterocycle, the term "nitrogen" includes a substituted nitrogen. As an example, in a saturated or partially unsaturated ring having 0-3 heteroatoms selected from oxygen, sulfur or nitrogen, the nitrogen may be N (as in 3,4-dihydro-2H-pyrrolyl), NH (as in pyrrolidinyl), or ^{+}NR (as in N-substituted pyrrolidinyl).

A heterocyclic ring can be attached to its pendant group at any heteroatom or carbon atom that results in a stable structure and any of the ring atoms can be optionally substituted. Examples of such saturated or partially unsaturated heterocyclic radicals include tetrahydrofuranyl, tetrahydrothiophenyl pyrrolidinyl, piperidinyl, pyrrolinyl, tetrahydroquinolinyl, tetrahydroisoquinolinyl, decahydroquinolinyl, oxazolidinyl, piperazinyl, dioxanyl, dioxolanyl, diazepinyl, oxazepinyl, thiazepinyl, morpholinyl, and quinuclidinyl. The terms "heterocycle," "heterocyclyl," "heterocyclyl ring," "heterocyclic group," "heterocyclic moiety," and "heterocyclic radical," are used interchangeably herein, and also include groups in which a heterocyclyl ring is fused to one or more aryl, heteroaryl, or cycloaliphatic rings, such as indolinyl, 3H-indolyl, chromanyl, phenanthridinyl, or tetrahydroquinolinyl, where the radical or point of attachment is on the heterocyclyl ring. A heterocyclyl group may be mono- or bicyclic. The term "heterocyclylalkyl" refers to an alkyl group substituted by a heterocyclyl, wherein the alkyl and heterocyclyl portions independently are optionally substituted.

As described herein, compounds may contain "optionally substituted" moieties. In general, the term "substituted", whether preceded by the term "optionally" or not, means that one or more hydrogens of the designated moiety of compounds are replaced with a suitable substituent. "Substituted" applies to one or more hydrogens that are either explicit or implicit from the structure (e.g.,



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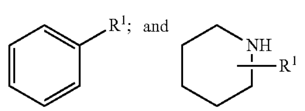

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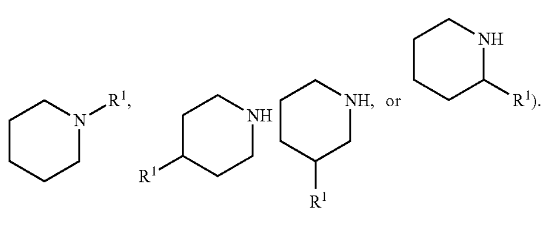


Unless otherwise indicated, an "optionally substituted" group may have a suitable substituent at each substitutable position of the group, and when more than one position in any given structure may be substituted with more than one substituent selected from a specified group, the substituent may be either the same or different at every position. Combinations of substituents envisioned by this disclosure are preferably those that result in the formation of stable or chemically feasible compounds. The term "stable", as used herein, refers to compounds that are not substantially altered when subjected to conditions to allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.

Suitable monovalent substituents on a substitutable carbon atom of an "optionally substituted" group are independently halogen; $—(CH_2)_{0-4}R^o$; $—(CH_2)_{0-4}OR^o$; $—O(CH_2)_{0-4}R^o$, $—O—(CH_2)_{0-4}C(O)OR^o$; $—(CH_2)_{0-4}CH(OR^o)_2$; $—(CH_2)_{0-4}SR^o$; $—(CH_2)_{0-4}Ph$, which may be substituted with R^o; $—(CH_2)_{0-4}O(CH_2)_{0-1}Ph$ which may be substituted with R^o; —CH═CHPh, which may be substituted with R^o; $—(CH_2)_{0-4}O(CH_2)_{0-1}$-pyridyl which may be substituted with R^o; $—NO_2$; —CN; $—N_3$; $—(CH_2)_{0-4}N(R^o)_2$; $—(CH_2)_{0-4}N(R^o)C(O)R^o$; $—N(R^o)C(S)R^o$; $—(CH_2)_{0-4}N(R^o)C(O)NR^o_2$; $—N(R^o)C(S)NR^o_2$; $—(CH_2)_{0-4}N(R^o)C(O)OR^o$; $—N(R^o)N(R^o)C(O)R^o$; $—N(R^o)N(R^o)C(O)NR^o_2$; $—N(R^o)N(R^o)C(O)OR^o$; $—(CH_2)_{0-4}C(O)R^o$; $—C(S)R^o$; $—(CH_2)_{0-4}C(O)OR^o$; $—(CH_2)_{0-4}C(O)SR^o$; $—(CH_2)_{0-4}C(O)OSiR^o_3$; $—(CH_2)_{0-4}OC(O)R^o$; $—OC(O)(CH_2)_{0-4}SR^o$, $SC(S)SR^o$; $—(CH_2)_{0-4}SC(O)R^o$; $—(CH_2)_{0-4}C(O)NR^o_2$; $—C(S)NR^o_2$; $—C(S)SR^o$; $—SC(S)SR^o$, $—(CH_2)_{0-4}OC(O)NR^o_2$; $—C(O)N(OR^o)R^o$; $—C(O)C(O)R^o$; $—C(O)CH_2C(O)R^o$; $—C(NOR^o)R^o$; $—(CH_2)_{0-4}SSR^o$; $—(CH_2)_{0-4}S(O)_2R^o$; $—(CH_2)_{0-4}S(O)_2OR^o$; $—(CH_2)_{0-4}OS(O)_2R^o$; $—S(O)_2NR^o_2$; $—(CH_2)_{0-4}S(O)R^o$; $—N(R^o)S(O)_2NR^o_2$; $—N(R^o)S(O)_2R^o$; $—N(OR^o)R^o$; $—C(NH)NR^o_2$; $—P(O)_2R^o$; $—P(O)R^o_2$; $—OP(O)R^o_2$; $—OP(O)(OR^o)_2$; SiR^o_3; $—(C_{1-4}$ straight or branched alkylene)$O—N(R^o)_2$; or $—(C_{1-4}$ straight or branched alkylene)$C(O)O—N(R^o)_2$, wherein each R^o may be substituted as defined below and is independently hydrogen, C_{1-6} aliphatic, $—CH_2Ph$, $—O(CH_2)_{0-1}Ph$, $—CH_2$-(5- to 6 membered heteroaryl ring), or a 5- to 6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R^o, taken together with their intervening atom(s), form a 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, which may be substituted as defined below.

Suitable monovalent substituents on R^o (or the ring formed by taking two independent occurrences of R^o together with their intervening atoms), are independently halogen, $—(CH_2)_{0-2}R^{\bullet}$, -(halo$R^{\bullet}$), $—(CH_2)_{0-2}OH$, $—(CH_2)_{0-2}OR^{\bullet}$, $—(CH_2)_{0-2}CH(OR^{\bullet})_2$; —O(halo$R^{\bullet}$), —CN, $—N_3$, $—(CH_2)_{0-2}C(O)R^{\bullet}$, $—(CH_2)_{0-2}C(O)OH$, $—(CH_2)_{0-2}C(O)OR^{\bullet}$, $—(CH_2)_{0-2}SR^{\bullet}$, $—(CH_2)_{0-2}SH$, $—(CH_2)_{0-2}NH_2$, $—(CH_2)_{0-2}NHR^{\bullet}$, $—(CH_2)_{0-2}NR^{\bullet}_2$, $—NO_2$, $—SiR^{\bullet}_3$, $—OSiR^{\bullet}_3$, $—C(O)SR^{\bullet}$, $—(C_{1-4}$ straight or branched alkylene)$C(O)OR^{\bullet}$, or $—SSR^{\bullet}$ wherein each $R^{\bullet}$ is unsubstituted or where preceded by "halo" is substituted only with one or more halogens, and is independently selected from C_{1-4} aliphatic, $—CH_2Ph$, $—O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents on a saturated carbon atom of R^o include ═O and ═S.

Suitable divalent substituents on a saturated carbon atom of an "optionally substituted" group include the following: ═O, ═S, $═NNR^*_2$, $═NNHC(O)R^*$, $═NNHC(O)OR^*$, $═NNHS(O)_2R^*$, $═NR^*$, $═NOR^*$, $—O(C(R^*_2))_{2-3}O—$, or $—S(C(R^*_2))_{2-3}S—$, wherein each independent occurrence of R^* is selected from hydrogen, C_{1-6} aliphatic which may be substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Suitable divalent substituents that are bound to vicinal substitutable carbons of an "optionally substituted" group of a compound of Formula I, and subgenera thereof, include: —$O(CR^*_2)_{2-3}O$—, wherein each independent occurrence of $R^\bullet$ is selected from hydrogen, C_{1-6} aliphatic which may be substituted as defined below, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

Suitable substituents on the aliphatic group of $R^\bullet$ include halogen, —$R^\bullet$, -(halo$R^\bullet$), —OH, —$OR^\bullet$, —O(halo$R^\bullet$), —CN, —C(O)OH, —C(O)$OR^\bullet$, —NH_2, —$NHR^\bullet$, —$NR^\bullet_2$, or —NO_2, wherein each $R^\bullet$ is unsubstituted or where preceded by "halo" is substituted only with one or more halogens, and is independently C_{1-4} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

Suitable substituents on a substitutable nitrogen of an "optionally substituted" group include —$R^\dagger$, —$NR^\dagger_2$, —C(O)$R^\dagger$, —C(O)O$R^\dagger$, —C(O)C(O)$R^\dagger$, —C(O)CH_2C(O)$R^\dagger$, —$S(O)_2R^\dagger$, —$S(O)_2NR^\dagger_2$, —C(S)$NR^\dagger_2$, —C(NH)$NR^\dagger_2$, or —$N(R^\dagger)S(O)_2R^\dagger$; wherein each $R^\dagger$ is independently hydrogen, C_{1-6} aliphatic which may be substituted as defined below, unsubstituted —OPh, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of $R^\dagger$, taken together with their intervening atom(s) form an unsubstituted 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

Suitable substituents on the aliphatic group of RT are independently halogen, —$R^\bullet$, -(halo$R^\bullet$), —OH, —$OR^\bullet$, —O(halo$R^\bullet$), —CN, —C(O)OH, —C(O)$OR^\bullet$, —NH_2, —$NHR^\bullet$, —$NR^\bullet_2$, or —NO_2, wherein each $R^\bullet$ is unsubstituted or where preceded by "halo" is substituted only with one or more halogens, and is independently C_{1-4} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

As used herein, the term "pharmaceutically acceptable salt" refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, S. M. Berge et al., describe pharmaceutically acceptable salts in detail in J. Pharmaceutical Sciences, 1977, 66, 1-19, incorporated herein by reference. Pharmaceutically acceptable salts of the compounds of this invention include those derived from suitable inorganic and organic acids and bases. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid and perchloric acid or with organic acids such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid or malonic acid or by using other methods used in the art such as ion exchange. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like.

Salts derived from appropriate bases include alkali metal, alkaline earth metal, ammonium and $N^+(C_{1-4}alkyl)_4$ salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, lower alkyl sulfonate and aryl sulfonate.

The term "about", when used herein in reference to a value, refers to a value that is similar, in context to the referenced value. In general, those skilled in the art, familiar with the context, will appreciate the relevant degree of variance encompassed by "about" in that context. For example, in some embodiments, the term "about" may encompass a range of values that within (i.e., ±) 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less of the referred value. For example, a dose that comprises "about 200 mg" of maribavir encompasses any amount of maribavir within a range of 180 mg to 220 mg.

As used herein, the term "administering" or "administration" typically refers to administration of a composition to a subject to achieve delivery of an active agent to a site of interest (e.g., a target site which may, in some embodiments, be a site of disease or damage, and/or a site of responsive processes, cells, tissues, etc.) As will be understood by those skilled in the art, reading the present disclosure, in some embodiments, one or more particular routes of administration may be feasible and/or useful in the practice of the present disclosure. In some embodiments, administration may be oral. In some embodiments, administration may involve only a single dose. In some embodiments, administration may involve application of a fixed number of doses. In some embodiments, administration may involve dosing that is intermittent (e.g., a plurality of doses separated in time) and/or periodic (e.g., individual doses separated by a common period of time) dosing.

The terms "treat" or "treating," as used herein, refers to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition. As used herein, the terms "treatment," "treat," and "treating" refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed. In some embodiments, the term "treating" includes halting the progression of a disease or disorder. Treatment may also be continued after symptoms have resolved, for example to prevent or delay their recurrence. Thus, in some embodiments, the term "treating" includes preventing relapse or recurrence of a disease or disorder.

It will be understood that unit ratios of L/kg, kg/kg, etc. are expressions that, for each reagent/component/etc. described, are relative to the mass (in kilograms) of the limiting reagent (e.g., in each of Steps 1, 2, or 3 as defined and described herein).

Provided compounds may be synthesized according to the schemes described herein. The reagents and conditions described are intended to be exemplary and not limiting. As one of skill in the art would appreciate, various analogs may be prepared by modifying the synthetic reactions such as using different starting materials, different reagents, and different reaction conditions (e.g., temperature, solvent, concentration, etc.).

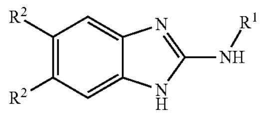
It will be appreciated that any intermediate depicted in Schemes 4 or 5 may be isolated and/or purified prior to each subsequent step. Alternatively, any intermediate depicted in Schemes 4 or 5 may be utilized in subsequent steps without isolation and/or purification. Such telescoping of steps is contemplated in the present disclosure.

In some embodiments, compounds described herein may be purified by any means known in the art. In some embodiments, purification of a compound described herein comprises filtration, chromatography, distillation, crystallization, or a combination thereof. In some embodiments, chromatography comprises high performance liquid chromatography (HPLC). In some embodiments, chromatography comprises normal phase, reverse phase, or ion-exchange elution over a cartridge comprising suitable sorbent media. Purification via chromatography methods typically utilizes one or more solvents, which are known to the skilled artisan or determined by routine experimentation.

Step 1

In some embodiments, the present disclosure provides an improved synthesis of compound 2a:

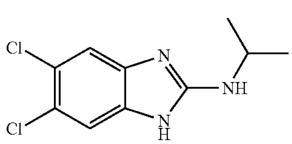
2a



or a salt thereof, wherein R^1 and R^2 are as defined and described herein.

In some embodiments, the present disclosure provides an improved synthesis of compound 2:

2

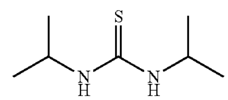


or a salt thereof.

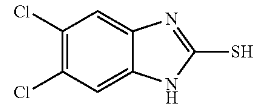
As shown in Schemes 4 and 5, Step 1 is the first of three steps for preparing maribavir as disclosed herein. Any byproducts formed within Step 1 may be present in a provided composition, may further react to form additional byproducts, and/or may lower the overall yield or quality of maribavir. In some embodiments, the present disclosure provides methods of preparing compound 2, or a salt thereof, with reduced and/or low levels of impurities. Without wishing to be bound by a particular theory, a lower volume of reaction solvent (e.g., 1,4-dioxane) can allow for a more efficient conversion to compound 2, resulting in reduced and/or low levels of impurities. Additionally or alternatively, without wishing to be bound by a particular theory, provided crystallization step(s) can result in compound 2 with reduced and/or low levels of impurities.

In some embodiments, the present disclosure provides compositions comprising compound 2, or a salt thereof, and one or more of the following compounds:

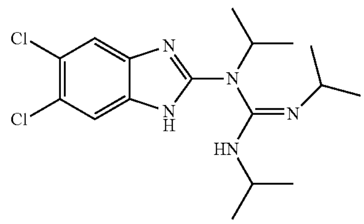
9



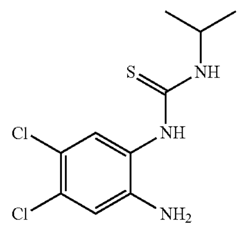
10



11



12



or salts thereof.

In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 9, 10, 11, and/or 12, or salts thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 9, 10, 11, and/or 12, or salts thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir.

In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 9, or a salt thereof. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 9, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 9, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, wherein compound 9, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 10, or a salt thereof. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 10, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 10, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, wherein compound 10, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

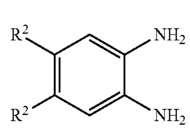
In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 11, or a salt thereof. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 11, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 11, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, wherein compound 11, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 12, or a salt thereof. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 12, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, and compound 12, or a salt thereof, in an amount of less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 2, or a salt thereof, wherein compound 12, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, such compositions comprising compound 2, or salt thereof are prepared as described herein. In some embodiments, the present disclosure also provides the recognition that certain reagents (and amounts thereof) and/or reaction conditions may provide improved quality compound 2, or a salt thereof (e.g., with higher purity and/or minimal byproducts), and/or improve the yield of compound 2, or a salt thereof. In some embodiments, such improvements in yield or quality of compound 2, or a salt thereof, may also improve the yield or quality of maribavir. Without wishing to be bound to a particular theory, the present disclosure provides the recognition that incorporation of a crystallization and/or recrystallization into Step 1 may improve the yield, purity, and/or reduce the formation of byproducts (e.g., compounds 9, 10, 11, and 12).

In some embodiments, compound 2a, or a salt thereof, is prepared as shown in Step 1 of Scheme 4. In some embodiments, at Step 1, compound 2a, or a salt thereof, is prepared by a method comprising:

(a) reacting compound 5a:

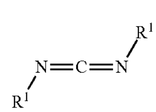


5a

or a salt thereof, wherein

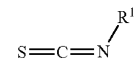
each R^2 is independently halogen;

with compounds 6a and 7a:



6a

-continued



7a

or salts thereof, wherein

each R^1 is independently an optionally substituted group selected from C_{1-6} aliphatic, 3- to 7-membered saturated or partially unsaturated carbocyclyl, 3- to 7-membered saturated or partially unsaturated heterocyclyl having 1-3 heteroatoms independently selected from nitrogen, oxygen, or sulfur, phenyl, or 5- to 6-membered heteroaryl having 1-3 heteroatoms independently selected from nitrogen, oxygen, or sulfur;

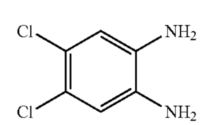
under suitable reaction conditions to provide compound 2a, or a salt thereof.

As generally defined above, each R^2 is independently halogen. In some embodiments, each R^2 is independently fluoro or chloro. In some embodiments, each R^2 is fluoro. In some embodiments, each R^2 is chloro.

As generally defined above, each R^1 is independently an optionally substituted group selected from C_{1-6} aliphatic, 3- to 7-membered saturated or partially unsaturated carbocyclyl, 3- to 7-membered saturated or partially unsaturated heterocyclyl having 1-3 heteroatoms independently selected from nitrogen, oxygen, or sulfur, phenyl, or 5- to 6-membered heteroaryl having 1-3 heteroatoms independently selected from nitrogen, oxygen, or sulfur. In some embodiments, each R^1 is independently an optionally substituted group selected from C_{1-6} aliphatic or 3- to 7-membered saturated or partially unsaturated carbocyclyl. In some embodiments, R^1 is an optionally substituted C_{1-6} aliphatic. In some embodiments, R^1 is an optionally substituted C_{1-3} aliphatic. In some embodiments, R^1 is an optionally substituted methyl, ethyl, n-propyl, isopropyl, n-butyl, s-butyl, or t-butyl. In some embodiments, R^1 is methyl, ethyl, n-propyl, isopropyl, n-butyl, s-butyl, or t-butyl. In some embodiments, R^1 is isopropyl. In some embodiments, R^1 is an optionally substituted group 3- to 7-membered saturated or partially unsaturated carbocyclyl. In some embodiments, R^1 is an optionally substituted cyclopropyl, cyclobutyl, cyclopentyl, or cyclohexyl. In some embodiments, R^1 is cyclopropyl, cyclobutyl, cyclopentyl, or cyclohexyl.

In some embodiments, compound 2, or a salt thereof, is prepared as shown in Step 1 of Scheme 5. In some embodiments, at Step 1, compound 2, or a salt thereof, is prepared by a method comprising:

(a) reacting compound 5:

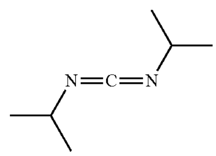


5

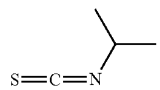
or a salt thereof,

with compounds 6 and 7:

6



7

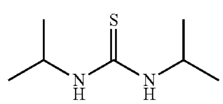


or salts thereof,

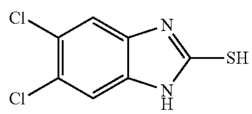
under suitable reaction conditions to provide compound 2, or a salt thereof.

In some embodiments, Step 1 further provides one or more of the following compounds:

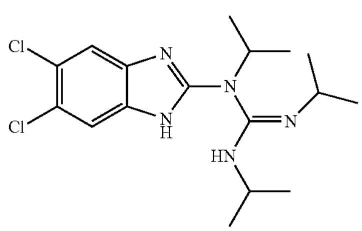
9



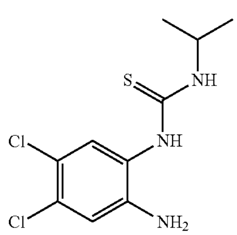
10



11



12



or salts thereof.

In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% w/w of compound 9, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR). In some embodiments, Step 1 provides a composition comprising about 1.1%, about 0.9%, about 0.8%, about 0.7%, about 0.6%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% w/w of compound 9, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR). In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% a/a of compound 9, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR). In some embodiments, Step 1 provides

a composition comprising about 1.1%, about 0.9%, about 0.8%, about 0.7%, about 0.6%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% a/a of compound 9, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR). In some embodiments, Step 1 provides a composition wherein compound 9, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 10, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 0.63%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.11%, about 0.1%, about 0.08%, or about 0.05% (w/w HPLC) of compound 10, relative to compound 2. In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 10, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 0.63%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.11%, about 0.1%, about 0.08%, or about 0.05% (a/a HPLC) of compound 10, relative to compound 2. In some embodiments, Step 1 provides a composition wherein compound 10, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 11, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 1.00%, about 0.5%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 11, relative to compound 2. In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 11, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 1.00%, about 0.5%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 11, relative to compound 2. In some embodiments, Step 1 provides a composition wherein compound 11, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 12, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 1.00%, about 0.5%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 12, relative to compound 2. In some embodiments, Step 1 provides a composition comprising less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 12, relative to compound 2. In some embodiments, Step 1 provides a composition comprising about 1.00%, about 0.5%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 12, relative to compound 2. In some embodiments, Step 1 provides a composition wherein compound 12, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 1 provides a composition as provided in any one of Tables 4-1 to 4-6 in Example 4.

In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.50 to 1.50 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.70 to 1.30 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.70 to 1.10 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.70 to 0.90 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.90 to 1.30 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 1.10 to 1.30 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.80 to 1.20 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.95 to 1.10 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 1.00 to 1.10 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present in an amount between about 0.95 to 1.05 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 0.90 to about 1.09 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 1.00 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 0.99, 1.00, 1.01, 1.02, 1.03, 1.04, 1.05, 1.06, 1.07, 1.08, or 1.09 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 1.02 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 1.04 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 6, or a salt thereof, is present at an amount of about 1.06 equivalents relative to compound 5, or a salt thereof.

In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.50 to 1.50 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.70 to 1.30 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.70 to 1.10 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.70 to 0.90 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.90 to 1.30 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.95 to 1.15 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 0.90 to 1.20 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 1.00 to 1.15 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present at an amount of about 1.03 to about 1.13 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present in an amount between about 1.05 to 1.10 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present at an amount of about 1.05, 1.06, 1.07, 1.08, 1.09, 1.10, 1.11, or 1.12 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present at an amount of about 1.06 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present at an amount of about 1.08 equivalents relative to compound 5, or a salt thereof. In some embodiments, compound 7, or a salt thereof, is present at an amount of about 1.10 equivalents relative to compound 5, or a salt thereof.

In some embodiments, reaction conditions comprise a solvent. In some embodiments, the solvent is or comprises acetone, acetonitrile, 1,2-dimethoxyethane, 1,4-dioxane, N,N-dimethylformamide, N,N-dimethylacetamide, N-methylpyrrolidinone, dimethylsulfoxide, ethyl acetate, isopropyl acetate, pyridine, sulfolane, tetrahydrofuran, 2-methyltetrahydrofuran, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, tert-butyl methyl ether, or cyclopentyl methyl ether. In some embodiments, the solvent is or comprises acetone. In some embodiments, the solvent is or comprises acetonitrile. In some embodiments, the solvent is or comprises 1,2-dimethoxyethane. In some embodiments, the solvent is or comprises N,N-dimethylformamide. In some embodiments, the solvent is or comprises N,N-dimethylacetamide. In some embodiments, the solvent is or comprises N-methylpyrrolidinone. In some embodiments, the solvent is or comprises dimethylsulfoxide. In some embodiments, the solvent is or comprises ethyl acetate. In some embodiments, the solvent is or comprises isopropyl acetate. In some embodiments, the solvent is or comprises pyridine. In some embodiments, the solvent is or comprises sulfolane. In some embodiments, the solvent is or comprises tetrahydrofuran. In some embodiments, the solvent is or comprises 2-methyltetrahydrofuran. In some embodiments, the solvent is or comprises toluene. In some embodiments, the solvent is or comprises benzene. In some embodiments, the solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the solvent is or comprises chlorobenzene. In some embodiments, the solvent is or comprises xylene. In some embodiments, the solvent is or comprises tert-butyl methyl ether. In some embodiments, the solvent is or comprises cyclopentyl methyl ether. In some embodiments, the solvent is or comprises 1,4-dioxane.

In some embodiments, the solvent is present in an amount between about 1.0 L/kg and about 20.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 20.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 15.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 12.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 10.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 8.0 L/kg. In some embodiments, the solvent is present in an amount between about 5.0 L/kg and about 6.0 L/kg. In some embodiments, the solvent is present in an amount between about 1.0 L/kg and about 5.0 L/kg. In some embodiments, the solvent is present in an amount between about 2.0 L/kg and about 5.0 L/kg. In some embodiments, the solvent is present in an amount between about 3.0 L/kg and about 5.0 L/kg. In some embodiments, the solvent is present in an amount between about 4.0 L/kg and about 5.0 L/kg. In some embodiments, the solvent is present in an amount between about 2.0 L/kg and about 8.0 L/kg. In some embodiments, the solvent is present in an amount between about 3.0 L/kg and about 7.0 L/kg. In some embodiments, the solvent is present in an amount between about 4.0 L/kg and about 6.0 L/kg. In some embodiments, the solvent is present in an amount of about 1.0 L/kg, 2.0 L/kg, 3.0 L/kg, 4.0 L/kg, 5.0 L/kg, 6.0 L/kg, 7 L/kg, 8 L/kg, 9 L/kg, or 10 L/kg. In some embodiments, the solvent is present in an amount of about 5.0 L/kg. In some embodiments, the solvent is present in an amount of about 4.0 L/kg. In some embodiments, the solvent is present in an amount of about 6.0 L/kg.

In some embodiments, reaction conditions comprise heating to a temperature T_7. In embodiments, the temperature T_7 is between about 50° C. and about 150° C. In embodiments, the temperature T_7 is between about 50° C. and about 130° C. In embodiments, the temperature T_7 is between about 50° C. and about 110° C. In embodiments, the temperature T_7 is between about 50° C. and about 100° C. In embodiments, the temperature T_7 is between about 60° C. and about 150° C. In embodiments, the temperature T_7 is between about 80° C. and about 150° C. In embodiments, the temperature T_7 is between about 100° C. and about 150° C. In some embodiments, the temperature T_7 is about 50° C., 55° C., 60° C., 65° C., 70° C., 75° C., 80° C., 85° C., 90° C., 95° C., 105° C., 110° C., 115° C., 120° C., 125° C., 130° C., 135° C., 140° C., 145° C., or 150° C. In some embodiments, the temperature T_7 is about 90° C. In some embodiments, the temperature T_7 is about 95° C. In some embodiments, the temperature T_7 is about 100° C. In some embodiments, the temperature T_7 is about 105° C. In some embodiments, the temperature T_7 is about 110° C. In some embodiments, the temperature T_7 is about 115° C. In some embodiments, the temperature T_7 is about 120° C.

In some embodiments, reaction conditions comprise maintaining the reaction at temperature T_7 for a period of time. In some embodiments, the period of time is at least 12 hrs. In some embodiments, the period of time is at least 24 hrs. In some embodiments, the period of time is at least 48 hrs. In some embodiments, the period of time is between about 2 hrs and about 20 hrs. In some embodiments, the period of time is between about 3 hrs and about 20 hrs. In some embodiments, the period of time is between about 4 hrs and about 20 hrs. In some embodiments, the period of time is between about 5 hrs and about 20 hrs. In some embodiments, the period of time is between about 7.5 hrs and about 20 hrs. In some embodiments, the period of time is between about 10 hrs and about 20 hrs. In some embodiments, the period of time is between about 11 hrs and about 20 hrs. In some embodiments, the period of time is between about 12 hrs and about 20 hrs. In some embodiments, the period of time is between about 13 hrs and about 20 hrs. In some embodiments, the period of time is between about 14 hrs and about 20 hrs. In some embodiments, the period of time is between about 14 hrs and about 19 hrs. In some embodiments, the period of time is between about 14 hrs and about 18 hrs. In some embodiments, the period of time is between about 14 hrs and about 17 hrs. In some embodiments, the period of time is between about 14 hrs and about 16 hrs. In some embodiments, the period of time is between about 14 hrs and about 15 hrs.

In some embodiments, e.g., after the reaction is performed, Step 1 further comprises (b) cooling the reaction mixture.

In some embodiments, after heating to a temperature T_7, the reaction conditions further comprise cooling to a temperature T_8. In some embodiments, the temperature T_8 is between about 50° C. and about 110° C. In some embodiments, the temperature T_8 is between about 50° C. and about 100° C. In some embodiments, the temperature T_8 is between about 50° C. and about 90° C. In some embodiments, the temperature T_8 is between about 50° C. and about 85° C. In some embodiments, the temperature T_8 is between about 50° C. and about 80° C. In some embodiments, the temperature T_8 is between about 65° C. and about 110° C. In some embodiments, the temperature T_8 is between about 75° C. and about 110° C. In some embodiments, the temperature T_8 is between about 60° C. and about 100° C. In some embodiments, the temperature T_8 is between about 70° C. and about 90° C. In some embodiments, the temperature T_8 is between about 75° C. and about 85° C. In some embodiments, the temperature T_8 is between about 75° C. and about 80° C. In some embodiments, the temperature T_8 is between about 80° C. and about 85° C. In some embodiments, the temperature T_8 is about 50° C., 55° C., 60° C., 65° C., 70° C., 75° C., 80° C., 85° C., 90° C., 95° C., 105° C., or 110° C. In some embodiments, the temperature T_8 is about 80° C. In some embodiments, the temperature T_8 is about 75° C. In some embodiments, the temperature T_8 is about 85° C.

In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 20° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 17.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 15° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 12.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 10° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 2° C./hr and about 7.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 5° C./hr and about 20° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 7.5° C./hr and about 20° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 10° C./hr and about 20° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 4° C./hr and about 17.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 4° C./hr and about 15° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 4° C./hr and about 12.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 5° C./hr and about 10° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of between about 6° C./hr and about 8° C./hr. In some embodiments, a reaction mixture is cooled to T_8 at a rate of about 2.0° C./hr, 3.0° C./hr, 4.0° C./hr, 5.0° C./hr, 6.0° C./hr, 7.0° C./hr, 8.0° C./hr, 9.0° C./hr, or 10.0° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of about 6.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of about 6.5° C./hr. In some embodiments, a reaction mixture is cooled to temperature T_8 at a rate of about 7.0° C./hr.

In some embodiments, a reaction (e.g., presence of compound 5 or compound 2) is monitored for completion by HPLC. In some embodiments, a reaction is monitored for presence of compound 5 or compound 2 (e.g., a/a by HPLC).

In some embodiments, a reaction is determined to be complete when less than 25% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 10% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 5% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 4% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 3% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 2% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC). In some embodiments, a reaction is determined to be complete when less than 1% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC).

In some embodiments, Step 1 further comprises a step of (c) crystallization of compound 2, or a salt thereof (e.g., from the reaction mixture). In some embodiments, the present disclosure provides the recognition that a crystallization may provide a higher yield and/or purity of compound 2, or a salt thereof, and/or provide compound 2, or a salt thereof, e.g., with low levels or undetectable levels of byproducts (e.g., compounds 9, 10, 11, or 12, or salts thereof), in particular when the reaction is scaled-up. Without wishing to be bound to a particular theory, it is thought that compound 2, or a salt thereof, crystalizes out at a different temperature than a byproduct (e.g., compounds 9, 10, 11, or 12, or salts thereof). Accordingly, without wishing to be bound to a particular theory, it is thought that performing the crystallization at particular temperatures and/or controlling the cooling rate of the reaction may provide a higher yield and/or purity of compound 2, or a salt thereof, and/or provide compound 2, or a salt thereof, e.g., with reduced or undetectable levels of byproducts (e.g., compounds 9, 10, 11, or 12, or salts thereof).

In some embodiments, crystallization of compound 2, or a salt thereof, comprises providing a primary mixture, wherein the primary mixture comprises compound 2, or a salt thereof, and a primary solvent. In some embodiments, the primary mixture is provided at temperature T_8.

In some embodiments, the primary solvent is or comprises acetone, acetonitrile, dichloromethane, 1,2-dichloroethane, 1,2-dimethoxyethane, 1,4-dioxane, N,N-dimethylformamide, N,N-dimethylacetamide, N-methylpyrrolidinone, dimethylsulfoxide, ethyl acetate, isopropyl acetate, pyridine, sulfolane, tetrahydrofuran, 2-methyltetrahydrofuran, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, tert-butyl methyl ether, or cyclopentyl methyl ether. In some embodiments, the primary solvent is or comprises acetone. In some embodiments, the primary solvent is or comprises acetonitrile. In some embodiments, the primary solvent is or comprises dichloromethane. In some embodiments, the primary solvent is or comprises 1,2-dichloroethane. In some embodiments, the primary solvent is or comprises 1,2-dimethoxyethane. In some embodiments, the primary solvent is or comprises N,N-dimethylformamide. In some embodiments, the primary solvent is or comprises N,N-dimethylacetamide. In some embodiments, the primary solvent is or comprises N-methylpyrrolidinone. In some embodiments, the primary solvent is or comprises dimethylsulfoxide. In some embodiments, the primary solvent is or comprises ethyl acetate. In some embodiments, the primary solvent is or comprises isopropyl acetate. In some embodiments, the solvent is or comprises pyridine. In some embodiments, the primary solvent is or comprises sulfolane. In some embodiments, the solvent is or comprises tetrahydrofuran. In some embodiments, the primary solvent is or comprises 2-methyltetrahydrofuran. In some embodiments, the primary solvent is or comprises toluene. In some embodiments, the primary solvent is benzene. In some embodiments, the primary solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the primary solvent is or comprises chlorobenzene. In some embodiments, the primary solvent is or comprises xylene. In some embodiments, the primary solvent is or comprises tert-butyl methyl ether. In some embodiments, the primary solvent is or comprises cyclopentyl methyl ether. In some embodiments, the primary solvent is or comprises 1,4-dioxane.

In some embodiments, the primary solvent is present at an amount of about 1.0 L/kg to 10.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 1.0 L/kg to 8.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 1.0 L/kg to 6.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 1.0 L/kg to 4.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 2.0 L/kg to 10.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 4.0 L/kg to 10 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 6.0 L/kg to 10 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 2.0 L/kg to 8.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 4.0 L/kg to 6.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 1.0 L/kg, 2.0 L/kg, 3.0 L/kg, 4.0 L/kg, 5.0 L/kg, 6.0 L/kg, 7.0 L/kg, 8.0 L/kg, 9.0 L/kg, or 10.0 L/kg. In some embodiments, the primary solvent is present at an amount of about 3.0 L/kg, 4.0 L/kg, 5.0 L/kg, 6.0, or 7.0 L/kg. In some embodiments, the primary solvent is present at an amount of about 3.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 4.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 4.9 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 5.0 L/kg of solute. In some embodiments, the primary solvent is present at an amount of about 6.0 L/kg of solute.

In some embodiments, crystallization of compound 2, or a salt thereof, comprises adding a seed crystal to the primary mixture. In some embodiments, the seed crystal is added in an amount between about 0.01 wt % and about 5.0 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 5.0 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 4.0 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 3.0 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 2.0 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 1.5 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.10 wt % and about 1.00 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 0.75 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 0.50 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 0.40 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 0.30 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.1 wt % and about 0.25 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.20 wt % and about

1.00 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.25 wt % and about 1.00 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.50 wt % and about 1.00 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.10 wt %, 0.15 wt %, 0.20 wt %, 0.25 wt %, 0.30 wt %, 0.35 wt %, 0.40 wt %, 0.45 wt %, or 0.50 wt % of compound 5, or a salt thereof.

In some embodiments, the seed crystal is added in an amount of about 0.05 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.10 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.15 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.20 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.25 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.30 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.35 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.45 wt % of compound 5, or a salt thereof. In some embodiments, the seed crystal is compound 2, or a salt thereof.

In some embodiments, crystallization of compound 2, or a salt thereof, comprises cooling the primary mixture to a temperature T_9. In some embodiments, temperature T_9 is between about 0° C. and about 50° C. In some embodiments, temperature T_9 is between about 0° C. and about 45° C. In some embodiments, temperature T_9 is between about 0° C. and about 40° C. In some embodiments, temperature T_9 is between about 0° C. and about 35° C. In some embodiments, temperature T_9 is between about 0° C. and about 30° C. In some embodiments, temperature T_9 is between about 0° C. and about 25° C. In some embodiments, temperature T_9 is between about 10° C. and about 50° C. In some embodiments, temperature T_9 is between about 15° C. and about 50° C. In some embodiments, temperature T_9 is between about 20° C. and about 50° C. In some embodiments, temperature T_9 is between about 25° C. and about 50° C. In some embodiments, temperature T_9 is between about 15° C. and about 40° C. In some embodiments, temperature T_9 is between about 20° C. and about 35° C. In some embodiments, temperature T_9 is between about 18° C. and about 32° C. In some embodiments, temperature T_9 is between about 20° C. and about 35° C. In some embodiments, temperature T_9 is between about 25° C. and about 30° C. In some embodiments, temperature T_9 is between about 26° C. and about 28° C. In some embodiments, the temperature T_9 is about 22° C., 23° C., 24° C., 25° C., 26° C., 27° C., 28° C., 29° C., 30° C., 31° C., or 32° C. In some embodiments, the temperature T_9 is about 24° C. In some embodiments, the temperature T_9 is about 25° C. In some embodiments, the temperature T_9 is about 26° C. In some embodiments, the temperature T_9 is about 27° C. In some embodiments, the temperature T_9 is about 29° C. In some embodiments, the temperature T_9 is about 30° C. In some embodiments, the temperature T_9 is about 31° C.

In some embodiments, the primary mixture is cooled at a rate of between about 2.0° C./hr and about 20° C./hr. In some embodiments, the primary mixture is cooled to temperature T_9 at a rate of between about 4.0° C./hr and about 20° C./hr. In some embodiments, the reaction mixture is cooled to

temperature T_9 at a rate of between about 6° C./hr and about 20° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 8° C./hr and about 20° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 2.0° C./hr and about 15° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 2.0° C./hr and about 10° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 2.0° C./hr and about 8.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 4.0° C./hr and about 12.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 5.0° C./hr and about 12.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 5.0° C./hr and about 10.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 5.0° C./hr and about 8.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of between about 6.0° C./hr and about 7.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of about 2.0° C./hr, 3.0° C./hr, 4.0° C./hr, 5.0° C./hr, 6.0° C./hr, 7.0° C./hr, 8.0° C./hr, 9.0° C./hr, or 10.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of about 5.5° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of about 6.0° C./hr. In some embodiments, the primary mixture is cooled to temperature T_9 at a rate of about 6.5° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of about 7.0° C./hr. In some embodiments, the reaction mixture is cooled to temperature T_9 at a rate of about 7.5° C./hr.

In some embodiments, Step 1 further comprises a step of (d) recrystallization of compound 2, or a salt thereof (e.g., from the reaction mixture or the primary mixture). In some embodiments, the present disclosure provides the recognition that a recrystallization may provide a higher yield and/or purity of compound 2, or a salt thereof, and/or provide compound 2, or a salt thereof, e.g., with reduced or undetectable levels of byproducts (e.g., compounds 9, 10, 11, and/or 12, or salts thereof), in particular when the reaction is scaled-up. In some embodiments, recrystallization conditions provided herein may provide a higher yield and/or purity of compound 2, or a salt thereof, and/or provide compound 2, or a salt thereof, e.g., with reduced or undetectable levels of byproducts (e.g., compounds 9, 10, 11, and/or 12, or salts thereof).

In some embodiments, recrystallization of compound 2, or a salt thereof, comprises (i) providing a primary mixture, wherein the primary mixture comprises compound 2, or a salt thereof, and a primary solvent, wherein the primary solvent is as described above and herein.

In some embodiments, the primary mixture is provided at the temperature T_9, wherein temperature T_9 is as described above and herein.

In some embodiments, the primary solvent is as described above and herein.

In some embodiments, recrystallization of compound 2, or a salt thereof, further comprises (ii) adding a secondary solvent to the primary mixture to form a secondary mixture. In some embodiments, the secondary solvent is or comprises chloroform, hexane, pentane, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the secondary solvent is or comprises chloroform. In some embodiments, the secondary solvent is or comprises hexane. In some embodiments, the secondary solvent is or comprises pentane. In some embodiments, the secondary solvent is or comprises diethyl ether. In some embodiments, the secondary solvent is or comprises ethyl acetate. In some embodiments, the secondary solvent is or comprises isopropyl acetate. In some embodiments, the secondary solvent is or comprises tert-butyl methyl ether. In some embodiments, the secondary solvent is or comprises cyclopentyl methyl ether. In some embodiments, the secondary solvent is or comprises toluene. In some embodiments, the secondary solvent is or comprises benzene. In some embodiments, the secondary solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the secondary solvent is or comprises chlorobenzene. In some embodiments, the secondary solvent is or comprises xylene. In some embodiments, the secondary solvent is or comprises dichloromethane.

In some embodiments, the secondary solvent is present at an amount between about 1 L/kg and about 20 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 2 L/kg and about 20 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 6 L/kg and about 20 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 10 L/kg and about 20 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 2 L/kg and about 18 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 2 L/kg and about 14 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 2 L/kg and about 10 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 4 L/kg and about 18 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 6 L/kg and about 14 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 8 L/kg and about 12 L/kg of solute. In some embodiments, the secondary solvent is present at an amount between about 9 L/kg and about 11 L/kg of solute. In some embodiments, the secondary solvent is present at an amount of about 5 L/kg, 6 L/kg, 7 L/kg, 8 L/kg, 9 L/kg, 10 L/kg, 11 L/kg, 12 L/kg, 13 L/kg, 14 L/kg, or 15 L/kg, of solute. In some embodiments, the secondary solvent is present at an amount of about 8 L/kg of solute. In some embodiments, the secondary solvent is present at an amount of about 9 L/kg of solute. In some embodiments, the secondary solvent is present at an amount of about 10 L/kg of solute. In some embodiments, the secondary solvent is present at an amount of about 11 L/kg of solute. In some embodiments, the secondary solvent is present at an amount of about 12 L/kg of solute.

In some embodiments, recrystallization of compound 2, or a salt thereof, further comprises (iii) holding and optionally agitating the secondary mixture at temperature T_9 for a period of time. In some embodiments, the secondary mixture is agitated at temperature T_9 for the period of time. In some embodiments, the period of time is at least 1 hr. In some embodiments, the period of time is at least 3 hrs. In some embodiments, the period of time is at least 6 hrs. In some embodiments, the period of time is between about 1 hr and about 8 hrs. In some embodiments, the period of time is between about 2 hrs and about 6 hrs. In some embodiments, the period of time is between about 2 hrs and about 5 hrs. In some embodiments, the period of time is between about 2 hrs and about 4 hrs. In some embodiments, the period of time is about 1 hr, 2 hrs, 3, hrs, 4 hrs, or 5 hrs. In some embodiments, the period of time is about 1 hrs. In some embodiments, the period of time is about 2 hrs. In some embodiments, the period of time is about 3 hrs. In some embodiments, the period of time is about 4 hrs. In some embodiments, the period of time is about 5 hrs.

In some embodiments, Step 1 further comprises a step of (e) filtering (e.g., the reaction mixture, primary mixture, or secondary mixture) to provide a filtered mixture.

In some embodiments, the secondary mixture is filtered at a temperature T_{10}. In some embodiments, temperature T_{10} is between about 0° C. and about 60° C. In some embodiments, temperature T_{10} is between about 0° C. and about 50° C. In some embodiments, temperature T_{10} is between about 0° C. and about 40° C. In some embodiments, temperature T_{10} is between about 0° C. and about 30° C. In some embodiments, temperature T_{10} is between about 10° C. and about 60° C. In some embodiments, temperature T_{10} is between about 20° C. and about 60° C. In some embodiments, temperature T_{10} is between about 10° C. and about 50° C. In some embodiments, temperature T_{10} is between about 15° C. and about 45° C. In some embodiments, temperature T_{10} is between about 20° C. and about 40° C. In some embodiments, temperature T_{10} is between about 25° C. and about 35° C. In some embodiments, temperature T_{10} is between about 24° C. and about 32° C. In some embodiments, temperature T_{10} is between about 26° C. and about 28° C. In some embodiments, temperature T_{10} is about 20° C., 21° C., 22° C., 23° C., 24° C., 25° C., 26° C., 27° C., 28° C., 29° C., 30° C., 31° C., or 32° C. In some embodiments, temperature T_{10} is about 24° C. In some embodiments, temperature T_{10} is about 26° C. In some embodiments, temperature T_{10} is about 27° C. In some embodiments, temperature T_{10} is about 28° C. In some embodiments, temperature T_{10} is about 30° C.

In some embodiments, Step 1 further comprises a step of (f) washing (e.g. the filtered mixture or secondary mixture), to provide a final mixture.

In some embodiments, washing comprises wash solvent 1. In some embodiments, wash solvent 1 comprises adding a wash solvent 1 to, e.g., the filtered mixture. In some embodiments, the wash solvent 1 is or comprises acetone, acetonitrile, dichloromethane, 1,2-dichloroethane, 1,2-dimethoxyethane, 1,4-dioxane, N,N-dimethylformamide, N,N-dimethylacetamide, N-methylpyrrolidinone, dimethylsulfoxide, ethyl acetate, isopropyl acetate, pyridine, sulfolane, tetrahydrofuran, 2-methyltetrahydrofuran, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, tert-butyl methyl ether, or cyclopentyl methyl ether. In some embodiments, wash solvent 1 is or comprises acetone. In some embodiments, wash solvent 1 is or comprises acetonitrile. In some embodiments, wash solvent 1 is or comprises dichloromethane. In some embodiments, wash solvent 1 is or comprises 1,2-dichloroethane. In some embodiments, wash solvent 1 is or comprises 1,2-dimethoxyethane. In some embodiments, wash solvent 1 is or comprises N,N-dimethylformamide. In some embodiments, wash solvent 1 is or comprises N,N-dimethylacetamide. In some embodiments, wash solvent 1 is or comprises N-methylpyrrolidinone. In some embodiments, wash solvent 1 is or comprises dimethylsulfoxide. In some embodiments, wash solvent 1 is or comprises ethyl acetate. In some embodiments, wash solvent 1 is or comprises isopropyl acetate. In some embodiments, wash solvent 1 is or comprises pyridine. In some embodiments, wash solvent 1 is or comprises sulfolane. In some embodiments, wash solvent 1 is or comprises tetrahydrofuran. In some embodiments, wash solvent 1 is or comprises 2-methyltetrahydrofuran. In some embodiments, wash solvent 1 is or comprises toluene. In some embodiments, wash solvent 1 is or comprises benzene. In some embodiments, wash solvent 1 is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent 1 is or comprises chlorobenzene. In some embodiments, wash solvent 1 is or comprises xylene. In some embodiments, wash solvent 1 is or comprises tert-butyl methyl ether. In some embodiments, wash solvent 1 is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent 1 is or comprises 1,4-dioxane. In some embodiments, wash solvent 1 is or comprises dichloromethane. In some embodiments, wash solvent 1 comprises 1,4-dioxane and dichloromethane.

In some embodiments, wash solvent 1 is in an amount between 0.30 L/kg and 5.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 1.0 L/kg and 5.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 1.50 L/kg and 5.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 2.50 L/kg and 5.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 0.30 L/kg and 4.50 L/kg. In some embodiments, wash solvent 1 is in an amount between 0.50 L/kg and 3.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 0.50 L/kg and 2.50 L/kg. In some embodiments, wash solvent 1 is in an amount between 0.50 L/kg and 2.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 1.50 L/kg and 4.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 2.25 L/kg and 3.00 L/kg. In some embodiments, wash solvent 1 is in an amount between 2.50 L/kg and 2.75 L/kg. In some embodiments, wash solvent 1 is in an amount of about 0.50 L/kg, 1.05 L/kg, 1.65 L/kg, 2.00 L/kg, 2.25 L/kg, 2.50 L/kg, or 2.70 L/kg. In some embodiments, wash solvent 1 is in an amount of about 0.50 L/kg. In some embodiments, wash solvent 1 is in an amount of about 1.65 L/kg. In some embodiments, wash solvent 1 is in an amount of about 2.00 L/kg. In some embodiments, wash solvent 1 is in an amount of about 2.25 L/kg. In some embodiments, wash solvent 1 is in an amount of about 2.50 L/kg. In some embodiments, wash solvent 1 is in an amount of about 2.75 L/kg.

In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.4 L/kg and about 1.5 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.4 L/kg and about 1.3 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.4 L/kg and about 1.0 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.6 L/kg and about 1.5 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.8 L/kg and about 1.5 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.6 L/kg and about 1.3 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.8 L/kg and about 0.9 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.60 L/kg, 0.70 L/kg, 0.80 L/kg, 0.90 L/kg, or 1.00 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.60 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.70 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.80 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.90 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 1.00 L/kg.

In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.0 L/kg and about 3.2 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.0 L/kg and about 3.0 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.0 L/kg and about 2.5 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.0 L/kg and about 2.0 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.5 L/kg and about 3.0 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.25 L/kg and about 2.0 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.5 L/kg and about 1.7 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.35 L/kg, 1.45 L/kg, 1.55 L/kg, 1.65 L/kg, 1.75 L/kg, or 1.85 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.35 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.45 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.55 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.64 L/kg. In some embodiments, wash solvent 1 comprises dichloromethane in an amount between about 1.75 L/kg.

In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.4 L/kg and about 1.5 L/kg, and dichloromethane in an amount between about 1.0 L/kg and about 3.2 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount between about 0.8 L/kg and about 0.9 L/kg, and dichloromethane in an amount between about 1.5 L/kg and about 1.7 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.60 L/kg, 0.70 L/kg, 0.80 L/kg, 0.90 L/kg, or 1.00 L/kg, and dichloromethane in an amount of about 1.40 L/kg, 1.50 L/kg, 1.60 L/kg, 1.70 L/kg, or 1.80 L/kg. In some embodiments, wash solvent 1 comprises 1,4-dioxane in an amount of about 0.86 L/kg and dichloromethane in an amount of about 1.64 L/kg.

In some embodiments, wash solvent 1 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 1 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 1 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 1 is provided at between about 10° C. and 50° C. In some embodiments, wash solvent 1 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 1 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 1 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 1 is provided at between about 18° C. and 32° C. In some embodiments, wash solvent 1 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 1 is provided at between about 23° C. and 27° C. In some embodiments, wash solvent 1 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 1 is provided at or about 23° C. In some embodiments, wash solvent 1 is provided at or about 25° C. In some embodiments, wash solvent 1 is provided at or about 27° C. In some embodiments, Wash 1 comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 1 further comprises removing wash solvent 1.

In some embodiments, washing comprises Wash 2. In some embodiments, Wash 2 comprises adding a wash solvent 2 to, e.g., the filtered mixture. In some embodiments, wash solvent 2 is or comprises chloroform, hexane, pentane, 1,4-dioxane, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, wash solvent 2 is or comprises chloroform. In some embodiments, wash solvent 2 is or comprises hexane. In some embodiments, wash solvent 2 is or comprises pentane. In some embodiments, wash solvent 2 is or comprises diethyl ether. In some embodiments, wash solvent 2 is or comprises ethyl acetate. In some embodiments, wash solvent 2 is or comprises isopropyl acetate. In some embodiments, wash solvent 2 is or comprises tert-butyl methyl ether. In some embodiments, wash solvent 2 is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent 2 is or comprises toluene. In some embodiments, wash solvent 2 is or comprises benzene. In some embodiments, wash solvent 2 is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent 2 is or comprises chlorobenzene. In some embodiments, wash solvent 2 is or comprises xylene. In some embodiments, wash solvent 2 is or comprises 1,4-dioxane.

In some embodiments, wash solvent 2 is provided in an amount between about 0.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 0.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 0.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 1.0 L/kg and 5.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 2.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 1.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 1.0 L/kg and 3.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 2.0 L/kg and 2.5 L/kg. In some embodiments, wash solvent 2 is provided in an amount between about 2.3 L/kg and 2.7 L/kg. In some embodiments, wash solvent 2 is provided in an amount of about 1.0 L/kg, 1.5 L/kg, 2.0 L/kg, 2.5 L/kg, 3.0, 3.5, 4.0, 4.5, or 5.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount of about 2.0 L/kg. In some embodiments, wash solvent 2 is provided in an amount of about 2.5 L/kg. In some embodiments, wash solvent 2 is provided in an amount of about 3.0 L/kg.

In some embodiments, wash solvent 2 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 2 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 2 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 2 is provided at between about 10° C. and 50° C. In some embodiments, wash solvent 2 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 2 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 2 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 2 is provided at between about 18° C. and 32° C. In some embodiments, wash solvent 2 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 2 is provided at between about 23° C. and 27° C. In some embodiments, wash solvent 2 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 2 is provided at or about 23° C. In some embodiments, wash solvent 2 is provided at or about 25° C. In some embodiments, wash solvent 2 is provided at or about 27° C. In some embodiments, Wash 2 further comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 2 further comprises removing wash solvent 2.

In some embodiments, washing comprises Wash 3 or Wash 4. In some embodiments, Wash 3 or Wash 4 is the same as Wash 2. In some embodiments, Wash 3 comprises adding a wash solvent 3 to, e.g., the filtered mixture. In some embodiments, wash solvent 3 is or comprises chloroform, hexane, pentane, 1,4-dioxane, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, chlorobenzene, or dichloromethane. In some embodiments, wash solvent 3 is or comprises chloroform. In some embodiments, wash solvent 3 is or comprises hexane. In some embodiments, wash solvent 3 is or comprises pentane. In some embodiments, wash solvent 3 is or comprises diethyl ether. In some embodiments, wash solvent 3 is or comprises ethyl acetate. In some embodiments, wash solvent 3 is or comprises isopropyl acetate. In some embodiments, wash solvent 3 is or comprises tert-butyl methyl ether. In some embodiments, wash solvent 3 is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent 3 is or comprises toluene. In some embodiments, wash solvent 3 is or comprises benzene. In some embodiments, wash solvent 3 is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent 3 is or comprises xylene. In some embodiments, wash solvent 3 is or comprises chlorobenzene. In some embodiments, wash solvent 3 is or comprises 1,4-dioxane.

In some embodiments, wash solvent 3 is provided in an amount between about 0.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 0.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 0.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 1.0 L/kg and 5.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 2.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 1.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 1.0 L/kg and 3.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 1.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 2.0 L/kg and 2.8 L/kg. In some embodiments, wash solvent 3 is provided in an amount between about 2.3 L/kg and 2.7 L/kg. In some embodiments, wash solvent 3 is provided in an amount of about 1.0 L/kg, 1.5 L/kg, 2.0 L/kg, 2.5 L/kg, 3.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount of about 2.0 L/kg. In some embodiments, wash solvent 3 is provided in an amount of about 2.5 L/kg. In some embodiments, wash solvent 3 is provided in an amount of about 3.0 L/kg.

In some embodiments, wash solvent 3 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 3 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 3 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 3 is provided at between about 10° C. and 50° C. In some embodiments, wash solvent 3 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 3 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 3 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 3 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 3 is provided at between about 22° C. and 28° C. In some embodiments, wash solvent 3 is provided at between about 24° C. and 26° C. In some embodiments, wash solvent 3 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 3 is provided at or about 25° C. In some embodiments, Wash 3 further comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 3 further comprises removing wash solvent 3.

In some embodiments, Wash 4 comprises adding a wash solvent 4 to, e.g., the filtered mixture. In some embodiments, wash solvent 4 is or comprises chloroform, hexane, pentane, 1,4-dioxane, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, wash solvent 4 is or comprises chloroform. In some embodiments, wash solvent 4 is or comprises hexane. In some embodiments, wash solvent 4 is or comprises pentane. In some embodiments, wash solvent 4 is or comprises diethyl ether. In some embodiments, wash solvent 4 is or comprises ethyl acetate. In some embodiments, wash solvent 4 is or comprises isopropyl acetate. In some embodiments, wash solvent 4 is or comprises tert-butyl methyl ether. In some embodiments, wash solvent 4 is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent 4 is or comprises toluene. In some embodiments, wash solvent 4 is or comprises benzene. In some embodiments, wash solvent 4 is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent 4 is or comprises chlorobenzene. In some embodiments, wash solvent 4 is or comprises xylene. In some embodiments, wash solvent 4 is or comprises 1,4-dioxane.

In some embodiments, wash solvent 4 is provided in an amount between about 0.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 0.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 0.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 1.0 L/kg and 5.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 2.5 L/kg and 5.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 1.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 1.0 L/kg and 3.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 1.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 2.0 L/kg and 2.8 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 2.3 L/kg and 2.7 L/kg. In some embodiments, wash solvent 4 is provided in an amount between about 2.4 L/kg and 2.5 L/kg. In some embodiments, wash solvent 4 is provided in an amount of about 1.0 L/kg, 1.5 L/kg, 2.0 L/kg, 2.5 L/kg, 3.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount of about 2.0 L/kg. In some embodiments, wash solvent 4 is provided in an amount of about 2.5 L/kg. In some embodiments, wash solvent 4 is provided in an amount of about 3.0 L/kg.

In some embodiments, wash solvent 4 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 4 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 4 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 4 is provided at between about 10° C. and 20° C. In some embodiments, wash solvent 4 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 4 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 4 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 4 is provided at between about 18° C. and 32° C. In some embodiments, wash solvent 4 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 4 is provided at between about 23° C. and 27° C. In some embodiments, wash solvent 4 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 4 is provided at or about 25° C. In some embodiments, Wash 4 further comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 4 further comprises removing wash solvent 4.

In some embodiments, washing comprises Wash 5. In some embodiments, Wash 5 comprises adding a wash solvent 5 to, e.g., the filtered mixture. In some embodiments, wash solvent 5 is or comprises chloroform, hexane, pentane, 1,4-dioxane, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, wash solvent 5 is or comprises chloroform. In some embodiments, wash solvent 5 is or comprises hexane. In some embodiments, wash solvent 5 is or comprises pentane. In some embodiments, wash solvent 5 is or comprises diethyl ether. In some embodiments, wash solvent 5 is or comprises ethyl acetate. In some embodiments, wash solvent 5 is or comprises isopropyl acetate. In some embodiments, wash solvent 5 is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent 5 is or comprises toluene. In some embodiments, wash solvent 5 is or comprises benzene. In some embodiments, wash solvent 5 is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent 5 is or comprises chlorobenzene. In some embodiments, wash solvent 5 is or comprises xylene. In some embodiments, wash solvent 5 is or comprises tert-butyl methyl ether.

In some embodiments, wash solvent 5 is provided at between 0.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 5 is provided at between 0.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 5 is provided at between 0.5 L/kg and 2.0 L/kg. In some embodiments, wash solvent 5 is provided at between 1.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 5 is provided at between 1.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 5 is provided at between 2.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 5 is provided at between 1.5 L/kg and 2.5 L/kg. In some embodiments, wash solvent 5 is provided at between 1.8 L/kg and 2.2 L/kg. In some embodiments, wash solvent 5 is provided in an amount of about 1.0 L/kg, 1.5 L/kg, 2.0 L/kg, 2.5 L/kg, 3.0, 3.5, or 4.0 L/kg. In some embodiments, wash solvent 5 is provided in an amount of about 1.5 L/kg. In some embodiments, wash solvent 5 is provided in an amount of about 2.0 L/kg. In some embodiments, wash solvent 5 is provided in an amount of about 2.5 L/kg.

In some embodiments, wash solvent 5 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 5 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 5 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 5 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 5 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 5 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 5 is provided at between about 18° C. and 32° C. In some embodiments, wash solvent 5 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 5 is provided at between about 23° C. and 37° C. In some embodiments, wash solvent 5 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 5 is provided at or about 25° C. In some embodiments, Wash 5 further comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 5 further comprises removing wash solvent 5.

In some embodiments, washing further comprises Wash 6. In some embodiments, Wash 6 comprises adding a wash solvent 6 to, e.g., the filtered mixture. In some embodiments, wash solvent 6 is or comprises pentane, hexane, or heptane. In some embodiments, wash solvent 6 is or comprises n-pentane, n-hexane, or n-heptane. In some embodiments, wash solvent 6 is or comprises pentane. In some embodiments, wash solvent 6 is or comprises n-pentane. In some embodiments, wash solvent 6 is or comprises hexane. In some embodiments, wash solvent 6 is or comprises n-hexane. In some embodiments, wash solvent 6 is or comprises heptane. In some embodiments, wash solvent 6 is or comprises n-heptane.

In some embodiments, wash solvent 6 is provided at between 0.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 6 is provided at between 0.5 L/kg and 3.0 L/kg. In some embodiments, wash solvent 6 is provided at between 1.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 6 is provided at between 1.5 L/kg and 4.0 L/kg. In some embodiments, wash solvent 6 is provided at between 2.0 L/kg and 4.0 L/kg. In some embodiments, wash solvent 6 is provided at between 2.25 L/kg and 3.0 L/kg. In some embodiments, wash solvent 6 is provided at between 2.3 L/kg and 2.75 L/kg. In some embodiments, wash solvent 6 is provided in an amount of about 1.0 L/kg, 1.5 L/kg, 2.0 L/kg, 2.5 L/kg, or 3.0 L/kg. In some embodiments, wash solvent 6 is provided in an amount of about 2.0 L/kg. In some embodiments, wash solvent 6 is provided in an amount of about 2.5 L/kg. In some embodiments, wash solvent 6 is provided in an amount of about 3.0 L/kg.

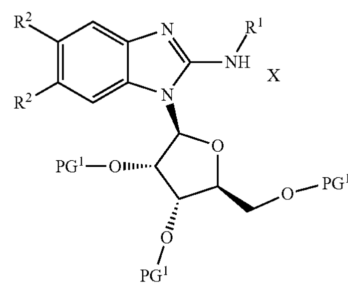
In some embodiments, wash solvent 6 is provided at between about 0° C. and 50° C. In some embodiments, wash solvent 6 is provided at between about 0° C. and 40° C. In some embodiments, wash solvent 6 is provided at between about 0° C. and 30° C. In some embodiments, wash solvent 6 is provided at between about 10° C. and 20° C. In some embodiments, wash solvent 6 is provided at between about 20° C. and 50° C. In some embodiments, wash solvent 6 is provided at between about 30° C. and 50° C. In some embodiments, wash solvent 6 is provided at between about 10° C. and 40° C. In some embodiments, wash solvent 6 is provided at between about 18° C. and 32° C. In some embodiments, wash solvent 6 is provided at between about 20° C. and 30° C. In some embodiments, wash solvent 6 is provided at between about 23° C. and 27° C. In some embodiments, wash solvent 6 is provided at about 20° C., 22° C., 24° C., 25° C., 26° C., 28° C., or 30° C. In some embodiments, wash solvent 6 is provided at or about 25° C. In some embodiments, Wash 6 further comprises agitating, e.g., the filtered mixture. In some embodiments, Wash 6 further comprises removing wash solvent 6.

In some embodiments, Step 1 further comprises a step of (g) deliquoring, drying, and/or isolating compound 2, or a salt thereof.

Step 2

In some embodiments, the present disclosure provides an improved synthesis of compound 3a:

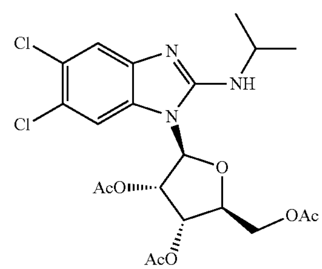
3a



or a salt thereof, wherein R^1, R^2, PG^1, and X are as defined and described herein.

In some embodiments, the present disclosure provides an improved synthesis of compound 3:

3

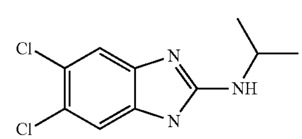


or a salt thereof.

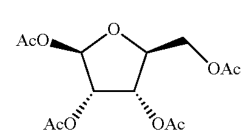
As shown in Schemes 4 and 5, Step 2 is the second of three steps for preparing maribavir as disclosed herein. Any byproducts formed within Step 2 may be present in a provided composition, may further react to form additional byproducts, and/or may lower the overall yield or quality of maribavir. In some embodiments, the present disclosure provides methods of preparing compound 3, or a salt thereof, with reduced and/or low levels of impurities.

In some embodiments, the present disclosure provides compositions comprising compound 3, or a salt thereof, and one or more of the following compounds:

2



8



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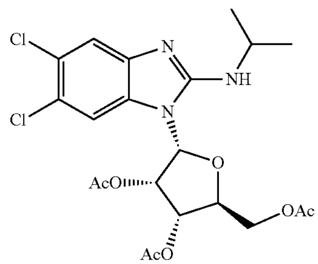
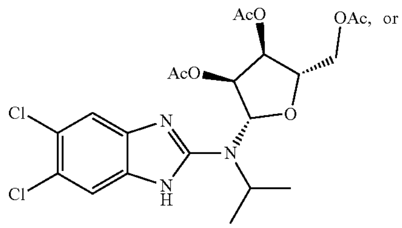
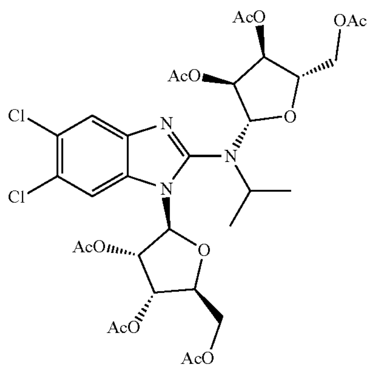
13



14



15



or salts thereof.

It will be appreciated that compound 14 may exist in a mixture of α and β isomers, with the β isomer being predominant. It will also be appreciated that compound 15 can be a mixture of four stereoisomers: β, β isomer; α, β isomer; β, α isomer; and α, α isomer.

In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 2, 8, 13, 14, and/or 15, or salts thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 2, 8, 13, 14, and/or 15, or salts thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir.

In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 13, 14, and/or 15, or salts thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 13, 14, and/or 15, or salts thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir.

In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 2, or a salt thereof. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 2, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 2, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, wherein compound 2, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 8, or a salt thereof. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 8, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 8, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, wherein compound 8, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

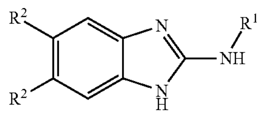
In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 13, or a salt thereof. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 13, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 13, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, wherein compound 13, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 14, or a salt thereof. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 14, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 14, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, wherein compound 14, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR). In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 15, or a salt thereof. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 15, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, and compound 15, or a salt thereof, in an amount of less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC), relative to maribavir. In some embodiments, provided compositions comprise compound 3, or a salt thereof, wherein compound 15, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, such compositions comprising compound 3, or salt thereof are prepared as described herein. In some embodiments, the present disclosure also provides the recognition that certain reagents (and amounts thereof) and/or reaction conditions may provide improved quality compound 3, or a salt thereof (e.g., with higher purity and/or minimal byproducts), and/or improve the yield and/or purity of compound 3, or a salt thereof. In some embodiments, such improvements in yield or quality of compound 3, or a salt thereof, may also improve the yield or quality of maribavir. Without wishing to be bound to a particular theory, the present disclosure provides the recognition that incorporation of a crystallization (e.g., comprising salt formation) into Step 2 may improve the yield and/or reduce the formation of byproducts (e.g., compounds 2, 8, 13, 14, and/or 15, or salts thereof).

In some embodiments, compound 3a, or a salt thereof, is prepared as shown in Step 2 of Scheme 4. In some embodiments, at Step 2, compound 3a, or a salt thereof, is prepared by a method comprising:

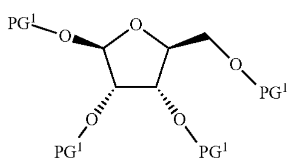
(a) providing compound 2a:



2a

or a salt thereof, wherein

each R^1 and R^2 is as defined above and described herein; and compound 8a:



8a

or a salt thereof, wherein

each PG^1 is independently a suitable oxygen protecting group;

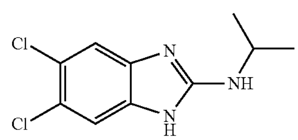
under suitable reaction conditions to afford compound 3a, or a salt thereof.

As generally defined above, each PG^1 is independently a suitable oxygen protecting group. Various methods and conditions for protecting and deprotecting alcohols are known in the chemical arts. For example, methods and conditions for protecting and deprotecting alcohols are well known in the art and include those described in detail in *Protecting Groups in Organic Synthesis*, T. W. Green and P. G. M. Wuts, 3rd edition, John Wiley & Sons, 1999, the entirety of which is incorporated herein by reference. In some embodiments, PG^1 is acetyl (Ac), benzoyl (Bz), benzyl (Bn), β-methoxyethoxymethyl ether (MEM), dimethoxytrityl (DMT), methoxymethyl ether (MOM), methoxytrityl (MMT), p-methoxybenzyl ether (PMB), p-methoxyphenyl ether (PMP), methylthiomethyl ether, pivaloyl (Piv), tetrahydropyranyl (THP), tetrahydrofuran (THF), a silyl ether (e.g., trimethylsilyl (TMS), tert-butyldimethylsilyl (TBS), tri-iso-propylsilyloxymethyl (TOM), and triisopropylsilyl (TIPS) ether), methyl ether, or ethoxyethyl ether. In some embodiments, each PG^1 is the same. In some embodiments, each PG^1 is acetyl (Ac).

In some embodiments, X represents a salt of compound 3a. Suitable salts are well known in the art, e.g., see generally, *March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, M. B. Smith and J. March, 5th Edition, John Wiley & Sons, 2001. In some embodiments, X is camphorsulfonic acid (CSA), benzenesulfonic acid ($PhSO_3H$), methanesulfonic acid (MsOH), toluenesulfonic acid (TsOH), 1,5-naphthalenedisulfonic acid (napsylic acid), 1,2-ethanedisulfonic acid (edisic acid), ethanesulfonic acid (esic acid), 4-chlorobenzenesulfonic acid (closic acid), 6,7-dihydroxycoumarin-4-methanesulfonic acid (cromesic acid), or trifluoromethanesulfonic acid (HOTf) salt. In some embodiments, X is a HF, HCl, HBr, or HI. In some embodiments, X is HCl. In some embodiments, X is HBr.

In some embodiments, compound 3, or a salt thereof, is prepared as shown in Step 2 of Scheme 5. In some embodiments, at Step 2, compound 3, or a salt thereof, is prepared by a method comprising:

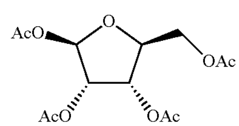
(a) providing compound 2:



2

or a salt thereof, and

compound 8:

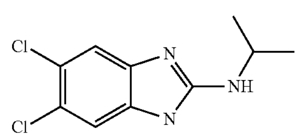


8

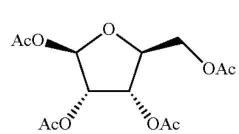
or a salt thereof,

under suitable reaction conditions to afford compound 3, or a salt thereof.

In some embodiments, Step 2 further provides one or more of the following compounds:



2



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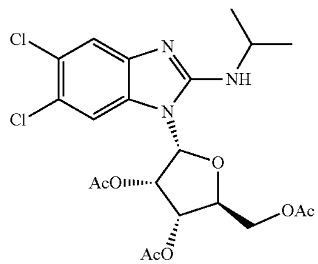
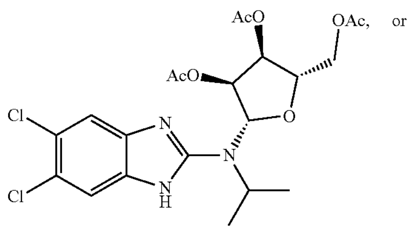
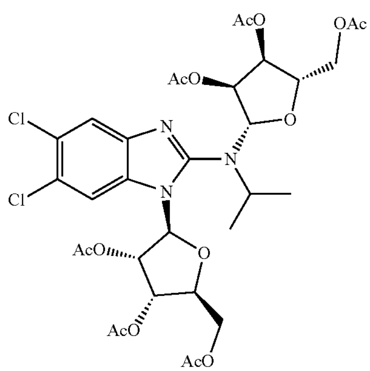


13



14



15

In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 2, relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 0.12%, about 0.1%, about 0.09%, about 0.08%, about 0.07%, about 0.05%, or about 0.02% (w/w HPLC) of compound 2, relative to compound 3. In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 2 relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 0.12%, about 0.1%, about 0.09%, about 0.08%, about 0.07%, about 0.05%, or about 0.02% (a/a HPLC) of compound 2, relative to compound 3. In some embodiments, Step 2 provides a composition wherein compound 2, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 8, relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 8, relative to compound 3. In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 8 relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 8, relative to compound 3. In some embodiments, Step 2 provides a composition wherein compound 8, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 13, relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 13, relative to compound 3. In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 13 relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 13, relative to compound 3. In some embodiments, Step 2 provides a composition wherein compound 13, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 14, relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 14, relative to compound 3. In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 14 relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 14, relative to compound 3. In some embodiments, Step 2 provides a composition wherein compound 14, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (w/w HPLC) of compound 15, relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (w/w HPLC) of compound 15, relative to compound 3. In some embodiments, Step 2 provides a composition comprising less than 7%, 5%, 3%, 2%, 1%, 0.75%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.02% (a/a HPLC) of compound 15 relative to compound 3. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.75%, about 0.5%, about 0.4%, about 0.3%, about 0.2%, about 0.1%, about 0.05%, or about 0.02% (a/a HPLC) of compound 15, relative to compound 3. In some embodiments, Step 2 provides a composition wherein compound 15, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition as provided in any one of Tables 4-7 to 4-11 in Example 4.

In some embodiments, Step 2 provides compound 3 as a camphorsulfonic acid (CSA), benzenesulfonic acid ($PhSO_3H$), methanesulfonic acid (MsOH), toluenesulfonic acid (TsOH), 1,5-naphthalenedisulfonic acid (napsylic acid), 1,2-ethanedisulfonic acid (edisic acid), ethanesulfonic acid (esic acid), 4-chlorobenzenesulfonic acid (closic acid), 6,7-dihydroxycoumarin-4-methanesulfonic acid (cromesic acid), or trifluoromethanesulfonic acid (HOTf) salt. In some embodiments, Step 2 provides compound 3 as a hydrohalide salt. In some embodiments, Step 2 provides compound 3 as an HCl salt. In some embodiments, Step 2 provides compound 3 as an HBr salt.

In some embodiments, compound 2, or a salt thereof, is provided in an amount between about 0.50 and about 1.50 equivalents relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount between about 0.75 and about 1.25 equivalents relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount between about 0.90 and about 1.51 equivalents relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount between about 0.99 and about 1.01 equivalents relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 0.90, 0.95, 1.0, 1.05, or 1.10 equivalent relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 0.90 equivalent relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 0.95 equivalent relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 1.0 equivalent relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 1.05 equivalent relative to compound 8, or a salt thereof. In some embodiments, compound 2, or a salt thereof, is provided in an amount of about 1.10 equivalent relative to compound 8, or a salt thereof.

In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 0.50 and about 2.0 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.00 and about 1.50 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.10 and about 1.40 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.1, 1.2, 1.3, 1.4 or 1.5 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.0 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.1 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.2 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.3 equivalents relative to compound 2, or a salt thereof. In some embodiments, compound 8, or a salt thereof, is provided in an amount between about 1.4 equivalents relative to compound 2, or a salt thereof.

In some embodiments, reaction conditions comprise a solvent. In some embodiments, the solvent is or comprises acetone, acetonitrile, dichloromethane, 1,2-dichloroethane, 1,2-dimethoxyethane, 1,4-dioxane, ethyl acetate, isopropyl acetate, tetrahydrofuran, 2-methyltetrahydrofuran, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, tert-butyl methyl ether, or cyclopentyl methyl ether. In some embodiments, the solvent is or comprises acetone. In some embodiments, the solvent is or comprises acetonitrile. In some embodiments, the solvent is or comprises dichloromethane. In some embodiments, the solvent is or comprises 1,2-dichloroethane. In some embodiments, the solvent is or comprises 1,2-dimethoxyethane. In some embodiments, the solvent is or comprises isopropyl acetate. In some embodiments, the solvent is or comprises tetrahydrofuran. In some embodiments, the solvent is or comprises 2-methyltetrahydrofuran. In some embodiments, the solvent is or comprises toluene. In some embodiments, the solvent is or comprises benzene. In some embodiments, the solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the solvent is or comprises chlorobenzene. In some embodiments, the solvent is or comprises xylene. In some embodiments, the solvent is or comprises tert-butyl methyl ether. In some embodiments, the solvent is or comprises cyclopentyl methyl ether. In some embodiments, the solvent is or comprises 1,4-dioxane. In some embodiments, the solvent is ethyl acetate.

In some embodiments, the solvent is present in an amount between about 2.8 kg/kg and about 16.0 kg/kg. In some embodiments, the solvent is present in an amount between about 2.8 kg/kg and about 14.0 kg/kg. In some embodiments, the solvent is present in an amount between about 2.8 kg/kg and about 10.0 kg/kg. In some embodiments, the solvent is present in an amount between about 2.8 kg/kg and about 8.0 kg/kg. In some embodiments, the solvent is present in an amount between about 3.6 kg/kg and about 10.0 kg/kg. In some embodiments, the solvent is present in an amount between about 4.4 kg/kg and about 10.0 kg/kg. In some embodiments, the solvent is present in an amount between about 5.2 kg/kg and about 10.0 kg/kg. In some embodiments, the solvent is present in an amount between about 5.2 kg/kg and about 8.0 kg/kg. In some embodiments, the solvent is present in an amount between about 5.5 kg/kg and about 7.5 kg/kg. In some embodiments, the solvent is present in an amount between about 6.0 kg/kg and about 7.4 kg/kg. In some embodiments, the solvent is present in an amount between about 6.5 kg/kg and about 7.3 kg/kg. In some embodiments, the solvent is present in an amount between about 6.0 kg/kg and about 7.5 kg/kg. In some embodiments, the solvent is present in an amount of about 6.5 kg/kg, 6.6 kg/kg, 6.7 kg/kg, 6.8 kg/kg, 6.89 kg/kg, 7.0 kg/kg, 7.1 kg/kg, or 7.2 kg/kg. In some embodiments, the solvent is present in an amount of about 6.7 kg/kg. In some embodiments, the solvent is present in an amount of about 6.8 kg/kg. In some embodiments, the solvent is present in an amount of about 6.89 kg/kg. In some embodiments, the solvent is present in an amount of about 7.0 kg/kg. In some embodiments, the solvent is present in an amount of about 7.1 kg/kg.

In some embodiments, the reaction conditions comprise a desiccant. In some embodiments, the desiccant is or comprises N,O-bis(trimethylsilyl)-acetamide, N,O-bis(trimethylsilyl)trifluoroacetamide, N-(trimethylsilyl)acetamide, molecular sieves. In some embodiments, the desiccant is or comprises N,O-bis(trimethylsilyl)-acetamide. In some embodiments, the desiccant is or comprises N,O-bis(trimethylsilyl)trifluoroacetamide. In some embodiments, the desiccant is or comprises N-(trimethylsilyl)acetamide. In some embodiments, the desiccant is or comprises molecular sieves.

In some embodiments, the desiccant is present in an amount between about 0.25 and about 1.50 equivalents. In some embodiments, the desiccant is present in an amount between about 0.55 and about 1.25 equivalents. In some embodiments, the desiccant is present in an amount between about 0.25 and about 1.00 equivalents. In some embodiments, the desiccant is present in an amount between about 0.25 and about 0.75 equivalents. In some embodiments, the desiccant is present in an amount between about 0.50 and about 1.00 equivalents. In some embodiments, the desiccant is present in an amount between about 0.45 and about 0.75 equivalents. In some embodiments, the desiccant is present in an amount between about 0.50 and about 0.65 equivalents. In some embodiments, the desiccant is present in an amount between about 0.55 and about 0.63 equivalents. In some embodiments, the desiccant is present in an amount of about 0.56, 0.57, 0.58, 0.59, 0.60, 0.61, 0.62, 0.63, 0.64 or 0.65 equivalents. In some embodiments, the desiccant is present in an amount of about 0.56 equivalents. In some embodiments, the desiccant is present in an amount of about 0.58 equivalents. In some embodiments, the desiccant is present in an amount of about 0.59 equivalents. In some embodiments, the desiccant is present in an amount of about 0.593 equivalents. In some embodiments, the desiccant is present in an amount of about 0.60 equivalents. In some embodiments, the desiccant is present in an amount of about 0.62 equivalents. In some embodiments, desiccant is present in an amount of about 0.65 equivalents.

In some embodiments, reaction conditions comprise agitation for a period of time. In some embodiments, the reaction is agitated for about 25 minutes to about 5 days. In some embodiments, the reaction is agitated for about 25 minutes to about 1 day. In some embodiments, the reaction is agitated for about 25 minutes to about 12 hrs. In some embodiments, the reaction is agitated for about 25 minutes to about 6 hrs. In some embodiments, the reaction is agitated for about 25 minutes to about 3 hrs. In some embodiments, the reaction is agitated for at least 15 minutes. In some embodiments, the reaction is agitated for at least 25 minutes. In some embodiments, the reaction is agitated for at least 45 minutes. In some embodiments, the reaction is agitated for at least 60 minutes. In some embodiments, the reaction is agitated for at least 120 minutes.

In some embodiments, reaction conditions comprise a catalyst. In some embodiments, the catalyst is or comprises tert-butyl dimethylsilyl triflate, triethylsilyl triflate, or trimethylsilyl triflate. In some embodiments, the catalyst is or comprises tert-butyl dimethylsilyl triflate. In some embodiments, the catalyst is or comprises triethylsilyl triflate. In some embodiments, the catalyst is or comprises trimethylsilyl triflate.

In some embodiments, the catalyst is present in a catalytic amount. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.90 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.80 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.70 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.60 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.50 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.05 and about 0.40 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.10 and about 0.60 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.20 and about 0.60 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.30 and about 0.60 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.25 and about 0.40 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.30 and about 0.38 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.32 and about 0.366 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount between about 0.34 and about 0.35 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.32, 0.33, 0.34, 0.35, 0.36, 0.37, or 0.38 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.32 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.33 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.34 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.349 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.35 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.36 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.37 equivalents relative to compound 2, or a salt thereof. In some embodiments, the catalyst is present in an amount of about 0.38 equivalents relative to compound 2, or a salt thereof.

In some embodiments, suitable reaction conditions comprise providing compounds 2 and 8 at a temperature T_a. In some embodiments, temperature T_a is between about 0° C. and about 50° C. In some embodiments, temperature T_a is between about 0° C. and about 35° C. In some embodiments, temperature T_a is between about 0° C. and about 25° C. In some embodiments, temperature T_a is between about 10° C. and about 50° C. In some embodiments, temperature T_a is between about 15° C. and about 50° C. In some embodiments, temperature T_a is between about 5° C. and about 35° C. In some embodiments, temperature T_a is between about 10° C. and about 30° C. In some embodiments, temperature T_a is between about 19° C. and about 25° C. In some embodiments, temperature T_a is between about 21° C. and about 23° C. In some embodiments, temperature T_a is about 19° C., 20° C., 21° C., 22° C., 23° C., 24° C., or 25° C. In some embodiments, temperature T_a is about 20° C. In some embodiments, temperature T_a is about 21° C. In some embodiments, temperature T_a is about 22° C. In some embodiments, temperature T_a is about 23° C. In some embodiments, temperature T_a is about 24° C. In some embodiments, temperature T_a is about 25° C.

In some embodiments, Step 2 further comprises a step of (b) heating to a temperature Tb. In some embodiments, the reaction conditions comprising heating to temperature Tb. In some embodiments, temperature Tb is between about 20° C. and about 150° C. In some embodiments, temperature Tb is between about 20° C. and about 120° C. In some embodiments, temperature Tb is between about 20° C. and about 100° C. In some embodiments, temperature Tb is between about 20° C. and about 80° C. In some embodiments, temperature Tb is between about 40° C. and about 150° C. In some embodiments, temperature Tb is between about 60° C. and about 150° C. In some embodiments, temperature Tb is between about 40° C. and about 120° C. In some embodiments, temperature Tb is between about 60° C. and about 90° C. In some embodiments, temperature Tb is between about 65° C. and about 80° C. In some embodiments, temperature Tb is between about 72° C. and about 79° C. In some embodiments, temperature Tb is about 73° C., 74° C., 75° C., 76° C., 77° C., 78° C., 79° C., or 80° C. In some embodiments, temperature Tb is about 74° C. In some embodiments, temperature Tb is about 75° C. In some embodiments, temperature Tb is about 76° C. In some embodiments, temperature Tb is about 77° C. In some embodiments, temperature Tb is about 78° C. In some embodiments, temperature Tb is about 79° C. In some embodiments, temperature Tb is about 80° C.

In some embodiments, reaction conditions comprising aging and/or stirring the reaction mixture at temperature Tb for a period of time. In some embodiments, the period of time between about 1 hrs and about 48 hrs. In some embodiments, the period of time between about 2 hrs and about 36 hrs. In some embodiments, the period of time between about 3 hrs and about 24 hrs. In some embodiments, the period of time between about 4 hrs and about 12 hrs. In some embodiments, the period of time of about 5 hours. In some embodiments, the period of time of at least 2 hours. In some embodiments, the period of time of at least 4 hours. In some embodiments, the period of time of at least 6 hours. In some embodiments, the period of time of at least 8 hours. In some embodiments, the period of time of at least 12 hours. In some embodiments, the period of time of at least 24 hours.

In some embodiments, e.g., after allowing the reaction to proceed at temperature Tb for the period of time, Step 2 further comprises a step of (c) cooling the reaction mixture to a temperature T_c. In some embodiments, temperature T_c is between about 0° C. and about 70° C. In some embodiments, temperature T_c is between about 0° C. and about 60° C. In some embodiments, temperature T_c is between about 0° C. and about 50° C. In some embodiments, temperature T_c is between about 0° C. and about 40° C. In some embodiments, temperature T_c is between about 0° C. and about 30° C. In some embodiments, temperature T_c is between about 10° C. and about 60° C. In some embodiments, temperature T_c is between about 20° C. and about 60° C. In some embodiments, temperature T_c is between about 10° C. and about 50° C. In some embodiments, temperature T_c is between about 19° C. and about 55° C. In some embodiments, temperature T_c is between about 15° C. and about 40° C. In some embodiments, temperature T_c is between about 16° C. and about 30° C. In some embodiments, temperature T_c is between about 20° C. and about 25° C. In some embodiments, temperature T_c is about 19° C., 20° C., 21° C., 22° C., 23° C., or 24° C. In some embodiments, temperature T_c is about 19° C. In some embodiments, temperature T_c is about 20° C. In some embodiments, temperature T_c is about 21° C. In some embodiments, temperature T_c is about 22° C. In some embodiments, temperature T_c is about 23° C. In some embodiments, temperature T_c is about 24° C.

In some embodiments, the reaction (e.g., presence of compound 3 or compound 2) is monitored for completion by HPLC. In some embodiments, the reaction is monitored for the presence of compound 3. In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 50.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 25.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 15.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 10.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 5.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 4.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 3.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 2.0% (a/a) (e.g., as measured by HPLC). In some embodiments, the reaction is monitored for the presence of compound 2 in an amount less than or equal to about 1.0% (a/a) (e.g., as measured by HPLC).

In some embodiments, Step 2 further comprises a step of (d) washing and/or quenching the reaction mixture, e.g., to provide a first resulting mixture. In some embodiments, Step 2 further comprises a step of washing the reaction mixture. In some embodiments, Step 2 further comprises a step of quenching the reaction mixture.

In some embodiments, washing and/or quenching comprises a base wash 1. In some embodiments, the base wash 1 is or comprises washing the reaction mixture with an aqueous base. In some embodiments, the aqueous base is or comprises LiOH, NaOH, KOH, Li_2CO_3, Na_2CO_3, K_2CO_3, $LiHCO_3$, $NaHCO_3$, or $KHCO_3$. In some embodiments, the aqueous base is or comprises $KHCO_3$.

In some embodiments, the aqueous base concentration is between about 5% and about 30% wt base. In some embodiments, the aqueous base concentration is between about 10% and 25% wt base. In some embodiments, the aqueous base concentration is between about 15% and about 25% base. In some embodiments, the aqueous base concentration is about 10%, 15%, 20%, 25%, 30% or 35% wt base. In some embodiments, the aqueous base concentration is about 10% wt base. In some embodiments, the aqueous base concentration is about 15% wt base. In some embodiments, the aqueous base concentration is about 20% wt base. In some embodiments, the aqueous base concentration is about 25% wt base. In some embodiments, the aqueous base concentration is about 30% wt base. In some embodiments, the aqueous base concentration is about 35% wt base. In some embodiments the aqueous base comprises potassium hydrogen carbonate (e.g., between about 0.811 kg/kg and 1.02 kg/kg, such as 1.02 kg/kg) and water (e.g., between about 3.672 kg/kg and about 4.488 kg/kg, such as 4.08 kg/kg). In some such embodiments, these amounts are split between two separate washes with the aqueous base.

In some embodiments, the base wash 1 is performed at temperature T_d. In some embodiments, temperature T_d is between about 0° C. and about 50° C. In some embodiments, temperature T_d is between about 0° C. and about 40° C. In some embodiments, temperature T_d is between about 10° C. and about 40° C. In some embodiments, temperature T_d is between about 10° C. and about 30° C. In some embodiments, temperature T_d is between about 15° C. and about 25° C. In some embodiments, temperature T_d is about 17° C., 18° C., 19° C., 20° C., 21° C., 22° C., or 23° C. In some embodiments, temperature T_d is about 18° C. In some embodiments, temperature T_d is about 20° C. In some embodiments, temperature T_d is about 22° C.

In some embodiments, base wash 1 further comprises addition of water to the reaction mixture. In some embodiments, additional water is provided in an amount of between about 2 kg/kg and about 6 kg/kg. In some embodiments, additional water is provided in an amount of between about 3.672 kg/kg and about 4.488 kg/kg. In some embodiments, additional water is provided in an amount of about 3, 3.5, 4, 4.5, 5, 5.5, or 6 kg/kg. In some embodiments, additional water is provided in an amount of about 4.488 kg/kg.

In some embodiments, washing and/or quenching comprises a brine wash 2. In some embodiments, brine wash 2 comprises washing the reaction mixture with brine (i.e., an aqueous NaCl solution). In some embodiments, the aqueous NaCl solution is between about 1% and 100% NaCl by weight. In some embodiments, the aqueous NaCl solution is between about 1% and 50% NaCl by weight. In some embodiments, the aqueous NaCl solution is between about 1% and 10% NaCl by weight. In some embodiments, the aqueous NaCl solution is about 0.1%, 0.5%, 1%, 5%, 10%, 15%, 20%, or 25% NaCl by weight. In some embodiments the aqueous NaCl solution is prepared from NaCl (e.g., between about 0.41 kg/kg and 0.69 kg/kg, such as 0.628 kg/kg) and water (e.g., between about 1.835 kg/kg and about 2.243 kg/kg, such as 2.039 kg/kg).

In some embodiments, brine wash 2 proceeds at temperature T_d as described above and herein.

In some embodiments, each of base wash 1 and brine wash 2 independently comprises agitation for a period of time. In some embodiments, the period of time is for between about 5 minutes to about 5 days. In some embodiments, the period of time is for between about 5 minutes to about 1 day. In some embodiments, the period of time is for between about 5 minutes to about 12 hrs. In some embodiments, the period of time is for between about 5 minutes to about 6 hrs. In some embodiments, the period of time is for between about 5 minutes to about 3 hrs. In some embodiments, the period of time is for at least 5 minutes. In some embodiments, the period of time is for at least 15 minutes. In some embodiments, the period of time is for at least 45 minutes. In some embodiments, the period of time is for at least 60 minutes. In some embodiments, the period of time is for at least 120 minutes.

In some embodiments, brine wash 2 comprises allowing the reaction mixture or the first resulting mixture to settle for a period of time. In some embodiments, the period of time is between about 1 minute and 60 minute. In some embodiments, the period of time is between about 10 minute and 30 minute. In some embodiments, the period of time is greater than 20 minutes.

In some embodiments, Step 2 further comprises a step of (e) distillation and/or solvent exchange of the reaction mixture or first resulting mixture to provide a second resulting mixture comprising compound 3, or a pharmaceutically acceptable salt thereof, and a solvent.

In some embodiments, distillation and/or solvent exchange comprises (i) removing a solvent (e.g., the reaction solvent as described above and herein) from the reaction mixture and/or the first resulting mixture. In some embodiments, removing the solvent (e.g., the reaction solvent as described above and herein) comprises heating the reaction mixture and/or the first resulting mixture to a temperature T_e. In some embodiments, the temperature T_e is between about 0° C. and about 100° C. In some embodiments, the temperature T_e is between about 0° C. and about 60° C. In some embodiments, the temperature T_e is between about 0° C. and about 40° C. In some embodiments, the temperature T_e is between about 10° C. and about 60° C. In some embodiments, the temperature T_e is between about 20° C. and about 60° C. In some embodiments, the temperature T_e is between about 30° C. and about 60° C. In some embodiments, the temperature T_e is between about 20° C. and about 50° C. In some embodiments, the temperature T_e is between about 30° C. and about 40° C. In some embodiments, the temperature T_e is between about 40° C. and about 50° C. In some embodiments, the temperature T_e is at an amount less than or equal to about 60° C. In some embodiments, the temperature T_e is at an amount less than or equal to about 45° C. In some embodiments, the temperature T_e is at an amount less than or equal to about 40° C.

In some embodiments, the reaction mixture and/or the first resulting mixture is maintained at temperature T_e for a period of time. In some embodiments, the period of time is until a certain volume of solvent is collected from the reaction mixture or first resulting mixture. For example, in some embodiments, the collected volume of solvent is between about 388 L and about 430 L, such as about 409 L.

In some embodiments, distillation and/or solvent exchange comprises (ii) adding a distillation solvent to the reaction mixture or first resulting mixture. In some embodiments, the distillation solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the distillation solvent is or comprises chloroform. In some embodiments, the distillation solvent is or comprises diethyl ether. In some embodiments, the distillation solvent is or comprises ethyl acetate. In some embodiments, the distillation solvent is or comprises isopropyl acetate. In some embodiments, the distillation solvent is or comprises cyclopentyl methyl ether. In some embodiments, the distillation solvent is or comprises toluene. In some embodiments, the distillation solvent is or comprises benzene. In some embodiments, the distillation solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the distillation solvent is or comprises chlorobenzene. In some embodiments, the distillation solvent is or comprises xylene. In some embodiments, the distillation solvent is or comprises dichloromethane. In some embodiments, the distillation solvent is or comprises tert-butyl methyl ether.

In some embodiments, the solution containing compound 3, or salt thereof is charged with an amount of distillation solvent. In some embodiments, the amount of distillation solvent is between about 100 kg and about 1000 kg. In some embodiments, the amount of distillation solvent is between about 100 kg and about 800 kg. In some embodiments, the amount of distillation solvent is between about 100 kg and about 600 kg. In some embodiments, the amount of distillation solvent is between about 100 kg and about 400 kg. In some embodiments, the amount of distillation solvent is between about 200 kg and about 1000 kg. In some embodiments, the amount of distillation solvent is between about 300 kg and about 1000 kg. In some embodiments, the amount of distillation solvent is between about 400 kg and about 1000 kg. In some embodiments, the amount of distillation solvent is between about 200 kg and about 600 kg. In some embodiments, the amount of distillation solvent is between about 300 kg and about 500 kg. In some embodiments, the amount of distillation solvent is between about 350 kg and about 450 kg. In some embodiments, the amount of distillation solvent is between about 367 kg and 407 kg.

In some embodiments, the amount of distillation solvent is between about 375 kg and about 400 kg. In some embodiments, the amount of distillation solvent is about 375 kg, 382 kg, 387 kg, 392 kg or 400 kg. In some embodiments, the amount of distillation solvent is about 375 kg. In some embodiments, the amount of distillation solvent is about 387 kg. In some embodiments, the amount of distillation solvent is about 400 kg.

In some embodiments, the distillation solvent is added at temperature T_f. In some embodiments, the temperature T_f is between about 0° C. and about 50° C. In some embodiments, the temperature T_f is between about 0° C. and about 40° C. In some embodiments, the temperature T_f is between about 0° C. and about 30° C. In some embodiments, the temperature T_f is between about 0° C. and about 20° C. In some embodiments, the temperature T_f is between about 10° C. and about 40° C. In some embodiments, the temperature T_f is between about 15° C. and about 40° C. In some embodiments, the temperature T_f is between about 20° C. and about 40° C. In some embodiments, the temperature T_f is between about 10° C. and about 35° C. In some embodiments, the temperature T_f is between about 15° C. and about 30° C. In some embodiments, the temperature T_f is between about 18° C. and about 23° C. In some embodiments, the temperature T_f is about 18° C., 19° C., 20° C., 21° C., 22° C., or 23° C. In some embodiments, the temperature T_f is about 18° C. In some embodiments, the temperature T_f is about 20° C. In some embodiments, the temperature T_f is about 21° C. In some embodiments, the temperature T_f is about 22° C.

In some embodiments, the second resulting mixture is agitated at temperature T_f (as described above and herein).

In some embodiments, Step 2 further comprises a step of (f) crystallization of compound 3, or a pharmaceutically acceptable salt thereof (e.g., from the reaction mixture, first resulting mixture, or second resulting mixture). In some embodiments, crystallization comprises formation of a salt of compound 3 (e.g. a salt as described above and herein).

In some embodiments, crystallization comprises providing a crystallization mixture comprising compound 3, or a pharmaceutically acceptable salt thereof, and a crystallization solvent. In some embodiments, the crystallization solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the crystallization solvent is or comprises chloroform. In some embodiments, the crystallization solvent is or comprises diethyl ether. In some embodiments, the crystallization solvent is or comprises ethyl acetate. In some embodiments, the crystallization solvent is or comprises cyclopentyl methyl ether. In some embodiments, the crystallization solvent is or comprises toluene. In some embodiments, the crystallization solvent is or comprises benzene. In some embodiments, the crystallization solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the crystallization solvent is or comprises chlorobenzene. In some embodiments, the crystallization solvent is or comprises xylene. In some embodiments, the crystallization solvent is or comprises dichloromethane. In some embodiments, the crystallization solvent is or comprises isopropyl acetate.

In some embodiments, the crystallization mixture further comprises an acid. In some embodiments, the acid is camphorsulfonic acid (CSA), benzenesulfonic acid ($PhSO_3H$), methanesulfonic acid (MsOH), toluenesulfonic acid (TsOH), 1,5-naphthalenedisulfonic acid (napsic acid), 1,2-ethanedisulfonic acid (edisic acid), ethanesulfonic acid (esic acid), 4-chlorobenzenesulfonic acid (closic acid), 6,7-dihydroxycoumarin-4-methanesulfonic acid (cromesic acid), or trifluoromethanesulfonic acid (HOTf). In some embodiments, the acid is hydrochloric acid ("HCl").

In some embodiments, the crystallization mixture comprises compound 3 as a salt of the acid described above and herein. In some embodiments, the crystallization mixture comprises compound 3 as a hydrohalide salt. In some embodiments, the crystallization mixture comprises compound 3 as an HCl salt. In some embodiments, the crystallization mixture comprises compound 3 as an HBr salt.

In some embodiments, the acid is provided in an amount between about 0.5 and 1.5 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.75 and 1.5 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.95 and 1.5 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 1.05 and 1.5 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.5 and 1.25 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.5 and 1.15 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.5 and 1.05 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.6 and 1.4 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.8 and 1.2 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is provided in an amount between about 0.95 and 1.05 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is present in an amount of about 0.95, 0.98, 1.01, 1.03, 1.04, 1.05, 1.06, or 1.07 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is present in an amount of about 1.01 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is present in an amount of about 1.037 equivalents relative to compound 3, or a salt thereof. In some embodiments, the acid is present in an amount of about 1.05 equivalents relative to compound 3, or a salt thereof.

In some embodiments, the acid provided is a hydrochloric acid solution. In some embodiments, the acid is provided as an ethyl acetate solution of hydrochloric acid. In some embodiments, the acid is provided as an aqueous solution of hydrochloric acid. In some embodiments, the acid is provided as a methanolic solution of hydrochloric acid. In some embodiments, the acid is provided as a dioxane solution of hydrochloric acid. In some embodiments, the acid is provided as a diethyl ether solution of hydrochloric acid.

In some embodiments, the crystallization mixture is provided at a temperature T_g. In some embodiments, temperature T_g is between about 0° C. and about 50° C. In some embodiments, temperature T_g is between about 10° C. and about 50° C. In some embodiments, temperature T_g is between about 20° C. and about 50° C. In some embodiments, temperature T_g is between about 0° C. and about 30° C. In some embodiments, temperature T_g is between about 5° C. and about 40° C. In some embodiments, temperature T_g is between about 10° C. and about 30° C. In some embodiments, temperature T_g is between about 15° C. and about 25° C. In some embodiments, temperature T_g is about 18° C., 20° C., 22° C., or 24° C. In some embodiments, temperature T_g is about 20° C. In some embodiments, temperature T_g is about 22° C. In some embodiments, temperature T_g is about 24° C.

In some embodiments, crystallization comprises adding the acid to the crystallization mixture over a period of time. In some embodiments, the period of time is between about 5 and about 120 minutes. In some embodiments, the period of time between about 5 and about 90 minutes. In some embodiments, the period of time between about 5 and about 60 minutes. In some embodiments, the period of time between about 5 and about 30 minutes. In some embodiments, the period of time between about 15 and about 120 minutes. In some embodiments, the period of time between about 30 and about 120 minutes. In some embodiments, the period of time of at least 15 minutes. In some embodiments, the period of time of at least 30 minutes. In some embodiments, the period of time of at least 60 minutes. In some embodiments, the period of time of at least 120 minutes. In some embodiments, the period of time of about 15 minutes. In some embodiments, the period of time of about 30 minutes. In some embodiments, the period of time of about 60 minutes. In some embodiments, crystallization comprising adding the acid to the crystallization mixture at temperature T_g, wherein temperature T_g is as described above and herein.

In some embodiments, a seed crystal is added to the crystallization mixture. In some embodiments, the seed crystal is compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.01% and about 0.12% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.01% and about 0.10% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.01% and about 0.08% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.01% and about 0.07% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.01% and about 0.06% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.03% and about 0.10% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.04% and about 0.10% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.05% and about 0.10% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.06% and about 0.10% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.03% and about 0.08% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount between about 0.05% and about 0.07% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.04%, 0.05%, 0.06%, 0.07%, or 0.08% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.04% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.05% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.06% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.07% w/w relative to compound 2, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.08% w/w relative to compound 2, or a salt thereof.

In some embodiments, the seed crystal is provided in a seed crystal solvent (e.g., the crystallization solvent, as described above and herein). In some embodiments, the seed crystal solvent is or comprises an amount between about 0 kg and about 0.41 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0 kg and about 0.31 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0 kg and about 0.21 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0 kg and about 0.16 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0 kg and about 0.11 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0 kg and about 0.60 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0.10 kg and about 0.21 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount between about 0.16 kg and about 0.21 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount of about 0.10 kg, 0.15 kg, 0.20 kg, or 0.25 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount of about 0.10 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount of about 0.15 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount of about 0.20 kg. In some embodiments, the seed crystal solvent suspension is or comprises an amount of about 0.25 kg.

In some embodiments, the crystallization mixture is aged for a period of time at temperature T_g, wherein temperature T_g is as described above and herein. In some embodiments, the period of time is between about 5 and about 300 minutes. In some embodiments, the period of time is between about 5 and about 240 minutes. In some embodiments, the period of time is between about 10 and about 180 minutes. In some embodiments, the period of time is between about 30 and about 120 minutes. In some embodiments, the period of time is between about 60 and about 100 minutes. In some embodiments, the period of time is about 30, 60, 90, or 120 minutes. In some embodiments, the period of time is about 90 minutes. In some embodiments, the period of time is at least 10 minutes. In some embodiments, the period of time is at least 30 minutes. In some embodiments, the period of time is at least 60 minutes. In some embodiments, the period of time is at least 90 minutes. In some embodiments, the crystallization mixture is aged for a period of time of at least 120 minutes.

In some embodiments, the crystallization mixture is cooled to temperature T_1 over a period of time. In some embodiments, the period of time is between about 15 minutes and about 240 minutes. In some embodiments, the period of time is between about 15 minutes and about 200 minutes. In some embodiments, the period of time is between about 15 minutes and about 180 minutes. In some embodiments, the period of time is between about 15 minutes and about 150 minutes. In some embodiments, the period of time is between about 15 minutes and about 120 minutes. In some embodiments, the period of time is between about 15 minutes and about 90 minutes. In some embodiments, the period of time is between about 30 minutes and about 180 minutes. In some embodiments, the period of time is between about 60 minutes and about 180 minutes. In some embodiments, the period of time is between about 90 minutes and about 180 minutes. In some embodiments, the period of time is between about 30 minutes and about 150 minutes. In some embodiments, the period of time is between about 60 minutes and about 120 minutes. In some embodiments, the period of time is between about 80 minutes and about 100 minutes. In some embodiments, the period of time is about 90 minutes. In some embodiments, the period of time is about 100 minutes. In some embodiments, the period of time is about 120 minutes. In some embodiments, the period of time is about 150 minutes. In some embodiments, the period of time is at least 30 minutes. In some embodiments, the period of time is at least 60 minutes. In some embodiments, the period of time is at least 90 minutes. In some embodiments, the period of time is at least 120 minutes.

In some embodiments, the temperature T_i is between about −40° C. and about 40° C. In some embodiments, the temperature T_i is between about −40° C. and about 30° C. In some embodiments, the temperature T_i is between about −40° C. and about 20° C. In some embodiments, the temperature T_i is between about −40° C. and about 10° C. In some embodiments, the temperature T_i is between about −20° C. and about 40° C. In some embodiments, the temperature T_i is between about −10° C. and about 40° C. In some embodiments, the temperature T_i is between about 0° C. and about 40° C. In some embodiments, the temperature T_i is between about −30° C. and about 30° C. In some embodiments, the temperature T_i is between about −20° C. and about 20° C. In some embodiments, the temperature T_i is between about −10° C. and about 10° C. In some embodiments, the temperature T_i is between about −5° C. and about 5° C. In some embodiments, the temperature T_i is about −2° C., 0° C., 2° C., 4° C., or 6° C. In some embodiments, the temperature T_i is about −2° C. In some embodiments, the temperature T_i is about 0° C. In some embodiments, the temperature T_i is about 2° C. In some embodiments, the temperature T_i is about 4° C.

In some embodiments, the crystallization mixture is aged at temperature T_i for a period of time. In some embodiments, the period of time is between about 30 minutes and 7 days. In some embodiments, the period of time is between about 30 minutes and 1 days. In some embodiments, the period of time is between about 30 minutes and 12 hrs. In some embodiments, the period of time is between about 30 minutes and 6 hrs. In some embodiments, the period of time is between about 30 minutes and 3 hrs. In some embodiments, the period of time is between about 30 minutes and 2 hrs. In some embodiments, the period of time is about 30 minutes. In some embodiments, the period time is about 60 minutes. In some embodiments, the crystallization mixture is aged for a period time is about 90 minutes. In some embodiments, the crystallization mixture is aged for a period time is about 120 minutes. In some embodiments, the period of time is at least 30 minutes. In some embodiments, the period of time is at least 60 minutes. In some embodiments, the period of time is at least 90 minutes. In some embodiments, the period of time is at least 120 minutes. In some embodiments, the period of time is at least 5 minutes. In some embodiments, the period of time is less than 7 days.

In some embodiments, Step 2 further comprises a step of (g) optionally filtering or washing the crystallization mixture to provide a filtered mixture. In some embodiments, Step 2 further comprises filtering the crystallization mixture to provide the filtered mixture. In some embodiments, Step 2 further comprises washing the crystallization mixture to provide the filtered mixture.

In some embodiments, filtering is performed at temperature T_i, wherein temperature T_i is as described above and herein.

In some embodiments, filtrating is performed over a period of time. In some embodiments, the period of time is between about 10 minutes and about 4 days. In some embodiments, the period of time is between about 10 minutes and about 1 day. In some embodiments, the period of time is between about 10 minutes and about 12 hrs. In some embodiments, the period of time is between about 10 minutes and about 6 hrs. In some embodiments, the period of time is between about 10 minutes and about 1 hr. In some embodiments, the period of time is between about 10 minutes and about 30 minutes. In some embodiments, the period of time is at least 10 minutes. In some embodiments, the period of time is at least 20 minutes. In some embodiments, the period of time is at least 30 minutes. In some embodiments, the period of time is at least 60 minutes.

In some embodiments, washing comprises Wash A. In some embodiments, Wash A comprises washing with wash solvent A. In some embodiments, wash solvent A is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, wash solvent A is or comprises chloroform. In some embodiments, the wash solvent A is or comprises diethyl ether. In some embodiments, wash solvent A is or comprises ethyl acetate. In some embodiments, wash solvent A is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent A is or comprises toluene. In some embodiments, wash solvent A is or comprises benzene. In some embodiments, wash solvent A is or comprises α,α,α-trifluorotoluene. In some embodiments, wash solvent A is or comprises chlorobenzene. In some embodiments, wash solvent A is or comprises xylene. In some embodiments, wash solvent A is or comprises dichloromethane. In some embodiments, wash solvent A is or comprises isopropyl acetate.

In some embodiments, wash solvent A is present in an amount between about 0.30 kg/kg and about 5.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.30 kg/kg and about 4.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.30 kg/kg and about 3.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.30 kg/kg and about 2.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.50 kg/kg and about 5.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 1.00 kg/kg and about 5.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 1.25 kg/kg and about 5.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.25 kg/kg and about 4.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.50 kg/kg and about 3.0 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.69 kg/kg and about 2.5 kg/kg. In some embodiments, wash solvent A is present in an amount between about 0.95 kg/kg and about 1.75 kg/kg. In some embodiments, wash solvent A is present in an amount between about 1.25 kg/kg and about 1.5 kg/kg. In some embodiments, wash solvent A is present in an amount of about 1.15 kg/kg, 1.25 kg/kg, 1.35 kg/kg, 1.55 kg/kg, or 1.75 kg/kg. In some embodiments, wash solvent A is present in an amount of about 1.15 kg/kg. In some embodiments, wash solvent A is present in an amount of about 1.35 kg/kg. In some embodiments, wash solvent A is present in an amount of about 1.55 kg/kg.

In some embodiments, wash solvent A is present at a temperature T_i, wherein Ti is as described above and herein.

In some embodiments, after the addition of wash solvent A, the filtered mixture is agitated for a period of time. In some embodiments, the period of time between about 5 minutes and about 60 minutes. In some embodiments, the period of time between about 5 minutes and about 45 minutes. In some embodiments, the period of time between about 5 minutes and about 30 minutes. In some embodiments, the period of time between about 5 minutes and about 15 minutes. In some embodiments, the period of time of about 5 minutes or more. In some embodiments, the period of time of about 10 minutes or more. In some embodiments, the period of time of about 15 minutes or more. In some embodiments, the period of time of about 20 minutes or more. In some embodiments, the period of time of about 30 minutes or more. In some embodiments, the period of time of about 60 minutes or more.

In some embodiments, washing comprises Wash B. In some embodiments, Wash B comprises washing with wash solvent B. In some embodiments, wash solvent B is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, wash solvent B is or comprises chloroform. In some embodiments, the wash solvent B is or comprises diethyl ether. In some embodiments, wash solvent B is or comprises ethyl acetate. In some embodiments, wash solvent B is or comprises cyclopentyl methyl ether. In some embodiments, wash solvent B is or comprises toluene. In some embodiments, wash solvent B is or comprises benzene. In some embodiments, wash solvent B is or comprises α,α,α-trifluorotoluene In some embodiments, wash solvent B is or comprises chlorobenzene. In some embodiments, wash solvent B is or comprises xylene. In some embodiments, wash solvent B is or comprises dichloromethane. In some embodiments, wash solvent B is or comprises isopropyl acetate. In some embodiments, wash solvent B is or comprises tert-butyl methyl ether.

In some embodiments, wash solvent B is present in an amount between about 0.50 kg/kg and 4.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 0.50 kg/kg and 3.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 0.50 kg/kg and 2.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.0 kg/kg and 4.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.50 kg/kg and 4.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.75 kg/kg and 4.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.0 kg/kg and 3.0 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.25 kg/kg and 2.55 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.50 kg/kg and 2.05 kg/kg. In some embodiments, wash solvent B is present in an amount between about 1.75 kg/kg and 1.85 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.65 kg/kg, 1.70 kg/kg, 1.75 kg/kg, 1.80 kg/kg, or 1.85 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.65 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.70 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.75 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.80 kg/kg. In some embodiments, wash solvent B is present in an amount of about 1.85 kg/kg.

In some embodiments, wash solvent B is present at a temperature T_i, wherein Ti is as described above and herein.

In some embodiments, after the addition of wash solvent B, the filtered mixture is agitated for a period of time, e.g., as described above and herein with respect to wash A.

In some embodiments, washing comprises Wash C. In some embodiments, Wash C is the same as Wash B, as described above and herein.

In some embodiments, Step 2 further comprises a step of (h) drying the filtered mixture.

In some embodiments, drying comprises heating the filtered mixture to temperature T_m for a period of time. In some embodiments, temperature T_m is between about 0° C. and about 60° C. In some embodiments, temperature T_m is between about 0° C. and about 50° C. In some embodiments, temperature T_m is between about 0° C. and about 40° C. In some embodiments, temperature T_m is between about 0° C. and about 30° C. In some embodiments, temperature T_m is between about 10° C. and about 40° C. In some embodiments, temperature T_m is between about 20° C. and about 40° C. In some embodiments, temperature T_m is between about 30° C. and about 40° C. In some embodiments, temperature T_m is less than or about 10° C. In some embodiments, temperature T_m is less than or about 20° C. In some embodiments, temperature T_m is less than or about 25° C. In some embodiments, temperature T_m is less than or about 30° C. In some embodiments, temperature T_m is less than or about 35° C. In some embodiments, temperature T_m is less than or about 40° C. In some embodiments, temperature T_m is less than or about 100° C. In some embodiments, the filtered mixture is agitated for the period of time.

In some embodiments, the filtered mixture is held at temperature T_m for a period of time. In some embodiments, the period of time is until minimal condensate is visible in the filter drier.

In some embodiments, the filtered mixture is agitated at temperature T_m (as described above and herein).

In some embodiments, the filtered mixture is dried at a temperature T_n between about 0° C. and about 80° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 20° C. and about 80° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 40° C. and about 80° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 60° C. and about 80° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 10° C. and about 60° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 20° C. and about 50° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 40° C. and about 50° C. In some embodiments, the filtered mixture is dried at a temperature T_n between about 25° C. and about 60° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 45° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 50° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 55° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 60° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 65° C. In some embodiments, the filtered mixture is dried at a temperature T_n less than or about 70° C.

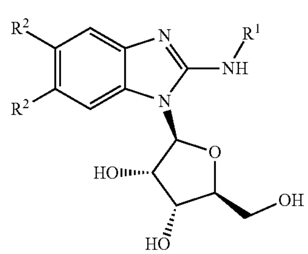
In some embodiments, the filtered mixture is held at temperature T_n for a period of time. In some embodiments, the period of time is until total solvent content is below 1.5% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 1.3% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 1.1% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 0.8% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 0.5% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 0.2% w/w (e.g., as measured by GCHS). In some embodiments, the period of time is until total solvent content is below 0.1% w/w (e.g., as measured by GCHS).

In some embodiments, the filtered mixture is held at temperature T_n for a period of time. In some embodiments, the period of time is until water content is below 1.5% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 1.3% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 1.1% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 0.8% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 0.5% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 0.2% w/w (e.g., as measured by Karl Fischer). In some embodiments, the period of time is until water content is below 0.1% w/w (e.g., as measured by Karl Fischer).

Step 3

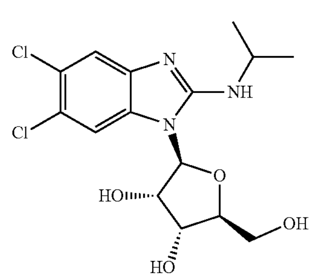
In some embodiments, the present disclosure provides an improved synthesis of compound 1:

1



or a pharmaceutically salt thereof, wherein each R^1 and R^2 is as defined above and described herein.

In some embodiments, the present disclosure provides an improved synthesis of maribavir:

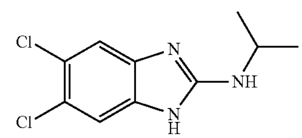


or a pharmaceutically salt thereof.

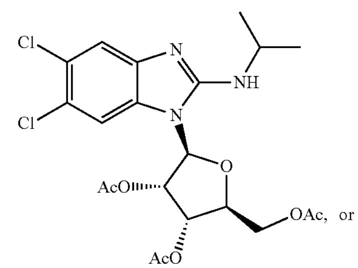
As shown in Scheme 4, Step 3 is the last of three steps for preparing maribavir as disclosed herein. Any byproducts formed within Steps 1, 2, or 3 (or derivatives thereof) may be present in a provided composition comprising maribavir and/or may lower the overall yield or quality of maribavir. In some embodiments, the present disclosure provides methods of preparing maribavir, or a pharmaceutically acceptable salt thereof with reduced and/or low levels of impurities (e.g., as described herein).

In some embodiments, the present disclosure provides a composition comprising maribavir, or a pharmaceutically acceptable salt thereof, and one or more of the following compounds:

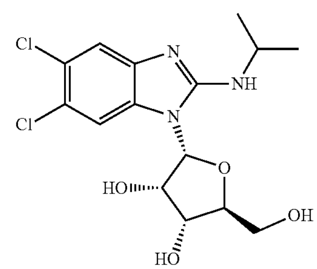
2



3



4



or salts thereof.

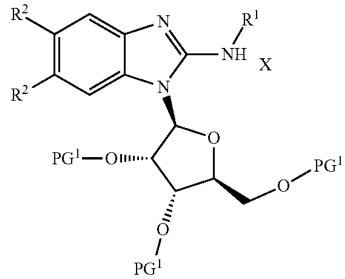
In some embodiments, provided compositions comprising maribavir are as described herein.

In some embodiments, the present disclosure also provides the recognition that certain reagents (and amounts thereof) and/or reaction conditions may provide improved quality maribavir, or a pharmaceutically salt thereof (e.g., with higher purity and/or minimal byproducts), and/or improve the yield of maribavir, or a pharmaceutically salt thereof. Without wishing to be bound by a particular theory, the present disclosure provides the recognition that incorporation of a crystallization into Step 3 may improve the yield and/or reduce the formation of byproducts (e.g., compounds 2, 3, or 4, or salts thereof). In some embodiments, the present disclosure provides the recognition that the amount of maribavir, or a salt thereof, and/or water in the crystallization mixture may affect the yield and/or quality of maribavir (e.g., particle size distribution). Additionally or alternatively, the present disclosure provides the recognition that incorporation of a phase separation and/or extraction into Step 3 at particular pH's may improve the yield and/or reduce the formation of byproducts (e.g., compounds 2, 3, or 4, or salts thereof).

In some embodiments, compound 1, or a pharmaceutically salt thereof, is prepared as shown in Step 3 of Scheme 4. In some embodiments, at Step 3, compound 1, or a pharmaceutically salt thereof, is prepared by a method comprising a step of

(a) reacting compound 3a:

3a



or a salt thereof, wherein

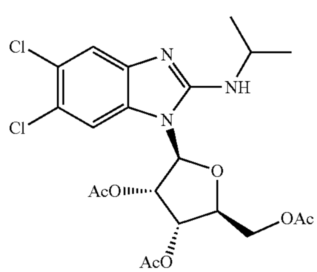
each R^1, R^2, X, and PG^1 is as defined above and described herein;

under suitable reaction conditions to provide maribavir, or a pharmaceutically acceptable salt thereof.

In some embodiments, maribavir, or a pharmaceutically salt thereof, is prepared as shown in Step 3 of Scheme 5. In some embodiments, at Step 3, maribavir, or a pharmaceutically salt thereof, is prepared by a method comprising a step of

(a) reacting compound 3:

3

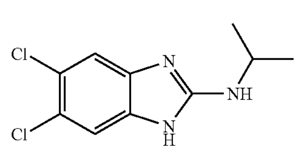


or a salt thereof,

under suitable reaction conditions to provide maribavir, or a pharmaceutically acceptable salt thereof.

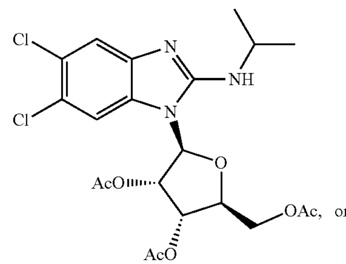
In some embodiments, Step 3 further provides one or more of the following compounds:

2

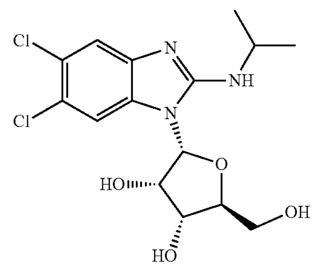


-continued

3



4



or salts thereof.

In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 2, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 3, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 2 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 4, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 2, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 3, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.10% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.01-0.05% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising 0.02-0.05% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (a/a HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition comprising about 1%, about 0.5%, about 0.1%, about 0.05%, about 0.02%, or about 0.01% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, Step 3 provides a composition, wherein compound 4, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, Step 3 provides a composition as provided in any of Tables 4-12 to 4-21 in Example 4.

In some embodiments, reaction conditions comprise a solvent. In some embodiments the solvent is or comprises methanol, ethanol, isoproprano1, tert-butanol, tert-butyl methyl ether, dimethyl sulfoxide, dimethylformamide, tetrahydrofuran, cyclopentyl methyl ether, 2-methyl tetrahydrofuran, or water. In some embodiments, the solvent is or comprises methanol. In some embodiments, the solvent is or comprises ethanol. In some embodiments, the solvent is or comprises isopropanol. In some embodiments, the solvent is or comprises tert-butanol. In some embodiments, the solvent is or comprises dimethyl sulfoxide. In some embodiments, the solvent is or comprises dimethylformamide. In some embodiments, the solvent is or comprises tetrahydrofuran. In some embodiments, the solvent is or comprises cyclopentyl methyl ether. In some embodiments, the solvent is or comprises 2-methyl tetrahydrofuran. In some embodiments, the solvent is or comprises water. In some embodiments, the solvent is or comprises tert-butyl methyl ether. In some embodiments, the solvent comprises methanol and tert-butyl methyl ether.

In some embodiments, the solvent is provided in an amount between about 1 L/kg and about 12 L/kg. In some embodiments, the solvent is provided in an amount between about 2 L/kg and about 12 L/kg. In some embodiments, the solvent is provided in an amount between about 4 L/kg and about 12 L/kg. In some embodiments, the solvent is provided in an amount between about 1 L/kg and about 8 L/kg. In some embodiments, the solvent is provided in an amount between about 1 L/kg and about 6 L/kg. In some embodiments, the solvent is provided in an amount between about 1 L/kg and about 4 L/kg. In some embodiments, the solvent is provided in an amount between about 3 L/kg and about 9 L/kg. In some embodiments, the solvent is provided in an amount between about 4 L/kg and about 8 L/kg. In some embodiments, the solvent is provided in an amount between about 5 L/kg and about 7 L/kg. In some embodiments, the solvent is provided in an amount of about 1 L/kg, 2 L/kg, 3 L/kg, 4 L/kg, 5 L/kg, 6 L/kg, 7 L/kg, 8 L/kg, 9 L/kg, 10 L/kg, 11 L/kg, or 12 L/kg. In some embodiments, the solvent is provided in an amount of about 3 L/kg. In some embodiments, the solvent is provided in an amount of about 4 L/kg. In some embodiments, the solvent is provided in an amount of about 5 L/kg. In some embodiments, the solvent is provided in an amount of about 6 L/kg. In some embodiments, the solvent is provided in an amount of about 7 L/kg.

In some embodiments, methanol is provided in an amount of between about 0.10 L/kg and about 2.0 L/kg. In some embodiments, methanol is provided in an amount of between about 0.25 L/kg and about 2.0 L/kg. In some embodiments, methanol is provided in an amount of between about 0.50 L/kg and about 2.0 L/kg. In some embodiments, methanol is provided in an amount of between about 0.10 L/kg and about 1.5 L/kg. In some embodiments, methanol is provided in an amount of between about 0.10 L/kg and about 1.0 L/kg. In some embodiments, methanol is provided in an amount of between about 0.10 L/kg and about 0.5 L/kg. In some embodiments, methanol is provided in an amount of between about 0.20 L/kg and about 1.0 L/kg. In some embodiments, methanol is provided in an amount of between about 0.3 L/kg and about 0.7 L/kg. In some embodiments, methanol is provided in an amount of between about 0.4 L/kg and about 0.6 L/kg. In some embodiments, methanol is provided in an amount of about 0.2 L/kg, 0.3 L/kg, 0.4 L/kg, 0.5 L/kg, 0.6 L/kg, 0.7 L/kg, or about 0.8 L/kg. In some embodiments, methanol is provided in an amount of about 0.3 L/kg. In some embodiments, methanol is provided in an amount of about 0.4 L/kg. In some embodiments, methanol is provided in an amount of about 0.5 L/kg. In some embodiments, methanol is provided in an amount of about 0.6 L/kg. In some embodiments, methanol is provided in an amount of about 0.7 L/kg.

In some embodiments, tert-butyl methyl ether is provided in an amount of between about 2 L/kg and about 12 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 4 L/kg and about 12 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 6 L/kg and about 12 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 2 L/kg and about 8 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 2 L/kg and about 6 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 3 L/kg and about 7 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of between about 4 L/kg and about 6 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 3 L/kg, 4 L/kg, 5 L/kg, 6 L/kg, or about 7 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 3 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 4 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 5 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 6 L/kg. In some embodiments, tert-butyl methyl ether is provided in an amount of about 7 L/kg.

In some embodiments, methanol is provided in an amount of between about 0.10 L/kg and about 2.0 L/kg, and tert-butyl methyl ether is provided in an amount of between about 2 L/kg and about 12 L/kg. In some embodiments, methanol is provided in an amount of between about 0.4 L/kg and about 0.6 L/kg, and tert-butyl methyl ether is provided in an amount of between about 4 L/kg and about 6 L/kg. In some embodiments, methanol is provided in an amount of between about 0.34 L/kg and about 0.56 L/kg, and tert-butyl methyl ether is provided in an amount of between about 3.65 L/kg and about 6.35 L/kg. In some embodiments, methanol is provided in an amount of about 0.48 L/kg, and tert-butyl methyl ether is provided in an amount of about 5.3 L/kg.

In some embodiments, reaction conditions comprise a base. In some embodiments, the base is or comprises LiOH, NaOH, or KOH. In some embodiments, the base is LiOH. In some embodiments, the base is NaOH. In some embodiments, the base is KOH. In some embodiments, the base is aqueous LiOH. In some embodiments, the base is aqueous NaOH. In some embodiments, the base is aqueous KOH. In some embodiments, the aqueous base is between about 5% and about 50%. In some embodiments, the aqueous base is between about 10% and about 50%. In some embodiments, the aqueous base is between about 15% and about 50%. In some embodiments, the aqueous base is between about 20% and about 50%. In some embodiments, the aqueous base is between about 25% and about 50%. In some embodiments, the aqueous base is between about 30% and about 50%. In some embodiments, the aqueous base is between about 5% and about 45%. In some embodiments, the aqueous base is between about 5% and about 40%. In some embodiments, the aqueous base is between about 5% and about 35%. In some embodiments, the aqueous base is between about 5% and about 30%. In some embodiments, the aqueous base is between about 20% and about 45%. In some embodiments, the aqueous base is between about 25% and about 40%. In some embodiments, the aqueous base is between about 28% and about 34%. In some embodiments, the aqueous base is about 5%, 10%, 15%, 20%, 25%, 30%, 35%, or 40% w/w. In some embodiments, the aqueous base is about 5% w/w. In some embodiments, the aqueous base is about 10% w/w. In some embodiments, the aqueous base is about 10% w/w. In some embodiments, the aqueous base is about 15% w/w. In some embodiments, the aqueous base is about 25% w/w. In some embodiments, the aqueous base is about 30% w/w. In some embodiments, the aqueous base is about 35% w/w. In some embodiments, the aqueous base is about 40% w/w. In some embodiments, the base is LiOR, NaOR, or KOR, wherein R is optionally substituted C_{1-6} aliphatic or aryl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is optionally substituted C_{1-6} aliphatic. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is methyl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is ethyl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is propyl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is butyl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is pentyl. In some embodiments, the base is an alkoxide such as LiOR, NaOR, or KOR, wherein R is hexyl. In some embodiments, the base is LiOMe. In some embodiments, the base is NaOMe. In some embodiments, the base is KOMe. In some embodiments, the base is LiOEt. In some embodiments, the base is NaOEt. In some embodiments, the base is KOEt. In some embodiments, the base is LiOtBu. In some embodiments, the base is NaOtBu. In some embodiments, the base is KOtBu.

In some embodiments, the reaction conditions comprise performing the reaction at a temperature T_1 (e.g., heating to a temperature T_1). In some embodiments, temperature T_1 is between about 0° C. and 60° C. In some embodiments, temperature T_1 is between about 20° C. and 60° C. In some embodiments, temperature T_1 is between about 24° C. and 45° C. In some embodiments, temperature T_1 is between about 24° C. and 40° C. In some embodiments, temperature T_1 is between about 24° C. and 35° C. In some embodiments, temperature T_1 is between about 24° C. and 30° C. In some embodiments, temperature T_1 is about 25° C., 26° C., 27° C., 28° C., 29° C., 30° C., 31° C., 32° C., 33° C., 34° C., or 35° C. In some embodiments, temperature T_1 is about 30° C.

In some embodiments, the reaction is allowed proceed for a period of time. In some embodiments, the reaction is agitated for a period of time. In some embodiments, the period of time is greater than 1 hr. In some embodiments, the period of time is greater than 4 hrs. In some embodiments, the period of time is greater than 8 hrs. In some embodiments, the period of time is between about 2 and about 8 hrs. In some embodiments, the period of time is between about 1 and about 12 hrs. In some embodiments, the period of time is between about 2.5 and about 4 hrs. In some embodiments, the period of time is less than 4 hrs. In some embodiments, the period of time is less than 2 hrs. In some embodiments, the period of time is less than 1 hr.

In some embodiments, the reaction conditions comprising cooling the reaction mixture to a temperature T_2. In some embodiments, temperature T_2 is between about 0° C. and about 60° C. In some embodiments, temperature T_2 is between about 0° C. and about 50° C. In some embodiments, temperature T_2 is between about 0° C. and about 40° C. In some embodiments, temperature T_2 is between about 0° C. and about 30° C. In some embodiments, temperature T_2 is between about 10° C. and about 60° C. In some embodiments, temperature T_2 is between about 20° C. and about 60° C. In some embodiments, temperature T_2 is between about 20° C. and about 30° C. In some embodiments, temperature T_2 is between about 15° C. and about 25° C. In some embodiments, temperature T_2 is between about 20° C. and about 25° C. In some embodiments, temperature T_2 is about 18° C., 19° C., 20° C., 21° C., 22° C., 23° C., 24° C., or 25° C. In some embodiments, temperature T_2 is about 21° C. In some embodiments, temperature T_2 is about 22° C. In some embodiments, temperature T_2 is about 23° C.

In some embodiments, the reaction is monitored for completion (e.g., presence of maribavir or compound 3). In some embodiments, the reaction is monitored for presence of maribavir or compound 3 (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 95.0% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 96.0% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 97.0% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 97.5% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.0% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.1% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.2% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.3% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.4% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 98.5% a/a (e.g., by HPLC). In some embodiments, the reaction is monitored for the presence of maribavir in an amount greater than 99.0% a/a (e.g., by HPLC).

In some embodiments, Step 3 further comprises a step of (b) phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, from the reaction mixture to provide a first resulting mixture.

In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adding a first phase separation solvent. In some embodiments, the first phase separation solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the first phase separation solvent is or comprises chloroform. In some embodiments, the first phase separation solvent is or comprises diethyl ether. In some embodiments, the first phase separation solvent is or comprises ethyl acetate. In some embodiments, the first phase separation solvent is or comprises isopropyl acetate. In some embodiments, the first phase separation solvent is or comprises cyclopentyl methyl ether. In some embodiments, the first phase separation solvent is or comprises toluene. In some embodiments, the first phase separation solvent is or comprises benzene. In some embodiments, the first phase separation solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the first phase separation solvent is or comprises chlorobenzene. In some embodiments, the first phase separation solvent is or comprises xylene. In some embodiments, the first phase separation solvent is or comprises dichloromethane. In some embodiments, the first phase separation solvent is or comprises tert-butyl methyl ether.

In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adding a second phase separation solvent. In some embodiments, the second phase separation solvent is aqueous sodium chloride, aqueous lithium chloride, aqueous calcium chloride, or aqueous potassium chloride. In some embodiments, the second phase separation solvent is aqueous sodium chloride. In some embodiments, the second phase separation solvent is aqueous lithium chloride. In some embodiments, the second phase separation solvent is aqueous calcium chloride. In some embodiments, the second phase separation solvent is aqueous potassium chloride.

In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 4.0 to about 14.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.0 to about 10.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.0 to about 9.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.0 to about 8.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.0 to about 7.5. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.0 to about 7.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.5 to about 10.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 7.5 to about 10.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to between about 6.8 to about 7.5. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 6.0, 6.2, 6.4, 6.6, 6.8, 7.0, 7.2, 7.4, 7.6, 7.8, or 8.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 6.8. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 7.0. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 7.2. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 7.4. In some embodiments, phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adjusting the pH of the reaction mixture to about 7.6.

In some embodiments, Step 3 further comprises a step of (c) distillation and/or solvent exchange of the reaction mixture or first resulting mixture to provide a second resulting mixture comprising maribavir, or a pharmaceutically acceptable salt thereof, and a solvent.

In some embodiments, the reaction mixture or first resulting mixture comprises a first solvent. In some embodiments, the first solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the first solvent is or comprises chloroform. In some embodiments, the first solvent is or comprises diethyl ether. In some embodiments, the first solvent is or comprises ethyl acetate. In some embodiments, the first solvent is or comprises isopropyl acetate. In some embodiments, the first solvent is or comprises cyclopentyl methyl ether. In some embodiments, the first solvent is or comprises toluene. In some embodiments, the first solvent is or comprises benzene. In some embodiments, the first solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the first solvent is or comprises chlorobenzene. In some embodiments, the first solvent is or comprises xylene. In some embodiments, the first solvent is or comprises dichloromethane. In some embodiments, the first solvent is tert-butyl methyl ether. In some embodiments, the first solvent is removed from the reaction mixture or first resulting mixture. In some embodiments, the first solvent is removed by distillation.

In some embodiments, a second solvent is added to the reaction mixture or first resulting mixture. In some embodiments, the second solvent is or comprises isopropyl alcohol, methanol, ethanol, n-propanol, n-butanol, sec-butanol, tert-butanol, chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the second solvent is or comprises methanol. In some embodiments, the second solvent is or comprises ethanol. In some embodiments, the second solvent is or comprises n-propanol. In some embodiments, the second solvent is or comprises n-butanol. In some embodiments, the second solvent is or comprises sec-butanol. In some embodiments, the second solvent is or comprises tert-butanol. In some embodiments, the second solvent is or comprises chloroform. In some embodiments, the second solvent is or comprises diethyl ether. In some embodiments, the second solvent is or comprises ethyl acetate. In some embodiments, the second solvent is or comprises isopropanol. In some embodiments, the second solvent is or comprises cyclopentyl methyl ether. In some embodiments, the second solvent is or comprises toluene. In some embodiments, the second solvent is or comprises benzene. In some embodiments, the second solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the second solvent is or comprises chlorobenzene. In some embodiments, the second solvent is or comprises xylene. In some embodiments, the second solvent is or comprises dichloromethane. In some embodiments, the second solvent is tert-butyl methyl ether. In some embodiments, the second solvent is or comprises isopropyl acetate.

In some embodiments, the second resulting mixture comprises less than 1.0% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.75% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.50% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.25% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 1.0% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.2% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.09% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises less than 0.05% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises an undetectable amount of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the amount of water within the second resulting mixture is measured by ^{1}H NMR. In some embodiments, the amount of water within the second resulting mixture is measured by Karl Fischer.

In some embodiments, the second resulting mixture comprises between about 10-30% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises between about 13-25% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises between about 15-22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises between about 16-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises between about 17-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises between about 17-19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 15%, 16%, 17%, 18%, 19%, 20%, 21%, or 22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 15% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 16% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 17% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 18% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 21% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the second resulting mixture comprises about 22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

In some embodiments, Step 3 further comprises a step of (d) crystallization of maribavir, or a pharmaceutically acceptable salt thereof (e.g., from the reaction mixture, first resulting mixture, or second resulting mixture).

In some embodiments, crystallization comprises providing a crystallization mixture comprising maribavir, or a pharmaceutically acceptable salt thereof, and a primary solvent. In some embodiments, the primary solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the primary solvent is or comprises chloroform. In some embodiments, the primary solvent is or comprises diethyl ether. In some embodiments, the primary solvent is or comprises ethyl acetate. In some embodiments, the primary solvent is or comprises cyclopentyl methyl ether. In some embodiments, the primary solvent is or comprises toluene. In some embodiments, the primary solvent is or comprises benzene. In some embodiments, the primary solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the primary solvent is or comprises chlorobenzene. In some embodiments, the primary solvent is or comprises xylene. In some embodiments, the primary solvent is or comprises dichloromethane. In some embodiments, the primary solvent is or comprises isopropyl acetate.

In some embodiments, the crystallization mixture comprises less than 0.2% w/w of water (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises less than 0.09% w/w of water (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises less than 1.0% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.75% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.50% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.25% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 1.0% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.2% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.09% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the crystallization mixture comprises less than 0.05% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the second resulting mixture comprises an undetectable amount of water (e.g., as measured by ^{1}H NMR or Karl Fischer). In some embodiments, the amount of water within the crystallization mixture is measured by ^{1}H NMR. In some embodiments, the amount of water within the crystallization mixture is measured by Karl Fischer.

Without wishing to be bound by a particular theory, it was thought that too low concentrations of maribavir, or a pharmaceutically acceptable salt thereof, may result in possible dissolution of the seed crystals and potentially impact yield and/or particle size of maribavir. It was also thought that too high concentrations of maribavir, or a pharmaceutically acceptable salt thereof, may result in spontaneous crystallization and/or may affect particle size of maribavir.

In some embodiments, the crystallization mixture comprises between about 10-30% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises between about 13-25% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises between about 15-22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises between about 16-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises between about 17-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises between about 17-19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 15%, 16%, 17%, 18%, 19%, 20%, 21%, or 22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 15% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 16% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 17% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 18% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 21% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR). In some embodiments, the crystallization mixture comprises about 22% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

In some embodiments, crystallization further comprising heating the crystallization mixture to a temperature T_3. In some embodiments, temperature T_3 is greater than about 60° C. In some embodiments, temperature T_3 is greater than about 25° C. In some embodiments, temperature T_3 is less than about 100° C. In some embodiments, temperature T_3 is between about 50° C. and about 150° C. In some embodiments, temperature T_3 is between about 50° C. and about 100° C. In some embodiments, temperature T_3 is between about 50° C. and about 90° C. In some embodiments, temperature T_3 is between about 50° C. and about 85° C. In some embodiments, temperature T_3 is between about 60° C. and about 110° C. In some embodiments, temperature T_3 is between about 70° C. and about 110° C. In some embodiments, temperature T_3 is between about 80° C. and about 110° C. In some embodiments, temperature T_3 is between about 60° C. and about 100° C. In some embodiments, temperature T_3 is between about 70° C. and about 90° C. In some embodiments, temperature T_3 is between about 74° C. and about 90° C. In some embodiments, temperature T_3 is between about 77° C. and about 88° C. In some embodiments, temperature T_3 is between about 82° C. and about 86° C. In some embodiments, temperature T_3 is between about 83° C. and about 85° C. In some embodiments, temperature T_3 is about 77° C., 78° C., 79° C., 80° C., 81° C., 82° C., 83° C., 84° C., 85° C., 86° C., 87° C., 88° C. In some embodiments, temperature T_3 is about 81° C. In some embodiments, temperature T_3 is about 82° C. In some embodiments, temperature T_3 is about 83° C. In some embodiments, temperature T_3 is about 84° C. In some embodiments, temperature T_3 is about 85° C. In some embodiments, temperature T_3 is about 86° C.

In some embodiments, a secondary solvent is added to the crystallization mixture. In some embodiments, the secondary solvent is or comprises chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the secondary solvent is or comprises chloroform. In some embodiments, the secondary solvent is or comprises diethyl ether. In some embodiments, the secondary solvent is or comprises ethyl acetate. In some embodiments, the secondary solvent is or comprises isopropyl acetate. In some embodiments, the secondary solvent is or comprises cyclopentyl methyl ether. In some embodiments, the secondary solvent is or comprises benzene. In some embodiments, the secondary solvent is or comprises α,α,α-trifluorotoluene. In some embodiments, the secondary solvent is or comprises chlorobenzene. In some embodiments, the secondary solvent is or comprises xylene. In some embodiments, the secondary solvent is or comprises dichloromethane. In some embodiments, the secondary solvent is or comprises toluene.

In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for an amount of time between 10 minutes and 2 hrs. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for an amount of time between 10 minutes and 1.5 hrs. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for an amount of time between 10 minutes and 1 hr. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for an amount of time between 10 minutes and 45 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for an amount of time between 10 minutes and 30 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for less than about two hrs. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for less than about 60 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for less than about 30 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for less than about 10 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for less than about 5 minutes. In some embodiments, after the addition of the secondary solvent, the crystallization mixture is aged for at least about 30 minutes.

In some embodiments, a seed crystal is added to the crystallization mixture. Without wishing to be bound by a particular theory, it was thought that too low amounts of seed crystal may permit spontaneous/uncontrolled crystallization, which may impact particle size. In addition, it was thought that higher amounts of seed crystals may adversely affect particle size.

In some embodiments, the seed crystal is maribavir, or a pharmaceutically acceptable salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.01% and 0.50% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.01% and 0.15% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.15% and 0.30% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.05% and 0.30% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.10% and 0.20% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.12% and 0.18% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of between about 0.14% and 0.16% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.13% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.14% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.15% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.16% w/w relative to compound 3, or a salt thereof. In some embodiments, the seed crystal is added in an amount of about 0.17% w/w relative to compound 3, or a salt thereof.

Without wishing to be bound by a particular theory, it was thought that the size of the seed crystals may affect particle size of maribavir (e.g., too small seed crystals may result in smaller particle size of maribavir, and too large seed crystals may result in larger particle size of maribavir). In some embodiments, the size of a seed crystal may be characterized by its d(50) value or Specific Surface Area ("SSA") value. In some embodiments, the size of a seed crystal may be characterized by its d(50) value. It will be understood that "d(50) value" refers to the median particle size of a sample (e.g., 50% of the particles are smaller than the d(50) value and 50% of the particles are larger than the d(50) value). In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 9.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 8.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 7.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 6.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 5.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 4.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.00 μm and about 3.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 2.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 3.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 4.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 5.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 6.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 7.00 μm and about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.25 μm and 9.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 1.75 μm and about 8.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 2.25 μm and about 7.00 μm. In some embodiments, seed crystals comprise a d(50) value between about 2.75 μm and about 6.25 μm. In some embodiments, seed crystals comprise a d(50) value between about 3.00 μm and 6.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 1.25 μm, about 1.75 μm, about 2.25 μm, about 2.75 μm, about 3.00 μm, about 4.00 μm, about 5.00 μm, about 6.00 μm, about 6.25 μm, about 7.00 μm, about 8.00 μm, about 9.00 μm, or about 10.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 1.25 μm. In some embodiments, seed crystals comprise a d(50) value of about 1.75 μm. In some embodiments, seed crystals comprise a d(50) value of about 2.25 μm. In some embodiments, seed crystals comprise a d(50) value of about 2.75 μm. In some embodiments, seed crystals comprise a d(50) value of about 3.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 4.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 5.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 6.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 6.25 μm. In some embodiments, seed crystals comprise a d(50) value of about 7.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 8.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 9.00 μm. In some embodiments, seed crystals comprise a d(50) value of about 10.00 μm.

In some embodiments, after the seed crystal is added, the crystallization mixture is aged between about 15 minutes and about 120 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged between about 30 minutes and about 90 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged between about 30 minutes and about 60 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged between about 60 minutes and about 90 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged between about 90 minutes and about 120 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 5 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 15 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 30 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 60 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 90 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for at least about 120 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for about 30 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for about 60 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for about 90 minutes. In some embodiments, after the seed crystal is added, the crystallization mixture is aged for about 120 minutes.

In some embodiments, additional secondary solvent is added to the crystallization mixture. In some embodiments, the additional secondary solvent is added in an amount of between about 5/1 and about 1/5 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of between about 2/1 and about 1/2 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of between about 24/7 and about 2/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of between about 16/7 and about 4/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 8/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 9/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 10/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 11/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 12/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 13/7 of total primary solvent (kg/kg). In some embodiments, the additional secondary solvent is added in an amount of about 15/7 of total primary solvent (kg/kg).

In some embodiments, after the additional secondary solvent is added, the crystallization mixture is agitated for a period of time. In some embodiments, the period of time is at least 30 minutes. In some embodiments, the period of time is at least 60 mins. In some embodiments, the period of time is between about 15 minutes and 120 minutes. In some embodiments, the period of time is between about 15 minutes and 90 minutes. In some embodiments, the period of time is between about 15 minutes and 60 minutes. In some embodiments, the period of time is between about 30 minutes and 90 minutes. In some embodiments, the period of time is between about 45 minutes and 90 minutes. In some embodiments, the period of time is between about 60 minutes and 90 minutes. In some embodiments, the period of time is about 15 minutes. In some embodiments, the period of time is about 30 minutes. In some embodiments, the period of time is about 45 minutes. In some embodiments, the period of time is about 60 minutes. In some embodiments, the period of time is about 90 minutes.

In some embodiments, after the additional secondary solvent is added, the crystallization mixture is agitated at temperature T_3. In some embodiments, temperature T_3 is between about 50° C. and about 150° C. In some embodiments, temperature T_3 is between about 50° C. and about 100° C. In some embodiments, temperature T_3 is between about 50° C. and about 90° C. In some embodiments, temperature T_3 is between about 50° C. and about 80° C. In some embodiments, temperature T_3 is between about 70° C. and about 115° C. In some embodiments, temperature T_3 is between about 80° C. and about 115° C. In some embodiments, temperature T_3 is between about 60° C. and about 105° C. In some embodiments, temperature T_3 is between about 70° C. and about 95° C. In some embodiments, temperature T_3 is between about 75° C. and about 90° C. In some embodiments, temperature T_3 is between about 77° C. and about 88° C. In some embodiments, temperature T_3 is between about 81° C. and about 87° C. In some embodiments, temperature T_3 is between about 81° C. and about 86° C. In some embodiments, temperature T_3 is about 80° C., 81° C., 82° C., 84° C., 85° C., 86° C., 87° C., 88° C., 89° C., or 90° C. In some embodiments, temperature T_3 is about 83° C. In some embodiments, temperature T_3 is about 84° C. In some embodiments, temperature T_3 is about 85° C.

In some embodiments, the crystallization mixture is cooled to temperature T_4. In some embodiments, temperature T_4 is between about −20° C. and about 25° C. In some embodiments, temperature T_4 is between about −15° C. and about 20° C. In some embodiments, temperature T_4 is between about −15° C. and about 15° C. In some embodiments, temperature T_4 is between about −15° C. and about 10° C. In some embodiments, temperature T_4 is between about −15° C. and about 5° C. In some embodiments, temperature T_4 is between about −15° C. and about 0° C. In some embodiments, temperature T_4 is between about −5° C. and about 25° C. In some embodiments, temperature T_4 is between about 0° C. and about 25° C. In some embodiments, temperature T_4 is between about 5° C. and about 25° C. In some embodiments, temperature T_4 is between about 10° C. and about 25° C. In some embodiments, temperature T_4 is between about −10° C. and about 20° C. In some embodiments, temperature T_4 is between about −5° C. and about 15° C. In some embodiments, temperature T_4 is between about 0° C. and about 10° C. In some embodiments, temperature T_4 is between about 3° C. and about 7° C. In some embodiments, temperature T_4 is about 0° C., 1° C., 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., 8° C., 9° C., or 10° C. In some embodiments, temperature T_4 is about 4° C. In some embodiments, temperature T_4 is about 5° C. In some embodiments, temperature T_4 is about 6° C.

In some embodiments, the crystallization mixture is cooled to temperature T_4 over between about 0.5 hrs and about 8 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over between about 1 hr and about 5 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over between about 2 hrs and about 4 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over at least 1 hr. In some embodiments, the crystallization mixture is cooled to temperature T_4 over at least 3 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 0.5, 1, 2, 3, 4, 5, 6, 7, or 8 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 2 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 3 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 4 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 6 hrs. In some embodiments, the crystallization mixture is cooled to temperature T_4 over about 8 hrs.

In some embodiments, the crystallization mixture is agitated at temperature T_4 for a period of time. In some embodiments, the period of time is between about 3 hrs and about 30 hrs. In some embodiments, the period of time is between about 3 hrs and about 18 hrs. In some embodiments, the period of time is between about 3 hrs and about 15 hrs. In some embodiments, the period of time is between about 3 hrs and about 12 hrs. In some embodiments, the period of time is between about 3 hrs and about 8 hrs. In some embodiments, the period of time is between about 3 hrs and about 5 hrs. In some embodiments, the period of time is less than about 18 hrs. In some embodiments, the period of time is less than about than 12 hrs. In some embodiments, the period of time is less than about 6 hrs. In some embodiments, the period of time is less than about 3 hrs.

In some embodiments, Step 3 further comprises a step of (e) optionally filtering or washing the crystallization mixture to provide a filtered mixture. In some embodiments, Step 3 further comprises filtering the crystallization mixture to provide the filtered mixture. In some embodiments, Step 3 further comprises washing the crystallization mixture to provide the filtered mixture. In some embodiments, the crystallization mixture is washed with chloroform, diethyl ether, ethyl acetate, isopropyl acetate, tert-butyl methyl ether, cyclopentyl methyl ether, toluene, benzene, α,α,α-trifluorotoluene, chlorobenzene, xylene, or dichloromethane. In some embodiments, the crystallization mixture is washed with chloroform. In some embodiments, the crystallization mixture is washed with diethyl ether. In some embodiments, the crystallization mixture is washed with ethyl acetate. In some embodiments, the crystallization mixture is washed with isopropyl acetate. In some embodiments, the crystallization mixture is washed with cyclopentyl methyl ether. In some embodiments, the crystallization mixture is washed with benzene. In some embodiments, the crystallization mixture is washed with α,α,α-trifluorotoluene. In some embodiments, the crystallization mixture is washed with chlorobenzene. In some embodiments, the crystallization mixture is washed with xylene. In some embodiments, the crystallization mixture is washed with dichloromethane. In some embodiments, the crystallization mixture is washed with toluene.

In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 2.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 1.50 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 1.25 g/g. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 1.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 0.9 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.50 L/kg and about 2.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.75 L/kg and about 2.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.85 L/kg and about 2.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.30 L/kg and about 2.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.40 L/kg and about 1.75 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.50 L/kg and about 1.50 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.60 L/kg and about 1.25 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.67 L/kg and about 1.00 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.75 L/kg and about 0.90 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount between about 0.80 L/kg and about 0.85 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.60 L/kg, 0.65 L/kg, 0.70 L/kg, 0.75 L/kg, 0.80 L/kg, 0.84 L/kg, 0.90 L/kg, or 0.95 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.75 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.80 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.84 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.90 L/kg. In some embodiments, the crystallization wash solvent is provided in an amount of about 0.95 L/kg.

In some embodiments, the crystallization wash solvent is provided at a temperature between about −20° C. and about 60° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −10° C. and about 30° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −10° C. and about 20° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −10° C. and about 10° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −5° C. and about 40° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 0° C. and about 40° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 5° C. and about 40° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −5° C. and about 30° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about −5° C. and about 20° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 0° C. and about 20° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 0° C. and about 15° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 0° C. and about 10° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 2° C. and about 8° C. In some embodiments, the crystallization wash solvent is provided at a temperature between about 4° C. and about 6° C. In some embodiments, the crystallization wash solvent is provided at a temperature of about 1° C., 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., 8° C., or 9° C. In some embodiments, the crystallization wash solvent is provided at a temperature of about 4° C. In some embodiments, the crystallization wash solvent is provided at a temperature of about 5° C. In some embodiments, the crystallization wash solvent is provided at a temperature of about 6° C.

In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is less than 5 days. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 5 days. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 24 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 12 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 8 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 6 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 4 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 2 hrs. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 1 hr. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 45 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 30 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is between about 1 minute and about 30 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is about 15 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is at least 15 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is at least 30 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is at least 60 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is at least 90 minutes. In some embodiments, the crystallization wash solvent is provided for a period of time, that period of time is at least 1 day.

In some embodiments, Step 3 further comprises a step of (f) drying the filtered mixture.

In some embodiments, drying comprises heating the filtered mixture to temperature T_5 without agitation for a period of time. In some embodiments, temperature T_5 is between about 20° C. and about 100° C. In some embodiments, temperature T_5 is between about 20° C. and about 70° C. In some embodiments, temperature T_5 is between about 20° C. and about 50° C. In some embodiments, temperature T_5 is between about 30° C. and about 80° C. In some embodiments, temperature T_5 is between about 40° C. and about 80° C. In some embodiments, temperature T_5 is between about 50° C. and about 80° C. In some embodiments, temperature T_5 is between about 35° C. and about 60° C. In some embodiments, temperature T_5 is between about 40° C. and about 50° C. In some embodiments, temperature T_5 is less than or equal to 20° C. In some embodiments, temperature T_5 is less than or equal to about 20° C., 30° C., 40° C., 50° C., 60° C., 70° C., 80° C., 90° C., or 100° C. In some embodiments, temperature T_5 is less than or equal to 40° C. In some embodiments, temperature T_5 is less than or equal to 50° C. In some embodiments, temperature T_5 is less than or equal to 60° C.

In some embodiments, the filtered mixture is held at temperature T_5 for a period of time. In some embodiments, the period of time is at least 1 hr. In some embodiments, the period of time is at least 2 hrs. In some embodiments, the period of time is at least 3 hrs. In some embodiments, the period of time is at least 4 hrs. In some embodiments, the period of time is at least 5 hrs. In some embodiments, the period of time is at least 6 hrs. In some embodiments, the period of time is at least 7 hrs. In some embodiments, the period of time is at least 8 hrs. In some embodiments, the period of time is at least 9 hrs. In some embodiments, the period of time is at least 10 hrs.

In some embodiments, drying further comprises heating the filtered mixture to temperature T_6 without agitation for a period of time. In some embodiments, temperature T_6 is between about 20° C. and about 100° C. In some embodiments, temperature T_6 is between about 40° C. and about 100° C. In some embodiments, temperature T_6 is between about 20° C. and about 80° C. In some embodiments, temperature T_6 is between about 20° C. and about 70° C. In some embodiments, temperature T_6 is between about 20° C. and about 60° C. In some embodiments, temperature T_6 is between about 30° C. and about 60° C. In some embodiments, temperature T_6 is between about 40° C. and about 60° C. In some embodiments, temperature T_6 is between about 45° C. and about 55° C. In some embodiments, temperature T_6 is between about 45° C. and about 50° C. In some embodiments, temperature T_6 is less than or equal to 25° C., 30° C., 40° C., 45° C., 50° C., 55° C., 60° C. In some embodiments, temperature T_6 is less than or equal to 90° C. In some embodiments, temperature T_6 is less than or equal to 70° C. In some embodiments, temperature T_6 is less than or equal to 60° C. In some embodiments, temperature T_6 is less than or equal to 55° C. In some embodiments, temperature T_6 is less than or equal to 50° C. In some embodiments, temperature T_6 is less than or equal to 45° C. In some embodiments, temperature T_6 is less than or equal to 40° C.

In some embodiments, the filtered mixture is held at temperature T_6 for a period of time. In some embodiments, the period of time is between about 1 hr and 5 days. In some embodiments, the period of time is between about 1 hr and 2 days. In some embodiments, the period of time is between about 1 hr and 1 day. In some embodiments, the period of time is between about 1 hr and 12 hrs. In some embodiments, the period of time is between about 1 hr and 6 hrs. In some embodiments, the period of time is at least 1 hr. In some embodiments, the period of time is at least 2 hrs. In some embodiments, the period of time is at least 3 hrs. In some embodiments, the period of time is at least 4 hrs. In some embodiments, the period of time is at least 5 hrs. In some embodiments, the period of time is at least 6 hrs. In some embodiments, the period of time is until the filtered mixture appears dry.

In some embodiments, the period of time is until between about 100 ppm and about 1600 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 100 ppm and about 1400 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 100 ppm and about 1200 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 100 ppm and about 1000 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 100 ppm and about 800 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 100 ppm and about 700 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 300 ppm and about 1400 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 500 ppm and about 1400 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 700 ppm and about 1400 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 300 ppm and about 1200 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 500 ppm and about 1000 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 600 ppm and about 800 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 650 ppm and about 750 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 1500 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 1200 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 1000 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 800 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 700 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 600 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 500 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 400 ppm of toluene is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 300 ppm of toluene is present (e.g., as measured by GCHS).

In some embodiments, the period of time is until between about 1,000 ppm and about 10,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 1,000 ppm and about 8,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 1,000 ppm and about 6,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 1,000 ppm and about 5,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 1,000 ppm and about 4,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 1,000 ppm and about 3,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 3,000 ppm and about 10,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 4,000 ppm and about 10,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 5,000 ppm and about 10,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 2,000 ppm and about 8,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 3,000 ppm and about 6,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until between about 4,000 ppm and about 5,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 10,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 8,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 6,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 5,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 4,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 3,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 2,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS). In some embodiments, the period of time is until less than 1,000 ppm of isopropyl acetate is present (e.g., as measured by GCHS).

In some embodiments, Step 3 provides unmilled maribavir, or a pharmaceutically acceptable salt thereof. In some embodiments, Step 3 further comprises a step of (g) milling the unmilled maribavir, or a pharmaceutically acceptable salt thereof, to provide milled maribavir, or a pharmaceutically acceptable salt thereof. In some embodiments, milling comprises formulating into a tablet (e.g., a 200 mg tablet).

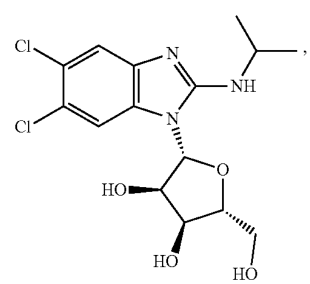
Formulations & Compositions

In some embodiments, provided compositions comprise at least 90% by weight of maribavir. In some embodiments, provided compositions comprise at least 95% by weight of maribavir. In some embodiments, provided compositions comprise at least 99% by weight of maribavir. In some embodiments, provided compositions comprise at least 99.5% by weight of maribavir. In some embodiments, provided compositions comprise at least 99.6% by weight of maribavir. In some embodiments, provided compositions comprise at least 99.7% by weight of maribavir. In some embodiments, provided compositions comprise at least 99.8% by weight of maribavir. In some embodiments, provided compositions comprise at least 99.9% by weight of maribavir.

In some embodiments, provided composition comprise maribavir substantially free of impurities. As used herein, the term "substantially free of impurities" means that the composition or compound contains no significant amount of extraneous matter. Such extraneous matter may include starting materials, residual solvents, or other impurities that may result from the preparation of, and/or isolation of maribavir.

In some embodiments, the present disclosure provides a composition comprising maribavir and one or more compound selected from compound 2, compound 3, and compound 4. In some embodiments, such compositions comprise 0.01-0.05% (w/w) of compound 2, 0.01-0.05% (w/w) compound 3, and/or 0.01-0.05% (w/w) compound 4. In some embodiments, such compositions comprise 0.01-0.05% (a/a) of compound 2, 0.01-0.05% (a/a) compound 3, and/or 0.01-0.05% (a/a) compound 4, as measured by HPLC. In some embodiments, the present disclosure provides a composition comprising maribavir and 0.01-0.5% ent-maribavir (i.e., D-maribavir):

D-maribavir



In some embodiments, the present disclosure provides a composition as set forth in any of Tables 4-12 to 4-20 in Example 4.

In some embodiments, Steps 1-3 provide a particular polymorphic form of maribavir. In some embodiments, Steps 1-3 provide Form VI of maribavir (e.g., at least 90%, 95%, or 99% by weight of Form VI of maribavir), as described in U.S. Pat. No. 6,482,939. In some embodiments, Steps 1-3 provide Form VI of maribavir substantially free of other polymorphic forms of maribavir. As used herein, the term "polymorphic forms" includes solvates and hydrates.

In some embodiments, provided compositions comprises Form VI of maribavir, as described in described in U.S. Pat. No. 6,482,939 (e.g., at least 90%, 95%, or 99% by weight of Form VI of maribavir). In some embodiments, provided compositions comprises Form VI of maribavir (e.g., at least 90%, 95%, or 99% by weight of Form VI of maribavir), as described in described in U.S. Pat. No. 6,482,939, and another polymorphic form of maribavir, e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777. In some embodiments, provided compositions comprises Form VI of maribavir, as described in described in U.S. Pat. No. 6,482,939, and an isopropyl acetate solvate of maribavir (e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777). In some embodiments, provided composition comprise Form VI of maribavir, and less than 10%, 5%, 1%, or 0.5% by weight of another polymorphic form of maribavir, e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777. In some embodiments, provided composition comprise Form VI of maribavir, and less than 10%, 5%, 1%, or 0.5% by weight of another polymorphic form of maribavir, e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777. In some embodiments, provided composition comprise Form VI of maribavir substantially free of other polymorphic forms, e.g., as described in U.S. Pat. Nos. 6,482,939, 8,546,344, or U.S. Pat. No. 11,130,777.

In some embodiments, maribavir (e.g., Form VI) is milled (e.g., hammer milled) and formulated into a tablet (e.g., a 200 mg tablet).

Particle size distribution ("PSD") is a measurement that defines the number of particles present according to their size. It will be appreciated that PSD is a valuable indicator of quality and performance and is necessary to ensure consistent drug product manufacturability. For example, PSD may impact ease of handling of a compound and excipients when formulating (e.g., into a tablet). Consistent formulations are important to ensure that the active drug is delivered to the correct location within the body, in the right concentration, and at the correct rate. In some embodiments, the present disclosure provides compositions of maribavir, and methods of making maribavir and compositions comprising maribavir, with PSDs that result in consistent formulation into tables. In some embodiments, Steps 1, 2, 3, and/or milling (e.g., hammer milling) provide maribavir with a PSD that permits manufacture of tablets with uniformity. In some embodiments, Steps 3 provides maribavir Form VI particle with a desirable PSD. In some embodiments, the present disclosure provides compositions comprising maribavir Form VI particle with a PSD of d(50) as between about 100 and about 400 μm. In some embodiments, the present disclosure provides compositions comprising maribavir Form VI particle with a PSD of d(50) as between about 100 and about 400 μm wherein such maribavir Form VI is manufactured by using maribavir seed with a PSD of d(50) between about between about 1 and about 10 μm. In some embodiments, the present disclosure provides methods of manufacturing maribavir Form VI particle by using maribavir Form VI seed with a PSD of d(50) between about between about 1 and about 10 μm.

In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 50 and about 500 μm. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 100 and about 400 μm. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 170 and about 350 μm. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 170 and about 226 μm. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 227 and about 280 μm. In some embodiments, maribavir, or a pharmaceutically acceptable salt thereof, has a PSD of d(50) as between about 281 and about 336 μm.

In some aspects, the present disclosure provides compositions comprising maribavir or a pharmaceutically acceptable salt thereof. In some embodiments, the present disclosure provides compositions made by a method disclosed here (e.g., Steps 1-3). In some embodiments, the present disclosure provides compositions comprising maribavir, or a pharmaceutically acceptable salt thereof, and one or more compounds selected from:

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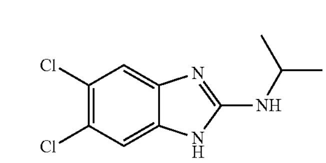

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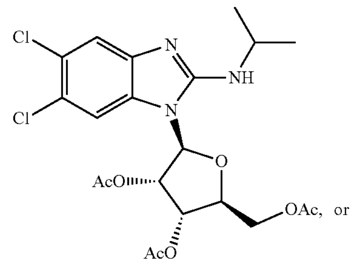

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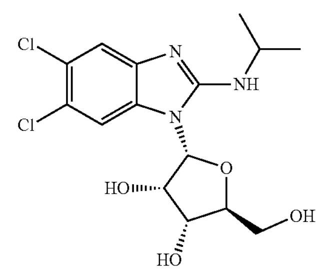


or pharmaceutically acceptable salts thereof.

In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% (w/w HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% w/w (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% w/w (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% a/a (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% a/a (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% a/a (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a (e.g., as measured by HPLC) of compound 2, relative to maribavir. In some embodiments, the present disclosure provides compositions, wherein compound 2, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% (w/w HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% w/w (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% w/w (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% a/a (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% a/a (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% a/a (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a (e.g., as measured by HPLC) of compound 3, relative to maribavir. In some embodiments, the present disclosure provides compositions, wherein compound 3, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% (w/w HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% w/w (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% w/w (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions composition comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.10% a/a (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.01-0.05% a/a (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising 0.02-0.05% a/a (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a (e.g., as measured by HPLC) of compound 4, relative to maribavir. In some embodiments, the present disclosure provides compositions, wherein compound 4, or a salt thereof, is not detectable (e.g., by HPLC or ^{1}H NMR).

Dosing Regimens

Provided herein are methods of treating a patient suffering from CMV, comprising administering maribavir (e.g., by administering a composition that comprises and/or delivers maribavir as described herein) to the patient. In some embodiments, provided methods comprise administering about 400 mg of maribavir orally to the patient twice daily. In some embodiments, provided methods comprise administering about 200 mg of maribavir orally to the patient twice daily.

Maribavir can administered with or without food. In some embodiments, maribavir is administered as 200 mg tablets. In some embodiments, maribavir is administered as 400 mg tablets. In some embodiments, maribavir is administered as 200 mg oral solid composition. In some embodiments, maribavir is administered as 400 mg oral solid composition.

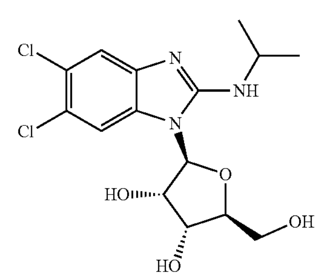
In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is intolerant to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, (v) less than 0.05% of D-maribavir or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients are administered immunosuppressant drugs that are CYP3A4 and/or P-gp substrates including tacrolimus, cyclosporine, sirolimus and everolimus. In some embodiments, the present disclosure provides methods of controlling drug concentrations of maribavir and immunosuppressant in patients with post-transplant CMV infection or disease comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of

Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients are administered immunosuppressant drugs. In some embodiments, the present disclosure provides methods of controlling drug concentrations of maribavir and immunosuppressant in patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients are administered immunosuppressant drugs that are CYP3A4 and/or P-gp substrates including tacrolimus, cyclosporine, sirolimus and everolimus. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients are administered immunosuppressant drugs that are CYP3A4 and/or P-gp substrates including tacrolimus, cyclosporine, sirolimus and everolimus, wherein immunosuppressant drug levels in said patients are monitored after administration of maribavir. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 800 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI particle having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients have received carbamazepine before the administration of maribavir. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 1200 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients, wherein said patients have received phenytoin or phenobarbital before the administration of maribavir. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day for more than 6 months, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day for more than 12 months, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) less than about 400 μm, (ii) less than 0.05% of Compound 2 or a salt thereof, (iii) less than 0.05% of Compound 3 or a salt thereof, (iv) less than 0.05% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) between about 50 and about 400 μm, (ii) less than 0.1% of Compound 2 or a salt thereof, (iii) less than 0.15% of Compound 3 or a salt thereof, (iv) less than 0.1% of Compound 4 or a salt thereof, and (v) pharmaceutically acceptable excipients. In some embodiments, the present disclosure provides methods of treating patients with post-transplant CMV infection or disease that is refractory to treatment with ganciclovir, valganciclovir, cidofovir or foscarnet comprising orally administering maribavir or a salt thereof to said patients in an amount of 400 mg twice a day, wherein maribavir is administered as an oral solid composition, wherein such composition comprises (i) maribavir Form VI having a PSD of d(50) between about 50 and about 400 μm, (ii) less than 0.1% of intermediate impurities, (iii) less than 0.1% of enantiomeric impurities, (iv) less than 0.1% of unspecified impurities, and (v) pharmaceutically acceptable excipients.

ENUMERATED EMBODIMENTS

1. A composition comprising

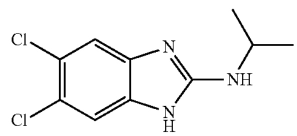
(i) maribavir:



or a pharmaceutically acceptable salt thereof, and

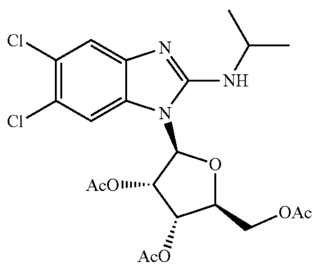
(ii) and one or more of the following:
Compound 2,

2



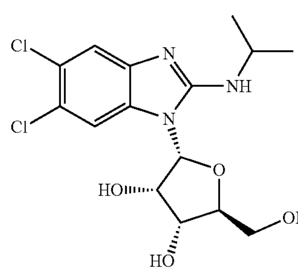
or a pharmaceutically acceptable salt thereof,
Compound 3,

3



or a pharmaceutically acceptable salt thereof, or
Compound 4:

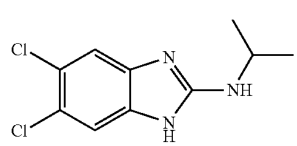
4



or a pharmaceutically accept salt thereof.

2. The composition of embodiment 1, comprising Compound 2:

2



or a pharmaceutically acceptable salt thereof.

3. The composition of embodiment 2, comprising 0.01-0.10% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

4. The composition of embodiment 2 or 3, comprising 0.01-0.05% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

5. The composition of any one of embodiments 2-4, comprising 0.02-0.05% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

6. The composition of embodiment 2, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

7. The composition of embodiment 2, comprising 0.01-0.10% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

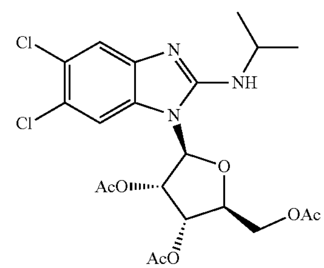
8. The composition of embodiment 2 or 7, comprising 0.01-0.5% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

9. The composition of any one of embodiments 2 or 7-8, comprising 0.02-0.05% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

10. The composition of embodiment 2, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

11. The composition of any one of embodiments 1-10, comprising Compound 3:

3



or a pharmaceutically acceptable salt thereof.

12. The composition of embodiment 11, comprising 0.01-0.10% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

13. The composition of embodiment 11 or 12, comprising 0.01-0.5% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

14. The composition of any one of embodiments 11-13, comprising 0.02-0.05% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

14. The composition of embodiment 11, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

15. The composition of embodiment 11, comprising 0.01-0.10% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

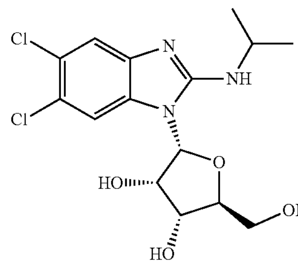
16. The composition of embodiment 11 or 15, comprising 0.01-0.5% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

17. The composition of any one of embodiments 11 or 15-16, comprising 0.02-0.05% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

18. The composition of embodiment 11, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

19. The composition of any one of embodiments 1-18, comprising Compound 4:

4



or a pharmaceutically acceptable salt thereof.

20. The composition of embodiment 19, comprising 0.01-0.10% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

21. The composition of embodiment 19 or 20, comprising 0.01-0.5% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

22. The composition of any one of embodiments 19-21, comprising 0.02-0.05% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

23. The composition of embodiment 19, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

24. The composition of embodiment 19, comprising 0.01-0.10% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

25. The composition of embodiment 19 or 24, comprising 0.01-0.5% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

26. The composition of any one of embodiments 19 or 24-25, comprising 0.02-0.05% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

27. The composition of embodiment 19, comprising less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

28. The composition of any one of embodiments 1-27, comprising maribavir having a particle size distribution (PSD) of between of d(50) of about 170 and about 350 μm (e.g., about 170 and about 226 μm, about 227 and about 280 μm, or about 281 and about 336 μm).

29. The composition of any one of embodiments 1-28, further comprising a pharmaceutically acceptable excipient.

30. An oral solid formulation of maribavir comprising the composition of any of embodiments 1-29.

31. The oral solid formulation of embodiment 31, wherein the formulation is a tablet.

32. The oral solid formulation of embodiment 30 or 31, comprising about 200 mg of maribavir.

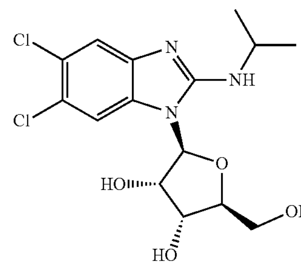
33. The composition of any one of embodiments 1-29 or the oral solid formulation of any one of embodiments 30-32, wherein maribavir is prepared by a process of any one of embodiments 37-186.

34. A method of treating a patient suffering from cytomegalovirus (CMV) infection comprising administering to the patient the pharmaceutical composition of any one of embodiments 1-29, or the oral solid formulation of any one of embodiments 30-33.

35. The method of embodiment 34, wherein the patient is a transplant recipient.

36. The method of embodiment 34 or 35, wherein the patient is refractory to treatment with ganciclovir, valganciclovir, cidofovir, or foscarnet.

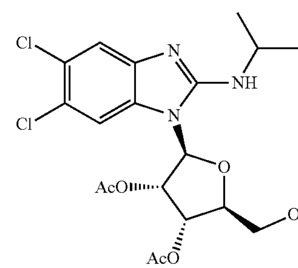
37. A method of preparing maribavir:



or a pharmaceutically acceptable salt thereof, comprising a step of

(a) reacting compound 3:

3



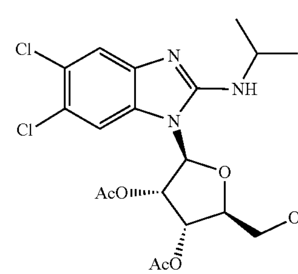
or a salt thereof,

under suitable reaction conditions to provide maribavir, or a pharmaceutically acceptable salt thereof.

38. A method of preparing a composition according to any one of embodiments 1-32, comprising a step of

(a) reacting compound 3:

3



or a salt thereof,

under suitable reaction conditions to provide maribavir, or a pharmaceutically acceptable salt thereof.

39. The method of embodiment 37 or 38, wherein the reaction conditions comprise a solvent.

40. The method of embodiment 39, wherein the solvent is or comprises methanol.

41. The method of embodiment 39, wherein the solvent is or comprises MTBE.

42. The method of any one of embodiments 39-41, wherein the solvent comprises methanol and MTBE.

43. The method of embodiment 42, wherein methanol is provided in an amount of between about 0.34 L/kg and about 0.56 L/kg, and MTBE is provided in an amount of between about 3.65 L/kg and about 6.35 L/kg.

44. The method of embodiment 42 or 43, wherein methanol is provided in an amount of about 0.48 L/kg, and MTBE is provided in an amount of about 5.3 L/kg.

45. The method of any one of embodiments 37-43, wherein the reaction conditions comprise a base.

46. The method of embodiment 45, wherein the base is NaOH.

47. The method of embodiment 45 or 46, wherein the base is aqueous NaOH.

48. The method of any one of embodiments 37-47, wherein the reaction conditions comprise heating to temperature T_1.

49. The method of embodiment 48, wherein temperature T_1 is between about 24° C. and 45° C.

50. The method of embodiment 48 or 49, wherein temperature T_1 is about 30° C.

51. The method of any one of embodiments 37-50, wherein the reaction is allowed proceed for a period of time.

52. The method of embodiment 51, wherein the reaction is agitated for a period of time.

53. The method of embodiment 51 or 52, wherein the period of time is between about 2.5 and about 4 hrs.

54. The method of any one of embodiments 37-53, wherein the reaction conditions comprise cooling the reaction mixture to a temperature T_2.

55. The method of embodiment 54, wherein temperature T_2 is about 22° C.

56. The method of any one of embodiments 37-55, wherein the reaction is monitored for the presence of maribavir in an amount greater than 98.3% a/a by HPLC.

57. The method of any one of embodiments 37-56, further comprising a step of (b) phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, from the reaction mixture to provide a first resulting mixture.

58. The method of embodiment 57, wherein phase separation and/or extraction of maribavir, or a pharmaceutically acceptable salt thereof, comprises adding a first phase separation solvent.

59. The method of embodiment 58, wherein the first phase separation solvent is MBTE.

60. The method of any one of embodiments 57-59, wherein phase separation and/or extraction comprises adding a second phase separation solvent.

61. The method of embodiment 60, wherein the second phase separation solvent is aqueous sodium chloride.

62. The method of any one of embodiments 57-61, wherein phase separation and/or extraction comprises adjusting the pH of the reaction mixture to between about 6.0 to about 10.0.

63. The method of any one of embodiments 57-62, wherein phase separation and/or extraction comprises adjusting the pH of the reaction mixture to between about 6.8 to about 7.5.

64. The method of any one of embodiments 37-63, further comprising a step of (c) distillation and/or solvent exchange of the reaction mixture or first resulting mixture to provide a second resulting mixture comprising maribavir, or a pharmaceutically acceptable salt thereof.

65. The method of embodiment 64, wherein the reaction mixture or first resulting mixture comprises a first solvent.

66. The method of embodiment 65, wherein the first solvent is TBME.

67. The method of any one of embodiments 65-67, wherein the first solvent is removed from the reaction mixture or first resulting mixture.

68. The method of embodiment 67, wherein the first solvent is removed by distillation.

69. The method of any one of embodiments 63-68, wherein a second solvent is added to the reaction mixture or first resulting mixture.

70. The method of embodiment 69, wherein the second solvent is isopropyl acetate.

71. The method of any one of embodiments 64-70, wherein the second resulting mixture comprises isopropyl acetate.

72. The method of any one of embodiments 64-71, wherein the second resulting mixture comprises less than 0.2% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer).

73. The method of any one of embodiments 64-72, wherein the second resulting mixture comprises less than 0.09% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer).

74. The method of any one of embodiments 64-73, wherein the second resulting mixture comprises between about 17-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

75. The method of any one of embodiments 64-74, wherein the second resulting mixture comprises between about 17-19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

76. The method of any one of embodiments 37-75, further comprising a step of (d) crystallization of maribavir, or a pharmaceutically acceptable salt thereof (e.g., from the reaction mixture, first resulting mixture, or second resulting mixture).

77. The method of embodiment 76, crystallization comprises providing a crystallization mixture comprising maribavir, or a pharmaceutically acceptable salt thereof, and a primary solvent.

78. The method of embodiment 77, wherein the primary solvent is isopropyl acetate.

79. The method of any one of embodiments 76-78, wherein the crystallization resulting mixture comprises less than 0.2% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer).

80. The method of any one of embodiments 76-79, wherein the crystallization mixture comprises less than 0.09% w/w of water (e.g., as measured by ^{1}H NMR or Karl Fischer).

81. The method of any one of embodiments 76-80, wherein the crystallization mixture comprises between about 17-20% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

82. The method of any one of embodiments 76-81, wherein the crystallization resulting mixture comprises between about 17-19% w/w of maribavir, or a pharmaceutically acceptable salt thereof (e.g., as measured by ^{1}H NMR).

83. The method of any one of embodiments 76-82, wherein crystallization further comprising heating the crystallization mixture to a temperature T_3.

84. The method of embodiment 83, wherein temperature T_3 is between about 77 and about 88° C.

85. The method of embodiment 83 or 84, wherein temperature T_3 is about 84° C.

86. The method of any one of embodiments 76-85, wherein a secondary solvent is added to the crystallization mixture.

87. The method of embodiment 86, wherein the secondary solvent is toluene.

88. The method of any one of embodiments 86-87, wherein, after the addition of the secondary solvent, the crystallization mixture is aged for less than about 30 minutes.

89. The method of any one of embodiments 76-88, wherein a seed crystal is added to the crystallization mixture.

90. The method of embodiment 89, the seed crystal is maribavir, or a pharmaceutically acceptable salt thereof.

91. The method of embodiment 89 or 90, wherein the seed crystal is added in an amount of between about 0.05% and 0.30 w/w relative to compound 3, or a salt thereof.

92. The method of any one of embodiments 89-91, wherein the seed crystal is added in an amount of about 0.15% w/w relative to compound 3, or a salt thereof.

93. The method of any one of embodiments 89-92, wherein the size of the seed crystals (d(50)) is between about 2.25-7.00 μm.

94. The method of any one of embodiments 89-93, wherein the size of the seed crystals (d(50)) is between about 2.75-6.25 μm.

95. The method of any one of embodiments 89-94, wherein after the seed crystal is added, the crystallization mixtures is aged for about 30 minutes.

96. The method of any one of embodiments 76-95, wherein additional secondary solvent is added to the crystallization mixture.

97. The method of embodiment 96, wherein the additional secondary solvent is added in an about of about 11/7 of total primary solvent (kg/kg).

98. The method of embodiment 96 or 97, wherein, after the additional secondary solvent is added, the crystallization mixture is agitated for a period of time.

99. The method of embodiment 98, wherein the period of time is about 1 hr.

100. The method of embodiments 98 or 99, wherein, after the additional secondary solvent is added, the crystallization mixture is agitated at temperature T_3.

101. The method of embodiment 100, wherein temperature T_3 is between about 77 and about 88° C.

102. The method of embodiment 100 or 101, wherein temperature T_3 is about 84° C.

103. The method of any one of embodiments 76-102, wherein the crystallization mixture is cooled to temperature T_4.

104. The method of embodiment 103, wherein temperature T_4 is between about 3° C. and about 7° C.

105. The method of embodiment 103 or 104, wherein temperature T_4 is about 5° C.

106. The method of any one of embodiments 103-105, wherein the crystallization mixture is cooled to temperature T_4 over about 3 hrs.

107. The method of any one of embodiments 103-106, wherein the crystallization mixture is agitated at temperature T_4 for a period of time.

108. The method of embodiment 107, wherein the period of time is between about 3 and about 18 hrs.

109. The method of any one of embodiments 76-108, further comprising a step of (e) optionally filtering or washing the crystallization mixture to provide a filtered mixture.

110. The method of embodiment 109, comprising a step of filtering the crystallization mixture to provide the filtered mixture.

111. The method of embodiment 109 or 110, comprising a step of washing the crystallization mixture to provide the filtered mixture.

112. The method of embodiment 111, wherein the crystallization mixture is washed with toluene.

113. The method of any one of embodiments 76-112, further comprising a step of (f) drying the filtered mixture.

114. The method of embodiment 113, wherein drying comprises heating the filtered mixture to temperature T_5 without agitation for a period of time.

115. The method of embodiment 114, wherein temperature T_5 is less than or equal to 40° C.

116. The method of embodiment 114 or 115, wherein the period of time is at least 5 hrs.

117. The method of any one of embodiments 113-116, wherein drying further comprises heating the filtered mixture to temperature T_6 without agitation for a period of time.

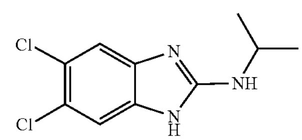
119. The method of embodiment 117, wherein temperature T_6 is less than or equal to 50° C.

120. The method of embodiment 117 or 118, wherein the period of time is at least 3 hrs.

121. The method of any one of embodiments 34-120, wherein compound 3, or a salt thereof, is prepared by a method comprising:

(a) providing compound

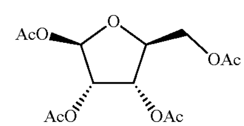
2



or a salt thereof, and

compound 8:

8



or a salt thereof,

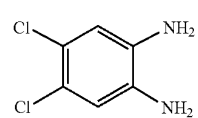
under suitable reaction conditions to afford compound 3, or a salt thereof.

122. The method of embodiment 121, wherein compound 3 is prepared as an HCl salt.

123. The method of embodiment 121 or 122, wherein compound 2, or a salt thereof, is prepared by a method comprising:

(a) reacting compound 5:

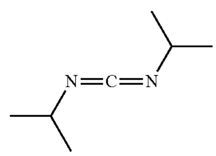
5



or a salt thereof,

with compounds 6 and 7:

6



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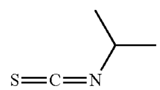
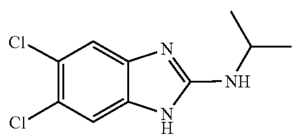


under suitable reaction conditions to provide compound 2, or a salt thereof.

124. A method of preparing compound 2:

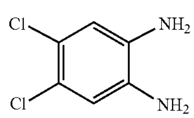
2



or a salt thereof, comprising the step of

(a) reacting compound 5:

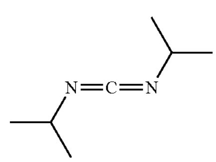
5



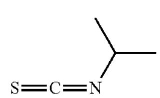
or a salt thereof,

with compounds 6 and 7:

6



7

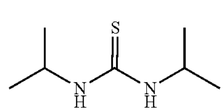


or salts thereof,

under suitable reaction conditions to provide compound 2, or a salt thereof.

125. The method of embodiment 123 or 124, further providing one or more of the following compounds:

9



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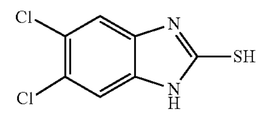
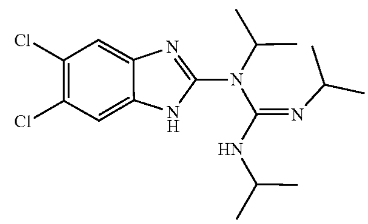
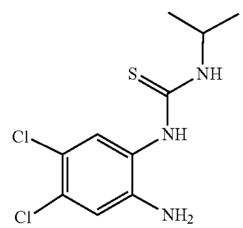
10



11



12



or salts thereof.

126. The method of embodiment 125, wherein compound 9 is provided in an amount less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% w/w, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR).

127. The method of embodiment 125, wherein compound 10 is provided in an amount less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% w/w, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR).

128. The method of embodiment 125, wherein compound 11 is provided in an amount less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% w/w, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR).

129. The method of embodiment 125, wherein compound 12 is provided in an amount less than 1.00%, 0.5%, 0.1%, 0.05%, or 0.02% w/w, relative to compound 2 (e.g., as measured by HPLC or ^{1}H NMR).

130. The method of any one of embodiments 123-129, wherein compound 6, or a salt thereof, is present at an amount of about 0.90 to about 1.09 equivalents relative to compound 5, or a salt thereof.

131. The method of embodiment 130, wherein compound 6, or a salt thereof, is present at an amount of about 1.00 equivalents relative to compound 5, or a salt thereof.

132. The method of embodiment 130, wherein compound 6, or a salt thereof, is present at an amount of about 1.04 equivalents relative to compound 5, or a salt thereof.

133. The method of any one of embodiments 123-132, wherein compound 7, or a salt thereof, is present at an amount of about 1.03 to about 1.13 equivalents relative to compound 5, or a salt thereof.

134. The method of embodiment 133, wherein compound 7, or a salt thereof, is present at an amount of about 1.08 equivalents relative to compound 5, or a salt thereof.

135. The method of any one of embodiments 123-134, wherein the reaction conditions comprise a solvent.

136. The method of embodiment 135, wherein the solvent is 1,4-dioxane.

137. The method of embodiment 136, wherein 1,4-dioxane is present in an amount of about 5.0 L/kg.

138. The method of any one of embodiments 123-137, wherein the reaction conditions comprise heating to a temperature T_7.

139. The method of embodiment 138, wherein the temperature T_7 is about 100° C.

140. The method of embodiment 138 or 139, wherein the reaction conditions comprise maintaining the reaction at temperature T_7 for a period of time.

141. The method of embodiment 140, wherein the period of time is between about 2 hrs and about 20 hrs.

142. The method of embodiment 140 or 141, wherein the period of time is between about 14 and about 15 hours.

143. The method of any one of embodiments 139-142, wherein, after heating to a temperature T_7, the reaction conditions further comprise (b) cooling the reaction mixture to a temperature T_8.

144. The method of embodiment 143, wherein the temperature T_8 is between about 60° C. and about 90° C.

145. The method of embodiment 143 or 144, wherein the temperature T_8 is about 80° C.

146. The method of any one of embodiments 138-145, wherein the reaction mixture is cooled to temperature T_8 at a rate of between about 5° C./hr and about 10° C./hr.

147. The method of any one of embodiments 138-145, wherein the reaction mixture is cooled to the second temperature at a rate of about 6.5° C./hr.

148. The method of any one of embodiments 123-147, wherein the reaction is monitored for completion by HPLC.

149. The method of embodiment 148, wherein the reaction is determined to be complete when less than 1.0% a/a of compound 5, or a salt thereof, is detected (e.g., by HPLC).

150. The method of any one of embodiments 123-149, further comprising a step of (c) crystallization of compound 2, or a salt thereof (e.g., from the reaction mixture).

151. The method of embodiment 150, wherein crystallization of compound 2, or a salt thereof, comprises providing a primary mixture, wherein the primary mixture comprises compound 2, or a salt thereof, and a primary solvent.

152. The method of embodiment 150 or 151, wherein the primary solvent is 1,4-dioxane.

153. The method of any one of embodiments 150-152, wherein the primary solvent is present at an amount of about 4.9 L/kg of solute.

154. The method of any one of embodiments 151-153, wherein the primary mixture is provided at temperature T_8.

155. The method of embodiment 154, wherein the temperature T_8 is between about 60° C. and about 90° C.

156. The method of embodiment 154 or 155, wherein the temperature T_8 is about 80° C.

157. The method of any one of embodiments 150-156, wherein crystallization of compound 2, or a salt thereof, comprises adding a seed crystal to the primary mixture.

158. The method of embodiment 157, wherein the seed crystal is added in an amount between about 0.1 wt % and about 1 wt % of compound 5, or a salt thereof.

159. The compound of embodiment 157 or 158, wherein the seed crystal is added in an amount of about 0.25 wt % of compound 5, or a salt thereof.

160. The compound of any one of embodiments 157-159, wherein the seed crystal is compound 2, or a salt thereof.

161. The method of any one of embodiments 150-160, wherein crystallization of compound 2, or a salt thereof, comprises cooling the primary mixture to a temperature T_9.

162. The method of embodiment 161, wherein temperature T_9 is between about 18 and about 32° C.

163. The method of embodiment 161 or 162, wherein the temperature T_9 is about 27° C.

164. The method of any one of embodiments 161-163, wherein the primary mixture is cooled at a rate of between about 5° C./hr and about 10° C./hr.

165. The method of any one of embodiments 161-164, wherein the primary mixture is cooled at a rate of about 6.5° C./hr.

166. The method of any one of embodiments 123-165, further comprising a step of (d) recrystallization of compound 2, or a salt thereof (e.g., from the reaction mixture or the primary mixture).

167. The method of embodiment 166, wherein recrystallization of compound 2, or a salt thereof, comprises (i) providing a primary mixture, wherein the primary mixture comprises compound 2, or a salt thereof, and a primary solvent.

168. The method of embodiment 167, wherein the primary mixture is provided at the temperature T_9.

169. The method of embodiment 168, wherein temperature T_9 is between about 18 and about 32° C.

170. The method of embodiment 168 or 169, wherein the temperature T_9 is about 27° C.

171. The method of any one of embodiments 167-170, wherein the primary solvent is 1,4-dioxane.

172. The method of any one of embodiments 167-171, wherein the primary solvent is present at an amount of about 4.9 L/kg of solute.

173. The method of any one of embodiments 166-172, wherein recrystallization of compound 2, or a salt thereof, further comprises (ii) adding a secondary solvent to the primary mixture to form a secondary mixture.

174. The method of embodiment 173, wherein the secondary solvent is dichloromethane.

175. The method of embodiment 173 or 174, wherein the secondary solvent is present at an amount between about 8 and about 12 L/kg of solute.

176. The method of any one of embodiments 173-175, wherein the secondary solvent is present at an amount of about 10 L/kg of solute.

177. The method of any one of embodiments 166-176, wherein recrystallization of compound 2, or a salt thereof, further comprises (iii) holding and optionally agitating the secondary mixture at temperature T_9 for a period of time.

178. The method of embodiment 177 wherein the secondary mixture is agitated at temperature T_9 for the period of time.

179. The method of embodiment 177 or 178, wherein the period of time is between about 2 and bout 6 hrs.

180. The method of any one of embodiments 159-161, wherein the period of time is about 3 hrs.

181. The method of any one of embodiments 123-180, further comprising a step of (e) filtering (e.g., the reaction mixture, primary mixture, or secondary mixture) to provide a filtered mixture.

182. The method of embodiment 181, wherein the secondary mixture is filtered at a temperature T_{10}.

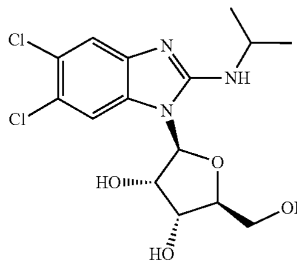
183. The method of embodiment 182, wherein temperature T_{10} is between about 24° C. and about 32° C.

184. The method of embodiment 182 or 183, wherein temperature T_{10} is about 27° C.

185. The method of any one of embodiments 123-184, further comprising a step of (f) washing, e.g. the filtered mixture or secondary mixture, to provide a final mixture.

186. The method of any one of embodiments 123-185, further comprising a step of (g) deliquoring, drying, and/or isolating compound 2, or a salt thereof.

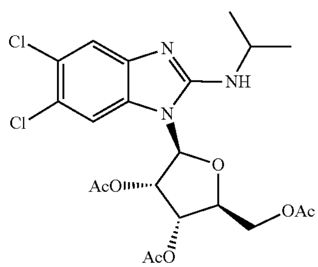
187. A method of preparing maribavir:



or a pharmaceutically acceptable salt thereof, comprising a step of

(a) reacting compound 3:

3



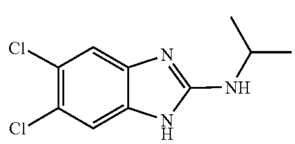
or a salt thereof,

under suitable reaction conditions to provide maribavir, or a pharmaceutically acceptable salt thereof,

wherein compound 3, or a salt thereof, is prepared by a method comprising a step of

(a) reacting compound 2:

2



or a salt thereof,

under suitable reaction conditions to provide compound 3, or a pharmaceutically acceptable salt thereof.

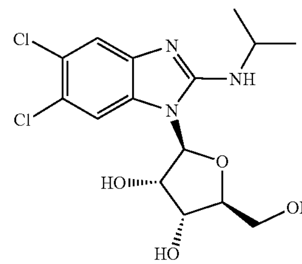
188. The composition of any one of embodiments 1-29 or the oral solid formulation of any one of embodiments 30-32, wherein maribavir is polymorphic Form VI which is crystalized by using isopropyl acetate/toluene as a crystallization solvent.

189. The composition of any one of embodiments 1-29 or the oral solid formulation of any one of embodiments 30-32, wherein maribavir is polymorphic Form VI which is crystalized by using isopropyl acetate/toluene as a crystallization solvent and maribavir seed crystals having a PSD of d(50) between about 1.00 μm and about 10.00 μm.

190. A method of treating a patient with post-transplant cytomegalovirus (CMV) infection and/or disease comprising orally administering maribavir or a pharmaceutically acceptable salt thereof to the patient in an amount of 400 mg twice a day, wherein maribavir is administered as the composition of any one of embodiments 1-29, 188 or 189 or the oral solid formulation of any one of embodiments 30-32, 188 or 189.

1A. A composition comprising a unit dose of maribavir, wherein the composition comprises

(i) 200 mg or 400 mg of maribavir:

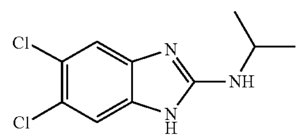


or a pharmaceutically acceptable salt thereof, and

(ii) one or more of the following:

Compound 2,

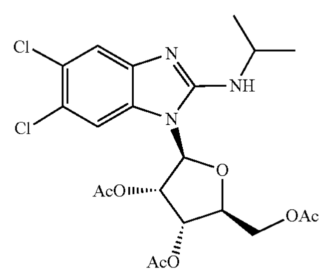
2



or a pharmaceutically acceptable salt thereof,

Compound 3,

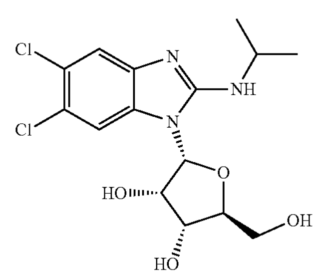
3



or a pharmaceutically acceptable salt thereof, or

Compound 4:

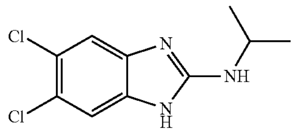
4



or a pharmaceutically accept salt thereof.

2A. The composition of embodiment 1A, wherein the composition comprises Compound 2:

2



or a pharmaceutically acceptable salt thereof.

3A. The composition of embodiment 2A, wherein the composition comprises 0.01-0.10% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

4A. The composition of embodiment 2A or 3A, wherein the composition comprises 0.01-0.05% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

5A. The composition of any one of embodiments 2A-4A, wherein the composition comprises 0.02-0.05% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

6A. The composition of embodiment 2A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 2, relative to maribavir (e.g., as measured by HPLC).

7A. The composition of embodiment 2A, wherein the composition comprises 0.01-0.10% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

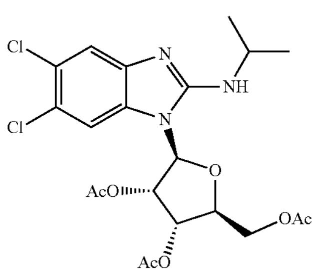
8A. The composition of embodiment 2A or 7A, wherein the composition comprises 0.01-0.5% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

9A. The composition of any one of embodiments 2A or 7A-8A, wherein the composition comprises 0.02-0.05% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

10A. The composition of embodiment 2A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 2, relative to maribavir (e.g., as measured by HPLC).

11A. The composition of any one of embodiments 1A-10A, wherein the composition comprises Compound 3:

3



or a pharmaceutically acceptable salt thereof.

12A. The composition of embodiment 11A, wherein the composition comprises 0.01-0.10% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

13A. The composition of embodiment 11A or 12A, wherein the composition comprises 0.01-0.5% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

14A. The composition of any one of embodiments 11A-13A, wherein the composition comprises 0.02-0.05% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

15A. The composition of embodiment 11A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 3, relative to maribavir (e.g., as measured by HPLC).

16A. The composition of embodiment 11A, wherein the composition comprises 0.01-0.10% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

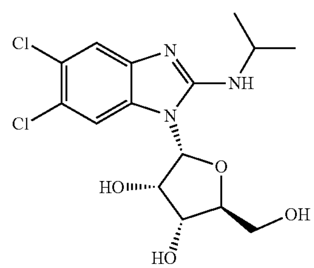
17A. The composition of embodiment 11A or 16A, wherein the composition comprises 0.01-0.5% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

18A. The composition of any one of embodiments 11A or 16A-17A, wherein the composition comprises 0.02-0.05% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

19A. The composition of embodiment 11A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

20A. The composition of any one of embodiments 1A-19A, wherein the composition comprises Compound 4:

4



or a pharmaceutically acceptable salt thereof.

21A. The composition of embodiment 20A, wherein the composition comprises 0.01-0.10% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

22A. The composition of embodiment 20A or 21A, wherein the composition comprises 0.01-0.5% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

23A. The composition of any one of embodiments 20A-22A, wherein the composition comprises 0.02-0.05% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

24A. The composition of embodiment 20A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% w/w of Compound 4, relative to maribavir (e.g., as measured by HPLC).

25A. The composition of embodiment 20A, wherein the composition comprises 0.01-0.10% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

26A. The composition of embodiment 20A or 25A, wherein the composition comprises 0.01-0.5% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

27A. The composition of any one of embodiments 20A or 25A-26A, wherein the composition comprises 0.02-0.05% a/a of Compound 4, relative to maribavir (e.g., as measured by HPLC).

28A. The composition of embodiment 20A, wherein the composition comprises less than 1%, 0.5%, 0.1%, 0.05%, 0.02%, or 0.01% a/a of Compound 3, relative to maribavir (e.g., as measured by HPLC).

29A. The composition of any one of embodiments 1A-28A, wherein the composition comprises maribavir having a particle size distribution (PSD) of between of d(50) of about 170 and about 350 μm (e.g., about 170 and about 226 μm, about 227 and about 280 μm, or about 281 and about 336 μm).

30A. The composition of any one of embodiments 1A-29A, further comprising a pharmaceutically acceptable excipient.

31A. An oral solid formulation of maribavir comprising the composition of any of embodiments 1A-30A.

32A. The oral solid formulation of embodiment 31A, wherein the formulation is a tablet.

33A. The oral solid formulation of embodiment 31A or 32A, comprising about 200 mg of maribavir.

34A. The composition of any one of embodiments 1A-30A or the oral solid formulation of any one of embodiments 31A-33A.

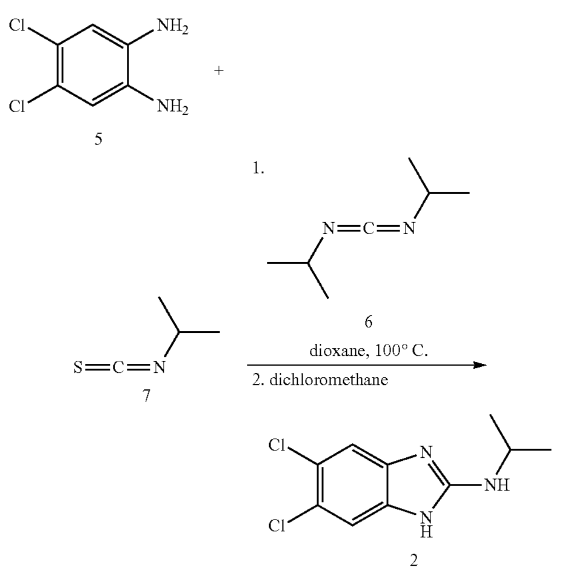
35A. The composition of any one of embodiments 1A-34A or the oral solid formulation of any one of embodiments 31A-33A, wherein the composition comprises less than 0.05% of D-maribavir.

36A. The composition of any one of embodiments 1A-35A, wherein maribavir has a particle size distribution (d(50)) of between about 50 and about 400 μm.

EXAMPLES

As depicted in the Examples below, in certain exemplary embodiments, compounds are prepared according to the following general procedures. It will be appreciated that, although the general methods depict the synthesis of certain compounds of the present invention, the following general methods, and other methods known to one of ordinary skill in the art, can be applied to all compounds and subclasses and species of each of these compounds, as described herein.

Example 1. Synthesis of 5,6-dichloro-N-isopropyl-1H-benzo[d]imidazol-2-amine (Compound 2)



Exemplary Synthesis of 5,6-dichloro-N-isopropyl-1H-benzo[d]imidazol-2-amine (Compound 2)

Batch size in Qualified Equipment. 100 kg±20 kg of Compound 2; the process description below outlines an example production using ~100 kg of Compound 5. Operating ranges have been established for all process parameters as described herein, and the description provides the target for each of the parameters as described herein.

Chemical Reaction. Compound 5 (100.07 kg, limiting reagent), 1,4-dioxane (488.9 L), Compound 7 (61.70 kg, 1.08 molar equivalents), and Compound 6 (74.10 kg, 1.04 molar equivalents) were charged to the reactor at ambient temperature. The resultant mixture was heated to 104.8° C. for at least 2 hours with agitation. The mixture was then kept at 104.8° C. for about 14 to 15 hours and was then cooled to 80.0° C. over 3 hours, 40 mins (cooling rate=6.8° C./hour). If the following criteria are not met: ≤1.0% a/a by HPLC of Compound 5, and Loss on Drying ("LOD") of ≤0.5%, the batch is reheated to about 103° C., held at that temperature a minimum of 2 hours, and then re-cooled to about 80° C. using about the same cooling rate as above and resampled. The process is repeated until criteria are met. These criteria were met.

Crystallization. Upon meeting the criteria, the batch was examined to ascertain if crystallization had started. If crystallization had not started, a Compound 2 seed slurry (Compound 2, 0.25 kg and 1,4-dioxane, 2 kg) would be charged to the batch. In this exemplary synthesis, Compound 2 seed slurry was not added as crystallization started spontaneously. After crystallization had started, the batch was cooled from 79.5° C. to a temperature of 27.0° C. over 8 hours, 31 mins (cooling rate 6.2° C./hour).

Recrystallization. DCM (1002.7 L) was charged into the reactor while maintaining a batch temperature of about 27.0° C., and the batch was agitated for a minimum of 3 hours. The resultant slurry was filtered with the filter dryer jacket set at 27.0° C.

Filtration and Wetcake Washing. DCM (164.5 L) and 1,4-dioxane (85.8 L) were charged to the reactor, and the resultant mixture was agitated while the reactor batch temperature was adjusted to 27.2-27.3° C. The jacket temperature of the filter/dryer was adjusted to about 27.0° C., and the resultant solvent mixture was transferred onto the Compound 2 wetcake in the filter/drier. Solvent was then pushed through the cake with inert gas pressure.

1,4-dioxane (249.8 L) was charged to the reactor. (If significant amounts of solids are observed on the vessel wall, the batch temperature is adjusted to about 50° C. and agitated for at least 1 hour). The reactor batch temperature and temperature of the filter/dryer were adjusted to about 27.2-27.3° C. and the resultant solvent was transferred onto the Compound 2 wetcake in the filter/drier in one portion. The slurry was agitated on the filter/drier for about 20 minutes, allowed to settle, the solvent was then pushed through the cake with inert gas pressure. This 1,4-dioxane wash was repeated two more times with 499.0 L total of 1,4-dioxane between the two additional washes.

Methyl-t-butyl ether (184.9 kg) was charged to the reactor. Batch temperature and the jacket temperature of the filter/dryer were adjusted to 27.0° C. and resultant solvent was transferred onto the Compound 2 wetcake in the filter/drier in one portion. The slurry was agitated on the filter/drier for about 20 minutes, allowed to settle, and the solvent was then pushed through the cake with inert gas pressure.

n-Heptane (250.0 L) was charged to the reactor. Batch temperature and the jacket temperature of the filter/dryer were adjusted to about 27.0° C. and resultant solvent was transferred onto the Compound 2 wetcake in the filter/drier in one portion. The solvent was then pushed through the cake with inert gas pressure.

Drying and Isolation. After deliquoring, the cake was dried with a partial vacuum and the filter/drier jacket temperature was set to ≤60° C. with intermittent agitation until condensate was no longer visible. The filter/drier jacket temperature was adjusted to ≤100° C., with intermittent agitation, and vacuum applied until the material met the LOD criteria. The yield was 72.6%, and the product had a purity of 100.0%, with 99.7% w/w by HPLC.

Process Characterization.

Process characterization studies were performed in order to ensure consistent production of Compound 2 with a defined impurity profile. A defined impurity profile of Compound 2 allows for more consistent impurity profiles in downstream reactions involved in the formation of maribavir.

In particular, multiple process characterization studies were conducted to assess certain parameters involved in the synthesis of Compound 2. The following exemplary studies were designed to examine impurity formation and retention of Compound 2 product for the following: (i) reaction conditions so as to maximize yield and minimize impurity formation, (ii) identify optimal conditions for the crystallization of Compound 2, (iii) identify optimal conditions for the recrystallization for Compound 2, (iv) identify optimal conditions for filtration and washing of the wetcake product Compound 2, and (v) identify optimal conditions for drying of the product Compound 2. The process characterization studies performed on a laboratory scale are summarized below in Table 1-2, and further details are provided herein in Examples 1-1 through 1-3.

At the completion of studies outlined in Table 1-2, further studies were conducted on larger laboratory scale to confirm findings from the laboratory studies, and process robustness studies were also performed on the recrystallization, in particular, focusing upon the slurry temperature for the recrystallization (see Examples 1-1 through 1-4). Based upon all of these studies, the tested ranges for the reaction provide control over the formation of impurities in Compound 2. Further, the tested ranges for the crystallization and recrystallization, and filtration and washing of Compound 2 product purge the impurities formed in the reaction process. In addition, the tested conditions for drying of the final Compound 2 do not produce new impurities nor are the levels of impurities increased. Hence, this example leads to the experimentally verified process parameter ranges presented in Table 1-1.

TABLE 1-1

Exemplary Process Parameters.

Unit of Operation	Step/Parameter	Lower value	Standard Value	Upper value
Charge chemicals/ Prepare for chemical reaction	Charge Compound 5	0.99 molar equivalent	1.0 molar equivalent	1.01 molar equivalent
	1,4-dioxane	2.7 L/kg[1]	4.9 L/kg	5.5 L/kg
	Charge isopropyl isothiocyanate (Compound 7)	1.04 molar equivalent[2]	1.08 molar equivalent	1.13 molar equivalent
	Charge 1,4-dioxane (transfer line rinse after Compound 7 charge)	5 L	5 L	13 L
	Charge N,N'-diisopropyl carbodiimide (Compound 6)	0.99 molar equivalent	1.04 molar equivalent[2]	1.10 molar equivalent
	Temperature before heating the batch	15° C.	20° C.	30° C.
	Hold time for unreacted reaction mixture	0	N/A	256 hrs
	Charge 1,4-dioxane (transfer line rise after Compound 6 charge)	5 L	5 L	13 L
Heat to effect chemical reaction	Heat to temperature	95° C.	103° C.	110° C.
	Ramp heating time	0.77 hrs	≥2 hrs	N/A
	Reaction time prior to taking first sample and test criteria	10 hrs	14.5 hrs	48 hrs
	Temperature during reaction	92° C.	103° C.	110° C.
Cool and sample for reaction criteria	Cool for criteria. Maintain temperature until criteria results available	75° C.	80° C.	85° C.
	Hold time at 80° C.	0 hr	Until criteria results are available	132 hrs
	Rate of cooling	4° C./hr	6.5° C./hr	10° C./hr
Crystallization	Hold time at 80° C. prior to optional seeding	0 hr	≥1.5 hrs	4 hrs
	Seeding (optional)	0 kg	0.25 kg	0.263 kg
	1,4-dioxane (for seed slurry)	0 kg	2 kg	4 kg

TABLE 1-1-continued

Exemplary Process Parameters.

Unit of Operation	Step/Parameter	Lower value	Standard Value	Upper value
	Jacket temperature	18° C.	27° C.	35° C.
	Hold time. 1,4-dioxane slurry at ~27° C.	0 hrs	N/A	168 hrs
	Rate of cooling	4° C./hr	6.5° C./hr	10° C./hr
Recrystallization	Charge DCM	8.0 L/kg	10 L/kg	12 L/kg
	Agitation time for DCM/1,4-dioxane	2 hrs	3.5 hrs	72 hrs
	slurry temperature	18° C.	27° C.	32° C.
Filtration Washing	Jacket temperature	18° C.	27° C.	32° C.
Wash 1	DCM for wash	0	1.64 L/kg	1.95 L/kg
	1,4-dioxane for wash	0	0.86 L/kg	1.05 L/kg
	Temperature of DCM/1,4-dioxane wash	18° C.	25° C.	32° C.
	DCM/1,4-dioxane wash contact time	N/A	N/A	N/A
Wash 2	1,4-dioxane for wash	2 L/kg	2.5 L/kg	3 L/kg
	Optional heat 1,4 dioxane	25° C.	50° C.	65° C.
	Optional agitation time	30 mins	≥60 mins	≥60 mins
	Optional cool prior to filtration	18° C.	25° C.	32° C.
	Wash 2 agitation time	10 mins	20 mins	79 mins
	Settlement time prior to filtration	≥10 mins	≥10 mins	≥10 mins
	Wash 2 temperature	18° C.	25° C.	32° C.
Wash 3	1,4-dioxane for wash 3	2 L/kg	2.5 L/kg	3 L/kg
	Wash 3 agitation time	10 mins	20 mins	63 mins
	Settlement time	≥10 mins	≥10 mins	≥10 mins
	Wash 3 temperature	18° C.	25° C.	32° C.
Wash 4	1,4-dioxane for wash 4	2 L/kg	2.5 L/kg	3 L/kg
	Wash 4 agitation time	10 mins	20 mins	59 mins
	Settlement time	≥10 mins	≥10 mins	≥10 mins
	Wash 4 temperature	18° C.	25° C.	32° C.
Wash 5	MBTE for wash 5	1.5 L/kg	2.5 L/kg	3 L/kg
	Wash 5 agitation time	10 mins	20 mins	77 mins
	Settlement time	≥10 mins	≥10 mins	≥10 mins
	Wash 5 temperature	18° C.	25° C.	32° C.
Wash 6	n-heptane for wash 6	1.66 L/kg	2.5 L/kg	3 L/kg
	Wash 6 agitation time	0 mins	0 mins	30 mins
	Wash 6 temperature	18° C.	25° C.	32° C.
Drying	Deliquoring	1 hr	2 hrs	6 hrs
	Preliminary drying under vacuum, temperature ≤60° C.	20° C.	≤60° C.	60° C.
	Preliminary drying time	N/A	Until no condensate visible in filter drier	N/A
	Temperature during drying	85° C.	100° C.	120° C.
	Drying time	8 hrs	Until IPC met LOD ≤0.5%	168 hrs at 100° C. or 72 hrs at 120° C.

[1]Expressed in L/kg of Compound 5 charged.

[2]Expressed in molar equivalents relative to the amount of Compound 5 charged.

TABLE 1-2

Process Characterization Studies.				
Example	Operation	Process Parameter	Lower Limit tested	Upper Limit tested
1-1	Reaction	Compound 6 molar equivalents	1.00	1.12
		Compound 7 molar equivalents	1.04	1.12
		Reaction Time (hours)	10	18
1-2	Crystallization	1,4-dioxane volumes (L/kg)	3.5	5.5
		Crystallization Temperature (° C.)	65	85
		Hold Time at Crystallization Temperature (hours)	0	4
		Cooling Rate (° C./hour)	5	10
	Recrystallization	Dichloromethane volumes (L/kg)	8	12
		Recrystallization Temperature (° C.)	18	32
		Agitation time (hours)	2	6
1-3	Filtration, washing, drying	Input quality (see Table 1-3)	Marginal	Good
		Filtration Temperature (° C.)	24	32
		Solvent Wash Volumes (L/kg)	2	3
		Wash Time (minutes)	10	30
		Wash Temperature (° C.)	18	32
		Deliquor Time (hours)	2	6
		Drying Temperature (° C.)	85	105

Example 1-1: Optimization of Compound 6 and Compound 7 Stoichiometries and Reaction Time

This example focused on the following three factors: (i) stoichiometry of Compound 6, (ii) stoichiometry of Compound 7, and (iii) reaction time (in hours). The reaction temperature was kept constant at about 100° C., since lower reaction temperatures resulted in considerably slower reaction kinetics and substantially increased reaction time.

The first samples examined equivalents of Compound 6 (1.04, 1.08, and 1.12); equivalents of Compound 7 (1.04, 1.08, and 1.12) and reaction time (10, 14, and 18 hrs). Analysis indicated that lower levels of reagents (i.e., Compounds 6 and 7 less than 1.12 equivalents) were superior to higher levels of reagents, and in particular the lower levels of Compound 6 resulted in lower levels of the impurity Compound 11, and that a reaction time of 14 hrs ensured complete conversion and lower levels of impurities.

Additional samples examined lower levels of Compound 6 (1.04, 1.06, and 1.08 equiv) and combined with higher levels of Compound 7 (1.1, 1.11, and 1.12 equiv), while keeping the reaction time constant (14 hrs). Reduced input stoichiometries of Compound 6 resulted in lower levels of the impurity Compound 11.

Additional samples examined further reduced levels of Compound 6 (1.00, 1.03, and 1.06 equiv) and lower levels of Compound 7 (1.05, 1.08, and 1.11 equiv), while keeping the reaction time constant at 14 hrs. These samples produced the lowest total levels of impurities of Compounds 10 and 11. These findings suggest employing: (i) 1.00 equiv of Compound 6, (ii) 1.08 equiv of Compound 7, and (iii) a reaction time of no longer than 14 hrs.

A pair of laboratory scale-up studies were performed on 21 g input (using 4 volumes of dioxane), which appear to confirm prior findings described here. However, without wishing to be bound to a particular theory, substoichiometric amounts of Compound 6 indicated that the level of Compound 10 would be substantially higher at the end of the reaction, and its level in isolated Compound 2 would be problematic. Alternatively or additionally, given this risk and the fact the Compound 7 is readily decomposed by moisture (offering challenges when a container of it is used for multiple batches of Compound 2), it was decided to set the stoichiometry of Compound 6 at 1.04 equivalents as product quality had a lower risk of failure and yields of isolated material were comparable.

Example 1-2: Optimization of Dioxane Volume, Crystallization Temperature, Crystallization Hold Time, Temperature Ramp Rate, DCM Volume, Recrystallization Temperature, and Agitation Time

Table 1-2 outlines the parameters examined in this example. After final recrystallized materials were filtered, they were washed with: (i) dioxane 1.5 volumes, (ii) dioxane 1.0 volumes, and (iii) MTBE 1.5 volumes, and the solid material was then analyzed. Samples were prepared according to Table 1-2.

It was desired to identify conditions that result in minimal production of Compound 9, e.g., of ≤1.0%. In particular, samples associated with a Recrystallization Temperature of 18° C. and a DCM volume of 8 L/kg generally resulted in Compound 9 values of greater than 10.00%, which indicates that these parameters are not acceptable for producing quality Compound 2 product. Such samples were omitted from any further analysis.

Furthermore, of the samples that utilized 12 volumes of DCM and a recrystallization temperature of 18° C., some produced acceptable quality Compound 2 and others produced unacceptable quality Compound 2. Closer examination of this subset revealed that if a temperature ramp rate of 5° C./hr is utilized, three of the four samples produced acceptable quality Compound 2, and one sample produced unacceptable quality Compound 2. In addition, closer examination of this subset revealed that if a temperature ramp rate of 10° C./hr was utilized, only one of the four samples produced acceptable quality Compound 2, wherein three of the four samples produced unacceptable quality Compound 2. This may indicate that when using marginal conditions (e.g., 12 volumes of DCM and a recrystallization temperature of 18° C.), a more controlled cooling temperature ramp rate of 5° C./hr can help improve Compound 2 quality. Notably, twelve of sixteen samples that utilized a recrystallization temperature of 18° C. produced unacceptable quality Compound 2, and therefore it was found that a recrystallization temperature of 18° C. should be avoided. When using higher recrystallization temperatures (i.e., 32° C.), there were only two samples which produced unacceptable quality Compound 2, indicating that higher recrystallization temperatures are preferred.

Two other transformations were incorporated to determine when determining acceptable conditions. First, given that there was a significant number of samples associated with 0% Compound 9, an adjustment factor was incorporated as a means of improving the model fit. The adjustment factor was defined as half the magnitude of the smallest non-zero response represented as a proportion. Specifically, an adjustment factor of 0.00052/2=0.0026 was added to samples associated with 0% Compound 9. Second, the log it transformation was applied to the adjusted Compound 9 values represented as a proportion. Specifically, the adjusted Compound 9 values were transformed via:

$$\text{Log it}(p)=\ln(p/1-p)$$

The corresponding predicted responses were then back-transformed via:

$$\text{Predicted Compound 9}=\exp(\text{predicted log it})/1+\exp(\text{predicted log it})$$

Example 1-3. Optimization of Washing and Drying Operation

Table 1-2 outlines the parameters examined in this Example. Parameters for "Marginal" and "Good" input are provided in Table 1-11. For the wetcake washes, the following wash sequence was performed: (i) wash 1: dioxane, (ii) wash 2: dioxane, (iii) wash 3: dioxane, (iv) wash 3: MBTE, and (v) wash 5: heptane. The solvent washes were kept constant for all samples. The solvent wash volumes were kept consistent and constant for all washes of a given sample (e.g., a solvent volume of two means that all five washes used two solvent volumes).

TABLE 1-3

Definition of Marginal and Good Quality Input Conditions

Operation	Marginal Quality Input Settings	Good Quality Input Settings
Reaction Stoichiometry	Compound 6-1.04 equiv; Compound 7-1.08 equiv	Compound 6-1.04 equiv; Compound 7-1.08 equiv
Reaction Dioxane Volume	4.0	5.5
Ramp Cooling Rate	10° C./hour	10° C./hour
Final crystallization, recrystallization, and first filtration temperatures	24° C.	32° C.
DCM Volumes added	8.5	12.0
Recrystallization agitation time	2.0 hrs	3.0 hrs

Notably, using "Good" input conditions (i.e., reaction, crystallization, and recrystallization conditions) afforded a wetcake with acceptable quality Compound 2 (e.g., greater than 99% HPLC area). However, using "Marginal" input conditions afforded wetcake with significantly lower in-process Compound 2 product quality (e.g., less than 99% HPLC area) with only one of the eight samples being capable of producing acceptable quality Compound 2.

However, even using "Marginal" input quality conditions with the full five wash sequence afforded acceptable quality product for all but one set of conditions examined. Notably, this failed sample used the minimal wash solvent volumes (2 volumes), the minimal wetcake washing temperature (18° C.), and the minimal wetcake washing hold times (10 minutes), which represented the lower limit settings for the wetcake washing operations.

FIG. **1** displays a plot of Compound 2 purity as a function of input quality, after filtration of the recrystallized Compound 2 (i.e., before the wash steps). "Good" quality input afforded high purity material in a fairly narrow range of Compound 2 purity between 80.89% and 90.84%, indicating that "Good" quality input conditions were robust. "Marginal" quality input afforded much lower quality material at this stage with a wide distribution of Compound 2 purity ranging from 18.13% to 72.19%, indicating that "Marginal" quality input was less robust.

Example 1-4. Process Robustness Studies: Recrystallization Temperatures

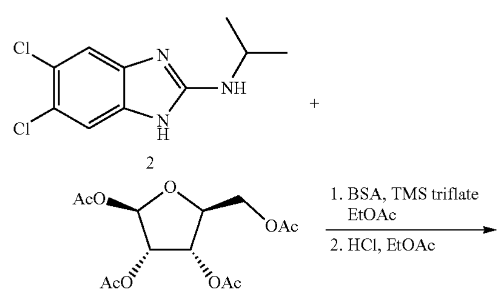
A sequence of robustness studies were performed to determine a lower temperature bound for the dioxane/DCM recrystallization, and temperatures between 18-30° C. were tested. For the reactions tested, 4.25 volumes of dioxane were used, and the mixtures (after meeting criteria) were cooled to the desired temperature (either 18, 22, 26, or 30° C.) with a ramp rate of 10° C./hour. DCM (10 volumes) was then added, and the mixtures were agitated at the requisite temperature for 3 hours, filtered, and subjected to an identical wetcake wash sequence (i.e., all washes performed with the wash solvent at 25° C.: wash 1 (dioxane 2 volumes), wash 2 (dioxane 2 volumes), wash 3 (dioxane 2 volumes), wash 4 (MTBE 2 volumes), wash 5 heptane (2 volumes)) and dry sequence (100° C. vacuum oven ~36 hours). A significant Compound 2 failure rate (i.e., Compound 2<99.0% by HPLC) was observed at the lower recrystallization temperatures of 18° C. and 22° C., whereas no product failure was observed when the recrystallization temperature was either 26° C. or 30° C.

Product quality as a function of observed crystallization temperature was analyzed as shown in FIG. **2**. Product quality was lower when Compound 2 spontaneously crystalized at lower temperatures (see FIG. **2**). In addition, an interval plot of Compound 2 versus the recrystallization temperature shows higher recrystallization temperatures (i.e., preferably above 24° C.) are preferred (see FIG. **3**).

Furthermore, a contour plot of Compound 2 purity looking at the Effects of Recrystallization Temperature and Observed Crystallization Temperature was generated in FIG. **4**, which indicates a lower bound for the Recrystallization Temperature should be about 23° C., and the Observed Crystallization Temperature should be at a temperature ≥75° C. For these studies the material crystallized spontaneously and was not controlled. In order to ensure the crystallization occur at 75° C. or higher, the batch could be seeded at 75° C. if spontaneous crystallization does not occur.

Furthermore, two studies were performed to examine seeding at relatively high temperatures (e.g., 70° C. and 85° C.) to assess if seeding would be viable. Both examples afforded material that would be of acceptable quality.

Example 2. Synthesis of (2S,3S,4S,5S)-2-(acetoxymethyl)-5-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)tetrahydrofuran-3,4-diyl diacetate hydrochloride (Compound 3 HCl)



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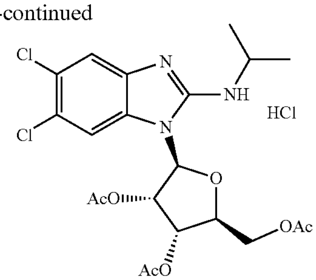


3-HCl

Exemplary Synthesis of (2S,3S,4S,5S)-2-(acetoxymethyl)-5-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)tetrahydrofuran-3,4-diyl diacetate hydrochloride (Compound 3 HCl)

Batch size in Qualified Equipment. 62 kg±12.4 kg of input Compound 2; the process description below outlines an example production using 62.0 kg of Compound 2. Operating ranges have been established for all process parameters as described herein, and the description provides the target for each of the parameters as described herein.

Chemical Reaction. Compound 2 (62.00 kg, limiting reagent), ethyl acetate (347.0 kg), Compound 8 (105.00 kg, 1.3 molar equivalents), N,O-Bis(TMS)-acetamide (30.60 kg, 0.593 molar equivalents) with a rinse of the transfer line using ethyl acetate (~20 kg), trimethylsilyl triflate (19.70 kg, 0.349 molar equivalents) with a rinse of the transfer line using ethyl acetate (~20 kg) were charged to the reactor at ambient temperature, and the mixture was stirred for at least 25 minutes at ambient temperature. The resultant mixture was heated to 78.8-80.1° C. (reflux) with agitation and kept for 5 hours, 1 min. The mixture was then cooled to 22° C.; when the mixture reached a temperature of <50° C. a sample was taken for reaction criteria. If the criteria are not met (Compound 2≤5% a/a by HPLC; Total solvent content ≤0.5% (w/w) or 5000 ppm by GC HS, and water content ≤0.5% w/w) by Karl Fischer), the batch is reheated to about 77° C., is held at that temperature for about 3 hours, and then is re-cooled to 22° C. and is sampled again for reaction criteria. These criteria were met.

Aqueous Quench and Workups. Upon meeting the reaction criteria, an aqueous solution of potassium hydrogen carbonate (31.60 kg) dissolved in water (126.4 kg) was added slowly to the reaction mixture while maintaining the temperature at about 20° C. The resultant mixture was agitated for at least 15 minutes and allowed to settle without agitation and the resultant denser aqueous phase was removed. This wash with aqueous potassium hydrogen carbonate was repeated one more time.

To the resultant Compound 3 rich organic phase, an aqueous solution of sodium chloride was added and agitated at about 20° C. for at least 15 minutes. The mixture was allowed to settle without agitation for at least 20 minutes and the resultant, denser aqueous, phase was removed.

Distillation. Solvent was removed via vacuum distillation at a jacket temperature of <45° C. (while maintaining a batch temperature of <40° C.). The distillation was stopped when about 408 L distillate had been collected. Ethyl acetate (386.8 kg) was charged, and the mixture was agitated while maintaining the internal temperature at about 20° C.

Hydrochloric Acid Salt Formation and Crystallization. To the mixture was added HCl in water (32% w/w, 30.0 kg) with agitation while maintaining an internal temperature of about 22° C. After about 10% of the HCl solution had been added, the batch was examined and crystallization had commenced. If crystallization has not commenced, the batch can be seeded by the addition of a slurry of Compound 3 HCl suspended in ethyl acetate (about 0.2 kg). Once crystals were evident in the batch, the remainder of the HCl was added commensurate with a total addition time of about 30 minutes. The mixture was agitated at about 25° C. for about 1.5 hours and then cooled to about 2° C. over a period of about 1.5 hours and was held for about 1.5 hours.

Filtration and Wetcake Washing. The resultant slurry was transferred to an agitated filter drier with the jacket set to a temperature of about 2° C.

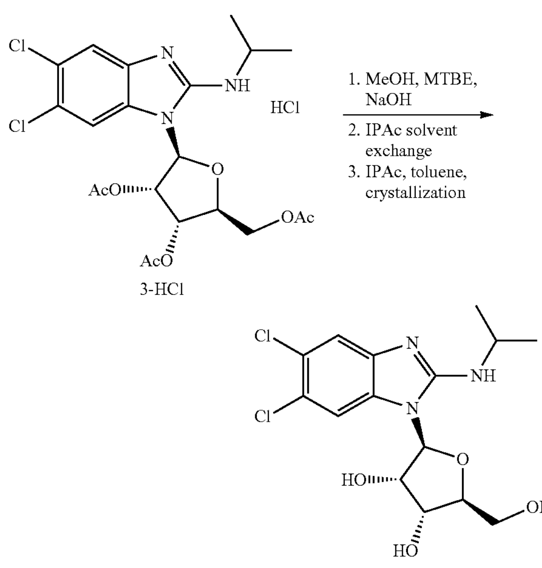
Wash 1. After filtration, ethyl acetate (49.9 kg) at a temperature of about 2° C. was transferred onto the wetcake and the mixture was agitated for at least ten minutes and then was filtered.

Wash 2. Methyl-t-butyl ether (108.9 kg) at a temperature of about 2° C. was transferred onto the wetcake and the mixture was agitated for at least ten minutes and then was filtered.

Wash 3. Methyl-t-butyl ether (108.9 kg) at a temperature of about 2° C. was transferred onto the wetcake and the mixture was agitated for at least ten minutes and then was filtered.

Drying and Isolation. The wet cake was dried under vacuum at a jacket temperature ≤40° C. until a minimal amount of condensate was visible. The cake was dried with agitation under vacuum and a jacket temperature ≤40° C. until IPC criteria are met (Compound 2≤5% a/a by HPLC; Total solvent content ≤0.5% (w/w) or 5000 ppm by GC HS, and water content ≤0.5% w/w) by Karl Fischer). The batch was cooled and then is discharged and was packaged. The yield was 74.2%, with a purity of 97.7% by HPLC.

Example 3. Synthesis and Characterization of Maribavir



Exemplary Synthesis of Maribavir (Maribavir)

Batch Size and Qualified Equipment. 64.0-96.0 kg Compound 3 HCl; the process description below outlines an example production using 80.0 kg of Compound 3 HCl. Operating ranges have been established for all process parameters as described herein and the description provides the target for each of the parameters as described herein.

Chemical Reaction. Compound 3 HCl (80.0 kg, limiting reagent), MeOH (38.6 L), and MTBE (419.9 L) were charged to the reactor, and 30% NaOH (103.3 kg) was added such that the temperature is maintained at ≤35° C. The resultant mixture was agitated at about 30° C. between 2.5-4 h. After cooling the contents to a temperature of about 22° C., a sample was taken to monitor reaction completion. If the reaction criteria are not met (≥99% (a/a) by RP-HPLC of maribavir), the batch is reheated to about 30° C., and stirred at that temperature for a minimum of 0.5-2 h, and then re-cooled to 22° C. and resampled for reaction criteria. These criteria were met.

Phase separation/Extraction. Upon meeting the reaction criteria, agitation was stopped, and the phases were allowed to separate. After transferring the aqueous layer into a separate reactor, MTBE (96 L) was added, and the mixture was stirred for at least 20 minutes. Agitation was stopped, the phases were allowed to separate, and the lower aqueous layer was discarded. A solution of sodium chloride was added to the combined organic phases and stirred. 1% w/w aqueous HCl (24.6 kg) was added while stirring to adjust the pH of the solution to 6.8-7.5. If needed, the pH was adjusted with 1% aqueous NaOH. The organic phase was washed two additional times with a solution of sodium chloride with adjustment of the pH after each wash. Aqueous phase was discarded, and the organic phase was transferred to a distillation reactor.

Distillation and solvent exchange. About 254 L of distillate was distilled under vacuum at a jacket temperature of ≤55° C. After distillation, the reactor was cooled and the vacuum was released with nitrogen. IPAc (374.7 L) was charged to the reactor with stirring, and vacuum distillation was resumed at a jacket temperature of ≤65° C. until additional about 268 L of distillate was distilled. The reactor was cooled again to about 30° C., and the vacuum is released with nitrogen. A second charge of IPAc (~1.32 L/kg) was added to the reactor with stirring in two portions (~0.83 L/kg and ~0.49 L/kg) to ensure homogeneous solution, and vacuum distillation was resumed at a jacket temperature of ≤65° C. and batch temperature of ≤60° C. to distill about 93 L of distillate. The reactor was cooled, and a sample to determine the water content is drawn. The process of charging IPAc (~1.32 L/kg) and azeotropic vacuum distillation and testing criteria is repeated until the water content is <0.09% w/w by Karl-Fischer, after which, the same sample is tested for concentration of crude maribavir in IPAc by ^{1}H NMR (17-20% w/w).

Crystallization. The concentration of crude maribavir in IPAc was adjusted to the required target of about 18% (w/w) [17-19% (w/w)] either by dilution with additional IPAc or by distillation to an amount of about 243.6 kg. The resulting solution was heated to a batch temperature of about 84° C., and toluene in the amount of about 3/7 of total IPAc (kg/kg) (104.4 kg) was charged to the reactor in about 30 minutes. A slurry of micronized maribavir seed crystals (about 121 g, 0.15% (w/w) of Compound 3) in about 185 g of toluene (seed slurry) was prepared. The seed slurry was added via a sampler to the reactor at batch temperature of about 84° C. The bottle was rinsed with about 185 g of toluene, and the rinsate was added to the reactor. Seeding was completed in not more than 30 minutes after addition of toluene. In about 30 minutes after seeding, a second charge of toluene in the amount of about 11/7 of total IPAc (kg/kg) (382.7 kg) was charged to the reactor. The contents of the reactor were stirred at medium speed for about 1 hour at a batch temperature of about 84° C. The contents were then cooled to a batch temperature of about 5° C. over a period of about 3 hours and were stirred at medium speed for 3-18 hours at about 5° C.

Filtration and Wetcake Washing. The jacket temperature of the filter/dryer was adjusted to about 5° C. and the contents of the crystallization reactor were transferred to the filter/dryer in one or more portions allowing the wetcake to settle, pushing the solvent through the cake with inert gas pressure, if necessary. Toluene (66.8 L) at a batch temperature of about 5° C. was charged to the reactor and agitated. The slurry was transferred to the filter/dryer, and the wet cake was deliquored until the filtrate stops running.

Drying and Isolation. The cake was pre-dried for at least 5 hours without stirring at a filter/drier jacket temperature of ≤40° C. and a partial vacuum such that no condensation was visible in the agitated pressure filter. For the main drying, jacket temperature of the filter/drier was set to ≤50° C. with best possible vacuum. The cake was dried without stirring for at least 3 hours or until the material meets the reaction criteria for Loss on Drying (≤0.4% by gravimetric). Afterwards, the filter cake was agitated for one minute at medium speed for homogenization and a sample for residual solvent by GCHS was taken. Intermittent agitation of the filter cake with a stirring time of 1 minute and pause time of 60 minutes was continued for at least 2.5 hours or until the material meets the reaction criteria for residual toluene and IPAc (≤700 ppm toluene and ≤4000 ppm IPAc by GC). The filter/drier was cooled to a jacket temperature of 10° C., and the filter cake was subjected to intermittent agitation with a stirring time of 1 minute and pause time of 20 minutes for at least 1 hour before discharging. The yield was 84.7%.

Process Characterization.

Process characterization studies were performed in order to ensure maribavir's consisting production with a defined impurity profile.

In particular, multiple studies, one variable at a time (OVAT) laboratory scale studies and full-scale production runs were conducted to optimize process control and assess the current operating space for operations within synthesis of maribavir from Compound 3 within the drug substance manufacturing process. The studies focused on assessing variability on the process parameters in 1) the saponification reaction; 2) extraction and work-up; 3) distillation; 4) crystallization; 5) filtration/washing and 6) drying steps. Table 3-2 outlines the parameters investigated either through studies and/or full scale production runs, and further details are provided herein in Examples 3-1 through 3-22. Based on these studies, the current operating range for each operation of the process provides adequate control over the quality of unmilled maribavir, as shown in Table 3-1. For each parameter, statistical evaluation was carried out using the program Modde®. The model was calculated using a negative logarithmic transformation and showed good robustness/predictability/validity/reproducibility. For determination of preferred ranges (i.e., lower and upper limits) for each parameter, design space plots were calculated. The preferred lower and upper limits were defined from the design space plots having three parameters fixed at the set-point while two parameters are flexible, and were defined using ~5% failure probability.

TABLE 3-1

Process Parameters.

Unit of Operation	Step/Parameter	Lower Value	Standard Value	Upper Value
Charge chemicals/prepare for chemical reaction	Charge of Compound 3	0.87 molar equivalents	1.00 molar equivalents	1.04 molar equivalents
	MeOH[1]	0.34 L/kg	0.48 L/kg	0.56 L/kg
	MTBE[1]	3.65 L/kg	5.26 L/kg	6.35 L/kg
	Deionized water[1]	2.70 L/kg	4.4 L/kg	5.10 L/kg
	30% w/w aqueous NaOH	5.00 molar equivalents	5.21 molar equivalents	5.90 molar equivalents
Chemical reaction	Reaction temperature	24° C.	30° C.	45° C.
	Reaction time	1.5 hrs	2.5-4 hrs	7 days
Cool and sample for reaction criteria	Cooling temperature of reaction mixture for IPC sampling	N/A	22° C.	N/A
	Reaction criteria (HPLC)	≥98.34% (a/a) of maribavir	≥99.0% (a/a) of maribavir	N/A
Phase separation/ extraction	Temperature of phase separation	10° C.	25° C.	40° C.
	Time for phase separation after reaction	2 mins	At least 20 mins	7 days
	Time for phase separation for extraction/ washing	2 mins	At least 5 mins to at least 20 mins	7 days
	MBTE for wash	0.96 L/kg	1.2 L/kg	1.44 L/kg
	Sodium Chloride	14.3 kg	17.8 kg	21.5 kg
	Deionized water	1.97 L/kg	2.46 L/kg	2.95 L/kg
	pH of reaction mixture during/after washing	6.0	6.8-7.5	10.0
Distillation/ Solvent change	Volume of distillate-first vacuum distillation	216 L	~254 L	292 L
	First IPAc charge[1]	3.98 L/kg	4.69 L/kg	5.39 L/kg
	Volume of distillate-second vacuum distillation	228 L	~268 L	322 L
	Second IPAc charge[1]	1.1 L/kg	1.32 L/kg	1.51 L/kg
	Volume of distillate-third vacuum distillation	79 L	~93 L	107 L
	Reaction criteria (KF) for water content	N/A	≤0.09% (w/w)	<0.2% (w/w)
	Reaction criteria for concentration of maribavir in IPAc (NMR)	17% (w/w)	17-19% (w/w) target: 18% w/w	20% w/w
	IPAc charge to adjust concentration[2]	0.93 L/kg	~1.10 L/kg	1.26 L/kg
Crystallization	Stirring rate during crystallization	56-87 rpm	56-87 rpm	88-120 rpm
	Temperature during dosage of toluene/seeding	77° C.	84° C.	88° C.
	First toluene charge[3]	2.55/7 of IPAc (kg/kg)	3/7 of IPAc (kg/kg)	3.45/7 of IPAc (kg/kg)
	Dosage time of toluene before seeding	5 mins	15-45 mins	90 mins
	Time between toluene addition and seeding	N/A	Not more than 30 mins	N/A

TABLE 3-1-continued

Process Parameters.				
Unit of Operation	Step/Parameter	Lower Value	Standard Value	Upper Value
	Amount of maribavir seed[4]	0.05% (w/w)	0.14-0.16% (w/w) target: 0.15% (w/w)	0.30% (w/w)
	maribavir seed size: d(50) in μm	2.25	2.75-6.25	7.00
	Stirring time after seeding	10 mins	30 mins	60 mins
	Second toluene charge[5]	9.35/7 of IPAc (kg/kg)	11/7 of IPAc (kg/kg)	12.65/7 of IPAc (kg/kg)
	Dosage time of second toluene charge	60 mins	~120 mins	240 mins
	Stirring time after second toluene addition	15 mins	60 mins	120 mins
	Final temperature of crystallization mixture	−5° C.	3-7° C.	20° C.
	Cooling time	1.5 hrs	2.5-3.5 hrs	4.5 hrs
	Stirring time of crystallization mixture at final temperature	0 mins	3.0-18.0 mins	7 mins
Filtration and washing	Temperature during dosage filtration/ washing	−5° C.	5° C.	20° C.
	Toluene for wash[1]	0.67 L/kg	0.84 L/kg	1.00 L/kg
	Duration of slurry wash	0 mins	≥15 mins	5 mins
Drying	Temperature for pre-drying	N/A	≤40° C.	70° C.
	Time for pre-drying	N/A	≥5 hrs	N/A
	Temperature for main drying	N/A	≤50° C.	70° C.
	Reaction criteria for LOD	N/A	≤0.4% (w/w)	N/A
	Reaction criteria for residual solvent by HC-HS	N/A	≤700 ppm toluene ≤4000 ppm IPAc	N/A

[1]Expressed as L/kg of Compound 3 HCl

[2]Actual amount to be charged is calculated based on the result from the concentration reaction criteria.

[3]Amount of toluene is calculated based on mass and assay of reaction mixture. Upper and lower values are determined in combination with amount of 2nd toluene charge.

[4]Expressed as (w/w) of Compound 3 HCl charged.

[5]Amount of toluene is calculated based on mass and assay of reaction mixture. Upper and lower values are determined in combination with amount of 1st toluene charge.

TABLE 3-2

Process Characterization studies.				
Example	Operation	Process Parameter	Lower Limit studied	Upper Limit studied
3-1	Reaction	Amount of MeOH	0.27 g/g	0.44 g/g
		Amount of MTBE	2.70 g/g	5.10 g/g
		Amount of water	2.70 g/g	5.10 g/g
		Amount of NaOH	5 eq.	5.90 eq.
		Reaction Temperature	15° C.	45° C.
		Reaction Time	1.5 h	7 days
3-2		Conversion	98.4% a/a	100% a/a
3-3	Workup	Temperature during extractions, washes, and phase separations	10° C.	40° C.
3-4		Amount of MBTE for extraction	0.713 g/g	1.069 g/g

TABLE 3-2-continued

Process Characterization studies.				
Example	Operation	Process Parameter	Lower Limit studied	Upper Limit studied
3-5		Extraction/washing time	2 minutes	7 days
3-6		Amount of NaCl solution for washing	NaCl: 2.44 eq Water: 1.97 g/g	NaCl: 3.66 eq Water: 2.954 g/g
3-7		pH after work-up	6.0	10.0
3-8	Distillation	Amount of distillates during first, second, and third distillations	216 mL 228 mL 79 mL	292 mL 322 mL 107 mL
		Amount of IPAc during first, second, and third feed	3.465 g/g 0.974 g/g 0.811 g/g	4.689 g/g 1.318 g/g 1.097 g/g

TABLE 3-2-continued

Process Characterization studies.				
Example	Operation	Process Parameter	Lower Limit studied	Upper Limit studied
3-9	Crystallization	Concentration of maribavir in IPAc before crystallization	15% w/w	20% w/w
3-10		Water content before crystallization	0% (w/w)	0.5% (w/w)
3-11		Hold time after 1st toluene addition and pre-seeding	0 minutes	1 hours
3-12		Temperature during toluene addition and seeding	60° C.	88° C.
3-13		Amounts of toluene during crystallization	32.5% (w/w)	67.5% (w/w)
3-14		Dosing times of toluene before and after seeding	First: 5 mins Second: 60 mins	First: 90 mins Second: 240 mins
3-15		Seed loading of maribavir seeds for crystallization	0.05% (w/w)	1.40% (w/w)
3-16		Time post-seeding before addition of 2nd toluene charge	10 mins	2 hours
3-17		Time after 2nd toluene addition before cooling	15 mins	120 mins
3-18		Cooling duration	1.5 h	5 h
3-19		Cooling Temperature	5° C.	20° C.
3-20	Filtration and washing	Temperature during filtration and washing	−5° C.	20° C.
3-21		Amount of toluene for washing	0.58 g/g	0.87 g/g
3-22		Stirring time during toluene wash	0 mins	5 days

Example 3-1. Optimization of Amount of MeOH, MTBE, Water, NaOH, and Reaction Temperature Time

This example focuses on the following six factors: (i) amount of MeOH, (ii) amount of MBTE, (iii) amount of water, (iv), amount of NaOH, (v) reaction time, and (vi) reaction temperature. The amounts of Compound 3 HCl were adapted accordingly. The reactions were carried until reaction criteria were met and not worked up. Samples were tested in accordance with the solvent and temperature ranges shown in Table 3-2. The six factors focused on within this example were determined to have no impact on the quality or yield of maribavir.

Example 3-2. Impact of Lower Conversion

Prior studies indicated that before extraction, maribavir was measured with a purity of 95.51% a/a by HPLC, but after extraction of the aqueous phase with TBME and stirring the organic layer (at pH ~11.7) at room temperature for about 10.25 hrs, the amount of maribavir rose to 99.71% a/a by HPLC. To justify obtaining >99.0% maribavir, an example with lower content of maribavir was conducted to track the fate and purge of residual Compound 3 and/or partially de-acetylated intermediates. The reaction times were set to 45 min and 2.5 h, and a sample was pulled from the reaction mixture, directly cooled down to 20° C., and worked-up without waiting for reaction criteria result. Stability of the reaction mixtures was also assessed at 40° C. over 7 days (see Example 3-23).

The amount of maribavir in the organic layer was checked by HPLC after every phase separation operation. The

amount of 99.23% a/a was reached after re-extraction of the alkaline aqueous layer and did not change any further during NaCl washes where the pH was adjusted to 6.8-7.5. The remaining 0.50% a/a of partially de-acetylated intermediate was identified as the 3-mono acetate of maribavir. Accordingly, this example demonstrates that >99.0% a/a of maribavir ensures delivery of maribavir in the desired yield and quality. This example also demonstrates that a failed reaction criteria can be corrected by a prolonged hold time of the biphasic alkaline reaction mixture at 20° C.

Example 3-3. Influence of Temperature During Extractions, Washes, and Phase Separations

The phase separations are generally performed at 25° C. for at least 20 minutes. Without wishing to be bound to a particular theory, it was thought that a lower temperature could possibly result in a lower yield or longer times for phase separations. To evaluate, a study with aqueous workups at 10 and 25° C. was conducted. In addition, to assess the influence of a higher temperature, the stability of the biphasic mixtures at 40° C. was tested (see Example 3-23).

Notably, the phase separation at 10° C. did not have an impact on purity of maribavir in solution. After aqueous work-up the water content and assay of the organic layer were comparable. The biphasic mixture before and after NaCl washes showed good stability over 7 days (see Example 3-23). Accordingly, it was determined that the temperature during phase separation has no impact on the quality or yield of maribavir.

Example 3-4. Influence of the Amount of Tert-Butylmethyl Ether for Extraction

With wishing to be bound to a particular theory, it was thought that a lower amount of TBME for the extraction of the first aqueous layer could lead to a diminished yield. Additionally, an altered solvent composition could impact the crystallization properties. Therefore, higher or lower amounts of TBME for extraction were evaluated, as shown in Table 3-2.

A lower or higher amount of TBME for extraction did not have an impact on the purity of maribavir in solution or the solvent composition after distillation. The water content after distillation was slightly higher, but still lower than the desired <0.1% w/w, and the altered amounts of TBME only had a minor impact on crude yield in solution after distillation, which is within the error range. A lower amount of TBME led to a higher concentration of crude maribavir in iPrOAc after distillation (e.g., 24.58% by NMR; typical concentrations in range of 19-23% w/w), but even with this higher concentration, the filtration was feasible without precipitation of maribavir. Accordingly, it was found that these amounts of TMBE for extraction did not impact the yield or quality of maribavir.

Example 3-5. Influence of Time for Extraction and Washing

Without wishing to be bound to a particular theory, it was thought that a shorter extraction time could impact the yield and purging of impurities. In addition, shorter NaCl-washings could potentially impact water content after washing steps. To evaluate this, a study with short extraction/washing times for all of these steps was performed (max. 5 mins each). Since the first NaCl washing will take ≥5 mins in the standard process, this washing was tested for ≤2 min. A longer contact with aqueous media could lead to formation of impurities. To assess the influence of a higher temperature, the stability of the biphasic mixtures at 40° C. was tested (see Example 3-23). It was determined that shorter washing times do not impact product quality. The product content in the organic layers was slightly lower at shorter washing times but still within standard error. In addition, the yield for crude maribavir after distillation was 93.1%, which is at the upper end of the expected range. The biphasic mixture before and after NaCl-washes showed good stability over 7 d (see Example 3-23). Accordingly, it was determined that the time for washing has no impact on the quality or yield of maribavir.

Example 3-6. Influence of the Amounts of NaCl-Solutions for Washings

Without wishing to be bound to a particular theory, it was thought that a lower or higher amount of 10% NaCl-solutions could lead to undesired effects during washings (e.g., dissolution of maribavir into aqueous layer). Additionally, an organic layer with different residual water content could be obtained, which could possibly impact the crystallization. Another possible impact could be dissolution of inorganics into the organic layer.

Various amounts of 10% NaCl-solution for washing were tested in accordance with Table 3-2. It was determined that the amount of NaCl-solutions for washing did not impact the purity or the concentration of maribavir in solution. The time for phase separation remained quick regardless of the amount of NaCl solution. Accordingly, it was determined that the amount of NaCl for washing has no impact on quality or yield of maribavir.

Example 3-7. Influence of Lower or Higher pH after Aqueous Work-Up

Within the work-up to afford maribavir, the pH is adjusted with either 1% HCl or NaOH solutions to 6.8-7.5. The impact of a pH outside of this range was unknown. Workups with lower or higher pH were performed. For this purpose, the reaction mixture was split up into two aliquots before NaCl washes. pH's were tested in accordance with Table 3-2.

A preliminary test was done to determine the lower pH to be tested: Three samples of the biphasic mixture during 1st NaCl-wash was taken and adjusted separately to pH 4, 5, and 6. These samples were refluxed overnight and sampled for HPLC purity at different time points. All three samples showed perfect stability. It was observed that at lower pH values, some white precipitate occurred. This precipitate dissolved upon raising the pH with NaOH. It was assumed that this was the hydrochloride salt of maribavir. Upon further review, at pH<5.5, the precipitate occurred. Therefore, the lowest pH tested was 6. The distilled reaction mixtures were subjected to crystallization on a 400 ml Easymax scale. Samples and results are shown in Table 3-3.

TABLE 3-3

Reaction Samples and Results.

Sample	(adjusted) pH after work-up	PSD	XRPD
3-7-1	~7.0	d(0.1) = 163 μm d(0.5) = 240 μm d(0.9) = 351 μm	Complies
3-7-2	6.04	d(0.1) = 195 μm d(0.5) = 327 μm d(0.9) = 482 μm	Complies
3-7-3	10.01	d(0.1) = 44.6 μm d(0.5) = 305 μm d(0.9) = 441 μm	Complies

TABLE 3-3-continued

[1] Approximate amount of product in organic layer after extraction.

The yield of both samples with altered pH after work-up gave slightly lower yields, but still within the expected range. There was no impact on purity or concentration of maribavir. The desired polymorph was obtained in all cases. The altered pH values impacted PSD: both PSD curves showed a small shoulder peak in the small particle size area. For the lower pH, overall particles became larger. For the higher pH d(0.1) became significantly smaller while d(0.5) and d(0.9) became larger. Since both samples matched the desired d(0.5)>176 μm, the results are generally acceptable. Accordingly, it was determined that the pH after aqueous work-up has no impact on yield but a minor impact on quality of maribavir.

Example 3-8. Influence of Lower or Higher Amounts of Feed/Distill During Solvent-Swap

Without wishing to be bound to a particular theory, the solvent composition and water content after the distillation steps were expected to be crucial for successful crystallization, For example, undesired solvent composition and water content could lead to a lower yield, lower purity, or undesired particle size distribution.

Here, ranges were tested in accordance with Table 3-2. For example, a sample was performed simulating a worst-case solvent-swap: using a lower amount of iPrOAc as feed and a lower amount of distillate. After the distillation, the concentration was adjusted as described in the exemplary synthesis provided above. This reaction mixture was subjected to a 400 mL scale crystallization. Concomitantly, another part of the reaction mixture was spiked with TBME and water to simulate a worst-case distillation outcome with 95:5 w:w ratio iPrOAc:TBME and 0.20% w/w water in total and crystallized accordingly. A distillation with a higher amount of feed and distillate will most likely lead to a solution of crude maribavir in iPrOAc with a low water content. The only effect would be a prolonged distillation time. This was evaluated in a stability test (see Example 3-23).

Reducing the feed and distill amounts led to higher contents of TBME. Additionally, the amount of water exceeds the internal process control of <0.10% w/w. While higher TBME and water led to a unimodal PSD curves with a bit higher values, no negative impact on PSD and XRPD was detected. Therefore, it was assessed that the amount of feed and distill has no impact on the quality or yield of maribavir.

Example 3-9. Influence of Lower or Higher Concentration of Maribavir in iPrOAc Before Crystallization

Samples were performed using lower or higher concentration of maribavir crude before crystallization. Without wishing to be bound to a particular theory, it was thought that a concentrated solution of the crude maribavir in iPrOAc might spontaneously crystallize during filtration/transfer operations, or sampling for NMR assay. This risk was evaluated with solubility data (see Example 3-24). Samples and results are shown in Table 3-4.

TABLE 3-4

Reaction Samples and Results.

Sample	maribavir (% w/w)	PSD	XRPD
3-9-1	17	d(0.1) = 163 μm d(0.5) = 240 μm d(0.9) = 351 μm	Complies
3-9-2	15	d(0.1) = 198 μm d(0.5) = 287 μm d(0.9) = 409 μm	Complies
3-9-3	20	d(0.1) = 178 μm d(0.5) = 284 μm d(0.9) = 452 μm	Complies

[1] Approximate amount of product in organic layer after extraction.

The variation of maribavir concentrations did not affect product purity and assay. The higher concentration of 20% w/w before toluene addition did not lead to spontaneous crystallization after addition of 1^{st} amount of toluene before seeding. The resulting PSD and polymorph for both 3-9-2 and 3-9-3 show slightly larger PSD. The values tested did not show an impact on quality or yield of maribavir. However, deviating from a range of 15-20% w/w may lead to an undesired PSD and/or a lower yield.

Example 3-10. Influence of Higher Water-Content Before Crystallization

Without wishing to be bound to particular theory, it was thought that the presence of water leads to higher solubility of maribavir, which may impact the yield and PSD. Therefore, higher water-content before crystallization was studied in accordance with Table 3-2.

In one sample, the higher amount of water led to a yield below 70%, although it was not determined whether this yield was representative.

Regardless, a higher water content (without a higher TBME content) is highly unrealistic for the following reasons, for example: a) an unsuccessful solvent exchange could lead to an elevated water content, and would also likely result in an elevated level of TBME; b) water up-take by the added solvents (e. g. iPrOAc for feed or dilution) is unlikely since water contents are controlled by raw material specifications (<0.05% water specified); and c) the water content is controlled and can be corrected by another distillation step. The higher water content by spiking had no impact on product purity, PSD, or XRPD. Therefore, it was assessed that the amount of water did not impact quality of maribavir, but could impact yield.

Example 3-11. Influence of Hold Time after 1^{st} Toluene Addition and Pre-Seeding

The stability of the supersaturated solution of maribavir with respect to self-nucleation was investigated by allowing the iPrOAc/toluene solution at 77° C. to stir for a period of time without seeding. When the solution was stirred before seeding for 1 hour, a very low trace amount of solid particles was spotted in the reaction mixture before the seeds were added. Nevertheless, the resulting PSD still matched the specifications. When the solution was stirred before seeding for 30 minutes, no solid particles were spotted in the reaction mixture before the seeds were added. Since seeding was assessed to be effective in this case, the reaction mixture was then used to test the impact of 0.5% wt seeds (see Example 3-15). The resulting PSD still matched the specifications.

The supersaturated solution of crude maribavir in iPrOAc: toluene 7:3 v:v was susceptible to self-seeding at T_i=77±3° C., since after 1 h of stirring pre-seeding led to the formation of small amounts of seeds. The self-seeding of the solution after 1 h didn't appear to impact the PSD negatively. After 30 min, no self-seeding occurred. Therefore, it is recommend to limit time between finished 1^{st} toluene dosage and seeding.

Example 3-12. Influence of Lower or Higher Temperature During Toluene Additions and Seeding

Samples were performed evaluating a lower and higher temperature during toluene addition and seeding (temperature was maintained during seeding, 2^{nd} addition of toluene and stirring time prior to cooling to 5° C.). Samples were prepared in according to Table 3-2.

The temperature had no impact on the yield or purity of maribavir. Seeding temperatures of 67 and 87° C. gave a similar PSD as the standard reaction. The desired polymorph was obtained in all cases. Therefore, it was assessed that the temperature during seeding had no impact on the quality or yield of maribavir.

Example 3-13. Influence of Lower or Higher Amounts of Toluene for Crystallization

The ideal volume ratios for iPrOAc:toluene prior to seeding was thought to be 7:3, and 7:11 for the final crystallization solution. Without wishing to be bound to a particular theory, it was thought that deviation from these ratios may lead to different PSD and may impact yield. Studies were performed using less or more toluene for crystallization, in accordance with Table 3-2.

The amount of toluene for crystallization had no impact on product yield and purity for the range examined. The PSD was not impacted significantly when the toluene amounts were altered and the correct polymorph was obtained. Therefore, it was assessed that the amount of toluene had no impact on the yield and quality for maribavir.

Example 3-14. Influence of Shorter or Longer Dosing Times of Toluene Before and After Seeding

Without wishing to be bound by a particular theory, it was thought that the dosing times of toluene might have an impact on crystallization properties, e.g., the resultant PSD. A quick dosage might even lead to spontaneous crystallization. Therefore, studies were performed for assessment of shorter and longer dosing times. The standard dosing times for toluene before seeding as 15-45 minutes, and after seeding was 120-140 mins. Samples were tested in accordance with Table 3-2.

The variation of the dosing times of toluene had no impact on product purity, yield, and PSD. Therefore, it was assessed that the dosing time for both toluene amounts had no impact on yield or quality of maribavir.

Example 3-15. Influence of Lower or Higher Amount of Seeds for Crystallization

Without wishing to be bound to a particular theory, it was thought that the amount of micronized seeds may have an impact on the crystallization. Typically, between 0.5-1.0% w/w of seed loading was used. Therefore, studies were performed for assessment of amount of seeds for crystallization in accordance with Table 3-2.

Lower or higher seed amounts had no impact on product purity and assay. The XRPD results matched the desired polymorph in all cases. The particles obtained with 0.3% seeds were larger than usual. All other PSDs were in good accordance to the standard procedure. Therefore, it was assessed that the amount of seeds has no impact on the quality or yield of maribavir. However, without wishing to be bound by a particular theory, there could be a risk that higher amounts of seed crystals can result in a product with decreased quality.

Example 3-16. Influence of Shorter or Longer Stirring Time Post-Seeding Before Addition of 2nd Amount of Toluene

Without wishing to be bound to a particular theory, it was thought that maintaining the seeded crystallization mixture at a fixed temperature and solvent composition may impact crystal growth and PSD. Studies were performed to assess the impact of a shorter or longer stirring time post-seeding before the addition of the 2nd amount of toluene (standard 25-35 mins). Samples and were prepared in accordance with Table 3-2.

Changing the stirring time post-seeding had no significant impact on product and crystal quality. Therefore, it was assessed that stirring time post-seeding had no impact on the quality or yield of maribavir.

Example 3-17. Influence of Shorter or Longer Stirring Time after Addition of 2nd Amount of Toluene

Typically, after addition of the 2nd amount of toluene to the seeded mixture, the batch was stirred over 45-75 min maintaining 77±3° C. before cooling down. Without wishing to be bound to a particular theory, it was thought that a shorter or longer hold time might impact the crystal quality. Therefore, assess the impact of a shorter or longer stirring time after addition of the 2nd amount of toluene were assessed. Sample were prepared in accordance with Table 3-2

Changing the stirring time after adding the second amount of toluene had no significant impact on product and crystal quality. Therefore, it was assessed that the stirring time after the second amount of toluene had no impact on the quality or yield of maribavir.

Example 3-18. Influence of Shorter or Longer Cooling Time During Crystallization

Studies were performed to assess if the cooling ramp affects the PSD. Samples and were prepared in accordance with Table 3-2.

The duration of the cooling time had no impact on product purity and yield. A shorter cooling led to a very similar PSD than the standard example. Therefore, it was assessed that the time for cooling had no impact on purity and yield of maribavir.

Example 3-19. Influence of Lower or Higher Final Temperature of Crystallization

The crystallization mixture is typically cooled to 3-7° C. Without wishing to be bound to a particular theory, it was thought that the temperature might impact the crystal quality or maribavir purity and yield: a lower temperature may lead to precipitation of impurities while a higher temperature could lead to dissolution of product and a lower yield. Therefore, examples were performed to assess the impact of a lower or higher final temperature in accordance with Table 3-2.

The final temperature after crystallization has no impact on product purity and yield. A lower final temperature leads to larger particles with a wider shape of the PSD. A microscope image showed the presence of agglomerates. A higher temperature leads to a slightly larger, but similar PSD than in the standard reaction. Therefore, it was assessed that the final temperature had no impact on quality or yield.

Example 3-20. Influence of Temperature During Filtration and Washing

The filtration/slurry wash is typically performed at 5° C. Without wishing to be bound to a particular theory, it was thought that a lower temperature could lead to precipitation of impurities while a higher temperature could lead to a lower yield. Therefore, samples with filtration/washing at a lower temperature or higher temperature where examined in accordance with Table 3-2.

A lower or higher temperature during washing the filter cake had no significant impact on product purity and yield. The results of this section are supported by the results of Example 3-19. Therefore, it was assessed that the temperature during washing has no impact on the quality or yield of maribavir.

Example 3-21. Influence of the Amount of Toluene for Washing

Without wishing to be bound to a particular theory, it was thought that the amount of toluene for washing the filter cake may possibly impact the yield and purity of the product. For example, a higher amount of toluene may reduce the yield of maribavir. This was evaluated based on the solubility of maribavir in toluene (see Example 3-24). A lower amount could have an influence on product quality. Therefore, washing with less toluene was performed in accordance with Table 3-2.

The lower amount of toluene for washing did not have an impact on product yield and quality. The solubility of maribavir in toluene at 20-25° C. is 0.2% w/w (see Example 3-24), and washing with higher amounts is not expected to lead to high product losses. Therefore, it was assessed that the amount of toluene for washing had no impact on the quality or yield of maribavir.

Example 3-22. Influence of Stirring Time During Toluene Wash

Without wishing to be bound by a particular theory, shorter stirring time during the wash could was that to result in suboptimal washing. Nevertheless, no additional studies are performed to test this, because the wash was done as displacement wash during this PAR study. The upper limit of the PAR was defined based on stability tests (see Example 3-23). The results were also supported by solubility data (see Example 3-24).

Example 3-23. Stability Data

The following stability data was generated and referenced in one or more of the prior examples. Stability of the following systems were studied and found to be stable for at least 7 days: Stability of Compound 3 HCl in MeOH/TBME at 40° C.; Stability of reaction mixture at 40° C. (amount of maribavir); Stability of biphasic mixture at 40° C. (before extraction); Stability of biphasic mixture at 40° C. (during last NaCl washing); Stability of reaction mixture after final distillation step at Stability of biphasic mixture at 75° C.; Stability of reaction mixture after final distillation step at 90° C.; Stability of crystallization mixture after dosage of 2^{nd} portion of toluene at reflux; Stability of crystallization at 20° C.; Stability of mixture during slurry wash at 20-25° C.; Stability of maribavir in the mother liquor at 20-25° C.; Stability of wet cake at 70° C.; and Stability of dry cake at 70° C.

Example 3-24. Solubility Data

The following solubility data was generated and referenced in one or more of the prior examples. A mixture of approx. 2 g of maribavir and 5 mL solvent was stirred overnight in sealed vials at different temperatures. The clear supernatant solutions were sampled for NMVR assay. All examples were done in triplicates. The samples and results are shown in Table 3-5. Table 3-5. Maribavir solubilities in iPrOAc and toluene.

TABLE 3-5

Maribavir solubilities in iPrOAc and toluene.

Solvent	Temperature (° C.)	Solubility (% w/w)
Toluene	20-25	0.2% w/w
iPrOAc	5	37.85
	10	39.71
	20	23.42
	25	19.66
	30	17.82
	35	15.89
	40	15.10
	60	3.05
	85	0.59

Maribavir showed a good solubility in iPrOAc below 30° C. leading to a low risk of spontaneous crystallization in solution.

Example 3-267 Scale-up Studies

Scaled-up reactions were performed as outlined in Table 3-2. Additional lab-scale runs were also performed and confirmed to be successful since the PSD followed the same trends at the scaled-up runs. The scaled-up samples are shown in Table 3-6.

TABLE 3-6

Scaled-Up Samples and Results.

	Sample:							
	1	2	3	4	5	6	7	8
Crystallization Temperature (° C.)	77	77	77	88	88	85	88	85
Concentration maribavir in IPAC (wt %)	17.5	17.5	17.5	17.5	17.5	17.5	19.5	17.5
Seed loading rel. to Compound 3 (wt %)	0.70	0.35	0.35	0.15	0.35	0.35	0.15	0.10
Seeds d(50) (μm)	2.72	2.72	4.47	4.47	4.47	2.25	2.25	2.25
Agitation after seeding (rpm)	100	65	60	65	65	65	65	65
Agitation cool down (rpm)	80	75	60	65	65	65	65	65
Agitation aging at 5° C.	80	75	60	65	65	65	65	65
PSD d(50) after crystallization (μm)	n/a	177	210	278	235	174	215	219
Acceptable	No	No	Yes	Yes	Yes	No	Yes	Yes

Among the tested parameters, seed loading and seed characteristics were demonstrated as having the greater impact on the resulting PSD. The crystallization temperature and concentration showed a lower significant as a single variant. The combined data between the laboratory and scaled-up runs was analyzed and preferred parameters shown in Table 3-1.

Example 4. Summary of Batch History. The Batch Histories are Summarized Below in Tables 4-1 to 4-20

Step I: Synthesis of Compound 2

TABLE 4-1

Batch Analysis				
		Batch No.		
		1	2	3
		Batch Size (kg)		
		81.48	89.02	89.02
Test		Results		
Color Appearance		Off-white Solid	Off-white Solid	Off-white Solid
HPLC (Purity) (a/a)	Compound 5[a]	ND	ND	ND
	Compound 10[b]	<0.1%	<0.1%	<0.1%
	Compound 9[c]	0.1%	0.1%	0.1%
	Unknown Impurities Each individual impurity	0.1%	0.1%	0.1%
	Sum of impurities	0.4%	0.4%	0.3%
HPLC Content (w/w)		99.1%	99.8%	99.4%

ND = not detected

[a]Limit of detection for Compound 5 = 0.01%

[b]Limit of detection for Compound 10 = 0.02%

[c]Limit of detection for Compound 9 = 0.02%

TABLE 4-2

Batch Analysis						
		Batch No.				
		4	5	6	7	8
	Batch Size (kg)	110.73	119.2	118.05	117.62	120.46
	Test	Result				
	Color Appearance	Off-white Solid	Off-white Solid	Off-white Solid	Off-white Solid	Off-white Solid
HPLC (Purity)	Compound 5 (w/w)	ND	ND	ND	ND	ND
	Compound 10 (w/w)	0.1%	0.1%	0.1%	0.1%	0.1%
	Compound 9 (w/w)	0.5%	0.5%	0.6%	1.1%	0.6%
	Compound 12 (w/w)[a]	ND	ND	ND	ND	ND
	Impurity m/z	ND	ND	ND	ND	ND
	Each unspecified (a/a)	0.1%	ND	0.1%	0.1%	0.1%
	Total unknown impurities (a/a)	0.1%	ND	0.1%	0.1%	0.1%
	Total known impurities (w/w)	0.5%	0.6%	0.7%	1.2%	0.6%
	HPLC Content (w/w)	99.4%	99.5%	99.6%	99.4%	99.5%

ND = not detected

[a]Limit of detection for Compound 12 = 0.01%

TABLE 4-3

Batch Analysis						
		Batch No.				
		9	10	11	12	13
	Batch Size (kg)	118.77	117.69	118.86	121.7	123.7
	Test	Result				
	Color Appearance	Off-white Solid	Off-white Solid	Off-white Solid	Off-white Solid	Off-white Solid
HPLC (Purity)	Compound 5 (w/w)	ND	ND	ND	ND	ND
	Compound 10 (w/w)	0.1%	0.1%	0.1%	ND	ND
	Compound 9 (w/w)	0.7%	0.6%	0.6%	0.1%	0.4%
	Compound 12 (w/w)[a]	ND	ND	ND	ND	ND
	Impurity m/z[b]	ND	ND	ND	0.1% (a/a) (RRT = 0.81)	ND

TABLE 4-3-continued

Batch Analysis					
	Batch No.				
	9	10	11	12	13
Each unspecified (a/a)	0.1%	0.1%	0.1%	0.1%	ND
Total unknown impurities (a/a)	0.1%	0.1%	0.1%	0.1%	ND
Total known impurities (w/w)	0.8%	0.7%	0.7%	0.1%	0.4%
HPLC Content (w/w)	99.4%	99.5%	99.6%	99.9%	99.6%

ND = Not detected
[a]Limit of detection for Compound 12 = 0.01%

TABLE 4-4

Batch Analysis					
		Batch No.			
		14	15	16	17
		Batch Size (kg)			
Test	Acceptance Criteria	116.64 Results	113.44 Results	107.78 Results	104.69 Results
Color Appearance	White to off-white solid	white solid	white solid	white solid	white solid
HPLC (Purity)	Compound 5 (w/w)[a]	<0.05%	<0.05%	<0.05%	<0.05%
	Compound 10 (w/w)[c]	0.08%	0.05%	0.11%	0.10%
	Compound 9 (w/w)[d]	<0.05%	<0.05%	<0.05%	<0.05%
	Compound 12 (w/w)[e]	<0.05%	<0.05%	<0.05%	<0.05%
	Compound 11 (w/w)[f]	<0.05%	<0.05%	<0.05%	<0.05%
	Sum unspecified impurities[b] (w/w)	0.14%[g]	0.15%[g]	0.13%[g]	0.16%[g]
	Sum specified impurities (w/w)	0.08%	0.05%	0.11%	0.1%
	Largest unspecified impurity[g] (w/w)	0.14%[g]	0.15%[g]	0.13%[g]	0.16%[g]
Assay HPLC (w/w)		98.6%	99.8%	98.3%	98.8%

[a]Limit of detection for Compound 5 = 0.01%
[b]% (a/a) Relative Response Factor Corrected.
[c]Limit of detection for Compound 10 = 0.02%
[d]Limit of detection for Compound 9 = 0.02%
[e]Limit of detection for Compound 12 = 0.01%
[f]Limit of detection for Compound 11 = 0.01% (a/a), Not Relative Response Factor Corrected
gThe unknown was identified as being 5,6-dichloro-N-ethyl-1H-benzo[d]imidazol-2-amine

TABLE 4-5

Batch Analysis						
		Batch No.				
		18	19	20	21	22
	Batch Size (kg)	18.82	15.05	16	17.75	30.34
	Test			Result		
	Color Appearance	White Solid	White Solid	White Solid	White Solid	White Solid
HPLC (Purity)	Compound 5[a]	ND	ND	ND	ND	ND
	Compound 10 (a/a)[c]	0.63%	0.1%	<0.05%	0.05%	<0.1%
	Compound 9 (a/a)[d]	ND	0.09%	<0.05%	ND	<0.1%
	Compound 12[e]	ND	ND	ND	ND	ND
	Compound 11[f]	ND	ND	ND	ND	ND
	Sum unspecified impurities (a/a)[b]	ND	ND	ND	<0.05%	0.1%
	Sum specified impurities (a/a)	0.63%	0.19%	<0.05%	0.05%	<0.10%

TABLE 4-5-continued

Batch Analysis					
	Batch No.				
	18	19	20	21	22
Largest unspecified impurity[g] (a/a)	ND	ND	ND	<0.05%	0.10%
Assay HPLC (w/w)	99.0%	99.4%	98.6%	99.3%	99.5%

ND = not detected
[a]Limit of detection for Compound 5 = 0.01%
[b]% (a/a) Relative Response Factor Corrected
[c]Limit of detection for Compound 10 = 0.02%
[d]Limit of detection for Compound 9 = 0.02%
[e]Limit of detection for Compound 12 = 0.01%
[f]Limit of detection for Compound 11 = 0.01% (a/a), Not Relative Response Factor Corrected
[g]The unknown was identified as being 5,6-dichloro-N-ethyl-1H-benzo[d]imidazol-2-amine

TABLE 4-6

Batch Analysis						
		Batch No.				
		23	24	25	26	27
	Batch Size (kg)	32.58	32.13	31.58	33.13	32.13
	Test			Result		
	Color Appearance	White Solid	White Solid	White Solid	White Solid	White Solid
HPLC (Purity)	Compound 5[a]	ND	ND	ND	ND	ND
	Compound 10 (a/a)[c]	<0.1%	0.1%	0.1%	<0.1%	<0.1%
	Compound 9 (a/a)[d]	<0.1%	<0.1%	0.1%	<0.1%	<0.1%
	Compound 12[e]	ND	ND	ND	ND	ND
	Compound 11[f]	ND	ND	ND	ND	ND
	Sum unspecified impurities[b] (a/a)	0.1%	0.1%	0.1%	0.1%	0.1%
	Total specified impurities (a/a)	<0.10%	0.10%	0.10%	<0.10%	<0.10%
	Largest unspecified impurity[g] (a/a)	0.10%	0.10%	0.10%	0.10%	0.10%
	Assay HPLC (w/w)	99.6%	99.5%	99.4%	99.6%	99.5%

ND = not detected
[a]Limit of detection for Compound 5 = 0.01%
[b]% (a/a) Relative Response Factor Corrected
[c]Limit of detection for Compound 10 = 0.02%
[d]Limit of detection for Compound 9 = 0.02%
[e]Limit of detection for Compound 12 = 0.01%
[f]Limit of detection for Compound 11 = 0.01% (a/a), Not Relative Response Factor Corrected
[g]The unknown was identified as being 5,6-dichloro-N-ethyl-1H-benzo[d]imidazol-2-amine

Step II: Synthesis of Compound 3

TABLE 4-7

Batch Analysis				
		Batch No.		
		1	2	3
		Batch Size (kg)		
	Test	175.6 Results	173.8 Results	180.5 Results
	Color Appearance	White Solid	White Solid	White Solid
HPLC (Purity)	Compound 3 (a/a)	100.0%	100.0%	100.0%
	Compound 2 (a/a)	ND[a]	ND	ND
	Maribavir	ND	ND	ND
	Compound 12	ND	ND	ND
	Sum unknown impurities	ND	ND	ND

TABLE 4-7-continued

Batch Analysis			
	Batch No.		
	1	2	3
	Batch Size (kg)		
Test	175.6 Results	173.8 Results	180.5 Results
Biggest unspecified impurity (a/a)	ND	ND	ND
Sum all impurities (a/a)	ND	ND	ND

ND = not detected
[a]Limit of detection = 0.01%

TABLE 4-8

Batch Analysis					
		Batch No.			
		4	5	6	7
		Batch Size (kg)			
		119.62	119.1	118.19	124.39
Test		Results			
Color		White	White	White	White
Appearance		Solid	Solid	Solid	Solid
Appearance Further Observations		Complies	Complies	Complies	Complies
HPLC (Purity)	Compound 3 (a/a)	98.96%	99.12%	99.04%	98.76%
	Compound 2 (a/a)	0.07%	0.12%	0.09%	0.08%
	Maribavir (a/a)	0.07%	ND	ND	ND
	Compound 12	ND	ND	ND	ND
	Biggest unspecified impurity (a/a)	RRT = 0.37, 0.12% (a/a)	RRT = 0.37, 0.09% (a/a)	RRT = 0.37, 0.09% (a/a)	RRT = 0.37, 0.13% (a/a)
		RRT = 0.40, 0.11% (a/a)	RRT = 0.40, 0.08% (a/a)	RRT = 0.40, 0.07% (a/a)	RRT = 0.40, 0.10% (a/a)
		RRT = 0.57, 0.27%, (a/a)	RRT = 0.57, 0.21%, (a/a)	RRT = 0.57, 0.28%, (a/a)	RRT = 0.57, 0.31%, (a/a)
		RRT = 0.60, 0.3% (a/a)	RRT = 0.60, 0.31% (a/a)	RRT = 0.60, 0.34% (a/a)	RRT = 0.60, 0.44% (a/a)
		RRT = 0.70, 0.09% (a/a)	RRT = 0.70, 0.08% (a/a)	RRT = 0.70, 0.09% (a/a)	RRT = 0.70, 0.11% (a/a)
		0.3%	0.31% (a/a)	0.34% (a/a)	0.44%, (a/a)
	Sum all impurities (a/a)	1.04%	0.88%	0.96%	1.24%

ND = not detected

TABLE 4-9

Batch Analysis			
		Batch No.	
		8	9
		Batch Size (kg)	
		99.45	99.1
Test		Result	
Color		White	White
Appearance		Solid	Solid
Appearance Further Observations		Complies	Complies
HPLC (Purity)	Compound 3 (a/a)	99.49%	99.11%
	Compound 2 (a/a)	<0.1%	<0.1%
	Biggest unspecified impurity (a/a).	N/A	RRT = 0.36, 0.11% (a/a)
		N/A	RRT = 0.40, 0.10% (a/a)
		RRT = 0.57, 0.21%, (a/a)	RRT = 0.57, 0.26%, (a/a)
		RRT = 0.59, 0.27% (a/a)	RRT = 0.59, 0.34% (a/a)
		0.27% (a/a)	0.34% (a/a)
	Sum all impurities (a/a)	0.48%	0.81%

N/A = not applicable

TABLE 4-10

Batch Analysis					
		Batch No.			
		10	11	12	13
		Batch Size (kg)			
		145.96	144	78.2	300.65
Test		Results			
Color		White solid	White solid	White solid	White solid
HPLC (Purity)	Compound 3 (a/a)	99.7%	98.5%	99.33%	99.50%
	Compound 2 (a/a)	<0.05%	<0.05%	<0.05%	<0.05%
	Largest unspecified impurity (a/a)	0.11%	0.42%	0.2%	0.15%
	Sum all impurities (a/a)	0.26%	1.6%	0.69%	0.50%

TABLE 4-11

Batch Analysis					
		Batch No.			
		14	15	16	17
		Batch Size (kg)			
		106.46	106.65	105.64	107.89
Test		Results			
Color		White	White	White	White
Appearance		Solid	Solid	Solid	Solid
Appearance		Complies	Complies	Complies	Complies
HPLC (Purity)	Compound 3 (a/a)	98.5%	98.5%	98.0%	99.5%
	Compound 2 (a/a)	<0.1%	<0.1%	<0.1%	<0.1%
	Maribavir (a/a)	0.18%	0.17%	0.30%	0.07%
	Biggest unspecified impurity (a/a)	RRT = 0.37, 0.22% (a/a)	RRT = 0.37, 0.21% (a/a)	RRT = 0.37, 0.31% (a/a)	N/A
		RRT = 0.41, 0.17% (a/a)	RRT = 0.41, 0.16% (a/a)	RRT = 0.41, 0.23% (a/a)	N/A
		RRT = 0.57, 0.37%, (a/a)	RRT = 0.57, 0.43%, (a/a)	RRT = 0.57, 0.61%, (a/a)	RRT = 0.57, 0.24%, (a/a)
		RRT = 0.60, 0.46% (a/a)	RRT = 0.60, 0.41% (a/a)	RRT = 0.60, 0.43% (a/a)	RRT = 0.60, 0.30% (a/a)
		RRT = 0.70, 0.12% (a/a)	RRT = 0.70, 0.12% (a/a)	RRT = 0.70, 0.12% (a/a)	N/A
		0.46% (a/a)	0.43% (a/a)	0.61% (a/a)	0.30%
	Sum all impurities (a/a)	1.5%	1.5%	2.0%	0.54%

Step III: Synthesis of Maribavir

TABLE 4-12

Unmilled Maribavir Batch Analysis							
		Batch No.					
		1	2	3	4	5	6
Size (kg)		22.5	22.6	25.5	26.9	44.2	49.4
Test		Results					
Description		Almost white solid	Almost white solid	White solid	White solid	White solid	White solid
Chromatographic Purity[a]	Compound 2 (w/w)	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Compound 3 (w/w)	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Compound 4 (w/w)	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Unspecified Impurities (w/w)	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Total Impurities	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
Assay (w/w), (as is basis)		100.2	100.2	100.1	100.4	99.8	99.8
XRPD		Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Thermal Analysis		Onset: 197.2° C. Peak: 199.9° C.	Onset: 197.1° C. Peak: 199.7° C.	Onset: 198.1° C. Peak: 199.3° C.	Onset: 199.5° C. Peak: 201.2° C.	Onset: 196.94° C. Peak: 200.3° C.	Onset: 197.01° C. Peak: 199.72° C.
Enantiomeric Purity[b]	D-maribavir	NT	NT	<0.025	<0.025	<0.025	<0.025
Particle Size	d(0.1) (um)	51	50	87	76	128	138
	d(0.5) (um)	214	205	182	174	279	245
	d(0.9) (um)	354	346	233	266	439	244

N/A = not applicable,

ND = not detected,

NT = not tested

[a]Reporting Limit (RL) = 0.05%

[b]Reporting Limit (RL) = 0.025%.

TABLE 4-13

Unmilled Maribavir Batch Analysis										
		Batch No.								
		7	8	9	10	11	12	13	14	15
	Size (kg)	41.1	41.4	41.4	41.4	40.9	53.7	21.24	21.3	21.5
	Test	Results								
	Description	White Solid	White Solid	White Solid	White Solid	White Solid	White Solid	White Solid	White Solid	White Solid
Purity HPLC[a]	Compound 2 (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Compound 3 (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Compound 4 (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Biggest Unknown Impurity (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Sum all Impurities (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	HPLC Identity	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Chiral Purity[b]	D-maribavir (a/a)	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Assay HPLC % (w/w)	99.0	99.0	99.2	99.1	99.1	99.6	100.4	100.5	100.4
Particle Size	d(0.1) (um)	109.2	64.2	66.7	71.4	59.8	70.3	141.4	161.2	147.4
	d(0.5) (um)	267.8	125.2	145	160.2	130.5	157.2	287.6	259.6	265.6
	d(0.9) (um)	481.2	238.9	292	346.8	283.9	368.5	503.6	405.7	451.8
	P-XRD Identity	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms

ND = not detected
[a]Reporting Limit (RL) = 0.05%
[b]Reporting Limit (RL) = 0.025%

TABLE 4-14

Unmilled Maribavir Batch Analysis							
		Batch No.					
		16	17	18	19	20	21
	Size (kg)	49.9	49.3	50.0	49.7	49.5	49.7
	Test	Results					
Description		Off-white solid	Off-white solid	Off-white solid	Off-white solid	Off-white solid	Off-white solid
Purity HPLC[a]	Compound 2 (a/a)	ND	ND	ND	ND	ND	ND
	Compound 3 (a/a)	ND	ND	ND	ND	ND	ND
	Compound 4 (a/a)	ND	ND	ND	ND	ND	ND
	Biggest Unknown Impurity (a/a)	ND	ND	ND	ND	ND	ND
	Sum all Impurities (a/a)	ND	ND	ND	ND	ND	ND
HPLC Identity	Retention time conforms to reference standard	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Chiral Purity[b]	D-maribavir (a/a)	NT[g]	ND	NT	NT	NT	NT
	Assay HPLC % (w/w)	100.2	100.1	99.9	100.0	99.8	99.9
Particle Size	d(0.1) (um)	164.5	147.5	140.2	170.5	165.8	166.4
	d(0.5) (um)	236.8	232.9	218.0	246.2	243.1	234.6
	d(0.9) (um)	340.7	350.0	330.5	348.4	352.8	330.0
P-XRD Identity	Conforms to Reference Standard	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms

ND = not detected,
NT = not tested
[a]Reporting Limit (RL) = 0.05%
[b]Reporting Limit (RL) = 0.025%.

TABLE 4-15

Unmilled Maribavir Batch Analysis					
		Batch No.			
		22	23	24	25
		Size (kg)			
		40.97	50.0	49.0	44.6
Test		Result			
Description		Off-white solid	Off-white solid	Off-white solid	White Solid
Purity HPLC[a]	Compound 2 (a/a)	<0.05	<0.05	<0.05	<0.05
	Compound 3 (a/a)	<0.05	<0.05	<0.05	<0.05
	Compound 4 (a/a)	<0.05	<0.05	<0.05	<0.05
	Biggest Unknown Impurity (a/a)	<0.05	<0.05	<0.05	<0.05
	Sum all Impurities (a/a)	<0.05	<0.05	<0.05	<0.05
HPLC Identity		Conforms	Conforms	Conforms	Conforms
Assay HPLC % (w/w)		100.2	99.9	100.1	99.9
Particle Size	d(0.1) (um)	143.0	107	134	129
	d(0.5) (um)	194.0	157	201	245
	d(0.9) (um)	265.0	223	291	401
P-XRD Identity	Conforms to Reference Standard	Conforms	Conforms	Conforms	Conforms

NT = not tested
[a]Reporting Limit (RL) = 0.05%

TABLE 4-16

Unmilled Maribavir Batch Analysis				
		Batch No.		
		26	27	28
		Size (kg)		
		54.25	3.35	60.2
Test		Result		
Description		White solid	White solid	White solid
Melting Point		197.1° C. (onset) 198.6° C. (peak)	197.1° C. (onset) 198.5° C. (peak)	196.9° C. (onset) 199.6° C. (peak)
Water—KF (w/w)		0.01	0.01	0.01
Purity HPLC[a]	Compound 2 (a/a)	ND	ND	ND
	Compound 3 (a/a)	ND	ND	ND
	Compound 4 (a/a)	ND	ND	ND
	Biggest Unknown Impurity (a/a)	ND	ND	ND
	Sum all Impurities (a/a)	ND	ND	ND
Assay HPLC % (w/w)		100.3	100.4	100.5
Particle Size	d(0.1) Report Result (um)	12.9	14.2	106.4
	d(0.5) Report Result (um)	62.6	69.9	251.2
	d(0.9) Report Result (um)	489.6	574.9	426.3

ND = not detected
[a]Reporting Limit (RL) = 0.05%

TABLE 4-17

Unmilled Maribavir Batch Analysis						
		Batch No.				
		29	30	31	32	33
	Size (kg)	12.3	12.9	12.7	12.7	12.9
	Test			Results		
	Description	White solid	White solid	White solid	White solid	White solid
Purity HPLC[a]	Compound 2 (a/a)	ND	ND	ND	ND	ND
	Compound 3 (a/a)	ND	ND	ND	ND	ND
	Compound 4 (a/a)	ND	ND	ND	ND	ND

TABLE 4-17-continued

Unmilled Maribavir Batch Analysis

		Batch No.				
		29	30	31	32	33
	Biggest Unknown Impurity (a/a)	ND	ND	ND	ND	ND
	Sum all Impurities (a/a)	ND	ND	ND	ND	ND
Assay HPLC % (w/w)		100.3	100.4	100.5	100.4	100.3
Particle Size	d(0.1) (um)	200.6	83.0	183.2	169.1	96.4
	d(0.5) (um)	285.9	195.4	268.6	239.3	192.7
	d(0.9) (um)	407.4	377.7	393.9	339.6	329.5

[a]Reporting Limit (RL) = 0.05%.

TABLE 4-18

Unmilled Maribavir Batch Analysis

		Batch No.			
		34	35	36	37
		Size (kg)			
		44.9	64.0	63.3	65.6
Test		Result			
Description		White solid	White solid	White solid	White solid
Purity HPLC[a]	Compound 2 (a/a)	ND	<0.05	<0.05	<0.05
	Compound 3 (a/a)	ND	<0.05	<0.05	<0.05
	Compound 4 (a/a)	ND	<0.05	<0.05	<0.05
	Biggest Unknown Impurity (a/a)	ND	<0.05	<0.05	<0.05
	Sum all Impurities (a/a)	ND	<0.05	<0.05	<0.05
	XRPD	Conform	Conform	Conform	Con-form
Assay HPLC (w/w) %		99.8	99.9	100.0	99.8
Particle Size	d(0.1) (um)	252.5	219.3	174.0	227.0
	d(0.5) (um)	362.5	317.9	331.3	336.2
	d(0.9) (um)	514.7	449.6	556.2	571.3

ND = not detected
[a]Reporting Limit (RL) = 0.05%.

TABLE 4-19

Unmilled Maribavir Batch Analysis

		Batch No.						
		38	39	40	41	42	43	44
Size (kg)		50.1	49.5	50.36	49.6	50.39	41.6	43.6
Test					Result			
Description		White solid	White solid	White solid	White solid	White solid	White solid	White solid
Purity HPLC[a] (a/a)	Compound 2	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Compound 3	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Compound 4	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Biggest Unknown Impurity	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
	Sum all Impurities	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
HPLC Identity		Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Assay HPLC (w/w)%)		99.8	99.9	99.8	99.4	99.7	100.5	101.3
Particle Size (μm)	d(0.1) (um)	140	140	124	24.6	44.1	101	112
	d(0.5) (um)	195	198	182	97.5	174	212	180
	d(0.9) (um)	268	275	263	280	346	384	27
P-XRD Identity		Conforms	Conforms.	Conforms	Conforms	Conforms	Conforms	Conforms.

[a]Reporting Limit (RL) = 0.05%

TABLE 4-20

Unmilled Maribavir Batch Analysis

		Batch No.	
		44	45
		Size (kg)	
		15.083	15.837
	Test	Result	
	Description	White solid	White solid
Purity HPLC[a]	Compound 2 (a/a)	<0.05	<0.05
	Compound 3 (a/a)	<0.05	<0.05
	Compound 4 (a/a)	<0.05	<0.05
	Biggest Unknown Impurity (a/a)	<0.05	<0.05
	Sum all Impurities (a/a)	<0.05	<0.05
	HPLC Identity	Conforms	Conforms
	Assay HPLC (w/w) %)	100.1	100.2
Particle Size	d(0.1) (um)	166.0	180.4
	d(0.5) (um)	240.3	261.8
	d(0.9) (um)	346.9	380.2
	P-XRD Identity	Conforms	Conforms

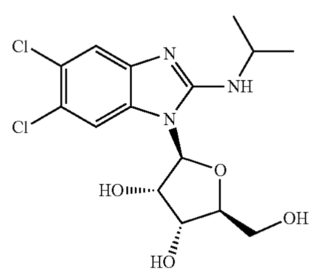
NT = not tested
[a]Reporting Limit (RL) = 0.05%.

While we have described a number of embodiments of this invention, it is apparent that our basic examples may be altered to provide other embodiments that utilize the compounds and methods of this invention. Therefore, it will be appreciated that the scope of this invention is to be defined by the appended claims rather than by the specific embodiments that have been represented by way of examples.

The invention claimed is:

1. A composition of maribavir, Compound 2, and Compound 4, the composition comprising maribavir, Compound 2, and Compound 4, having the following structures:

maribavir



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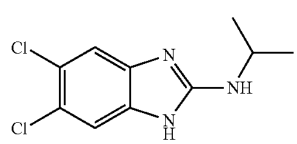

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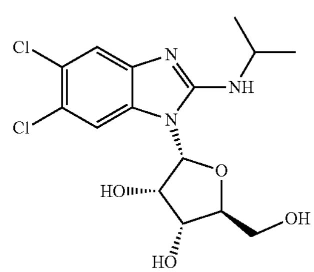
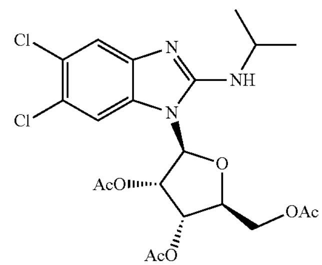


wherein Compound 2 is present in an amount of 0.1% w/w or less relative to maribavir and Compound 4 is present in an amount of 0.1% w/w or less relative to maribavir.

2. The composition of claim 1, wherein Compound 4 is present in an amount of from 0.01% to 0.1% w/w relative to maribavir.

3. The composition of claim 2, further comprising Compound 3,

3



wherein Compound 3 is present in an amount of from 0.01% to 0.1% w/w relative to maribavir.

4. The composition of claim 3, wherein Compound 2 is present in an amount of from 0.01% to 0.1% w/w relative to maribavir.

5. The composition of claim 4, further comprising D-maribavir, wherein D-maribavir is present in an amount of 0.1% w/w or less relative to maribavir.

6. A pharmaceutical composition comprising a composition of claim 5 and one or more pharmaceutically acceptable excipients.

7. A pharmaceutical composition comprising a composition of claim 4 and one or more pharmaceutically acceptable excipients.

8. A pharmaceutical composition comprising a composition of claim 3 and one or more pharmaceutically acceptable excipients.

9. The composition of claim 2, wherein Compound 2 is present in an amount of from 0.01% to 0.1% w/w relative to maribavir.

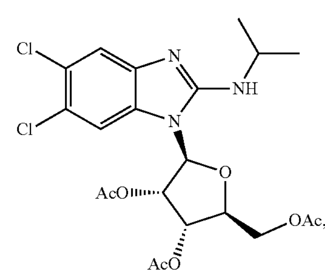
10. A pharmaceutical composition comprising a composition of claim 2 and one or more pharmaceutically acceptable excipients.

11. The composition of claim 1, wherein Compound 4 is present in an amount of 0.05% w/w or less relative to maribavir.

12. The composition of claim 11, wherein Compound 4 is present in an amount of from 0.01% to 0.05% w/w relative to maribavir.

13. The composition of claim 1, further comprising Compound 3,

3



wherein Compound 3 is present in an amount of 0.1% w/w or less relative to maribavir.

14. The composition of claim **1**, further comprising D-maribavir, wherein D-maribavir is present in an amount of 0.1% w/w or less relative to maribavir.

15. The composition of claim **1**, wherein the maribavir has a particle size distribution (PSD) with a d (50) of between about 170 and about 350 μm.

16. The composition of claim **15**, wherein the d (50) is between about 170 and about 226 μm.

17. The composition of claim **15**, wherein the d (50) is between about 227 and about 280 μm.

18. The composition of claim **15**, wherein the d (50) is between about 281 and about 336 μm.

19. A pharmaceutical composition comprising a composition of claim **15** and one or more pharmaceutically acceptable excipients.

20. A pharmaceutical composition comprising a composition of claim **1** and one or more pharmaceutically acceptable excipients.

* * * * *

EXHIBIT D

US012433907B2

(12) **United States Patent**
Song et al.

(10) **Patent No.: US 12,433,907 B2**
(45) **Date of Patent: *Oct. 7, 2025**

(54) **USE OF MARIBAVIR IN TREATMENT REGIMENS**

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(73) Assignee: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/990,585**

(22) Filed: **Dec. 20, 2024**

(65) **Prior Publication Data**

US 2025/0120995 A1 Apr. 17, 2025

Related U.S. Application Data

(63) Continuation of application No. 18/646,035, filed on Apr. 25, 2024, now Pat. No. 12,213,989, which is a continuation of application No. 18/400,304, filed on Dec. 29, 2023, now abandoned, which is a continuation of application No. PCT/US2022/050341, filed on Nov. 18, 2022.

(60) Provisional application No. 63/281,206, filed on Nov. 19, 2021.

(51) **Int. Cl.**
A61K 31/7056 (2006.01)
A61K 31/4166 (2006.01)
A61K 31/513 (2006.01)
A61K 31/55 (2006.01)

(52) **U.S. Cl.**
CPC ***A61K 31/7056*** (2013.01); ***A61K 31/4166*** (2013.01); ***A61K 31/513*** (2013.01); ***A61K 31/55*** (2013.01)

(58) **Field of Classification Search**
CPC A61K 31/7056; A61K 31/4166; A61K 31/513; A61K 31/55
USPC 514/43
See application file for complete search history.

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Primary Examiner — Yih-Horng Shiao
(74) *Attorney, Agent, or Firm* — Honigman LLP; Andrew S. Chipouras; Jonathan P. O'Brien

(57) **ABSTRACT**

Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

20 Claims, 4 Drawing Sheets

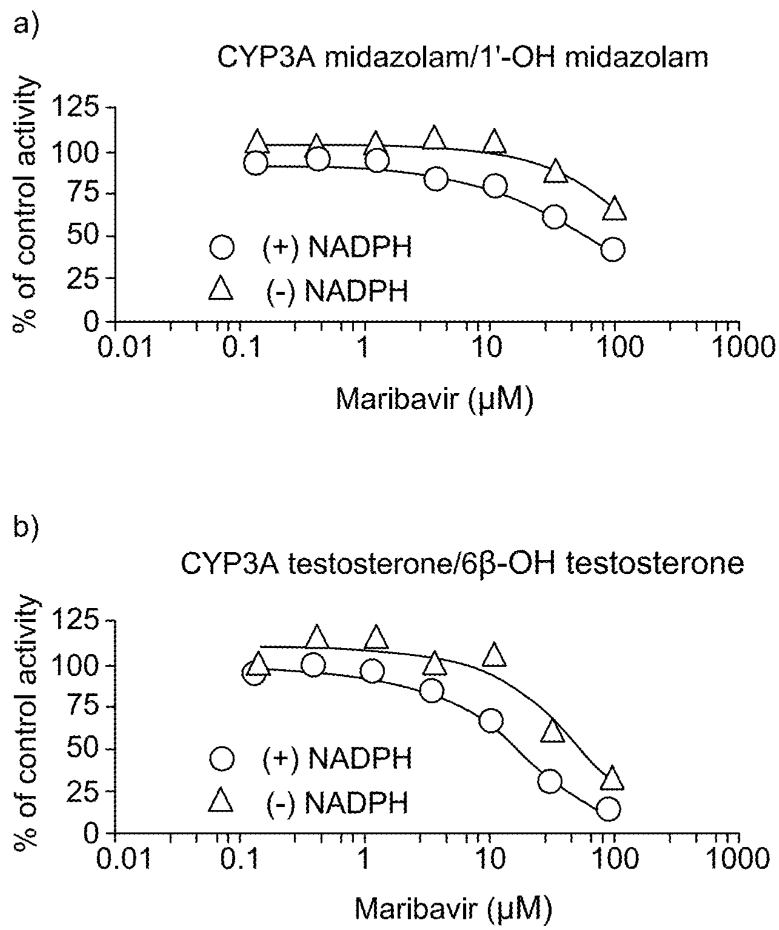

a)
CYP3A midazolam/1'-OH midazolam
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (µM)
b)
CYP3A testosterone/6β-OH testosterone
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (µM)

Figure 1

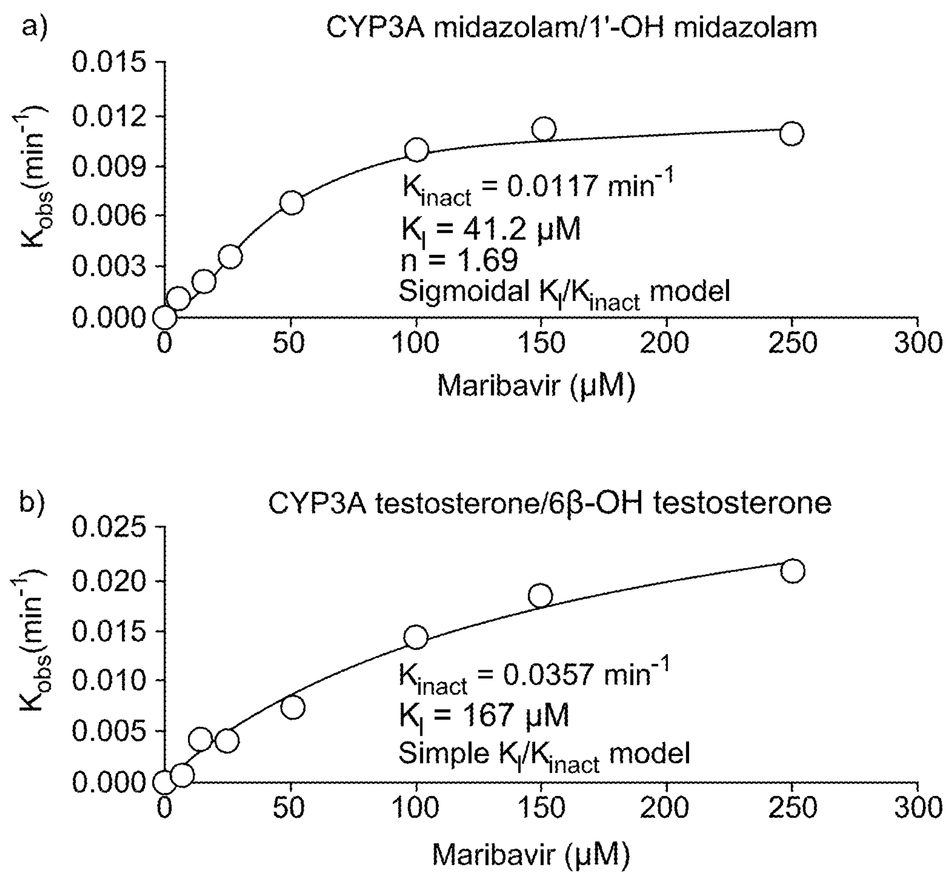

a)
CYP3A midazolam/1'-OH midazolam
K_{obs}(min^{-1})
0.015
0.012
0.009
0.006
0.003
0.000
K_{inact} = 0.0117 min^{-1}
K_I = 41.2 μM
n = 1.69
Sigmoidal K_I/K_{inact} model
0 50 100 150 200 250 300
Maribavir (μM)
b)
CYP3A testosterone/6β-OH testosterone
K_{obs}(min^{-1})
0.025
0.020
0.015
0.010
0.005
0.000
K_{inact} = 0.0357 min^{-1}
K_I = 167 μM
Simple K_I/K_{inact} model
0 50 100 150 200 250 300
Maribavir (μM)

Figure 2

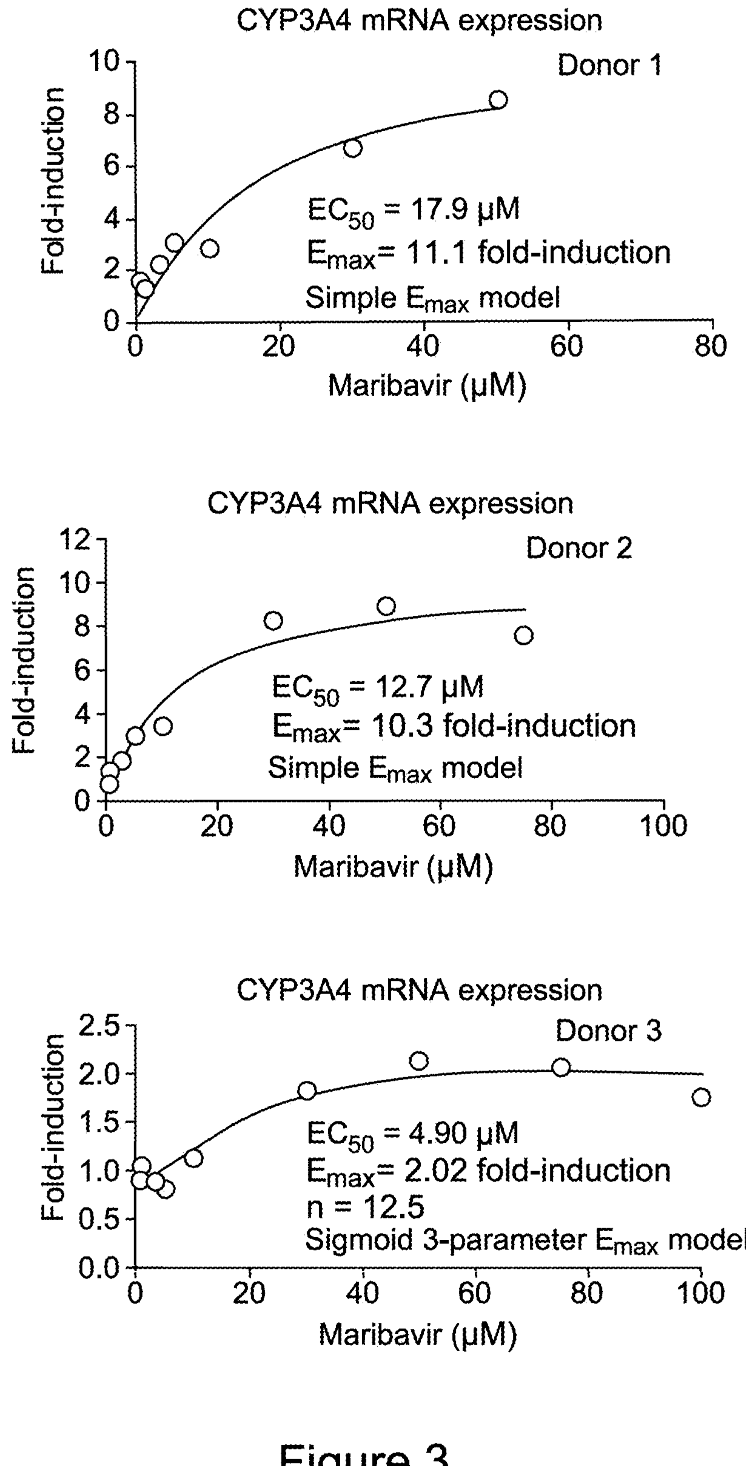

CYP3A4 mRNA expression
Donor 1
Fold-induction
EC50 = 17.9 μM
Emax= 11.1 fold-induction
Simple Emax model
Maribavir (μM)
CYP3A4 mRNA expression
Donor 2
Fold-induction
EC50 = 12.7 μM
Emax= 10.3 fold-induction
Simple Emax model
Maribavir (μM)
CYP3A4 mRNA expression
Donor 3
Fold-induction
EC50 = 4.90 μM
Emax= 2.02 fold-induction
n = 12.5
Sigmoid 3-parameter Emax model
Maribavir (μM)

Figure 3

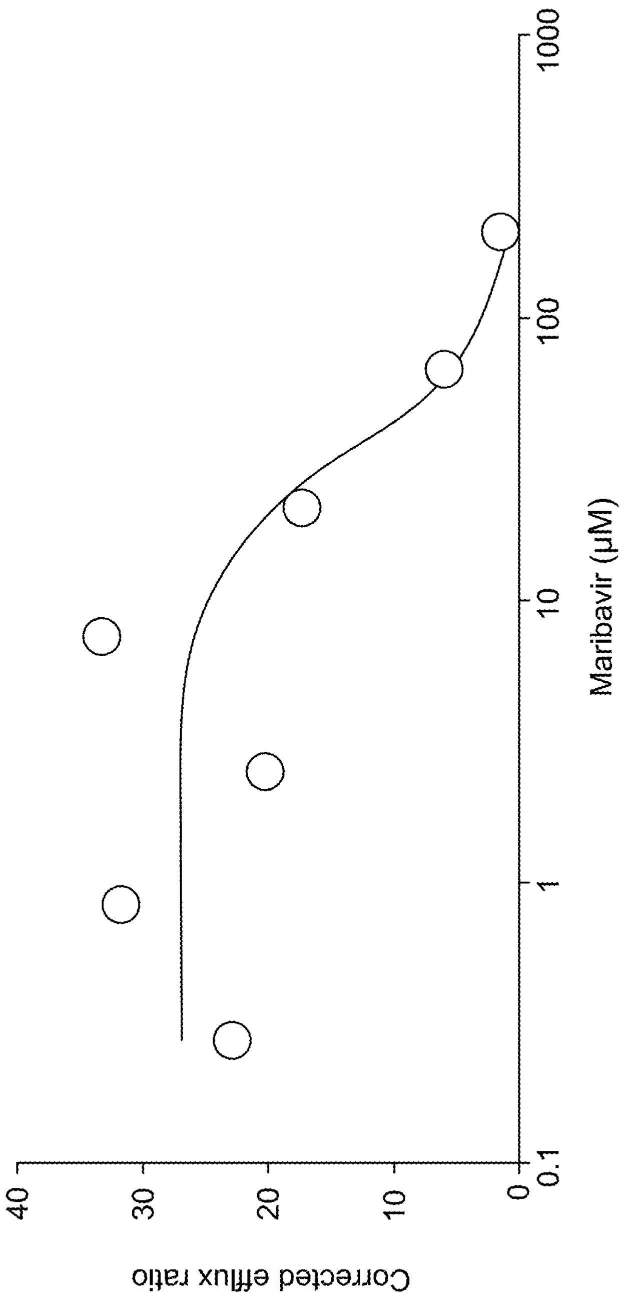

Corrected efflux ratio
0
10
20
30
40
Maribavir (μM)
0.1
1
10
100
1000

Figure 4

USE OF MARIBAVIR IN TREATMENT REGIMENS

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation application of U.S. application Ser. No. 18/646,035, filed Apr. 25, 2024, which is a continuation application of U.S. application Ser. No. 18/400,304, filed Dec. 29, 2023, which is a continuation application of PCT Application No. PCT/US22/50341, filed Nov. 11, 2022, which claims priority to U.S. Provisional Application No. 63/281,206, filed Nov. 19, 2021, each of which is incorporated herein by reference in its entirety.

BACKGROUND

Cytomegalovirus (CMV) infection and disease are significant post-transplant complications and are associated with substantial morbidity and reduced long-term survival among transplant recipients. There are currently no approved therapies for the treatment of CMV infection in transplant recipients.

Transplant patients often receive numerous concomitant medications to manage their comorbidities. Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

SUMMARY

Maribavir is a benzimidazole riboside, and is an orally available antiviral medication against cytomegalovirus (CMV). Maribavir is also known under its trade name LIVTENCITY™. Typically, patients (e.g., adults and/or children over 12 years of age and weighing at least 35 kg) are administered 400 mg if maribavir orally, twice daily. However, in some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a cytochrome P450 3A4 (CYP3A4) inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or a CYP3A4 inducer is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

- administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir; and
- discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

- administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a CYP3A4 inducer; and
- increasing the amount of maribavir administered to the patient.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

- administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments a CYP3A4 inducer decreases maribavir exposure. In some embodiments, an initial therapeutically effective amount of maribavir is administered to the patient, and the amount of maribavir is increased when co-administering a CYP3A4 inducer. In some embodiments, an initial therapeutically effective amount of maribavir is about 400 mg, orally twice daily. In some embodiments, an amount of maribavir is increased to an amount between about 800 mg and about 1200 mg, orally twice daily. In some embodiments, the provided methods further comprising a step of monitoring patient blood (e.g., whole blood or plasma) concentration of maribavir and/or the CYP3A4 inducer.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an immunosuppressant is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, provided methods further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL.

In some embodiments, tacrolimus is administered at an initial dose of between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax.

In some embodiments, a dose of tacrolimus is adjusted when co-administered with maribavir. In some embodiments, maribavir is co-administered at a concentration of 400 mg, 800 mg, or 1200 mg, administered twice daily. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an antiarrhythmic (e.g., digoxin) is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an antiarrhythmic (e.g., digoxin). In some embodiments, provided methods further comprise monitoring the levels of digoxin in patient serum. In some embodiments, provided methods further comprise the step of reducing the amount of digoxin administered to the patient (e.g., to the amount of antiarrhythmic that was administered prior to commencing administration of maribavir).

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or ganciclovir or valganciclovir is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

- administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and
- discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.

In some embodiments, concentration of maribavir and/or the other drug being co-administered (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration.

In some embodiments, the patient is refractory to treatment with one or more other drugs to treat CMV infection (e.g., ganciclovir, valganciclovir, cidofovir, or foscarnet). In some embodiments, maribavir is administered with or without food. In some embodiments, the patient is a transplant recipient (e.g., a hematopoietic stem cell transplant recipient or a solid organ transplant recipient). In some embodiments, the patient is an adult or a child over 12 years old and greater than 35 kg.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**. IC_{50} shift of maribavir in a concentration-dependent assay with HLM for CYP3A with midazolam (A) or testosterone (B) as a probe substrate. CYP, cytochrome P450; HLM, human liver microtomes; IC_{50}, half maximal inhibitory concentration; NADPH, nicotinamide-adenine dinucleotide phosphate.

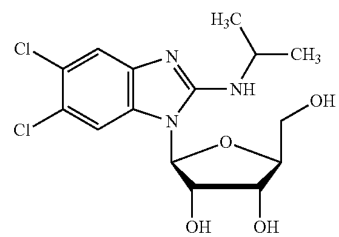
FIG. **2**. Observed enzyme inactivation rate constant vs. inhibitor concentration for TDI of CYP3A by maribavir with (A) midazolam or (B) testosterone as probe substrate. CYP, cytochrome P450; K_I, inhibitor concentration at half maximal enzyme inactivation; k_{inact}, maximum enzyme inactivation rate constant; k_{obs}, observed enzyme inactivation rate constant; TDI, time-dependent inhibition.

FIG. **3**. Estimation of induction E_{max} and EC_{50} of CYP3A4 mRNA expression by maribavir through non-linear regression of data obtained from three donors of human hepatocytes. For donor 3, n is the sigmoidicity coefficient that accommodates the shape of the curve; EC_{50}, half maximal effective concentration; E_{max}, maximum effect.

FIG. **4**. Estimation of IC_{50} of maribavir on P-gp efflux of digoxin (as determined by the corrected efflux ratio) in cultured Caco-2 cells. IC_{50}, half maximal inhibitory concentration; P-gp, P-glycoprotein.

DETAILED DESCRIPTION

Maribavir ((2S,3S,4R,5S)-2-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)-5-(hydroxymethyl)tetrahydrofuran=3,4-diol), a compound having the chemical structure:



is a potent and orally bioavailable antiviral for the treatment of cytomegalovirus (CMV) infection and disease in transplant recipients. Transplant recipients are at significant risk of CMV infection and also often receive numerous concomitant medications to manage their comorbidities. Thus, evaluating the drug-drug interaction potential between maribavir and prospective therapies is useful. Further, understanding the clinical pharmacology of maribavir is necessary to define the optimal dosing strategy in transplant recipients who often have multiple comorbidities and require complex concurrent medication regimens.

1. Definitions

As used herein, the term "about," when used in reference to a numerical value, means±10% of that value. For example, a dose that comprises "about 100 mg" of maribavir encompasses any amount of maribavir within a range of 90 mg to 110 mg.

As used herein, the term "reference" describes a standard or control relative to which a comparison is performed. For example, in some embodiments, an agent, animal, individual, population, sample, sequence or value of interest is compared with a reference or control agent, animal, individual, population, sample, sequence or value. In some embodiments, a reference or control is tested and/or determined substantially simultaneously with the testing or determination of interest. In some embodiments, a reference or control is a historical reference or control, optionally embodied in a tangible medium. Typically, as would be understood by those skilled in the art, a reference or control is determined or characterized under comparable conditions or circumstances to those under assessment. Those skilled in the art will appreciate when sufficient similarities are present to justify reliance on and/or comparison to a particular possible reference or control.

The terms "treat" or "treating," as used herein, refers to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition. As used herein, the terms "treatment," "treat," and "treating" refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed. In some embodiments, the term "treating" includes halting the progression of a disease or disorder. Treatment may also be continued after symptoms have resolved, for example to prevent or delay their recurrence. Thus, in some embodiments, the term "treating" includes preventing relapse or recurrence of a disease or disorder.

2. Dose Regimens

In some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some embodiments, maribavir is administered to patients at about 400 mg orally, twice daily ("standard dose"). In some embodiments, a patient has received an initial therapeutically effective amount of maribavir, e.g., prior to receiving the additional co-administered drug. In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, maribavir is administered as a tablet. In some embodiments, maribavir is administered as a tablet comprising 200 mg of maribavir. In some embodiments, provided methods comprise administering maribavir to a patient with or without food.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir is altered (e.g., increased or decreased) from the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of the co-administered drug is altered (e.g., increased or decreased) from its recommended amount provided by its label. In some embodiments, the dose of the co-administered drug is decreased as compared to its standard dosing regimen. In some embodiments, provided methods further comprising the step of monitoring the levels of a coadministered drug after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of a coadministered drug administered to the patient (e.g., to the amount of a coadministered drug that was administered prior to commencing administration of maribavir).

a. Maribavir and CYP3A4 Inducers

In some embodiments, the present disclosure provides the recognition that maribavir is primarily metabolized by CYP3A4. Additionally or alternatively, the present disclosure provides the recognition that drugs that are CYP3A4 inducers decrease maribavir plasma concentrations and may result in a reduced virologic response. In some embodiments, coadministration of maribavir with certain CYP3 A4 inducers is not recommended, and/or dosing regimens of one or both drugs should be altered.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir. In some embodiments, methods further comprise discontinuing administration of the CYP3A4 inducer prior to administration of maribavir. In some embodiments, method further comprise increasing the amount of maribavir administered to the patient.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital. In some embodiments, a CYP3A4 inducer is a strong CYP3A4 inducer. Strong CYP3A4 inducer include, e.g., rifampin, rifabutin, and St. John's wort. In some embodiments, CYP3A4 inducers may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for a given indication).

In some embodiments, a CYP3A4 inducer is also a p-Gp inducer. In some embodiments, a CYP3A4 inducer also decreases maribavir exposure.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and

discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering an initial therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and

increasing the amount of maribavir administered to the patient.

In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is carbamazepine, the amount of maribavir is increased to about 800 mg of maribavir orally twice daily. In some embodiments, wherein the CYP3A4 inducer is phenytoin or phenobarbital, the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient (e.g., 800 mg or about 1200 mg orally, twice daily), wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.

In some embodiments, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir, a therapeutically effective amount of maribavir is about 800 mg or about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is carbamazepine, the amount of maribavir administered is about 800 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is phenytoin or phenobarbital, the amount of maribavir administered is about 1200 mg orally, twice daily.

Drug-drug interaction potential of maribavir with cytochrome P450 enzymes and P-glycoprotein (P-gp) was thoroughly characterized. The reversible inhibition, time-dependent inhibition, and induction of CYPs were evaluated in the presence of maribavir using human cells or human-derived cell lines. The inhibition of P-gp was also assessed.

Maribavir was not a reversible inhibitor of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, or CYP3A4 but was a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 (the half maximal inhibitory concentration, or IC_{50}, was 40, 18, and 35 μM, respectively).

Maribavir was not a time-dependent inhibitor of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. However, it was a time-dependent inhibitor of CYP3A4, evident from IC_{50} shift values presented in FIG. **1**. With midazolam and testosterone as probe substrates, IC_{50} values for maribavir were greater than 1.9 and 3.2 μM, respectively. Maribavir's concentration at half-maximal enzyme inactivation, or K_I, of CYP3A was 41.2 μM and 167 μM with midazolam and testosterone, respectively, as shown in FIG. **2**.

Maribavir was not an inducer of CYP1A2 or CYP2B6 mRNA, but was a weak in vitro inducer for CYP3A4 mRNA, exhibiting a greater than 14-fold increase in mRNA induction and a greater than 30% increase in positive control levels. However, effects varied across donors, as FIG. **3** shows. The half-maximal effective concentration, or EC_{50}, of maribavir was greater than 5 μM for all donors.

As shown in FIG. **4**, maribavir demonstrated concentration-dependent inhibition of P-gp-mediated digoxin efflux. The IC_{50} of maribavir in this case was 33.8 μM.

The data presented herein, specifically in Example 1, demonstrate the concentrations of maribavir required for the observed levels of in vitro inhibition or induction of P-gp and/or some CYPs.

b. Maribavir and Immunosuppressants

In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of certain immunosuppressant drugs that are CYP3A4 and/or P-gp substrates, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, immunosuppressant drug levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of maribavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received an immunosuppressant. In some embodiments, methods further comprise monitoring the levels of immunosuppressant (e.g., in whole blood or plasma). In some embodiments, methods further comprise reducing the amount of immunosuppressant administered to a patient. In some embodiments, maribavir also increases immunosuppressant exposure. In some embodiments, provided methods further comprise the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprise the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is everolimus. In some embodiments, an immunosuppressant is cyclosporine. In some embodiments, an immunosuppressant is sirolimus.

In some embodiments, immunosuppressants may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for tacrolimus). In some embodiments, tacrolimus is administered as a stable dose, twice daily, with a total daily dose between about 0.5 mg and about 16 mg.

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from the initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL. In some embodiments, an initial dose of tacrolimus administered is between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some embodiments, the present disclosure provides methods of administering to a patient in need thereof a therapeutically effective amount of an immunosuppressant for the prophylaxis or treatment of organ rejection and/or graft versus host disease, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of an immunosuppressant, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of maribavir, the improvement comprising administering a therapeutically effective amount of an immunosuppressant to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

c. Maribavir and Digoxin

Digoxin is an antiarrhythmic drug most frequently used for atrial fibrillation, atrial flutter, and heart failure. In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of digoxin, which is a P-gp substrate, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, digoxin levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of mariabavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin. In some embodiments, provided methods further comprise monitoring the levels of digoxin (e.g., in whole blood or plasma). In some embodiments, provided methods further comprise reducing the amount of digoxin administered to a patient, e.g., as compared to the amount prior to co-administering maribavir, such as the US FDA approved dosage. In some embodiments, digoxin is administered as 0.5 mg, once daily.

d. Maribavir and Ganciclovir or Valganciclovir

In some embodiments, the present disclosure recognizes that maribavir may antagonize the antiviral activity of ganciclovir and/or valganciclovir by inhibiting human CMV pUL97 kinase, which is required for activation/phosphorylation of ganciclovir and valganciclovir.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir and/or valganciclovir. In some embodiments, provided methods further comprise discontinuing administration of ganciclovir and/or valganciclovir prior to administration of maribavir. In some embodiments, provided methods further comprising the step of monitoring the levels of ganciclovir and/or valganciclovir (e.g., in whole blood or plasma) after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of ganciclovir and/or valganciclovir administered to the patient (e.g., to the amount of ganciclovir and/or valganciclovir that was administered prior to commencing administration of maribavir).

3. Patients

As described above and herein, the present methods provide administering maribavir and a co-administered drug (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) to a patient in need thereof. In some embodiments, a patient suffers from CMV infection. In some embodiments, a patient suffers from post-transplant CMV infection. In some embodiments, a patient is a transplant recipient. In some embodiments, a patient is a hematopoietic stem cell transplant recipient. In some embodiments, a patient is a solid organ transplant recipient (e.g., liver, kidney, lung, heart, pancreas, intestine).

In some embodiments, a patient is refractory to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory with genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory without genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet. In some embodiments, a patient is refractory to treatment with ganciclovir or valganciclovir. In some embodiments, a patient is refractory to treatment with ganciclovir. In some embodiments, a patient is refractory to treatment with valganciclovir.

In some embodiments, a patient is an adult or a child over 12 years old and greater than 35 kg. In some embodiments, a patient is an adult. In some embodiments, a patient is a child. In some embodiments, a patient is a child over 12 years old. In some embodiments, a patient is a child over 35 kg. In some embodiments, a patient is a child over 12 years old and over 35 kg.

4. Evaluating Maribavir Drug-Drug Interaction Potential Using Data from Phase 1 Clinical Studies and Nonclinical In Vitro Studies

Maribavir's drug-drug interaction potential are described in Example 3.

Accumulated data have shown that maribavir is metabolized primarily in the liver, and renal clearance is a minor route (representing less than 5 percent). VP 44469 (N-dealkylated Maribavir) was the principal metabolite of maribavir in both urine and feces. In plasma, unchanged maribavir and VP 44469 represented approximately 69% and 9.8% of total radioactivity, respectively. The hepatic metabolism of maribavir was primarily driven by cytochrome P450s. CYP3A4 and CYP1A2 were responsible for 70 to 85% and 15 to 30% of the CYP-driven pathways, respectively. Glucuronidation accounted for less than 20% of metabolism.

Clinically significant drug interactions are summarized in Table 1-A of Example 3. Potent inducers of CYP34A and P-gp decrease maribavir exposure, necessitating maribavir dose increases. Inhibitors of CYP3A4 and/or P-gp may increase maribavir exposure. However, previous safety and tolerability data indicate that dose reduction is not necessary with CYP3A4 and/or P-gp inhibitors. Maribavir may increase the exposure of immunosuppressants, so monitoring of concomitant immunosuppressants should be considered.

5. Clinical Pharmacology of Maribavir

Following oral administration, maribavir was rapidly and well absorbed, with peak concentrations generally achieved within 1 to 3 hours; exposure was unaffected by food; and bioavailability wasn't affected by crushing the tablet or when it was taken with antacids.

Maribavir was metabolized in the liver through the cytochrome P450 3A4 and 1A2 pathways. Less than 5 percent of maribavir was eliminated through the kidneys. Maribavir demonstrated a half-life of approximately 5 to 7 hours.

Maribavir was found to present a low risk for drug-to-drug interactions. Co-administration of potent CYP3A4 inducers was found to decrease the exposure of maribavir, necessitating a maribavir dose increase. Some immunosuppressants, such as tacrolimus, may be affected by maribavir, which increased tacrolimus exposure by 51%.

In conclusion, maribavir is a suitable treatment for CMV infections in a broad range of transplant recipients. It can be taken with or without food, and no dose adjustment is needed in patients with mild-to-moderate hepatic impairment or renal impairment. Maribavir has a low drug-interaction potential and there are minimal dose adjustments required (for example, only when taken concurrently with CYP3A4 inducers or certain immunosuppressants) and there are no effects on QT interval.

Exemplary Embodiments

The following numbered embodiments, while non-limiting, are exemplary of certain aspects of the present disclosure:

1. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and
   discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

2. The method of embodiment 1, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

3. The method of embodiments 1 or 2, wherein the CYP3A4 inducer is a strong CYP3A4 inducer.

4. The method of any one of embodiments 1-3, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort.

5. The method of any one of embodiments 1-4, comprising administering about 400 mg of maribavir orally to the patient twice daily.

6. The method of any one of embodiments 1-5, comprising administering maribavir to the patient with or without food.

7. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and
   increasing the amount of maribavir administered to the patient.

8. The method of embodiment 7, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

9. The method of embodiment 7 or 8, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

10. The method of any one of embodiments 7-9, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.

11. The method of any one of embodiments 7-10, wherein the patient had received a therapeutically effective amount of maribavir prior to receiving a CYP3A4 inducer ("an initial therapeutically effective amount") of about 400 mg of maribavir orally twice daily.

12. The method of any one of embodiments 7-11, wherein the amount of maribavir is increased to about 800 or about 1200 mg of maribavir orally twice daily.

13. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir is increased to about 800 mg of maribavir orally twice daily.

14. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.

15. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.

16. The method of embodiment 15, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

17. The method of embodiment 15 or 16, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

18. The method of any one of embodiments 15-17, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.
19. The method of any one of embodiments 15-18, wherein the amount of maribavir administered is about 800 or about 1200 mg orally twice daily.
20. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir administered is about 800 mg orally twice daily.
21. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir administered is about 1200 mg orally twice daily.
22. The method of any one of embodiments 1-21, wherein the CYP3A4 inducer decreases maribavir exposure.
23. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant.
24. The method of embodiment 23, further comprising the step of monitoring the levels of immunosuppressant after commencing administration of Maribavir (e.g., with comparison to a reference or standard level).
25. The method of embodiment 23 or 24, further comprising the step of reducing the amount of the immunosuppressant administered to the patient.
26. The method of any one of embodiments 23-25, further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level).
27. The method of embodiment 26, further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).
28. The method of any one of embodiments 23-27, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate.
29. The method of any one of embodiments 23-28, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus.
30. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin.
31. The method of embodiment 30, further comprising the step of monitoring the levels of digoxin.
32. The method of embodiment 30 or 31, further comprising the step of reducing the amount of the digoxin administered to the patient.
33. The method of one of embodiments 1-32, wherein the patient is refractory to treatment with one or more other drugs to treat CMV infection.
34. The method of one of embodiments 1-33, wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.
35. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment with genotypic resistance.
36. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment without genotypic resistance.
37. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and
   discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.
38. The method of any one of embodiments 7-37, comprising administering maribavir to the patient with or without food.
39. The method of any one of embodiments 1-38, wherein the patient is a transplant recipient.
40. The method of any one of embodiments 1-39, wherein the patient is a hematopoietic stem cell transplant recipient.
41. The method of any one of embodiments 1-39, wherein the patient is a solid organ transplant recipient.
42. The method of any one of embodiments 1-41, wherein the patient is an adult or a child over 12 years old and greater than 35 kg.

EXAMPLES

Example 1—In Vitro Profiling for Potential Cytochrome P450 Drug-Drug Interaction by Maribavir

Inhibition or induction of the cytochrome P450 enzymes (CYPs) are among the most commonly observed mechanisms of drug-drug interactions.

In vitro systems such as human liver microsomes (HLM), recombinant enzymes, human hepatocytes, and other human-derived cells lines have long been used as validation systems for characterization of potential drug-drug interactions prior to clinical studies.

Reversible CYP Inhibition

HLM were used to evaluate maribavir's half maximal inhibitory concentration (IC_{50}) on inhibition of activities of nine CYP isoforms. Probe substrates were phenacetin (CYP1A2), coumarin (CYP2A6), bupropion (CYP2B6), paclitaxel (CYP2C8), tolbutamide (CYP2C9), (S)-mephenytoin (CYP2C19), dextromethorphan (CYP2D6), chlorzoxazone (CYP2E1), and midazolam and testosterone (CYP3A4).

CYP activities with 0, 0.1, 0.3, 1, 3, 10, 30, and 100 μM of maribavir were evaluated.

Incubation of maribavir in HLM at up to 100 μM resulted in no significant inhibition of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, and CYP3A4. Maribavir is a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 with IC_{50} of 40, 18, and 35 μM, respectively.

Time-Dependent CYP Inhibition

Time-dependent inhibition (TDI) potentials for various CYP enzymes were evaluated by a 30 minute pre-incubation of maribavir with HLM in the presence and absence of nicotinamide-adenine dinucleotide diphosphate (NADPH) followed by a CYP enzyme activity assay.

Enzyme inactivation kinetics of maribavir for the TDI of CYP3A, with midazolam and testosterone as probe substrates, was evaluated by pre-incubation of HLM with maribavir at various concentrations with NADPH for six different pre-incubation time frames. CYP3A activity was measured by determining the formation of CYP3A probe metabolite. Inhibitor concentration at half maximal enzyme inactivation (K_I), and maximum enzyme inactivation rate constant (K_{inact}), were estimated using non-linear least-square regression.

The IC_{50} shift values of maribavir were <1 μM for CYP1A2, CYP2C8, CYP2C9, CYP2C19, and CYP2D6 in

HLM. The IC_{50} shift values of maribavir for CYP3A were greater than 1.9 and 3.2 μM, with midazolam and testosterone as probe substrate, respectively (FIG. **1**). Therefore, maribavir, in concentrations of up to 100 μM, is unlikely a TDI of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6; but it is likely a TDI of CYP3A.

K_I and K_{intact} of maribavir for TDI of CYP3A with midazolam as substrate (FIG. **2**A) were 41.2 μM and 0.0117 min=1, respectively. K_I and K_{intact} of maribavir for TDI of CYP3A with testosterone as a substrate (FIG. **2**B) were 167 μM and 0.0357 min=1, respectively.

CYP Induction

Fresh human hepatocytes were treated with culture medium with maribavir concentrations of 36 μM, 144 μM, and 480 μM.

Positive controls were 50 μM omeprazole (CYP1A2), 1 mM phenobarbital (CYP2B6), and 50 μM rifampicin (CYP3A).

Incubation was conducted at 37° C. for 72 hours, with daily replacement of medium with compound. Total RNA was isolated, and cDNA was synthesized from up to 1 μg of total isolated RNA. Analysis of CYP expression was performed using qPCR with CYP-specific probes.

Half maximal effective concentration (EC_{50}) and maximum effect (E_{max}) values for CYP3A4 mRNA induction were further determined by a concentration-response curve in three donors with maribavir at a concentration between 0-100 μM.

Maribavir displayed varying degrees of CYP1A2 and CYP2B6 mRNA induction in human hepatocytes. The fold-increase in mRNA was not concentration-dependent, it was variable among donors, and it did not exceed 11% of the levels of positive controls. Therefore, maribavir is likely not an inducer of CYP1A2 or of CYP2B6 mRNA at clinically relevant concentrations.

At 36 μM, maribavir displayed a greater than 14-fold increase in CYP3A4 mRNA induction and greater than 30% of positive control (rifampicin) levels. The EC_{50} and E_{max} values for CYP3A4 induction were then further determined by a concentration-response curve in three donors.

Maribavir also displayed donor-dependent induction of CYP3A4 mRNA upon determination of its induction E_{max} and EC_{50} properties (FIG. **3**).

Example 2—Quantitative Prediction of Exposure of Maribavir Using Prior In Vitro and In Vivo Data

A Physiologically-based Pharmacokinetic ("PBPK") model based on prior in vitro and in vivo information on the metabolism and kinetics of maribavir was constructed with the aim of predicting plasma concentration-time profiles of maribavir and to evaluate the likely impact of coadministration of CYP3A4 inhibitors and inducers on the kinetics of maribavir in healthy subjects.

Model Development

A combination of in vitro data and clinical pharmacokinetic data obtained following a single dose of 400 mg maribavir were used to develop the PBPK model. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 2A. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data. The C_{max} was marginally under predicted, which could be expected because the PBPK model was optimized for better prediction of C_{12h}.

TABLE 2A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (400 mg).

	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L
PREDICTED		
Mean	97.98	11.98
Median	91.97	11.48
Geometric Mean	90.40	11.56
90% CI around geometric mean (lower limit)	84.45	11.06
90% CI around geometric mean (upper limit)	96.76	12.08
5^{th} percentile	46.52	7.61
95^{th} percentile	165.11	18.08
SD	39.33	3.26
% CV	40	27
OBSERVED		
Mean	97.8	16.7
SD	28.6	5.72
% CV	29.2	34.3
RATIO MEAN PREDICTED/OBSERVED	1.00	0.72

A renal clearance of 0.051 L/H was obtained following oral administration of 50 mg to 1600 mg single doses of maribavir. In vitro human liver microsomal and recombinant CYP data combined with oral clearance values reported from two studies (n=46 individuals; Ma et al., 2006) were used to assign the relative contribution of CYP3A4 to the clearance of maribavir. Distribution models evaluated included a full PBPK model and a minimal PBPK model, both of which consider liver and intestinal metabolism. Mass balance data from the [^{14}C] ADME study indicated a fraction absorbed of 0.83 or greater following oral administration of 400 mg dose of maribavir. Absorption models evaluated included a simple first-order absorption model and a more mechanistic "Advanced Dissolution Absorption and Metabolism" (ADAM) model. Although maribavir has been shown to be a P-gp substrate in vitro, relatively linear pharmacokinetics across the dose range of interest, (400 mg to 1600 mg) informed the selection of the simpler first order absorption model for the use in this PBPK study.

Model Verification

A comparison of observed and predicted plasma concentrations of maribavir following single oral doses of 800 mg and 1600 mg to healthy patients were. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Tables 3A and 4A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 3A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (800 mg).

	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L
PREDICTED		
Mean	195.71	24.05
Median	179.48	23.22
Geometric Mean	180.00	23.23
90% CI around geometric mean (lower limit)	167.95	22.23
90% CI around geometric mean (upper limit)	192.91	24.27
5^{th} percentile	93.03	15.22
95^{th} percentile	330.22	36.17

TABLE 3A-continued

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (800 mg).

	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L
SD	80.04	6.47
% CV	41	27
OBSERVED		
Mean	183	26.4
SD	69.1	6.85
% CV	38.0	26.0
RATIO MEAN PREDICTED/OBSERVED	1.07	0.91

TABLE 4A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (1600 mg).

	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L
PREDICTED		
Mean	391.43	48.10
Median	358.95	46.45
Geometric Mean	359.99	46.45
90% CI around geometric mean (lower limit)	335.89	44.46
90% CI around geometric mean (upper limit)	385.82	48.54
5^{th} percentile	186.06	30.44
95^{th} percentile	660.44	72.34
SD	106.09	12.94
% CV	41	27
OBSERVED		
Mean	437	48.8
SD	163	7.88
% CV	37.4	16.1
RATIO MEAN PREDICTED/OBSERVED	0.90	0.99

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral 400 mg BID doses of maribavir was performed. Mean concentrations for the total virtual population (n=150) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 5A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 5A

Predicted and Observed C_{12h}, C_{max} and $AUC_{(0-\infty)}$ values for maribavir after multiple oral dose of maribavir (400 mg BBID, 5 doses).

	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L
PREDICTED			
Mean	96.49	14.31	2.99
Median	90.13	14.05	2.44
Geometric Mean	89.31	13.78	2.13
90% CI around geometric mean (lower limit)	85.01	13.31	1.89
90% CI around geometric mean (upper limit)	93.82	14.26	2.15
5^{th} percentile	46.35	8.80	0.36
95^{th} percentile	162.88	20.59	7.76
SD	37.93	3.96	2.2
% CV	39	28	73
OBSERVED			
Mean	89.9	16.9	2.54
SD	24.5	5.22	1.37
% CV	27.2	30.8	53.8
RATIO MEAN PREDICTED/OBSERVED	1.07	0.85	1.18

A comparison of observed and predicted plasma concentrations of maribavir following a single oral dose of 400 mg administered in the absence of ketoconazole (a CYP3A4 inhibitor) and 1 hour after a single 400 mg dose of ketoconazole to healthy subjects was performed. Mean simulated and observed plasma maribavir concentrations are compared, and associated geometric mean C_{max} and $AUC_{(0-\infty)}$ values for maribavir in the presence and absence of ketoconazole are shown in Table 6A and are within 1.25-fold of the observed data.

TABLE 6A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L	$AUC_{(0-\infty)}$ mg/L * h	C_{max} mg/L	$AUC_{(0-\infty)}$	C_{max}
PREDICTED						
Mean	98.96	12.27	150.60	14.31	1.54	1.17
Median	93.51	11.91	140.38	13.75	1.49	1.15
Geometric Mean	92.29	11.81	141.16	13.77	1.53	1.17
90% CI around geometric mean (lower limit)	88.30	11.43	135.28	13.32	1.51	1.16
90% CI around geometric mean (upper limit)	96.45	12.20	147.30	14.23	1.55	1.17
5^{th} percentile	48.52	7.50	77.98	8.85	1.27	1.09
95^{th} percentile	169.37	17.86	246.30	21.20	1.97	1.29
SD	37.19	3.4	54.7	4.0	0.21	0.06
% CV	38	28	36	28	14	5

TABLE 6A-continued

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$	C_{max}	$AUC_{(0-\infty)}$	C_{max}		
	mg/L * h	mg/L	mg/L * h	mg/L	$AUC_{(0-\infty)}$	C_{max}
OBSERVED						
Mean	126	19.8	194	21.8	1.55	1.12
SD	41.6	3.9	64.3	4.4	0.23	0.22
% CV	33	20	33	20	15	20
Geometric mean	119	19.4	183	21.3	1.53	1.10
90% CI around geometric mean (lower limit)					1.44	1.01
90% CI around geometric mean (upper limit)					1.63	1.19
MEAN RATIO PREDICTED/OBSERVED	0.79	0.62	0.78	0.66	0.99	1.05
GEOMEAN RATIO PREDICTED/OBSERVED	0.78	0.61	0.77	0.65	1.00	1.06

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral doses of 400 mg administered in the absence of rifampicin and during concurrent treatment with rifampicin was performed. Subjects took maribavir 400 mg BID on Days 1-3 (5 doses), then rifampicin 600 mg QD on Days 4 through 12, followed by both maribavir 400 mg (BID) and rifampicin 600 mg QD on Days 13 and 14, with final doses of maribavir and rifampicin on the morning of Day 15. Associated geometric mean C_{12h}, C_{max}, and $AUC_{(0-12h)}$ values for maribavir in the presence or absence of rifampicin are shown in Table 7A. Predicted values for C_{max} and AUC are within 1.25-fold of the observed parameter values.

TABLE 7A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-12h)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of rifampicin with associated variability in healthy adults.

	Maribavir 400 mg BID			400 mg BID Maribavir + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	95.28	14.26	2.88	35.13	7.97	0.340	0.37	0.56	0.09
Median	90.02	13.98	2.61	30.52	7.57	0.145	0.35	0.55	0.06
Geometric Mean	88.77	13.78	2.07	30.53	7.44	0.088	0.34	0.54	0.04
90% CI around geometric mean (lower limit)	84.25	13.30	2.03	28.36	7.07	0.083	0.33	0.52	0.04
90% CI around geometric mean (upper limit)	93.54	14.29	2.10	32.86	7.83	0.093	0.36	0.56	0.04
5^{th} percentile	41.15	8.76	0.39	12.44	3.95	0.002	0.17	0.33	0.01
95^{th} percentile	151.23	20.23	6.49	67.37	13.43	1.507	0.62	0.79	0.33
SD	35.28	3.7	2.1	18.64	2.93	0.53	0.14	0.14	0.11
% CV	37	26	73	53	37	151	37	25	112
OBSERVED									
Mean	89.9	16.9	2.54	35.1	10.2	0.41	0.41	0.65	0.17
SD	24.5	5.22	1.37	7.07	2.3	0.21	0.08	0.24	0.07
% CV	86.6	16.2	2.24	34.5	9.88	0.41	0.40	0.61	0.18
RATIO MEAN PREDICTED/OBSERVED	1.06	0.84	1.14	1.00	0.78	0.83	0.91	0.87	0.56
RATIO GEOMETRIC MEAN PREDICTED/OBSERVED	1.03	0.85	0.92	0.88	0.75	0.22	0.85	0.89	0.24

Model Verification

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 8A. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for this interaction is compared with the PK parameters for a 400 mg BID dose of maribavir in Table 9A.

Maribavir C_{max} from 800 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone, however, $AUC_{(0-12)}$ was 23% lower and C_{12h} was 75% lower. This indicates that increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of rifampicin on the reduced therapeutic exposure to maribavir.

TABLE 8A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	194.99	28.96	5.95	73.25	16.40	0.73	0.37	0.56	0.10
Median	189.24	28.37	5.34	68.57	15.57	0.31	0.37	0.56	0.06
Geometric Mean	182.49	27.95	4.39	63.52	15.22	0.20	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	174.67	27.09	4.36	59.54	14.54	0.18	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	190.66	28.85	4.43	67.77	15.94	0.21	0.36	0.56	0.05
5^{th} percentile	94.23	17.97	0.93	24.65	7.93	0.00	0.18	0.35	0.00
95^{th} percentile	311.05	43.06	13.85	136.31	27.80	2.82	0.62	0.79	0.34
SD	68.94	7.68	4.00	38.65	6.25	1.09	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 9A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	95.28	14.26	2.88	73.25	16.40	0.73	0.77	1.15	0.25
Median	90.02	13.98	2.61	68.57	15.57	0.31	0.76	1.11	0.12
Geometric Mean	88.77	13.78	2.07	63.52	15.22	0.20	0.72	1.10	0.10
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	59.54	14.54	0.18	0.71	1.09	0.09
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	67.77	15.94	0.21	0.72	1.12	0.10
5^{th} percentile	41.15	8.76	0.39	24.65	7.93	0.00	0.60	0.91	0.00
95^{th} percentile	151.23	20.34	6.49	136.31	27.80	2.82	0.90	1.37	0.43
SD	35.28	3.7	2.1	38.65	6.25	1.09	1.10	1.69	0.52
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 10A. Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 11A.

Maribavir AUC from 1200 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone while the C_{max} was slightly higher. However, the geometric mean C_{12h} was 86% lower, indicating that increasing maribavir dose from 400 mg BID to 1200 mg BID cannot counteract the effect of rifampicin from an efficacy perspective (mainly driven by C_{12h}). Increasing the maribavir dose to 1600 mg BID had a minimal effect on the geometric mean C_{12h}, which was 81% lower than that with a 400 mg BID dose of maribavir.

TABLE 10A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	292.48	43.44	8.93	109.87	24.59	1.10	0.37	0.56	0.10
Median	283.87	42.55	8.00	102.85	23.35	0.47	0.37	0.56	0.06
Geometric Mean	273.73	41.93	6.59	95.28	22.84	0.30	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	262.00	40.63	6.55	89.31	21.81	0.28	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	285.99	43.27	6.62	101.65	23.91	0.31	0.36	0.56	0.05
5^{th} percentile	141.35	26.96	1.40	36.97	11.90	0.00	0.18	0.35	0.00
95^{th} percentile	466.58	64.59	20.77	204.46	41.71	4.23	0.62	0.79	0.34
SD	108.42	11.45	6.00	57.57	9.37	1.63	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 11A

Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	95.28	14.26	2.89	109.87	24.59	1.10	1.15	1.72	0.38
Median	90.02	13.98	2.61	102.85	23.35	0.47	1.14	1.67	0.18
Geometric Mean	88.77	13.78	2.07	95.28	22.84	0.30	1.07	1.66	0.14
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	89.31	21.81	0.28	1.06	1.64	0.14
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	101.65	23.91	0.31	1.09	1.67	0.15
5^{th} percentile	41.15	8.76	0.39	36.97	11.90	0.00	0.90	1.36	0.00
95^{th} percentile	151.23	20.34	6.49	204.46	41.71	4.23	1.35	2.05	0.65
SD	35.28	3.7	2.1	57.57	9.37	1.63	1.63	2.53	0.78
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 12A. Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 39%, 27% and 63% in $AUC_{(0-12h)}$, C_{max} and C_{12h}, respectively.

absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h} for the population are shown in Table 13A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 14A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 100 mg phenobarbital QD were marginally lower than those from maribavir 800 mg BID alone. However, C_{min} was 63% lower than with maribavir 800 mg alone.

TABLE 12A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	61.20	10.66	1.33	0.61	0.73	0.37
Median	93.26	14.23	2.40	55.41	10.44	0.85	0.63	0.75	0.37
Geometric Mean	92.65	14.09	2.29	55.66	10.21	0.72	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	52.88	9.86	0.70	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	58.60	10.57	0.73	0.62	0.74	0.31
5^{th} percentile	47.52	9.12	0.50	27.75	6.18	0.07	0.39	0.56	0.07
95^{th} percentile	169.70	21.94	7.30	112.27	16.47	4.17	0.78	0.86	0.65
SD	38.6	4.1	2.26	26.96	3.12	1.25	0.12	0.09	0.18
% CV	38	28	72	44	29	101	20	13	47

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the

Increasing the maribavir dose given with phenobarbital from 400 mg BID to 800 mg BID results in a mean C_{min} that is 15% lower than that with 400 mg BID maribavir alone. Thus, by increasing the dose of maribavir to 800 mg BID, the interaction is counteracted significantly.

TABLE 13A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	199.43	29.32	6.24	122.40	21.32	2.66	0.61	0.73	0.37
Median	186.51	28.46	4.81	110.83	20.88	1.70	0.63	0.75	0.37
Geometric Mean	185.31	28.19	4.58	111.33	20.42	1.43	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	105.75	19.72	1.41	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	117.20	21.14	1.45	0.62	0.74	0.31
5^{th} percentile	95.04	18.24	1.00	55.49	12.35	0.14	0.39	0.56	0.07
95^{th} percentile	339.40	43.89	14.60	224.54	32.93	8.33	0.78	0.86	0.65

TABLE 13A-continued

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
SD	36.3	8.2	4.53	53.9	6.24	2.7	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	48

TABLE 14A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	122.40	21.32	2.66	1.23	1.45	0.85
Median	93.26	14.23	2.40	110.83	20.88	1.70	1.19	1.47	0.71
Geometric Mean	92.65	14.09	2.29	111.33	20.42	1.43	1.20	1.45	0.62
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	105.75	19.72	1.41	1.19	1.45	0.62
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	117.20	21.14	1.45	1.21	1.45	0.63
5^{th} percentile	47.52	9.12	0.50	55.49	12.35	0.14	1.17	1.35	0.28
95^{th} percentile	169.70	21.94	7.30	224.54	32.93	8.33	1.32	1.50	1.14
SD	38.6	4.1	2.26	53.9	6.24	2.7	1.40	1.52	1.19
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 15A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 16A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 100 mg phenobarbital QD were about 2 times higher those from maribavir 400 mg BID alone while C_{min} is about 28% higher. Increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenobarbital. If the higher exposure (as seen with AUC and C_{max}) is not detrimental to treatment, it can be concluded that the maribavir dose should be adjusted from 400 mg BID to 1200 mg BID when co-administration with phenobarbital is needed.

TABLE 15A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	299.15	43.98	9.36	183.60	31.98	3.99	0.61	0.73	0.37
Median	279.77	42.69	7.21	166.24	31.31	2.55	0.63	0.75	0.37

TABLE 15A-continued

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
Geometric Mean	277.96	42.28	6.88	166.99	30.63	2.15	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	158.62	29.58	2.11	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	175.80	31.71	2.18	0.62	0.74	0.31
5^{th} percentile	142.56	27.35	1.50	83.24	18.53	0.21	0.39	0.56	0.07
95^{th} percentile	509.10	65.83	21.89	336.81	49.40	12.50	0.78	0.86	0.65
SD	114.5	12.3	6.8	80.9	9.36	4.04	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	47.97

TABLE 16A

Predicted PK parameters for marvibavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	183.60	31.98	3.99	1.84	2.18	1.28
Median	93.26	14.23	2.40	166.24	31.31	2.55	1.78	2.20	1.06
Geometric Mean	92.65	14.09	2.29	166.99	30.63	2.15	1.80	2.17	0.94
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	158.62	29.58	2.11	1.79	2.17	0.93
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	175.80	31.71	2.18	1.81	2.18	0.94
5^{th} percentile	47.52	9.12	0.50	83.24	18.53	0.21	1.75	2.03	0.42
95^{th} percentile	169.70	21.94	7.30	336.81	49.40	12.50	1.98	2.25	1.71
SD	38.6	4.1	2.26	80.9	9.36	4.04	2.10	2.28	1.79
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 17A.

Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 42%, 31% and 64% in $AUC_{(0-12)}$, C_{max} and C_{12h}, respectively.

TABLE 17A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	56.81	10.04	1.16	0.58	0.69	0.36
Median	93.26	14.23	2.40	56.31	9.95	0.81	0.58	0.69	0.35
Geometric Mean	92.65	14.09	2.29	52.42	9.63	0.68	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	49.99	9.30	0.67	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	54.98	9.96	0.69	0.58	0.70	0.30
5^{th} percentile	47.52	9.12	0.50	26.75	5.59	0.09	0.35	0.50	0.09
95^{th} percentile	169.70	21.94	7.30	99.79	14.78	3.33	0.82	0.88	0.68
SD	38.6	4.1	2.26	23.0	2.91	1.1	0.14	0.11	0.18
% CV	38	28	72	40	29	95	24	17	51.07

Simulation of Plasma Concertation-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 18A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 19A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 300 mg phenytoin QD was marginally higher than those from maribavir 400 mg BID alone, however, C_{min} was 26% lower. Increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}).

TABLE 18A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	199.43	29.32	6.24	113.63	20.08	2.32	0.58	0.69	0.36
Median	186.51	28.46	4.81	112.62	19.90	1.62	0.58	0.69	0.35
Geometric Mean	185.31	28.19	4.58	104.85	19.26	1.36	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	99.97	18.61	1.34	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	109.96	19.93	1.38	0.58	0.70	0.30
5^{th} percentile	95.04	18.24	1.00	53.51	11.18	0.18	0.35	0.50	0.09
95^{th} percentile	339.40	43.89	14.60	199.59	29.57	6.65	0.82	0.88	0.68
SD	76.33	8.2	4.53	46.1	5.83	2.2	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	51

TABLE 19A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	113.63	20.08	2.32	1.14	1.37	0.74
Median	93.26	14.23	2.40	112.62	19.90	1.62	1.21	1.40	0.68
Geometric Mean	92.65	14.09	2.29	104.85	19.26	1.36	1.13	1.37	0.59
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	99.97	18.61	1.34	1.13	1.37	0.59
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	109.96	19.93	1.38	1.13	1.37	0.60
5^{th} percentile	47.52	9.12	0.50	53.51	11.18	0.18	1.13	1.23	0.36
95^{th} percentile	169.70	21.94	7.30	199.59	29.57	6.65	1.18	1.35	0.91
SD	38.6	4.1	2.26	46.1	5.83	2.2	1.21	1.42	0.97
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-24h)}$ and C_{12h}) for the population are shown in Table 20A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 21A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 300 mg phenytoin QD were about 2-fold higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}). The impact of the higher exposure (AUC and C_{max}) will have to be evaluated.

TABLE 20A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	299.15	43.98	9.36	170.44	30.13	3.48	0.58	0.69	0.36
Median	279.77	42.69	7.21	168.93	29.85	2.42	0.58	0.69	0.35
Geometric Mean	277.96	42.28	6.88	157.27	28.88	2.05	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	149.96	27.91	2.02	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	164.94	29.89	2.08	0.58	0.70	0.30
5^{th} percentile	142.56	27.35	1.50	80.26	16.76	0.27	0.35	0.50	0.09
95^{th} percentile	509.10	65.83	21.89	299.38	44.35	9.98	0.82	0.88	0.68
SD	114.5	12.3	6.8	69.01	8.7	3.29	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	52

TABLE 21A

Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	99.72	14.66	3.12	170.44	30.13	3.48	1.71	2.06	1.12
Median	93.26	14.23	2.40	168.93	29.85	2.42	1.81	2.10	1.01
Geometric Mean	92.65	14.09	2.29	157.27	28.88	2.05	1.70	2.05	0.90
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	149.96	27.91	2.02	1.69	2.05	0.89
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	164.94	29.89	2.08	1.70	2.05	0.90
5^{th} percentile	47.52	9.12	0.50	80.26	16.76	0.27	1.69	1.84	0.54
95^{th} percentile	169.70	21.94	7.30	299.38	44.35	9.98	1.76	2.02	1.37
SD	38.2	4.1	2.26	69.01	8.7	3.29	1.81	2.12	1.46
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles for Maribavir 400 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 400 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 22A.

Due to CYP3A4 induction by carbamazepine, a mean decrease of 46% in maribavir C_{12h} (efficacy marker) and a 29% and 23% decrease in AUC and C_{max} respectively, is predicted when 400 mg BID maribavir is combined with carbamazepine.

TABLE 22A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of carbamazepine.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12h)}$	C_{max}	C_{12h}	$AUC_{(0-12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0-12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	98.93	14.60	3.11	67.85	11.12	1.65	0.71	0.77	0.54
Median	88.34	13.91	2.34	63.37	10.80	1.29	0.71	0.79	0.56
Geometric Mean	91.25	14.01	2.18	63.82	10.74	1.15	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	61.25	10.41	1.14	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	66.49	11.08	1.16	0.71	0.78	0.53
5^{th} percentile	48.05	8.74	0.52	36.23	7.02	0.25	0.54	0.61	0.36
95^{th} percentile	176.30	21.99	7.77	108.25	16.29	4.07	0.84	0.90	0.70
SD	40.96	4.23	2.53	24.40	2.99	1.34	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.20
3^{rd} quartile	123.18	17.26	4.22	82.72	12.92	2.34	0.77	0.09	0.61
1^{st} quartile	69.63	11.43	1.20	51.60	8.95	0.63	0.65	0.84	0.46
10^{th} percentile	55.45	9.76	0.75	40.08	7.78	0.42	0.58	0.72	0.39
90^{th} percentile	154.24	20.66	6.58	100.38	15.22	3.34	0.82	0.65	0.67

Simulation of Plasma Concentration-Time Profile Maribavir 800 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 800 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 23A. Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 24A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 400 mg carbamazepine QD were marginally higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID can counteract the effect of carbamazepine from an efficacy perspective (mainly driven by C_{min}).

TABLE 23A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of carbamazepine.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	197.86	29.19	6.21	135.70	22.25	3.30	0.71	0.77	0.54
Median	176.68	27.81	4.68	126.73	21.61	2.58	0.71	0.79	0.56
Geometric Mean	182.50	28.01	4.37	127.63	21.48	2.30	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	174.10	27.08	4.35	122.50	20.82	2.29	0.69	0.76	0.52
90% CI around geometric mean (upper limit)	191.31	28.97	4.39	132.99	22.16	2.31	0.71	0.78	0.53
5^{th} percentile	96.09	17.47	1.03	72.46	14.03	0.50	0.54	0.61	0.36
95^{th} percentile	352.60	43.98	15.54	216.50	32.57	8.13	0.84	0.90	0.70
SD	81.92	8.47	5.05	48.79	5.97	2.67	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3^{rd} quartile	246.37	34.52	8.44	165.43	25.84	4.67	0.77	0.84	0.61
1^{st} quartile	139.25	22.86	2.41	103.20	17.89	1.27	0.65	0.72	0.46
10^{th} percentile	110.89	19.53	1.50	80.17	15.56	0.85	0.58	55.45	0.39
90^{th} percentile	308.48	41.32	13.15	200.76	30.43	6.68	0.82	154.24	0.67

TABLE 24A

Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}			
	mg/L * h	mg/L	mg/L	mg/L * h	mg/L	mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	98.93	14.60	3.11	135.70	22.25	3.30	1.37	1.52	1.06
Median	88.34	13.91	2.34	126.73	21.61	2.58	1.43	1.55	1.10
Geometric Mean	91.25	14.01	2.18	127.63	21.48	2.30	1.40	1.53	1.06
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	122.50	20.82	2.29	1.41	1.54	1.06
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	132.99	22.16	2.31	1.39	1.53	1.05
5^{th} percentile	48.05	8.74	0.52	72.46	14.03	0.50	1.51	1.61	0.96
95^{th} percentile	176.30	21.99	7.77	216.50	32.57	8.13	1.23	1.48	1.05
SD	40.96	4.23	2.53	48.79	5.97	2.67	1.19	1.41	1.06
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
3^{rd} quartile	123.18	17.26	4.22	165.43	25.84	4.67	1.34	1.50	1.11
1^{st} quartile	69.63	11.43	1.20	103.20	17.89	1.27	1.48	1.57	1.06
10^{th} percentile	55.45	9.76	0.75	80.17	15.56	0.85	1.45	1.59	1.13
90^{th} percentile	154.24	20.66	6.58	200.76	30.43	6.68	1.30	1.47	1.02

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 1200 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h} for the population are shown in Table 25A. Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 26A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 400 mg carbamazepine QD were significantly higher (about 2-fold) than those from maribavir 400 mg BID alone. The C_{min} was about 50% higher than that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID might be preferred to a 1200 mg BID maribavir dose to counteract the effect of carbamazepine from an efficacy perspective.

TABLE 25A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of carbamazepine.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0\text{-}12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	296.79	43.79	9.32	203.54	33.37	4.95	0.71	0.77	0.54
Median	265.02	41.72	7.02	190.10	32.41	3.88	0.71	0.79	0.56
Geometric Mean	273.75	42.02	6.55	191.45	32.22	3.45	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	261.14	40.62	6.48	183.75	31.23	3.42	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	286.97	43.46	6.62	199.48	33.24	3.49	0.71	0.78	0.53
5th percentile	144.14	26.21	1.55	108.68	21.05	0.76	0.54	0.61	0.36
95th percentile	528.90	65.97	23.31	324.74	48.86	12.20	0.84	0.90	0.70
SD	122.87	12.70	7.58	73.19	8.96	4.01	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3rd quartile	369.55	51.78	12.65	248.14	38.76	7.01	0.77	0.84	0.61
1st quartile	208.88	34.29	3.61	154.80	26.84	1.90	0.65	0.72	0.46
10th percentile	166.34	29.29	2.26	120.25	23.34	1.27	0.58	55.45	0.39
90th percentile	462.72	61.98	19.73	301.14	45.65	10.02	0.82	154.24	0.67

TABLE 26A

Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0\text{-}12h)}$ mg/L * h	C_{max} mg/L	C_{12h} mg/L	$AUC_{(0\text{-}12h)}$	C_{max}	C_{12h}
PREDICTED									
Mean	98.93	14.60	3.11	203.54	33.37	4.95	2.06	2.29	1.59
Median	88.34	13.91	2.34	190.10	32.41	3.88	2.15	2.33	1.66
Geometric Mean	91.25	14.01	2.18	191.45	32.22	3.45	2.10	2.30	1.58
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	183.75	31.23	3.42	2.11	2.31	1.58
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	199.48	33.24	3.49	2.09	2.29	1.58
5th percentile	48.05	8.74	0.52	108.68	21.05	0.76	2.26	2.41	1.46
95th percentile	176.30	21.99	7.77	324.74	48.86	12.20	1.84	2.22	1.57
SD	40.96	4.23	2.53	73.19	8.96	4.01	1.79	2.12	1.58
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
	123.18	17.26	4.22	248.14	38.76	7.01	2.01	2.25	1.66
	69.63	11.43	1.20	154.80	26.84	1.90	2.22	2.35	1.58
	55.45	9.76	0.75	120.25	23.34	1.27	2.17	2.39	1.69
	154.24	20.66	6.58	301.14	45.65	10.02	1.95	2.21	1.52

Discussion

Prospective use of the model to predict the likely outcomes of interaction with rifampicin (800 mg and 1200 mg BID maribavir), phenobarbital (400 mg, 800 mg and 1200 mg BID maribavir), phenytoin (400 mg, 800 mg and 1200 mg BID maribavir), and carbamazepine (400 mg, 800 mg and 1200 mg BID maribavir) indicated geometric mean (arithmetic mean) C_{max} ratios of 0.54 (0.56), 0.72 (0.73), 0.68 (0.69), and 0.77 (0.77), respectively, for maribavir. Corresponding predicted geometric mean (arithmetic mean) $AUC_{0\text{-}12h}$ ratios were 0.35 (0.37), 0.60 (0.61), 0.57 (0.58), and 0.70 (0.71), respectively. Corresponding predicted geometric mean (arithmetic mean) C_{12h} ratios were 0.04 (0.10), 0.31 (0.37), 0.30 (0.36), and 0.53 (0.54) respectively. As there was no non-linearity in the PBPK model for maribavir, the changes in exposure as a consequence of CYP3A4-mediated induction by the perpetrators were not dependent on the dose of maribavir.

Efficacy of maribavir is correlated to the trough concentrations (C_{12h}) of the drug. Significant reductions in C_{12h} were predicted when the standard dose of 400 mg BID maribavir was combined with CYP3A4 inducers such as rifampicin, phenobarbital, phenytoin, and carbamazepine. Simulations with higher doses of maribavir (i.e. 800 mg BID, 1200 mg BID and 1600 mg BID) were performed to determine whether the reduction in C_{12h}, (and hence reduction in maribavir efficacy) could be overcome by dose increase. The reduction in C_{12h} due to the rifampicin induction of CYP3A4 could not be corrected by the use of up to 1600 mg BID maribavir. Use of a dose of 800 mg BID maribavir with phenobarbital resulted in a C_{12h} that was 15% lower than that with 400 mg BID maribavir alone while 1200 mg BID resulted a 1.28-fold increase in C_{12h}. A 1200 mg BID dose of maribavir with phenytoin was also predicted to overcome the reduction in C_{12h}, due to CYP3A4 induction. Increasing the maribavir dose to 800 or 1200 mg BID can be used to overcome the reduction in C_{12h} due to carbamazepine. Maribavir demonstrated acceptable safety/tolerability profiles with no limiting toxicity at doses up to 1200 mg BID in Phase 2 studies for the treatment of CMV infection/diseases and the predicted mean $AUC_{(0\text{-}12h)}$ and C_{max} values are 292 mg/L*h and 43.4 mg/L for 1200 mg BID maribavir. The predicted $AUC_{(0\text{-}12h)}$ and C_{max} values seen with maribavir dose increases, as discussed above, in the presence of inducers are below the exposure predicted for 1200 mg BID maribavir alone, therefore there is no safety concern with increasing maribavir dose from 400 mg to 800 mg, or 1200 mg in the presence of inducer. Overall, CYP3A4 inducers significantly reduce systemic exposure of maribavir; therefore maribavir dose increase is necessary when co-administration with CYP3A4 inducers is necessary. When co-administration with carbamazepine or phenobartital is needed, maribavir dose increase to 800 or 1200 mg BID is recommended. When co-administration with phenytoin is needed, maribavir dose increase to 1200 mg BID recommendation. Rifampicin caused significant reduction in maribavir exposure which cannot be overcome by maribavir dose increase up to 1600 mg BID, therefore coadministration with rifampicin should be prohibited or alternative antibacterial therapy of lower CYP3A4 induction potential should be considered.

General note: The dose recommendations for inducers were based on arithmetic mean ratios due to the bias in the geometric mean estimates with coadministration. However, geometric mean ratios were used for dose recommendation for inhibitors.

Example 3—Co-Administration of Maribavir with Other Drugs

Drug interaction studies were summarized based on Example 2 and other available data (e.g., in vitro and clinical data). Considerations for each coadministered drug are provided in Table 1-A, and the effects of co-administration of other drugs on the pharmacokinetics of maribavir are summarized in Tables 1-B and 1-C.

TABLE 1-A

Summary of established and other potentially significant drug interactions (observed and predicted).

Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Antiarrhythmics		
digoxin	↑digoxin	Use caution when maribavir and digoxin are co-administered. Monitor serum digoxin concentrations. The dose of digoxin may need to be reduced with co-administered with maribavir
Anticonvulsants		
Carbamazepine	↓Maribavir	A dose adjustment of maribavir to 800 mg, twice daily when co-administered with carbamazepine
Phenobarbital	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenobarbital
Phenytoin	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenytoin
Antimycobacterials		
Rifabutin	↓Maribavir	Co-administration of maribavir and rifamutin may decrease efficacy of maribavir
Rifampin	↓Maribavir	Co-administration of maribavir and rifampin may decrease efficacy of maribavir
Herbal Products		
St. John's wort	↓Maribavir	Co-administration of maribavir and St. John's wort may decrease efficacy of maribavir

TABLE 1-A-continued

Summary of established and other potentially significant drug interactions (observed and predicted).

Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Immunosuppressants		
Cyclosporine	↑Cyclosporine	Frequently monitor cyclosporine levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Everolimus	↑Everolimus	Frequently monitor everolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Sirolimus	↑Sirolimus	Frequently monitor sirolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Tacrolimus	↑Tacrolimus	Frequently monitor tacrolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed

TABLE 1-B

Summary of Changes in Pharmacokinetics of Maribavir in the Presence of Co-administered Drugs.

Co-administered Drug and Regimen		Maribavir Regimen	N	Geometric Mean Ratio (90% CI] of Maribavir PK with/without Co-administered Drug (No Effect = 1.00) AUC	C_{max}	$C_{tau}{}^{C}$
Anticonvulsants						
Carbamazepine[a]	400 mg once daily	800 mg twice daily/ 400 mg twice daily	200	1.40 (1.09, 1.67)	1.53 (1.22, 1.79)	1.05 (0.71, 1.40)
Phenobarbital[a]	100 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.80 (1.18, 2.35)	2.17 (1.69, 2.57)	0.94 (0.22, 1.97)
Phenytoin[a]	300 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.70 (1.06, 2.46)	2.05 (1.49, 2.63)	0.89 (0.26, 2.04)
Antimycobacterials						
Rifampin	600 mg once daily	400 mg twice daily	14	0.40 (0.36, 0.44)	0.61 (0.52, 0.72)	0.18 (0.14, 0.25)
Antifungals						
Ketoconazole	400 mg single dose	400 mg single dose	19	1.53 (1.44, 1.63)	1.10 (1.01, 1.19)	—
Aluminum hydroxide and magnesisum hydroxide antacid	20 mL[b] single dose	100 mg single dose	15	0.89 (0.83, 0.96)	0.84 (0.75, 0.94)	

[a]Based on physiologically based pharmacokinetic modelings results from 10 trials of 20 subjects each.
The maribavir dosing regimen and geometric mean ratios (5^{th} percentile, 95 percentile) correspond to dose-adjusted maribavir with inducer vs. 400 mg twice daily without inducer.
[b]Containing 800 mg aluminum hydroxide and 800 mg magnesium hydroxide.
[C]tau is maribavir dosing interval: 12 hours.

TABLE 1-C

Changes of Pharmacokinetics for Co-administered Drug in the Presence of 400 mg Twice Daily Maribavir.

Co-administered Drug and Regimen		N	Geometric Mean Ratio [90% CI] of Maribavir PK with/without Co-maribavir (No Effect = 1.00) AUC	C_{max}	C_{trough}
Immunosuppressants					
Tacrolimus	stable dose, twice daily (total daily dose: 0.5-16 mg)	20	1.51 (1.39, 1.65)	1.38 (1.20, 1.57)	1.57 (1.41, 1.74)

TABLE 1-C-continued

Changes of Pharmacokinetics for Co-administered Drug in the Presence of 400 mg Twice Daily Maribavir.

Co-administered Drug and Regimen		N	Geometric Mean Ratio [90% CI] of Maribavir PK with/without Co-maribavir (No Effect = 1.00) AUC	C_{max}	C_{trough}
P-gp substrate					
digoxin	0.5 mg single dose	18	1.21 (1.10, 1.32)	1.25 (1.13, 1.38)	—

While we have described a number of embodiments of this invention, it is apparent that our basic examples may be altered to provide other embodiments that utilize the compounds and methods of this invention. Therefore, it will be appreciated that the scope of this invention is to be defined by the appended claims rather than by the specific embodiments that have been represented by way of example.

The invention claimed is:

1. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom, the method comprising administering maribavir in an amount of 1200 mg orally twice daily, wherein the patient is a transplant recipient concomitantly exposed to or receiving an immunosuppressant and an anticonvulsant selected from phenytoin or phenobarbital.

2. The method of claim **1**, wherein the anticonvulsant is phenytoin.

3. The method of claim **1**, wherein the anticonvulsant is phenobarbital.

4. The method of claim **1**, wherein the patient is an adult or a child of 12 years of age or older, and wherein the patient weighs at least 35 kg.

5. The method of claim **1**, wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.

6. The method of claim **1**, wherein the patient is a hematopoietic stem cell transplant recipient.

7. The method of claim **1**, wherein the patient is a solid organ transplant recipient.

8. The method of claim **1**, wherein the immunosuppressant is cyclosporine.

9. The method of claim **1**, wherein the immunosuppressant is everolimus.

10. The method of claim **1**, wherein the immunosuppressant is sirolimus.

11. The method of claim **1**, wherein the immunosuppressant is tacrolimus.

12. The method of claim **11**, wherein the tacrolimus is administered as a stable dose, twice daily, with a total daily dose between 0.5 to 16 mg.

13. The method of claim **11**, wherein the anticonvulsant is phenobarbital.

14. The method of claim **13**, wherein the phenobarbital is administered to the patient in an amount of 100 mg once daily.

15. The method of claim **11**, wherein the anticonvulsant is phenytoin.

16. The method of claim **15**, wherein the phenytoin is administered to the patient in an amount of 300 mg once daily.

17. The method of claim **1**, further comprising monitoring immunosuppressant drug levels throughout treatment with maribavir.

18. The method of claim **17**, wherein the monitoring of immunosuppressant drug levels is performed following initiation, and after discontinuation, of treatment with maribavir in the patient.

19. The method of claim **17**, further comprising adjusting the dose of the immunosuppressant.

20. The method of claim **18**, further comprising adjusting the dose of the immunosuppressant.

* * * * *

EXHIBIT E

US012447169B2

(12) **United States Patent**
Peabody, III

(10) **Patent No.: US 12,447,169 B2**
(45) **Date of Patent: *Oct. 21, 2025**

(54) **MARIBAVIR ISOMERS, COMPOSITIONS, METHODS OF MAKING AND METHODS OF USING**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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Related U.S. Application Data

(60) Continuation of application No. 18/597,197, filed on Mar. 6, 2024, now abandoned, which is a continuation of application No. 18/200,636, filed on May 23, 2023, now abandoned, which is a division of application No. 16/983,310, filed on Aug. 3, 2020, now Pat. No. 11,684,632, which is a division of application No. 16/663,743, filed on Oct. 25, 2019, now Pat. No. 10,765,692, which is a division of application No. 15/291,639, filed on Oct. 12, 2016, now Pat. No. 10,485,813, which is a division of application No. 15/055,043, filed on Feb. 26, 2016, now abandoned, which is a continuation of application No. 14/595,548, filed on Jan. 13, 2015, now abandoned, which is a continuation of application No. 13/282,501, filed on Oct. 27, 2011, now Pat. No. 8,940,707.

(60) Provisional application No. 61/407,637, filed on Oct. 28, 2010.

(51) **Int. Cl.**

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G09B 19/00	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/7056*** (2013.01); ***A61K 45/06*** (2013.01); ***G01N 33/94*** (2013.01); ***G09B 19/00*** (2013.01); *G01N 2430/00* (2013.01); *G01N 2800/52* (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to novel compositions and methods of using maribavir which enhance its effectiveness in medical therapy, as well as to maribavir isomers and methods of use thereof for counteracting the potentially adverse effects of maribavir isomerization in vivo in the event it occurs.

8 Claims, 4 Drawing Sheets

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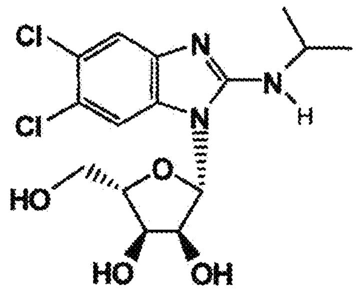
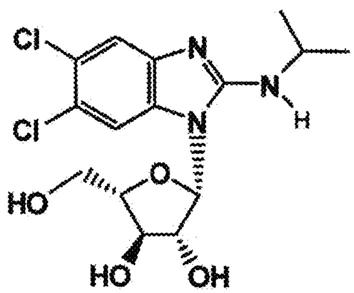


Maribavir



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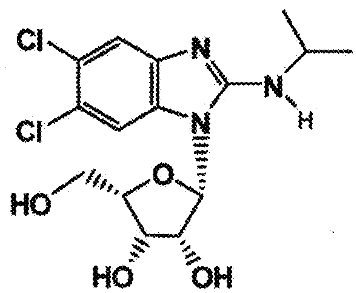

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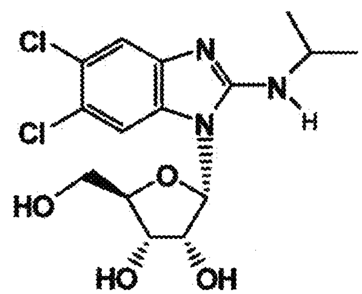

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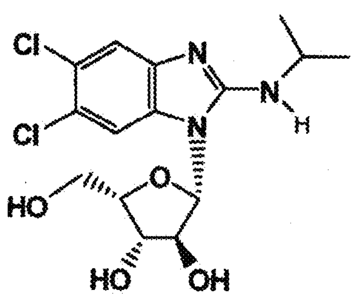

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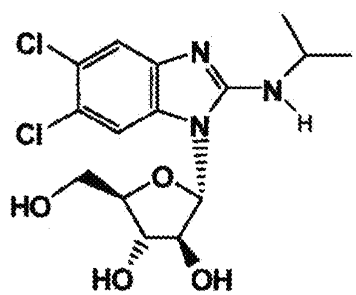

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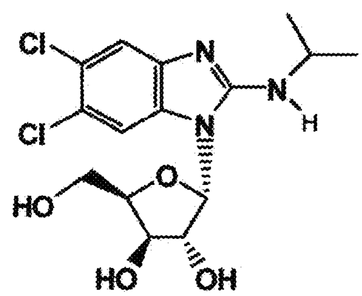

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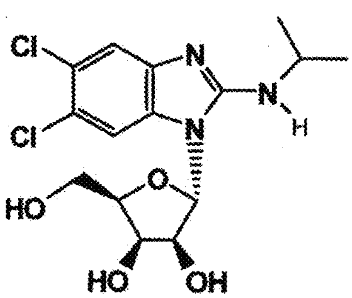

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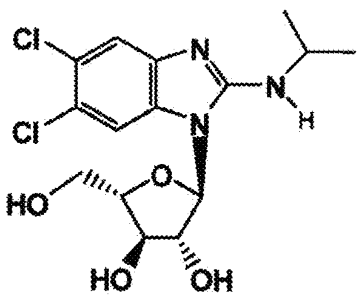
FIG. 1



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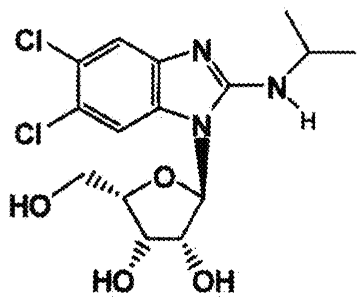

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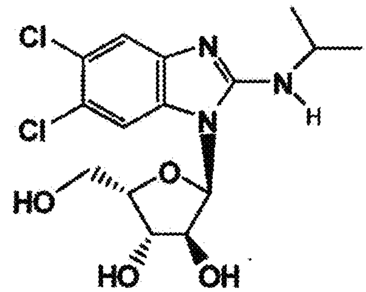

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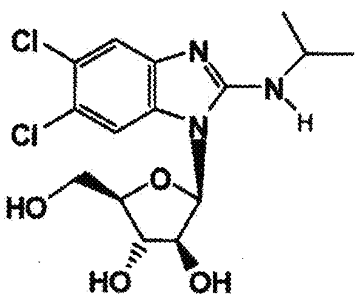

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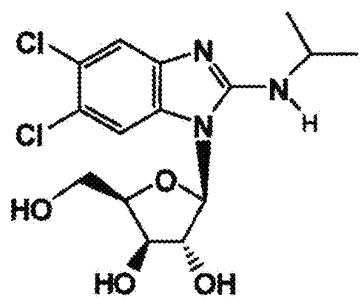

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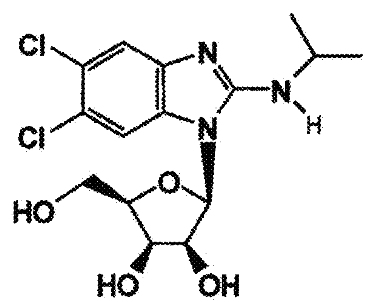

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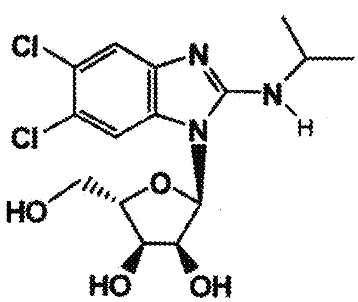

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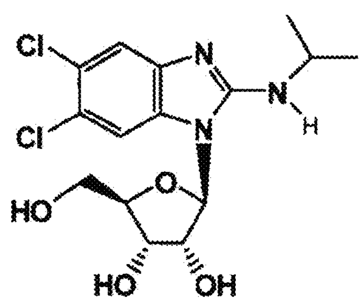

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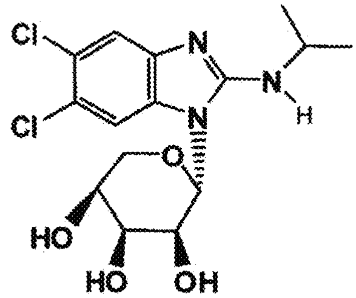
FIG. 2



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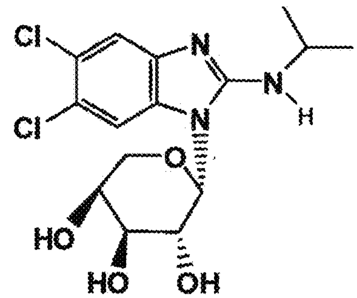

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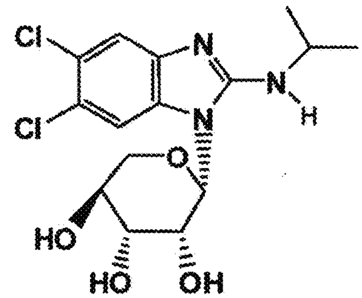

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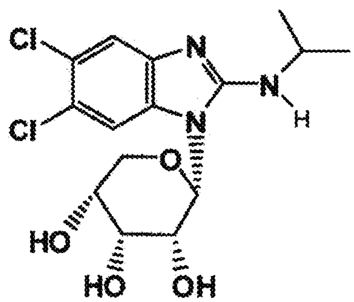

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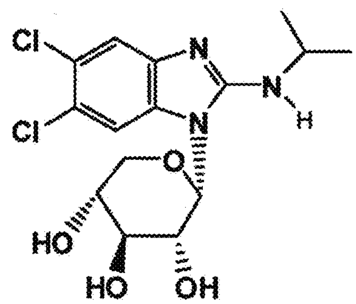

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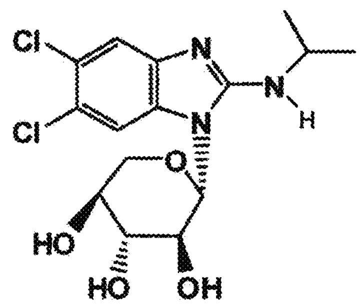

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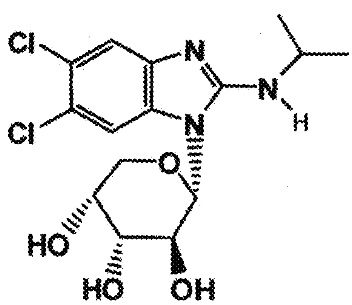

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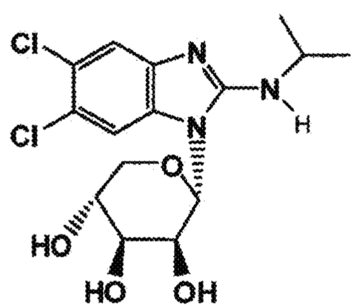

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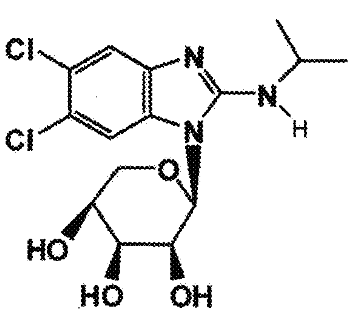

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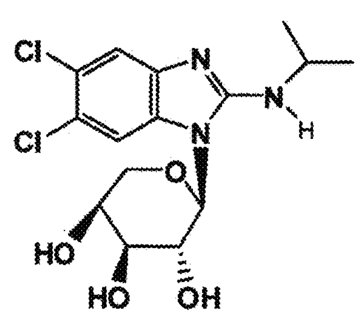

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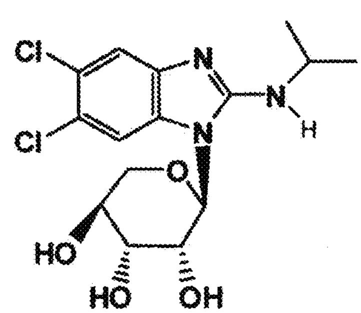

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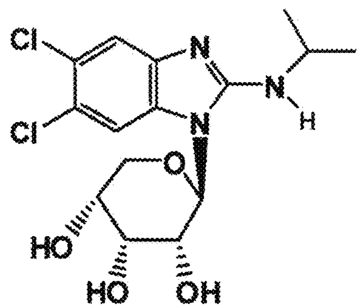
FIG. 3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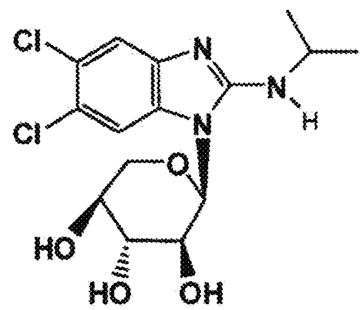
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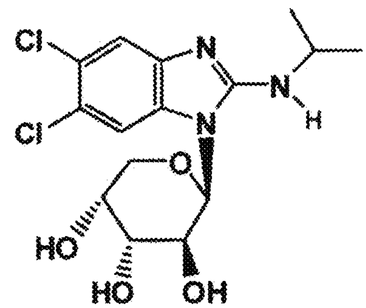
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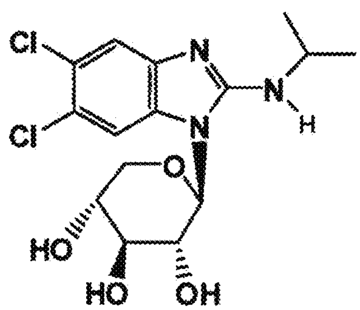
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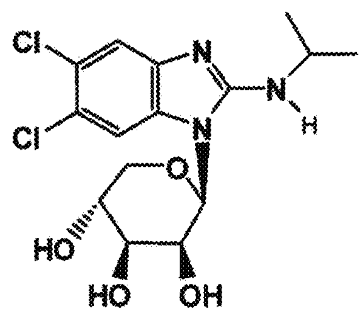
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FIG. 3 (CONT'D)

MARIBAVIR ISOMERS, COMPOSITIONS, METHODS OF MAKING AND METHODS OF USING

CROSS-REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of U.S. application Ser. No. 18/597,197 filed on Mar. 6, 2024, which is a Continuation of U.S. application Ser. No. 18/200,636 filed on May 23, 2023, which is a divisional of U.S. application Ser. No. 16/983,310 filed on Aug. 3, 2020, now U.S. Pat. No. 11,684,632, issued Jun. 27, 2023, which is a divisional of U.S. application Ser. No. 16/663,743 filed on Oct. 25, 2019, now U.S. Pat. No. 10,765,692, issued Sep. 8, 2020, which is a divisional of U.S. application Ser. No. 15/291,639 filed on Oct. 12, 2016, now U.S. Pat. No. 10,485,813, issued Nov. 26, 2019, which is a divisional of U.S. application Ser. No. 15/055,043 filed on Feb. 26, 2016, which is a continuation of U.S. application Ser. No. 14/595,548 filed on Jan. 13, 2015, which is a continuation of U.S. application Ser. No. 13/282,501 filed on Oct. 27, 2011, now U.S. Pat. No. 8,940,707, issued Jan. 27, 2015, which claims the benefit of priority to U.S. Provisional Patent Application No. 61/407,637 filed on Oct. 28, 2010, the contents of these applications are each incorporated herein by reference in their entirety.

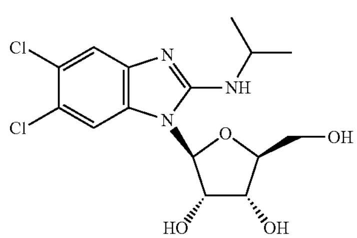
BACKGROUND OF THE INVENTION

The present invention relates to compositions and methods for enhancing the therapeutic efficacy of the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole (also known as 1263W94 and maribavir), as well as to maribavir isomers and a method of making such isomers.

5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole is a benzimidazole derivative useful in medical therapy. U.S. Pat. No. 6,077,832 discloses 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole and its use for the treatment or prophylaxis of viral infections such as those caused by herpes viruses. The compound as disclosed in U.S. Pat. No. 6,077,832 is an amorphous, non-crystalline material.

The structure of 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole is:

(I)



The preparation of certain unique crystalline forms and solvate forms of maribavir, as well as pharmaceutical formulations thereof and their use in therapy are described in U.S. Pat. Nos. 6,469,160 and 6,482,939.

The present invention has arisen out of the unexpected discovery that maribavir may isomerize under in vivo conditions to one or more configurational stereoisomers or constitutional isomers. The maribavir compound contains 4 (four) chiral carbon centers in the ribofuranosyl moiety and

therefore maribavir is just one of 16 (sixteen) potential stereoisomers that may be formed under various in vivo conditions.

Under in vivo conditions, maribavir can isomerize to other compounds that may (or may not) have the same or similar chemical, physical and biological properties. The in vivo isomerization of maribavir results in conversion of maribavir to other isomers that have the same molecular formula but a different molecular structure. The different molecular structures can be grouped into isomers that have different connectivity of the constituent atoms (constitutional isomers) or grouped into isomers that have the same "connectivity" but differ in the way the atoms and groups of atoms are oriented in space (configurational stereoisomers). Such molecular conversion in vivo is believed to result in the dilution of the effective maribavir concentration in the host that was treated with maribavir. The in vivo isomerization transforms and distributes dosed maribavir material into other molecular entities that do not necessarily have the same or similar biological activity. If the isomerization results in the formation of isomers that have a lower degree of corresponding biological activity relative to the activity of maribavir, then the isomerization will decrease the effective biological activity of maribavir dose administered to a host.

A recent maribavir Phase 3 clinical trial conducted by ViroPharma Incorporated (the 300 Study) that evaluated maribavir for cytomegalovirus (CMV) prophylaxis in allogeneic stem cell, or bone marrow, transplant (SCT) patients did not achieve its primary endpoint. In the primary analysis, there was no statistically significant difference between maribavir and placebo in reducing the rate of CMV disease. In addition, the study failed to meet its key secondary endpoints (ViroPharma Press Release dated Feb. 9, 2009). The 300 Study result appeared at first blush to be inconsistent with an earlier proof-of-concept (POC) maribavir Phase 2 clinical trial (the 200 Study) wherein ViroPharma reported positive preliminary results that showed that maribavir inhibited CMV reactivation in SCT patients. The data from this study demonstrate that prophylaxis with maribavir displays strong antiviral activity, as measured by significant reduction in the rate of reactivation of CMV in recipients of allogeneic stem cell (bone marrow) transplants, and that administration of maribavir for up to 12 weeks has a favorable tolerability profile in this very sick patient population (ViroPharma Press Release date Mar. 29, 2006).

However, the 300 Study result can be explained in terms of the instant maribavir isomerization theory/discovery. An unrecognized key difference between the 200 Study and the 300 Study was that the former provided for a fasted dosing protocol of maribavir, whereas the latter allowed the dosing protocol to be either under fasted or fed conditions (at the discretion of the clinician). The nature of the patient population in the 300 Study suggests that probably very few patients were dosed under the strict fasted dosing protocol that was previously used in the 200 Study. The change in dosing protocol in the 300 Study changed not only the in vivo dosing conditions for maribavir, but also the nature and/or degree of isomerization of maribavir that occurs in vivo, so that more maribavir was isomerized to other less effective compounds, thereby reducing the effective bioavailable concentration of maribavir drug below levels necessary to adequately prevent CMV infection and/or CMV re-activation in the host.

The degree and nature of the isomerization of maribavir depends on the particular in vivo conditions to which the drug is exposed, which are variable. The potential mechanisms for isomerizing maribavir in vivo are by chemical isomerization (acid, base and/or metal catalyzed isomerization), microbially-mediated isomerization, and/or host metabolism induced isomerization. See, for example, Okano, Kazuya, Tetrahedron, 65:1937-1949 (2009); Kelly, James A. et al., J. Med. Chem., 29:2351-2358 (1986); and Ahmed, Zakaria et al., Bangladesh J. Sci. Ind. Res., 25 (1-4): 90-104 (2000).

Thus, maribavir should be formulated, administered, packaged and promoted in ways that will prevent or at least reduce the unwanted occurrence of maribavir isomerization in vivo, thereby enhancing the drug's bioavailablity and efficacy, and/or counteract the potential adverse effect(s) of maribavir isomerization in vivo, if it occurs.

SUMMARY OF THE INVENTION

The invention generally relates to the unexpected discovery that maribavir may isomerize under in vivo conditions to one or more configurational stereoisomers and/or constitutional isomers. Aspect of the invention is illustrated below.

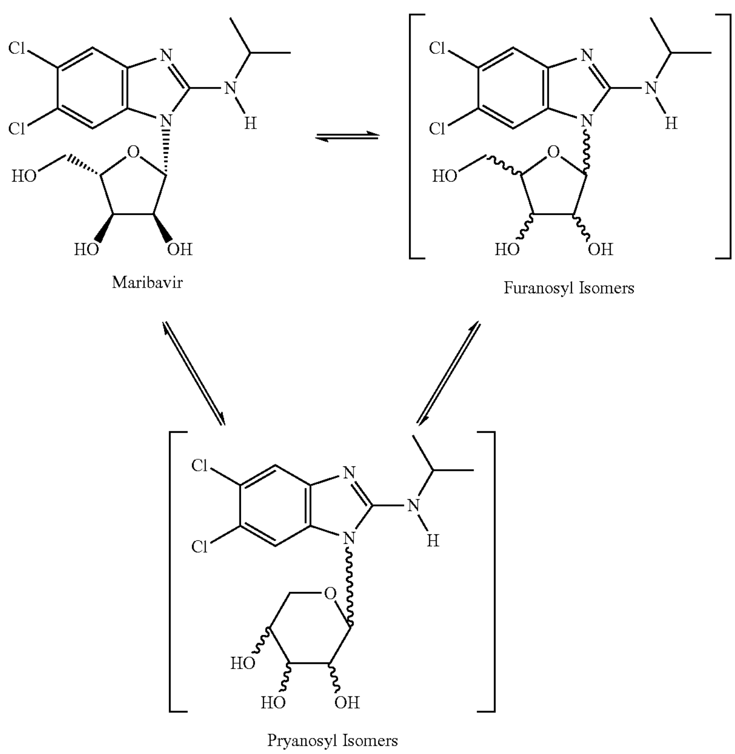
maribavir isomerization (quick release formulations, delayed/controlled release formulations, combination with antacids, intravenous (IV) formulations, combination formulations with antibiotics to prevent microbial isomerization).

(4) The invention includes methods of using one or more maribavir isomers to prevent or treat disease in host (as antivirals, for example for treating/preventing CMV, EBV, HCV).

(5) The invention includes methods of using one or more maribavir isomers as reference standards in analytical methods for monitoring the blood plasma concentrations of maribavir and related isomers.

(6) The invention includes methods of monitoring maribavir and maribavir isomers, and using the information to adjust treatment protocols (increase/decrease dose, change dosing regimen fed/fasted, discontinue maribavir therapy, start other therapy). The analytical methods for monitoring in vivo maribavir concentrations must

Maribavir In Vivo Isomerization Scheme



The practical applications of the instant discovery and related inventions are as follows:

(1) The invention includes methods of making maribavir isomers under in vivo conditions (method of administering maribavir as a prodrug to make other maribavir isomers).

(2) The invention includes methods dosing maribavir so as to mitigate the impact of in vivo maribavir isomerization (such as dosing under fasted conditions, or increasing the dose of maribavir).

(3) The invention includes maribavir formulations that effectively mitigate the adverse effects of in vivo

be able discriminate between maribavir and maribavir isomers (for example chiral chromatography, and in particular LC-MS-MS using a chiral sorbent material in the LC column).

(7) The invention includes a method of more safely and effectively using maribavir to treat or prevent disease in humans by including information and guidance in the product label and/or promotional materials to inform the public as to how to use the maribavir product so as to avoid or to mitigate the adverse impact of in vivo maribavir isomerization, and thus optimize therapeutic efficacy and safety.

(8) The invention includes the corresponding inventions related to the maribavir isomers that are pyranosyl constitutional isomers of maribavir.

The present invention also relates to a package or kit comprising therapeutically effective dosage forms of maribavir, prescribing information and a container for the dosage form. The prescribing information includes advice to a patient receiving maribavir therapy regarding the administration of maribavir without food to improve bioavailability.

DETAILED DESCRIPTION OF THE INVENTION

The following table contains list of useful dosing protocols that may be used to improve the efficacy and safety for treating a patient with maribavir.

Maribavir dosing Protocol	Dosing amount	Fasting conditions (before/after dosing)	Route of administration and dosage form.
01	3200 mg/2× day	None	Oral-tablet-immediate release
02	3200 mg/2× day	None	IV
03	1600 mg/2× day	None	Oral-tablet-w/antacids
04	1600 mg/1× day	None	Oral-tablet-w/antibiotics
05	800 mg/3× day	None	Oral-tablet-delayed release
06	800 mg/2× day	None	Oral-tablet-immediate release
07	800 mg/1× day	None	IV
08	400 mg/4× day	None	Oral-tablet-w/antacids
09	400 mg/3× day	None	Oral-tablet-w/antibiotics
10	400 mg/2× day	None	Oral-tablet-delayed release
11	400 mg/1× day	None	Oral-tablet-immediate release
12	3200 mg/2× day	12 hrs/3 hrs	IV
13	3200 mg/2× day	12 hrs/3 hrs	Oral-tablet-w/antacids
14	1600 mg/2× day	12 hrs/3 hrs	Oral-tablet-w/antibiotics
15	1600 mg/1× day	12 hrs/3 hrs	Oral-tablet-delayed release
16	800 mg/3× day	12 hrs/3 hrs	Oral-tablet-immediate release
17	800 mg/2× day	12 hrs/3 hrs	IV
18	800 mg/1× day	12 hrs/3 hrs	Oral-tablet-w/antacids
19	400 mg/4× day	12 hrs/3 hrs	Oral-tablet-w/antibiotics
20	400mg/3× day	12 hrs/3 hrs	Oral-tablet-delayed release
21	400 mg/2× day	12 hrs/3 hrs	Oral-tablet-immediate release
22	400 mg/1× day	12 hrs/3 hrs	Oral-tablet-immediate release
23	3200 mg/2× day	6 hrs/2 hrs	Oral-tablet-immediate release
24	3200 mg/2× day	6 hrs/2 hrs	IV
25	1600 mg/2× day	6 hrs/2 hrs	Oral-tablet-w/antacids
26	1600 mg/1× day	6 hrs/2 hrs	Oral-tablet-w/antibiotics
27	800 mg/3× day	6 hrs/2 hrs	Oral-tablet-delayed release
28	800 mg/2× day	6 hrs/2 hrs	Oral-tablet-immediate release
29	800 mg/1× day	6 hrs/2 hrs	IV
30	400 mg/4× day	6 hrs/2 hrs	Oral-tablet-w/antacids
31	400 mg/3× day	6 hrs/2 hrs	Oral-tablet-w/antibiotics
32	400 mg/2× day	6 hrs/2 hrs	Oral-tablet-delayed release
33	400 mg/1× day	6 hrs/2 hrs	Oral-tablet-immediate release
34	3200 mg/2× day	3 hrs/1 hr	IV
35	3200 mg/2× day	3 hrs/1 hr	Oral-tablet-w/antacids
36	1600 mg/2× day	3 hrs/1 hr	Oral-tablet-w/antibiotics
37	1600 mg/1× day	3 hrs/1 hr	Oral-tablet-delayed release
38	800 mg/3× day	3 hrs/1 hr	Oral-tablet-immediate release
39	800 mg/2× day	3 hrs/1 hr	IV
40	800 mg/1× day	3 hrs/1 hr	Oral-tablet-w/antacids
41	400 mg/4× day	3 hrs/1 hr	Oral-tablet-w/antibiotics
42	400 mg/3× day	3 hrs/1 hr	Oral-tablet-delayed release
43	400 mg/2× day	3 hrs/1 hr	Oral-tablet-immediate release
44	400 mg/1× day	3 hrs/1 hr	Oral-tablet-immediate release

In carrying out the method of the invention, it is preferably to determine the presence and/or concentration of maribavir isomers, especially isomers of diminished therapeutic efficacy in patient blood plasma samples as part of the method.

As used herein, the terms "fasted conditions", "fasting conditions" and "without food" are defined to mean, in general, the condition of not having consumed food during the period between from at least about 3 to 12 hours prior to the administration of maribavir to at least about 1 to 3 hours after the administration of maribavir. Other narrower "fasted conditions" are also contemplated herein and described below.

The term "with food" is defined to mean, in general, the condition of having consumed food prior to, during and/or after the administration of maribavir that is consistent with the relevant intended definition of "fasted conditions" (which may be narrow or broad depending on the circumstances). Preferably, the food is a solid food sufficient bulk and fat content that it is not rapidly dissolved and absorbed in the stomach. More preferably, the food is a meal, such as breakfast, lunch or dinner.

The term "isomers" means compounds that have the same molecular formula but a different molecular structure.

The term "constitutional isomers" is defined to mean isomers that have the same molecular formula but a different molecular structure wherein the molecular structures of the isomers have different connectivity of the constituent atoms.

The term "configurational stereoisomers" is defined to mean isomers that have the same "connectivity" but differ in the molecular structure in the way the atoms and groups of atoms are oriented in space.

The term "immediate release" is defined to mean release of drug from drug formulation by dissolution is less than 60 minutes or is otherwise release from the drug formulation in less than 60 minutes.

The term "IV" is defined to mean intravenous.

The chemical structure of maribavir and some maribavir isomers are shown below. The instant invention contemplates novel formulations, dosage levels and methods of use of maribavir, the maribavir isomers MFI-01 to MFI-015 (configurational stereoisomers), as well as the maribavir isomers MPI-01 to MPI-016 (constitutional isomers). The invention also contemplates the corresponding acyclic constitutional isomers wherein the sugar moiety is an open chain and attached to the benzimidazole.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows chemical structures of maribavir and maribavir configurational stereoisomers that have the same configuration at the furanoysl anomer carbon.

FIG. **2** shows chemical structures of maribavir configurational stereoisomers that have the opposite configuration at the furanosyl anomer carbon.

FIG. **3** shows chemical structures of maribavir "pyranosyl" constitutional isomers)

A number of patent and non-patent documents are cited in the foregoing specification in order to describe the state of the art to which this invention pertains. The entire disclosure of each of these citations is incorporated by reference herein.

While certain of the preferred embodiments of the present invention have been described and specifically exemplified above, it is not intended that the invention be limited to such embodiments. Various modifications may be made thereto without departing from the scope and spirit of the present invention, as set forth in the following claims. Furthermore, the transitional terms "comprising", "consisting essentially of" and "consisting of" define the scope of the appended claims, in original and amended form, with respect to what unrecited additional claim elements or steps, if any, are excluded from the scope of the claims. The term "comprising" is intended to be inclusive or open-ended and does not exclude additional, unrecited elements, methods step or materials. The phrase "consisting of" excludes any element, step or material other than those specified in the claim, and, in the latter instance, impurities ordinarily associated with the specified materials. The phrase "consisting essentially of" limits the scope of a claim to the specified elements, steps or materials and those that do not materially affect the basic and novel characteristic(s) of the claimed invention. All compositions or formulations identified herein can, in alternate embodiments, be more specifically defined by any of the transitional phases "comprising", "consisting essentially of" and "consisting of".

What is claimed is:

1. A method of treating a cytomegalovirus (CMV) infection in a patient having a CMV infection, the method comprising orally administering to the patient the compound 5,6-dichloro-2-(isopropylamino)-1-(β-L-ribofuranosyl)-1H-benzimidazole in an amount of 400 mg twice a day.

2. The method of claim **1**, wherein said compound is administered to said patient with food.

3. The method of claim **1**, wherein said compound is administered to said patient without food.

4. The method of claim **1**, wherein said compound is administered as a composition comprising a therapeutically acceptable adjuvant, excipient, or carrier medium.

5. The method of claim **4**, wherein said composition is an immediate release formulation.

6. The method of claim **1**, wherein said compound is administered to said patient in a fasted condition wherein the patient does not consume food during the period from 12 hours before to 3 hours after the administration.

7. The method of claim **1**, wherein said compound is administered to said patient in a fasted condition wherein the patient does not consume food during the period from 6 hours before to 2 hours after the administration.

8. The method of claim **1**, wherein said compound is administered to said patient in a fasted condition wherein the patient does not consume food during the period from 3 hours before to 1 hour after the administration.

* * * * *

EXHIBIT F

US012447170B2

(12) **United States Patent**
Song et al.

(10) **Patent No.: US 12,447,170 B2**
(45) **Date of Patent: *Oct. 21, 2025**

(54) **USE OF MARIBAVIR IN TREATMENT REGIMENS**

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(73) Assignee: **Takeda Pharmaceutical Company Limited**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/990,486**

(22) Filed: Dec. 20, 2024

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 18/646,035, filed on Apr. 25, 2024, now Pat. No. 12,213,989, which is a continuation of application No. 18/400,304, filed on Dec. 29, 2023, now abandoned, which is a continuation of application No. PCT/US2022/050341, filed on Nov. 18, 2022.

(60) Provisional application No. 63/281,206, filed on Nov. 19, 2021.

(51) **Int. Cl.**

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A61K 31/513	(2006.01)
A61K 31/55	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/7056*** (2013.01); ***A61K 31/4166*** (2013.01); ***A61K 31/513*** (2013.01); ***A61K 31/55*** (2013.01)

(58) **Field of Classification Search**

CPC A61K 31/7056; A61K 31/4166; A61K 31/513; A61K 31/55

USPC 514/43

See application file for complete search history.

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Primary Examiner — Yih-Horng Shiao

(74) *Attorney, Agent, or Firm* — Honigman LLP; Andrew S. Chipouras; Jonathan P. O'Brien

(57) **ABSTRACT**

Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

25 Claims, 4 Drawing Sheets

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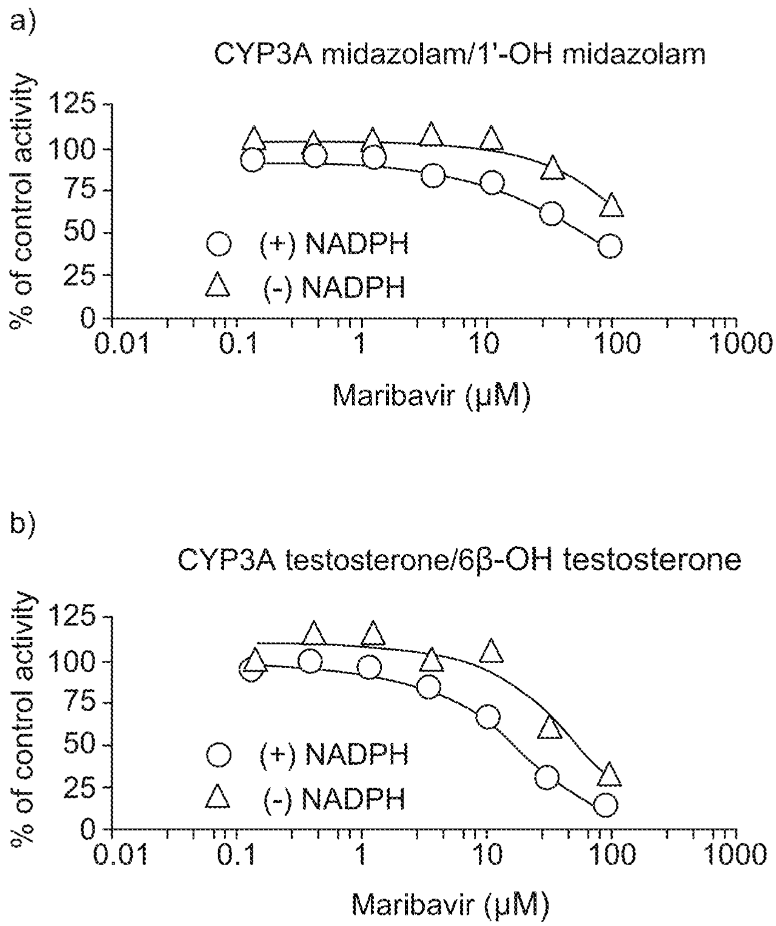

a)
CYP3A midazolam/1'-OH midazolam
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (μM)
b)
CYP3A testosterone/6β-OH testosterone
% of control activity
125
100
75
50
25
0
(+) NADPH
(-) NADPH
0.01
0.1
1
10
100
1000
Maribavir (μM)

Figure 1

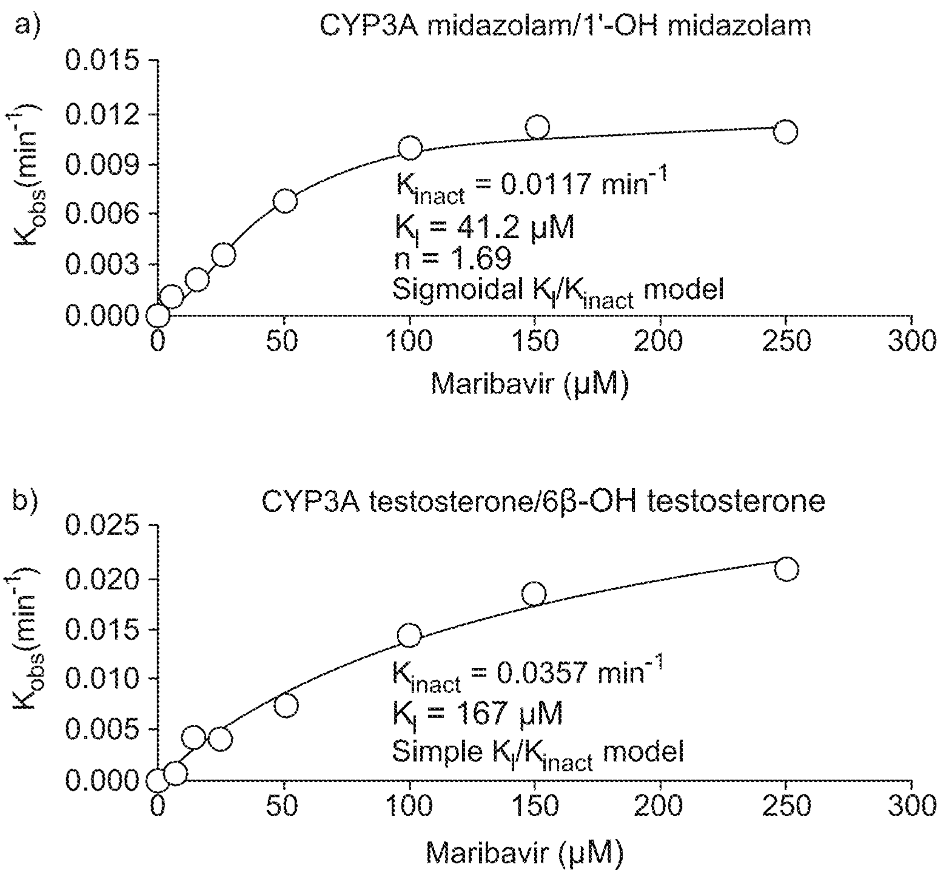

a)
CYP3A midazolam/1'-OH midazolam
0.015
0.012
0.009
0.006
0.003
0.000
$K_{obs}(min^{-1})$
$K_{inact} = 0.0117\ min^{-1}$
$K_I = 41.2\ \mu M$
n = 1.69
Sigmoidal K_I/K_{inact} model
0
50
100
150
200
250
300
Maribavir (μM)
b)
CYP3A testosterone/6β-OH testosterone
0.025
0.020
0.015
0.010
0.005
0.000
$K_{obs}(min^{-1})$
$K_{inact} = 0.0357\ min^{-1}$
$K_I = 167\ \mu M$
Simple K_I/K_{inact} model
0
50
100
150
200
250
300
Maribavir (μM)

Figure 2

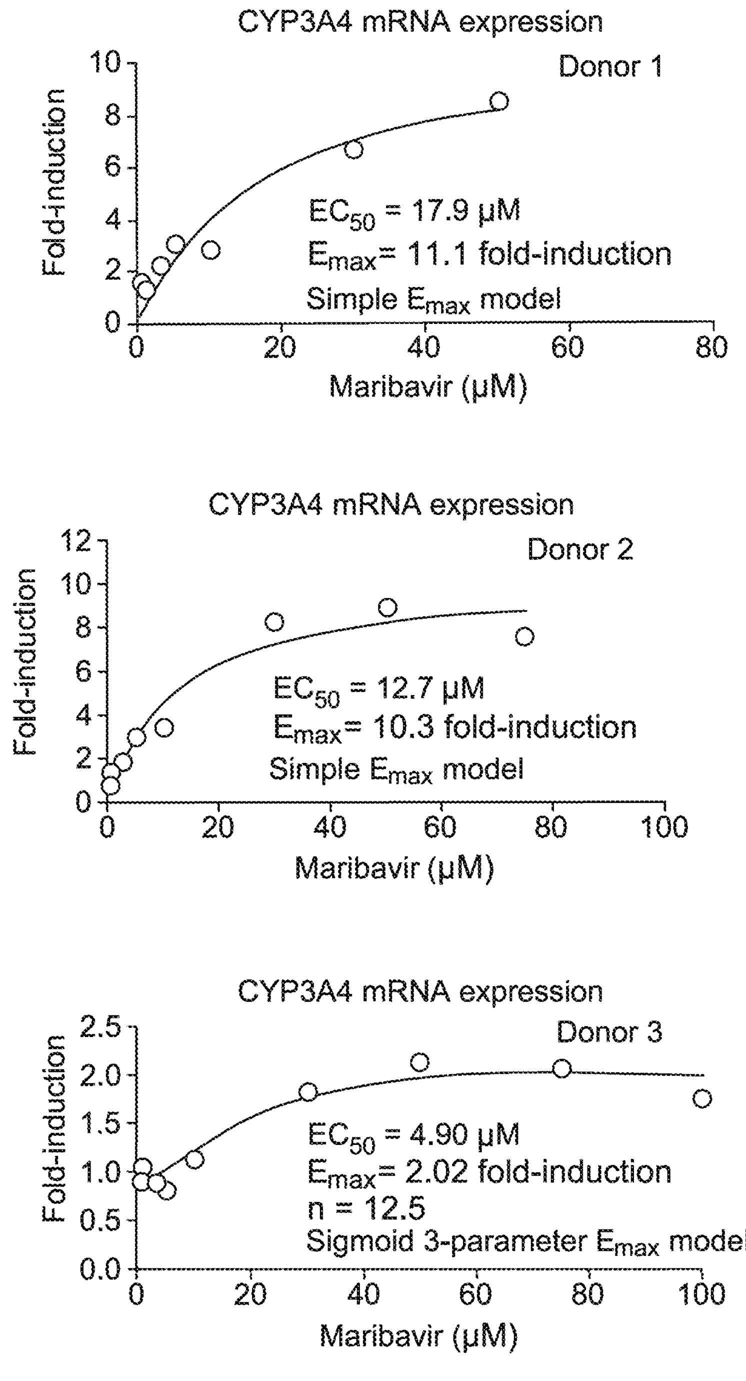

CYP3A4 mRNA expression
Donor 1
Fold-induction
Maribavir (μM)
EC50 = 17.9 μM
Emax= 11.1 fold-induction
Simple Emax model
CYP3A4 mRNA expression
Donor 2
Fold-induction
Maribavir (μM)
EC50 = 12.7 μM
Emax= 10.3 fold-induction
Simple Emax model
CYP3A4 mRNA expression
Donor 3
Fold-induction
Maribavir (μM)
EC50 = 4.90 μM
Emax= 2.02 fold-induction
n = 12.5
Sigmoid 3-parameter Emax model

Figure 3

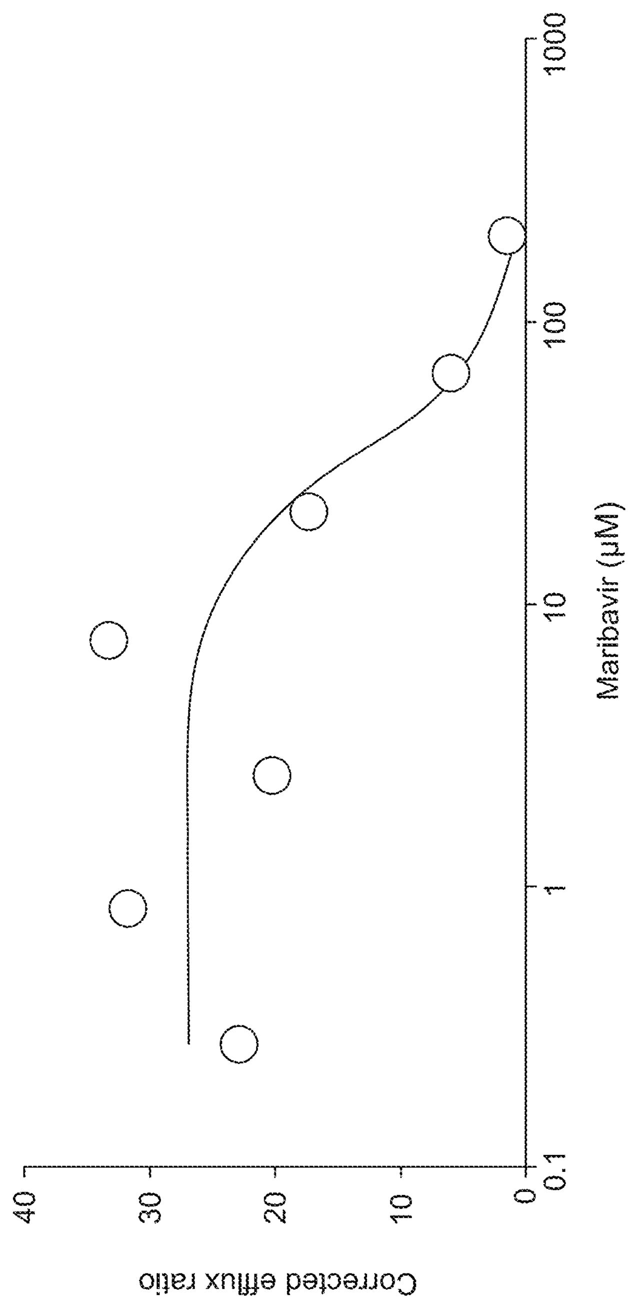

Corrected efflux ratio
40
30
20
10
0
0.1
1
10
100
1000
Maribavir (μM)

Figure 4

USE OF MARIBAVIR IN TREATMENT REGIMENS

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation application of U.S. application Ser. No. 18/646,035, filed Apr. 25, 2024, which is a continuation application of U.S. application Ser. No. 18/400,304, filed Dec. 29, 2023, which is a continuation application of PCT Application No. PCT/US22/50341, filed Nov. 11, 2022, which claims priority to U.S. Provisional Application No. 63/281,206, filed Nov. 19, 2021, each of which is incorporated herein by reference in its entirety.

BACKGROUND

Cytomegalovirus (CMV) infection and disease are significant post-transplant complications and are associated with substantial morbidity and reduced long-term survival among transplant recipients. There are currently no approved therapies for the treatment of CMV infection in transplant recipients.

Transplant patients often receive numerous concomitant medications to manage their comorbidities. Characterization of drug-drug interaction properties and pharmacological properties of maribavir is useful to inform potential drug-drug interactions and dosing strategies when administering with co-medications.

SUMMARY

Maribavir is a benzimidazole riboside, and is an orally available antiviral medication against cytomegalovirus (CMV). Maribavir is also known under its trade name LIVTENCITY™. Typically, patients (e.g., adults and/or children over 12 years of age and weighing at least 35 kg) are administered 400 mg if maribavir orally, twice daily. However, in some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a cytochrome P450 3A4 (CYP3A4) inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or a CYP3A4 inducer is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir; and

discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a CYP3A4 inducer; and

increasing the amount of maribavir administered to the patient.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments a CYP3A4 inducer decreases maribavir exposure. In some embodiments, an initial therapeutically effective amount of maribavir is administered to the patient, and the amount of maribavir is increased when co-administering a CYP3A4 inducer. In some embodiments, an initial therapeutically effective amount of maribavir is about 400 mg, orally twice daily. In some embodiments, an amount of maribavir is increased to an amount between about 800 mg and about 1200 mg, orally twice daily. In some embodiments, the provided methods further comprising a step of monitoring patient blood (e.g., whole blood or plasma) concentration of maribavir and/or the CYP3A4 inducer.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an immunosuppressant is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, the immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, provided methods further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL.

In some embodiments, tacrolimus is administered at an initial dose of between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax.

In some embodiments, a dose of tacrolimus is adjusted when co-administered with maribavir. In some embodiments, maribavir is co-administered at a concentration of 400 mg, 800 mg, or 1200 mg, administered twice daily. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or an antiarrhythmic (e.g., digoxin) is affected. In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an antiarrhythmic (e.g., digoxin). In some embodiments, provided methods further comprise monitoring the levels of digoxin in patient serum. In some embodiments, provided methods further comprise the step of reducing the amount of digoxin administered to the patient (e.g., to the amount of antiarrhythmic that was administered prior to commencing administration of maribavir).

In some aspects, the present disclosure demonstrates, among other things, potential drug-drug interactions with maribavir and how exposure to maribavir and/or ganciclovir or valganciclovir is affected. In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:

- administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and
- discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.

In some embodiments, concentration of maribavir and/or the other drug being co-administered (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration.

In some embodiments, the patient is refractory to treatment with one or more other drugs to treat CMV infection (e.g., ganciclovir, valganciclovir, cidofovir, or foscarnet). In some embodiments, maribavir is administered with or without food. In some embodiments, the patient is a transplant recipient (e.g., a hematopoietic stem cell transplant recipient or a solid organ transplant recipient). In some embodiments, the patient is an adult or a child over 12 years old and greater than 35 kg.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**. IC_{50} shift of maribavir in a concentration-dependent assay with HLM for CYP3A with midazolam (A) or testosterone (B) as a probe substrate. CYP, cytochrome P450; HLM, human liver microtomes; IC_{50}, half maximal inhibitory concentration; NADPH, nicotinamide-adenine dinucleotide phosphate.

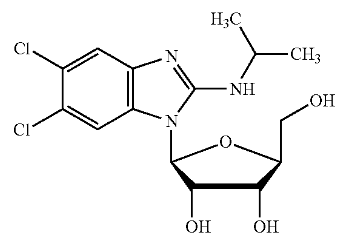
FIG. **2**. Observed enzyme inactivation rate constant vs. inhibitor concentration for TDI of CYP3A by maribavir with (A) midazolam or (B) testosterone as probe substrate. CYP, cytochrome P450; K_I, inhibitor concentration at half maximal enzyme inactivation; k_{inact}, maximum enzyme inactivation rate constant; k_{obs}, observed enzyme inactivation rate constant; TDI, time-dependent inhibition.

FIG. **3**. Estimation of induction E_{max} and EC_{50} of CYP3A4 mRNA expression by maribavir through non-linear regression of data obtained from three donors of human hepatocytes. For donor 3, n is the sigmoidicity coefficient that accommodates the shape of the curve; EC_{50}, half maximal effective concentration; E_{max}, maximum effect.

FIG. **4**. Estimation of IC_{50} of maribavir on P-gp efflux of digoxin (as determined by the corrected efflux ratio) in cultured Caco-2 cells. IC_{50}, half maximal inhibitory concentration; P-gp, P-glycoprotein.

DETAILED DESCRIPTION

Maribavir ((2S,3S,4R,5S)-2-(5,6-dichloro-2-(isopropylamino)-1H-benzo[d]imidazol-1-yl)-5-(hydroxymethyl)tetrahydrofuran-3,4-diol), a compound having the chemical structure:



is a potent and orally bioavailable antiviral for the treatment of cytomegalovirus (CMV) infection and disease in transplant recipients. Transplant recipients are at significant risk of CMV infection and also often receive numerous concomitant medications to manage their comorbidities. Thus, evaluating the drug-drug interaction potential between maribavir and prospective therapies is useful. Further, understanding the clinical pharmacology of maribavir is necessary to define the optimal dosing strategy in transplant recipients who often have multiple comorbidities and require complex concurrent medication regimens.

1. Definitions

As used herein, the term "about," when used in reference to a numerical value, means±10% of that value. For example, a dose that comprises "about 100 mg" of maribavir encompasses any amount of maribavir within a range of 90 mg to 110 mg.

As used herein, the term "reference" describes a standard or control relative to which a comparison is performed. For example, in some embodiments, an agent, animal, individual, population, sample, sequence or value of interest is compared with a reference or control agent, animal, individual, population, sample, sequence or value. In some embodiments, a reference or control is tested and/or determined substantially simultaneously with the testing or determination of interest. In some embodiments, a reference or control is a historical reference or control, optionally embodied in a tangible medium. Typically, as would be understood by those skilled in the art, a reference or control is determined or characterized under comparable conditions or circumstances to those under assessment. Those skilled in the art will appreciate when sufficient similarities are present to justify reliance on and/or comparison to a particular possible reference or control.

The terms "treat" or "treating," as used herein, refers to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition. As used herein, the terms "treatment," "treat," and "treating" refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a disorder or condition, or one or more symptoms of the disorder or condition, as described herein. In some embodiments, treatment may be administered after one or more symptoms have developed. In some embodiments, the term "treating" includes halting the progression of a disease or disorder. Treatment may also be continued after symptoms have resolved, for example to prevent or delay their recurrence. Thus, in some embodiments, the term "treating" includes preventing relapse or recurrence of a disease or disorder.

2. Dose Regimens

In some aspects, the present disclosure recognizes that, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir and/or the co-administered drug must be monitored, altered (e.g., increased or decreased), or discontinued.

In some embodiments, maribavir is administered to patients at about 400 mg orally, twice daily ("standard dose"). In some embodiments, a patient has received an initial therapeutically effective amount of maribavir, e.g., prior to receiving the additional co-administered drug. In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, maribavir is administered as a tablet. In some embodiments, maribavir is administered as a tablet comprising 200 mg of maribavir. In some embodiments, provided methods comprise administering maribavir to a patient with or without food.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of maribavir is altered (e.g., increased or decreased) from the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily.

In some embodiments, when co-administered with certain drugs (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir), the dose of the co-administered drug is altered (e.g., increased or decreased) from its recommended amount provided by its label. In some embodiments, the dose of the co-administered drug is decreased as compared to its standard dosing regimen. In some embodiments, provided methods further comprising the step of monitoring the levels of a coadministered drug after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of a coadministered drug administered to the patient (e.g., to the amount of a coadministered drug that was administered prior to commencing administration of maribavir).

a. Maribavir and CYP3A4 Inducers

In some embodiments, the present disclosure provides the recognition that maribavir is primarily metabolized by CYP3A4. Additionally or alternatively, the present disclosure provides the recognition that drugs that are CYP3A4 inducers decrease maribavir plasma concentrations and may result in a reduced virologic response. In some embodiments, coadministration of maribavir with certain CYP3A4 inducers is not recommended, and/or dosing regimens of one or both drugs should be altered.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a CYP3A4 inducer prior to administration of maribavir. In some embodiments, methods further comprise discontinuing administration of the CYP3A4 inducer prior to administration of maribavir. In some embodiments, method further comprise increasing the amount of maribavir administered to the patient.

In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort. In some embodiments, a CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital. In some embodiments, a CYP3A4 inducer is a strong CYP3A4 inducer. Strong CYP3A4 inducer include, e.g., rifampin, rifabutin, and St. John's wort. In some embodiments, CYP3A4 inducers may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for a given indication).

In some embodiments, a CYP3A4 inducer is also a p-Gp inducer. In some embodiments, a CYP3A4 inducer also decreases maribavir exposure.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and

discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering an initial therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and

increasing the amount of maribavir administered to the patient.

In some embodiments, an initial therapeutically effective amount of maribavir is the standard dose (400 mg orally, twice daily). In some embodiments, the amount of maribavir is increased to about 800 mg or about 1200 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 800 mg orally, twice daily. In some embodiments, the amount of maribavir is increased to about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is carbamazepine, the amount of maribavir is increased to about 800 mg of maribavir orally twice daily. In some embodiments, wherein the CYP3A4 inducer is phenytoin or phenobarbital, the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising:

administering a therapeutically effective amount of maribavir to the patient (e.g., 800 mg or about 1200 mg orally, twice daily), wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.

In some embodiments, wherein the patient has received a CYP3A4 inducer prior to administration of maribavir, a therapeutically effective amount of maribavir is about 800 mg or about 1200 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is carbamazepine, the amount of maribavir administered is about 800 mg orally, twice daily. In some embodiments, wherein the CYP3A4 inducer is phenytoin or phenobarbital, the amount of maribavir administered is about 1200 mg orally, twice daily.

Drug-drug interaction potential of maribavir with cytochrome P450 enzymes and P-glycoprotein (P-gp) was thoroughly characterized. The reversible inhibition, time-dependent inhibition, and induction of CYPs were evaluated in the presence of maribavir using human cells or human-derived cell lines. The inhibition of P-gp was also assessed.

Maribavir was not a reversible inhibitor of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, or CYP3A4 but was a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 (the half maximal inhibitory concentration, or IC_{50}, was 40, 18, and 35 μM, respectively).

Maribavir was not a time-dependent inhibitor of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. However, it was a time-dependent inhibitor of CYP3A4, evident from IC_{50} shift values presented in FIG. **1**. With midazolam and testosterone as probe substrates, IC_{50} values for maribavir were greater than 1.9 and 3.2 μM, respectively. Maribavir's concentration at half-maximal enzyme inactivation, or K_I, of CYP3A was 41.2 μM and 167 μM with midazolam and testosterone, respectively, as shown in FIG. **2**.

Maribavir was not an inducer of CYP1A2 or CYP2B6 mRNA, but was a weak in vitro inducer for CYP3A4 mRNA, exhibiting a greater than 14-fold increase in mRNA induction and a greater than 30% increase in positive control levels. However, effects varied across donors, as FIG. **3** shows. The half-maximal effective concentration, or EC_{50}, of maribavir was greater than 5 μM for all donors.

As shown in FIG. **4**, maribavir demonstrated concentration-dependent inhibition of P-gp-mediated digoxin efflux. The IC_{50} of maribavir in this case was 33.8 μM.

The data presented herein, specifically in Example 1, demonstrate the concentrations of maribavir required for the observed levels of in vitro inhibition or induction of P-gp and/or some CYPs.

b. Maribavir and Immunosuppressants

In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of certain immunosuppressant drugs that are CYP3A4 and/or P-gp substrates, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, immunosuppressant drug levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of maribavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating CMV infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient (e.g., 400 mg orally, twice daily), wherein the patient is receiving or has received an immunosuppressant. In some embodiments, methods further comprise monitoring the levels of immunosuppressant (e.g., in whole blood or plasma). In some embodiments, methods further comprise reducing the amount of immunosuppressant administered to a patient. In some embodiments, maribavir also increases immunosuppressant exposure. In some embodiments, provided methods further comprise the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprise the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).

In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, prednisone, and mycophenolate. In some embodiments, an immunosuppressant is tacrolimus. In some embodiments, an immunosuppressant is everolimus. In some embodiments, an immunosuppressant is cyclosporine. In some embodiments, an immunosuppressant is sirolimus.

In some embodiments, immunosuppressants may be administered in accordance with approved dosing regimens for the patient to be treated (e.g., a US FDA approved dosage for tacrolimus). In some embodiments, tacrolimus is administered as a stable dose, twice daily, with a total daily dose between about 0.5 mg and about 16 mg.

In some embodiments, tacrolimus is administered at an initial dose and whole blood trough concentrations are monitored. In some embodiments, an initial dose (e.g., prior to administration of maribavir or upon co-administration of maribavir) of tacrolimus administered is between about 0.075 mg/kg/day and about 0.3 mg/kg/day. In some embodiments, tacrolimus is administered orally in capsules of 0.5 mg, 1.0 mg, or 5.0 mg each. In some embodiments, tacrolimus is administered by injection in a concentration of 5.0 mg/mL. In some embodiments, tacrolimus is administered by oral suspension in 1 mg unit-dose granule packets. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored over a period of time between about 0 months and about 12 months from the initial administration of tacrolimus. In some embodiments, observed whole blood trough concentration of tacrolimus is monitored at initial administration, during co-administration, and at discontinuation of maribavir administration. In some embodiments, observed whole blood trough concentration of tacrolimus is between about 4 and about 20 ng/mL. In some embodiments, an initial dose of tacrolimus administered is between about 0.03% and about 0.1% (w/w) per gram of ointment. In some embodiments, tacrolimus is administered topically in a base selected from the group consisting of mineral oil, paraffin, propylene carbonate, white petrolatum and white wax. In some embodiments, co-administration of maribavir increases exposure to tacrolimus by about 50%.

In some embodiments, the present disclosure provides methods of administering to a patient in need thereof a therapeutically effective amount of an immunosuppressant for the prophylaxis or treatment of organ rejection and/or graft versus host disease, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of an immunosuppressant, the improvement comprising administering a therapeutically effective amount of maribavir to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

In some embodiments, the present disclosure provides methods for prophylaxis or treatment of organ rejection and/or graft versus host disease a patient in need thereof, wherein the patient is receiving or has received a therapeutically effective amount of maribavir, the improvement comprising administering a therapeutically effective amount of an immunosuppressant to the patient. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received. In some embodiments, the improvement further comprises a step of reducing the amount of immunosuppressant received, wherein the immunosuppressant is tacrolimus.

c. Maribavir and Digoxin

Digoxin is an antiarrhythmic drug most frequently used for atrial fibrillation, atrial flutter, and heart failure. In some embodiments, the present disclosure recognizes that maribavir has the potential to increase the drug concentrations of digoxin, which is a P-gp substrate, where minimal concentration changes may lead to serious adverse events. Additionally or alternatively, digoxin levels should be frequently monitored through treatment with maribavir, and especially following initiation and after discontinuation of mariabavir, and adjust the immunosuppressant dose, as needed.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin. In some embodiments, provided methods further comprise monitoring the levels of digoxin (e.g., in whole blood or plasma). In some embodiments, provided methods further comprise reducing the amount of digoxin administered to a patient, e.g., as compared to the amount prior to co-administering maribavir, such as the US FDA approved dosage. In some embodiments, digoxin is administered as 0.5 mg, once daily.

d. Maribavir and Ganciclovir or Valganciclovir

In some embodiments, the present disclosure recognizes that maribavir may antagonize the antiviral activity of ganciclovir and/or valganciclovir by inhibiting human CMV pUL97 kinase, which is required for activation/phosphorylation of ganciclovir and valganciclovir.

In some embodiments, the present disclosure provides methods of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir and/or valganciclovir. In some embodiments, provided methods further comprise discontinuing administration of ganciclovir and/or valganciclovir prior to administration of maribavir. In some embodiments, provided methods further comprising the step of monitoring the levels of ganciclovir and/or valganciclovir (e.g., in whole blood or plasma) after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level). In some embodiments, provided methods further comprising the step of increasing the amount of ganciclovir and/or valganciclovir administered to the patient (e.g., to the amount of ganciclovir and/or valganciclovir that was administered prior to commencing administration of maribavir).

3. Patients

As described above and herein, the present methods provide administering maribavir and a co-administered drug (e.g., a CYP3A4 inducer such as carbamazepine, phenytoin, or phenobarbital, a p-glycoprotein inducer (P-gp), an immunosuppressant, an antiarrhythmic such as digoxin, ganciclovir, or valganciclovir) to a patient in need thereof. In some embodiments, a patient suffers from CMV infection. In some embodiments, a patient suffers from post-transplant CMV infection. In some embodiments, a patient is a transplant recipient. In some embodiments, a patient is a hematopoietic stem cell transplant recipient. In some embodiments, a patient is a solid organ transplant recipient (e.g., liver, kidney, lung, heart, pancreas, intestine).

In some embodiments, a patient is refractory to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory with genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory without genotypic resistance to treatment with one or more other drugs to treat CMV infection. In some embodiments, a patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet. In some embodiments, a patient is refractory to treatment with ganciclovir or valganciclovir. In some embodiments, a patient is refractory to treatment with ganciclovir. In some embodiments, a patient is refractory to treatment with valganciclovir.

In some embodiments, a patient is an adult or a child over 12 years old and greater than 35 kg. In some embodiments, a patient is an adult. In some embodiments, a patient is a child. In some embodiments, a patient is a child over 12 years old. In some embodiments, a patient is a child over 35 kg. In some embodiments, a patient is a child over 12 years old and over 35 kg.

4. Evaluating Maribavir Drug-Drug Interaction Potential Using Data From Phase 1 Clinical Studies and Nonclinical In Vitro Studies

Maribavir's drug-drug interaction potential are described in Example 3.

Accumulated data have shown that maribavir is metabolized primarily in the liver, and renal clearance is a minor route (representing less than 5 percent). VP 44469 (N-dealkylated Maribavir) was the principal metabolite of maribavir in both urine and feces. In plasma, unchanged maribavir and VP 44469 represented approximately 69% and 9.8% of total radioactivity, respectively. The hepatic metabolism of maribavir was primarily driven by cytochrome P450s. CYP3A4 and CYP1A2 were responsible for 70 to 85% and 15 to 30% of the CYP-driven pathways, respectively. Glucuronidation accounted for less than 20% of metabolism.

Clinically significant drug interactions are summarized in Table 1-A of Example 3. Potent inducers of CYP34A and P-gp decrease maribavir exposure, necessitating maribavir dose increases. Inhibitors of CYP3A4 and/or P-gp may increase maribavir exposure. However, previous safety and tolerability data indicate that dose reduction is not necessary with CYP3A4 and/or P-gp inhibitors. Maribavir may increase the exposure of immunosuppressants, so monitoring of concomitant immunosuppressants should be considered.

5. Clinical Pharmacology of Maribavir

Following oral administration, maribavir was rapidly and well absorbed, with peak concentrations generally achieved within 1 to 3 hours; exposure was unaffected by food; and bioavailability wasn't affected by crushing the tablet or when it was taken with antacids.

Maribavir was metabolized in the liver through the cytochrome P450 3A4 and 1A2 pathways. Less than 5 percent of maribavir was eliminated through the kidneys. Maribavir demonstrated a half-life of approximately 5 to 7 hours.

Maribavir was found to present a low risk for drug-to-drug interactions. Co-administration of potent CYP3A4 inducers was found to decrease the exposure of maribavir, necessitating a maribavir dose increase. Some immunosuppressants, such as tacrolimus, may be affected by maribavir, which increased tacrolimus exposure by 51%.

In conclusion, maribavir is a suitable treatment for CMV infections in a broad range of transplant recipients. It can be taken with or without food, and no dose adjustment is needed in patients with mild-to-moderate hepatic impairment or renal impairment. Maribavir has a low drug-interaction potential and there are minimal dose adjustments required (for example, only when taken concurrently with CYP3A4 inducers or certain immunosuppressants) and there are no effects on QT interval.

Exemplary Embodiments

The following numbered embodiments, while non-limiting, are exemplary of certain aspects of the present disclosure:

1. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir; and
   discontinuing administration of the CYP3A4 inducer prior to administration of maribavir.
2. The method of embodiment 1, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.
3. The method of embodiments 1 or 2, wherein the CYP3A4 inducer is a strong CYP3A4 inducer.
4. The method of any one of embodiments 1-3, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, rifabutin, and St. John's wort.
5. The method of any one of embodiments 1-4, comprising administering about 400 mg of maribavir orally to the patient twice daily.
6. The method of any one of embodiments 1-5, comprising administering maribavir to the patient with or without food.
7. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering an initial therapeutically effective amount of maribavir to the patient, wherein the patient is receiving a cytochrome P450 3A4 (CYP3A4) inducer; and
   increasing the amount of maribavir administered to the patient.
8. The method of embodiment 7, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.
9. The method of embodiment 7 or 8, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.
10. The method of any one of embodiments 7-9, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.
11. The method of any one of embodiments 7-10, wherein the patient had received a therapeutically effective amount of maribavir prior to receiving a CYP3A4 inducer ("an initial therapeutically effective amount") of about 400 mg of maribavir orally twice daily.
12. The method of any one of embodiments 7-11, wherein the amount of maribavir is increased to about 800 or about 1200 mg of maribavir orally twice daily.
13. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir is increased to about 800 mg of maribavir orally twice daily.
14. The method of any one of embodiments 7-12, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir is increased to about 1200 mg of maribavir orally twice daily.
15. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
   administering a therapeutically effective amount of maribavir to the patient, wherein the patient has received a cytochrome P450 3A4 (CYP3A4) inducer prior to administration of maribavir.
16. The method of embodiment 15, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, avasimibe, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.

17. The method of embodiment 15 or 16, wherein the CYP3A4 inducer is selected from the group consisting of rifampin, carbamazepine, phenytoin, rifabutin, phenobarbital, and St. John's wort.
18. The method of any one of embodiments 15-17, wherein the CYP3A4 inducer is selected from the group consisting of carbamazepine, phenytoin, and phenobarbital.
19. The method of any one of embodiments 15-18, wherein the amount of maribavir administered is about 800 or about 1200 mg orally twice daily.
20. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is carbamazepine, and the amount of maribavir administered is about 800 mg orally twice daily.
21. The method of any one of embodiments 15-19, wherein the CYP3A4 inducer is phenytoin or phenobarbital, and the amount of maribavir administered is about 1200 mg orally twice daily.
22. The method of any one of embodiments 1-21, wherein the CYP3A4 inducer decreases maribavir exposure.

23. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received an immunosuppressant.

24. The method of embodiment 23, further comprising the step of monitoring the levels of immunosuppressant after commencing administration of Maribavir (e.g., with comparison to a reference or standard level).
25. The method of embodiment 23 or 24, further comprising the step of reducing the amount of the immunosuppressant administered to the patient.
26. The method of any one of embodiments 23-25, further comprising the step of monitoring the levels of immunosuppressant after discontinuing administration of maribavir (e.g., with comparison to a reference or standard level).
27. The method of embodiment 26, further comprising the step of increasing the amount of immunosuppressant administered to the patient (e.g., to the amount of immunosuppressant that was administered prior to commencing administration of maribavir).
28. The method of any one of embodiments 23-27, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, sirolimus, prednisone, and mycophenolate.
29. The method of any one of embodiments 23-28, wherein immunosuppressant is selected from the group consisting of tacrolimus, cyclosporine, everolimus, and sirolimus.
30. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received digoxin.
31. The method of embodiment 30, further comprising the step of monitoring the levels of digoxin.
32. The method of embodiment 30 or 31, further comprising the step of reducing the amount of the digoxin administered to the patient.
33. The method of one of embodiments 1-32, wherein the patient is refractory to treatment with one or more other drugs to treat CMV infection.
34. The method of one of embodiments 1-33, wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.
35. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment with genotypic resistance.
36. The method of any one of embodiments 1-34, wherein the patient is refractory to treatment without genotypic resistance.
37. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom comprising:
    administering a therapeutically effective amount of maribavir to the patient, wherein the patient is receiving or has received ganciclovir or valganciclovir prior to administration of maribavir; and
    discontinuing administration of the ganciclovir or valganciclovir prior to administration of maribavir.
38. The method of any one of embodiments 7-37, comprising administering maribavir to the patient with or without food.
39. The method of any one of embodiments 1-38, wherein the patient is a transplant recipient.
40. The method of any one of embodiments 1-39, wherein the patient is a hematopoietic stem cell transplant recipient.
41. The method of any one of embodiments 1-39, wherein the patient is a solid organ transplant recipient.
42. The method of any one of embodiments 1-41, wherein the patient is an adult or a child over 12 years old and greater than 35 kg.

EXAMPLES

Example 1—In Vitro Profiling for Potential Cytochrome P450 Drug-Drug Interaction by Maribavir

Inhibition or induction of the cytochrome P450 enzymes (CYPs) are among the most commonly observed mechanisms of drug-drug interactions.

In vitro systems such as human liver microsomes (HLM), recombinant enzymes, human hepatocytes, and other human-derived cells lines have long been used as validation systems for characterization of potential drug-drug interactions prior to clinical studies.

Reversible CYP Inhibition

HLM were used to evaluate maribavir's half maximal inhibitory concentration (IC_{50}) on inhibition of activities of nine CYP isoforms. Probe substrates were phenacetin (CYP1A2), coumarin (CYP2A6), bupropion (CYP2B6), paclitaxel (CYP2C8), tolbutamide (CYP2C9), (S)-mephenytoin (CYP2C19), dextromethorphan (CYP2D6), clorzoxazone (CYP2E1), and midazolam and testosterone (CYP3A4).

CYP activities with 0, 0.1, 0.3, 1, 3, 10, 30, and 100 μM of maribavir were evaluated.

Incubation of maribavir in HLM at up to 100 μM resulted in no significant inhibition of CYP2A6, CYP2B6, CYP2C8, CYP2D6, CYP2E1, and CYP3A4. Maribavir is a weak inhibitor of CYP1A2, CYP2C9, and CYP2C19 with IC_{50} of 40, 18, and 35 μM, respectively.

Time-Dependent CYP Inhibition

Time-dependent inhibition (TDI) potentials for various CYP enzymes were evaluated by a 30 minute pre-incubation of maribavir with HLM in the presence and absence of nicotinamide-adenine dinucleotide diphosphate (NADPH) followed by a CYP enzyme activity assay.

Enzyme inactivation kinetics of maribavir for the TDI of CYP3A, with midazolam and testosterone as probe substrates, was evaluated by pre-incubation of HLM with maribavir at various concentrations with NADPH for six different pre-incubation time frames. CYP3A activity was measured by determining the formation of CYP3A probe metabolite. Inhibitor concentration at half maximal enzyme inactivation (K_I), and maximum enzyme inactivation rate constant (k_{inact}), were estimated using non-linear least-square regression.

The IC_{50} shift values of maribavir were <1 μM for CYP1A2, CYP2C8, CYP2C9, CYP2C19, and CYP2D6 in HLM. The IC_{50} shift values of maribavir for CYP3A were greater than 1.9 and 3.2 μM, with midazolam and testosterone as probe substrate, respectively (FIG. **1**). Therefore, maribavir, in concentrations of up to 100 μM, is unlikely a TDI of CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6; but it is likely a TDI of CYP3A.

K_I and k_{intact} of maribavir for TDI of CYP3A with midazolam as substrate (FIG. **2**A) were 41.2 μM and 0.0117 min^{-1}, respectively. K_I and K_{intact} of maribavir for TDI of CYP3A with testosterone as a substrate (FIG. **2**B) were 167 μM and 0.0357 min^{-1}, respectively.

CYP Induction

Fresh human hepatocytes were treated with culture medium with maribavir concentrations of 36 μM, 144 μM, and 480 μM.

Positive controls were 50 μM omeprazole (CYP1A2), 1 mM phenobarbital (CYP2B6), and 50 μM rifampicin (CYP3A).

Incubation was conducted at 37° C. for 72 hours, with daily replacement of medium with compound. Total RNA was isolated, and cDNA was synthesized from up to 1 μg of total isolated RNA. Analysis of CYP expression was performed using qPCR with CYP-specific probes.

Half maximal effective concentration (EC_{50}) and maximum effect (E_{max}) values for CYP3A4 mRNA induction were further determined by a concentration-response curve in three donors with maribavir at a concentration between 0-100 μM.

Maribavir displayed varying degrees of CYP1A2 and CYP2B6 mRNA induction in human hepatocytes. The fold-increase in mRNA was not concentration-dependent, it was variable among donors, and it did not exceed 11% of the levels of positive controls. Therefore, maribavir is likely not an inducer of CYP1A2 or of CYP2B6 mRNA at clinically relevant concentrations.

At 36 μM, maribavir displayed a greater than 14-fold increase in CYP3A4 mRNA induction and greater than 30% of positive control (rifampicin) levels. The EC_{50} and E_{max} values for CYP3A4 induction were then further determined by a concentration-response curve in three donors.

Maribavir also displayed donor-dependent induction of CYP3A4 mRNA upon determination of its induction E_{max} and EC_{50} properties (FIG. **3**).

Example 2—Quantitative Prediction of Exposure of Maribavir Using Prior In Vitro and In Vivo Data

A Physiologically-based Pharmacokinetic ("PBPK") model based on prior in vitro and in vivo information on the metabolism and kinetics of maribavir was constructed with the aim of predicting plasma concentration-time profiles of maribavir and to evaluate the likely impact of coadministration of CYP3A4 inhibitors and inducers on the kinetics of maribavir in healthy subjects.

Model Development

A combination of in vitro data and clinical pharmacokinetic data obtained following a single dose of 400 mg maribavir were used to develop the PBPK model. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 2A. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data. The C_{max} was marginally under predicted, which could be expected because the PBPK model was optimized for better prediction of C_{12h}.

TABLE 2A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (400 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	97.98	11.98
Median	91.97	11.48
Geometric Mean	90.40	11.56
90% CI around geometric mean (lower limit)	84.45	11.06
90% CI around geometric mean (upper limit)	96.76	12.08
5^{th} percentile	46.52	7.61
95^{th} percentile	165.11	18.08
SD	39.33	3.26
% CV	40	27
OBSERVED		
Mean	97.8	16.7
SD	28.6	5.72
% CV	29.2	34.3
RATIO MEAN PREDICTED/OBSERVED	1.00	0.72

A renal clearance of 0.051 L/H was obtained following oral administration of 50 mg to 1600 mg single doses of maribavir. In vitro human liver microsomal and recombinant CYP data combined with oral clearance values reported from two studies (n=46 individuals; Ma et al., 2006) were used to assign the relative contribution of CYP3A4 to the clearance of maribavir. Distribution models evaluated included a full PBPK model and a minimal PBPK model, both of which consider liver and intestinal metabolism. Mass balance data from the [^{14}C] ADME study indicated a fraction absorbed of 0.83 or greater following oral administration of 400 mg dose of maribavir. Absorption models evaluated included a simple first-order absorption model and a more mechanistic "Advanced Dissolution Absorption and Metabolism" (ADAM) model. Although maribavir has been shown to be a P-gp substrate in vitro, relatively linear pharmacokinetics across the dose range of interest, (400 mg to 1600 mg) informed the selection of the simpler first order absorption model for the use in this PBPK study.

Model Verification

A comparison of observed and predicted plasma concentrations of maribavir following single oral doses of 800 mg and 1600 mg to healthy patients were. Mean concentrations for the total virtual population (n=100) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Tables 3A and 4A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 3A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (800 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	195.71	24.05
Median	179.48	23.22
Geometric Mean	180.00	23.23
90% CI around geometric mean (lower limit)	167.95	22.23
90% CI around geometric mean (upper limit)	192.91	24.27
5^{th} percentile	93.03	15.22
95^{th} percentile	330.22	36.17
SD	80.04	6.47
% CV	41	27
OBSERVED		
Mean	183	26.4
SD	69.1	6.85
% CV	38.0	26.0
RATIO MEAN PREDICTED/OBSERVED	1.07	0.91

TABLE 4A

Predicted and Observed C_{max} and $AUC_{(0-\infty)}$ values for maribavir after a single oral dose of maribavir (1600 mg).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L
PREDICTED		
Mean	391.43	48.10
Median	358.95	46.45
Geometric Mean	359.99	46.45
90% CI around geometric mean (lower limit)	335.89	44.46
90% CI around geometric mean (upper limit)	385.82	48.54
5^{th} percentile	186.06	30.44
95^{th} percentile	660.44	72.34
SD	106.09	12.94
% CV	41	27
OBSERVED		
Mean	437	48.8
SD	163	7.88
% CV	37.4	16.1
RATIO MEAN PREDICTED/OBSERVED	0.90	0.99

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral 400 mg BID doses of maribavir was performed. Mean concentrations for the total virtual population (n=150) are displayed, and associated mean C_{max} and $AUC_{(0-\infty)}$ values are compared in Table 5A. The simulated and observed maribavir concentrations were in reasonable agreement. In addition, the predicted mean maribavir AUC values were within 1.25-fold of the observed data.

TABLE 5A

Predicted and Observed $C_{12\ h}$, C_{max} and $AUC_{(0-\infty)}$ values for maribavir after multiple oral dose of maribavir (400 mg BBID, 5 doses).

	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L
PREDICTED			
Mean	96.49	14.31	2.99
Median	90.13	14.05	2.44
Geometric Mean	89.31	13.78	2.13
90% CI around geometric mean (lower limit)	85.01	13.31	1.89
90% CI around geometric mean (upper limit)	93.82	14.26	2.15
5^{th} percentile	46.35	8.80	0.36
95^{th} percentile	162.88	20.59	7.76
SD	37.93	3.96	2.2
% CV	39	28	73
OBSERVED			
Mean	89.9	16.9	2.54
SD	24.5	5.22	1.37
% CV	27.2	30.8	53.8
RATIO MEAN PREDICTED/OBSERVED	1.07	0.85	1.18

A comparison of observed and predicted plasma concentrations of maribavir following a single oral dose of 400 mg administered in the absence of ketoconazole (a CYP3A4 inhibitor) and 1 hour after a single 400 mg dose of ketoconazole to healthy subjects was performed. Mean simulated and observed plasma maribavir concentrations are compared, and associated geometric mean C_{max} and $AUC_{(0-\infty)}$ values for maribavir in the presence and absence of ketoconazole are shown in Table 6A and are within 1.25-fold of the observed data.

TABLE 6A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$AUC_{(0-\infty)}$ mg/L*h	C_{max} mg/L	$AUC_{(0-\infty)}$	C_{max}
PREDICTED						
Mean	98.96	12.27	150.60	14.31	1.54	1.17
Median	93.51	11.91	140.38	13.75	1.49	1.15
Geometric Mean	92.29	11.81	141.16	13.77	1.53	1.17

TABLE 6A-continued

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-\infty)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of ketoconazole with associated variability in healthy adults.

	Maribavir		Maribavir + Ketoconazole		Ratio	
	$AUC_{(0-\infty)}$	C_{max}	$AUC_{(0-\infty)}$	C_{max}		
	mg/L*h	mg/L	mg/L*h	mg/L	$AUC_{(0-\infty)}$	C_{max}
90% CI around geometric mean (lower limit)	88.30	11.43	135.28	13.32	1.51	1.16
90% CI around geometric mean (upper limit)	96.45	12.20	147.30	14.23	1.55	1.17
5^{th} percentile	48.52	7.50	77.98	8.85	1.27	1.09
95^{th} percentile	169.37	17.86	246.30	21.20	1.97	1.29
SD	37.19	3.4	54.7	4.0	0.21	0.06
% CV	38	28	36	28	14	5
OBSERVED						
Mean	126	19.8	194	21.8	1.55	1.12
SD	41.6	3.9	64.3	4.4	0.23	0.22
% CV	33	20	33	20	15	20
Geometric mean	119	19.4	183	21.3	1.53	1.10
90% CI around geometric mean (lower limit)					1.44	1.01
90% CI around geometric mean (upper limit)					1.63	1.19
MEAN RATIO PREDICTED/OBSERVED	0.79	0.62	0.78	0.66	0.99	1.05
GEOMEAN RATIO PREDICTED/OBSERVED	0.78	0.61	0.77	0.65	1.00	1.06

A comparison of observed and predicted plasma concentrations of maribavir following multiple oral doses of 400 mg administered in the absence of rifampicin and during concurrent treatment with rifampicin was performed. Subjects took maribavir 400 mg BID on Days 1-3 (5 doses), then rifampicin 600 mg QD on Days 4 through 12, followed by both maribavir 400 mg (BID) and rifampicin 600 mg QD on Days 13 and 14, with final doses of maribavir and rifampicin on the morning of Day 15. Associated geometric mean C_{12h}, C_{max}, and $AUC_{(0-12h)}$ values for maribavir in the presence or absence of rifampicin are shown in Table 7A. Predicted values for Cmax and AUC are within 1.25-fold of the observed parameter values.

TABLE 7A

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-12\ h)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of rifampicin with associated variability in healthy adults.

	Maribavir 400 mg BID			400 mg BID Maribavir + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
PREDICTED									
Mean	95.28	14.26	2.88	35.13	7.97	0.340	0.37	0.56	0.09
Median	90.02	13.98	2.61	30.52	7.57	0.145	0.35	0.55	0.06
Geometric Mean	88.77	13.78	2.07	30.53	7.44	0.088	0.34	0.54	0.04
90% CI around geometric mean (lower limit)	84.25	13.30	2.03	28.36	7.07	0.083	0.33	0.52	0.04
90% CI around geometric mean (upper limit)	93.54	14.29	2.10	32.86	7.83	0.093	0.36	0.56	0.04
5^{th} percentile	41.15	8.76	0.39	12.44	3.95	0.002	0.17	0.33	0.01
95^{th} percentile	151.23	20.23	6.49	67.37	13.43	1.507	0.62	0.79	0.33
SD	35.28	3.7	2.1	18.64	2.93	0.53	0.14	0.14	0.11
% CV	37	26	73	53	37	151	37	25	112
OBSERVED									
Mean	89.9	16.9	2.54	35.1	10.2	0.41	0.41	0.65	0.17
SD	24.5	5.22	1.37	7.07	2.3	0.21	0.08	0.24	0.07

TABLE 7A-continued

Simulated and observed PK parameters and geometric mean C_{max} and $AUC_{(0-12\ h)}$ ratios for maribavir following a single oral dose of maribavir in the presence and absence of rifampicin with associated variability in healthy adults.

	Maribavir 400 mg BID			400 mg BID Maribavir + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
% CV	86.6	16.2	2.24	34.5	9.88	0.41	0.40	0.61	0.18
RATIO MEAN PREDICTED/OBSERVED	1.06	0.84	1.14	1.00	0.78	0.83	0.91	0.87	0.56
RATIO GEOMETRIC MEAN PREDICTED/OBSERVED	1.03	0.85	0.92	0.88	0.75	0.22	0.85	0.89	0.24

Model Verification

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 8A. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for this interaction is compared with the PK parameters for a 400 mg BID dose of maribavir in Table 9A.

Maribavir C_{max} from 800 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone, however, $AUC_{(0-12)}$ was 23% lower and C_{12h} was 75% lower. This indicates that increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of rifampicin on the reduced therapeutic exposure to maribavir.

TABLE 8A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$ mg/L* h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	194.99	28.96	5.95	73.25	16.40	0.73	0.37	0.56	0.10
Median	189.24	28.37	5.34	68.57	15.57	0.31	0.37	0.56	0.06
Geometric Mean	182.49	27.95	4.39	63.52	15.22	0.20	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	174.67	27.09	4.36	59.54	14.54	0.18	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	190.66	28.85	4.43	67.77	15.94	0.21	0.36	0.56	0.05
5^{th} percentile	94.23	17.97	0.93	24.65	7.93	0.00	0.18	0.35	0.00
95^{th} percentile	311.05	43.06	13.85	136.31	27.80	2.82	0.62	0.79	0.34
SD	68.94	7.68	4.00	38.65	6.25	1.09	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 9A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	95.28	14.26	2.88	73.25	16.40	0.73	0.77	1.15	0.25
Median	90.02	13.98	2.61	68.57	15.57	0.31	0.76	1.11	0.12
Geometric Mean	88.77	13.78	2.07	63.52	15.22	0.20	0.72	1.10	0.10

TABLE 9A-continued

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	59.54	14.54	0.18	0.71	1.09	0.09
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	67.77	15.94	0.21	0.72	1.12	0.10
5^{th} percentile	41.15	8.76	0.39	24.65	7.93	0.00	0.60	0.91	0.00
95^{th} percentile	151.23	20.34	6.49	136.31	27.80	2.82	0.90	1.37	0.43
SD	35.28	3.7	2.1	38.65	6.25	1.09	1.10	1.69	0.52
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 600 mg Rifampicin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 600 mg rifampicin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 10A. Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 11A.

Maribavir AUC from 1200 mg BID in the presence of 600 mg rifampicin QD was similar to that from maribavir 400 mg BID alone while the C_{max} was slightly higher. However, the geometric mean C_{12h} was 86% lower, indicating that increasing maribavir dose from 400 mg BID to 1200 mg BID cannot counteract the effect of rifampicin from an efficacy perspective (mainly driven by C_{12h}). Increasing the maribavir dose to 1600 mg BID had a minimal effect on the geometric mean C_{12h}, which was 81% lower than that with a 400 mg BID dose of maribavir.

TABLE 10A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of rifampicin 600 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	292.48	43.44	8.93	109.87	24.59	1.10	0.37	0.56	0.10
Median	283.87	42.55	8.00	102.85	23.35	0.47	0.37	0.56	0.06
Geometric Mean	273.73	41.93	6.59	95.28	22.84	0.30	0.35	0.54	0.04
90% CI around geometric mean (lower limit)	262.00	40.63	6.55	89.31	21.81	0.28	0.33	0.53	0.04
90% CI around geometric mean (upper limit)	285.99	43.27	6.62	101.65	23.91	0.31	0.36	0.56	0.05
5^{th} percentile	141.35	26.96	1.40	36.97	11.90	0.00	0.18	0.35	0.00
95^{th} percentile	466.58	64.59	20.77	204.46	41.71	4.23	0.62	0.79	0.34
SD	108.42	11.45	6.00	57.57	9.37	1.63	0.14	0.14	0.11
% CV	35	26	67	53	38	149	37	25	108

TABLE 11A

Predicted PK parameters for maribavir 1200 mg BID in the presence of rifampicin 600 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 600 mg QD Rifampicin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	95.28	14.26	2.89	109.87	24.59	1.10	1.15	1.72	0.38
Median	90.02	13.98	2.61	102.85	23.35	0.47	1.14	1.67	0.18
Geometric Mean	88.77	13.78	2.07	95.28	22.84	0.30	1.07	1.66	0.14
90% CI around geometric mean (lower limit)	84.25	13.3	2.03	89.31	21.81	0.28	1.06	1.64	0.14
90% CI around geometric mean (upper limit)	93.54	14.29	2.1	101.65	23.91	0.31	1.09	1.67	0.15
5^{th} percentile	41.15	8.76	0.39	36.97	11.90	0.00	0.90	1.36	0.00
95^{th} percentile	151.23	20.34	6.49	204.46	41.71	4.23	1.35	2.05	0.65
SD	35.28	3.7	2.1	57.57	9.37	1.63	1.63	2.53	0.78
% CV	37	26	73	53	38	149	1.43	1.46	2.04

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 20 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 12A. Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 39%, 27% and 63% in $AUC_{(0-12h)}$, C_{max} and C_{12h}, respectively.

TABLE 12A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	61.20	10.66	1.33	0.61	0.73	0.37
Median	93.26	14.23	2.40	55.41	10.44	0.85	0.63	0.75	0.37
Geometric Mean	92.65	14.09	2.29	55.66	10.21	0.72	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	52.88	9.86	0.70	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	58.60	10.57	0.73	0.62	0.74	0.31
5^{th} percentile	47.52	9.12	0.50	27.75	6.18	0.07	0.39	0.56	0.07
95^{th} percentile	169.70	21.94	7.30	112.27	16.47	4.17	0.78	0.86	0.65
SD	38.6	4.1	2.26	26.96	3.12	1.25	0.12	0.09	0.18
% CV	38	28	72	44	29	101	20	13	47

Simulation of Plasma Concentration-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h} for the population are shown in Table 13A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 14A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 100 mg phenobarbital QD were marginally lower than those from maribavir 800 mg BID alone. However, C_{min} was 63% lower than with maribavir 800 mg alone. Increasing the maribavir dose given with phenobarbital from 400 mg BID to 800 mg BID results in a mean C_{min} that is 15% lower than that with 400 mg BID maribavir alone. Thus, by increasing the dose of maribavir to 800 mg BID, the interaction is counteracted significantly.

TABLE 13A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	199.43	29.32	6.24	122.40	21.32	2.66	0.61	0.73	0.37
Median	186.51	28.46	4.81	110.83	20.88	1.70	0.63	0.75	0.37
Geometric Mean	185.31	28.19	4.58	111.33	20.42	1.43	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	105.75	19.72	1.41	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	117.20	21.14	1.45	0.62	0.74	0.31
5^{th} percentile	95.04	18.24	1.00	55.49	12.35	0.14	0.39	0.56	0.07
95^{th} percentile	339.40	43.89	14.60	224.54	32.93	8.33	0.78	0.86	0.65
SD	36.3	8.2	4.53	53.9	6.24	2.7	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	48

TABLE 14A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	122.40	21.32	2.66	1.23	1.45	0.85
Median	93.26	14.23	2.40	110.83	20.88	1.70	1.19	1.47	0.71
Geometric Mean	92.65	14.09	2.29	111.33	20.42	1.43	1.20	1.45	0.62
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	105.75	19.72	1.41	1.19	1.45	0.62
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	117.20	21.14	1.45	1.21	1.45	0.63
5^{th} percentile	47.52	9.12	0.50	55.49	12.35	0.14	1.17	1.35	0.28
95^{th} percentile	169.70	21.94	7.30	224.54	32.93	8.33	1.32	1.50	1.14
SD	38.6	4.1	2.26	53.9	6.24	2.7	1.40	1.52	1.19
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 100 mg Phenobarbital QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 100 mg phenobarbital once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 15A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 16A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 100 mg phenobarbital QD were about 2 times higher those from maribavir 400 mg BID alone while Cmin is about 28% higher. Increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenobarbital. If the higher exposure (as seen with AUC and Cmax) is not detrimental to treatment, it can be concluded that the maribavir dose should be adjusted from 400 mg BID to 1200 mg BID when co-administration with phenobarital is needed.

TABLE 15A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenobarbital 100 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L			
Mean	299.15	43.98	9.36	183.60	31.98	3.99	0.61	0.73	0.37
Median	279.77	42.69	7.21	166.24	31.31	2.55	0.63	0.75	0.37
Geometric Mean	277.96	42.28	6.88	166.99	30.63	2.15	0.60	0.72	0.31
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	158.62	29.58	2.11	0.59	0.71	0.31
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	175.80	31.71	2.18	0.62	0.74	0.31
5th percentile	142.56	27.35	1.50	83.24	18.53	0.21	0.39	0.56	0.07
95th percentile	509.10	65.83	21.89	336.81	49.40	12.50	0.78	0.86	0.65
SD	114.5	12.3	6.8	80.9	9.36	4.04	0.12	0.09	0.18
% CV	38	28	73	44	29	101	20	13	47.97

TABLE 16A

Predicted PK parameters for marvibavir 1200 mg BID in the presence of phenobarbital 100 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 100 mg QD Phenobarbital			Ratio		
	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L			
Mean	99.72	14.66	3.12	183.60	31.98	3.99	1.84	2.18	1.28
Median	93.26	14.23	2.40	166.24	31.31	2.55	1.78	2.20	1.06
Geometric Mean	92.65	14.09	2.29	166.99	30.63	2.15	1.80	2.17	0.94
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	158.62	29.58	2.11	1.79	2.17	0.93
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	175.80	31.71	2.18	1.81	2.18	0.94
5th percentile	47.52	9.12	0.50	83.24	18.53	0.21	1.75	2.03	0.42
95th percentile	169.70	21.94	7.30	336.81	49.40	12.50	1.98	2.25	1.71
SD	38.6	4.1	2.26	80.9	9.36	4.04	2.10	2.28	1.79
% CV	38	28	72	44	29	101	1.16	1.04	1.40

Simulation of Plasma Concentration-Time Profiles of Maribavir 400 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 400 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 17A.

Induction of CYP3A4 by phenobarbital resulted in a mean decrease of 42%, 31% and 64% in $AUC_{(0\text{-}12)}$, C_{max} and C_{12h}, respectively.

TABLE 17A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	56.81	10.04	1.16	0.58	0.69	0.36
Median	93.26	14.23	2.40	56.31	9.95	0.81	0.58	0.69	0.35
Geometric Mean	92.65	14.09	2.29	52.42	9.63	0.68	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	49.99	9.30	0.67	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	54.98	9.96	0.69	0.58	0.70	0.30
5^{th} percentile	47.52	9.12	0.50	26.75	5.59	0.09	0.35	0.50	0.09
95^{th} percentile	169.70	21.94	7.30	99.79	14.78	3.33	0.82	0.88	0.68
SD	38.6	4.1	2.26	23.0	2.91	1.1	0.14	0.11	0.18
% CV	38	28	72	40	29	95	24	17	51.07

Simulation of Plasma Concertation-Time Profiles of Maribavir 800 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 800 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 18A. Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 19A.

Maribavir AUC and Cmax from 800 mg BID in the presence of 300 mg phenytoin QD was marginally higher than those from maribavir 400 mg BID alone, however, C_{min} was 26% lower. Increasing the maribavir dose from 400 mg BID to 800 mg BID cannot counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}).

TABLE 18A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	199.43	29.32	6.24	113.63	20.08	2.32	0.58	0.69	0.36
Median	186.51	28.46	4.81	112.62	19.90	1.62	0.58	0.69	0.35
Geometric Mean	185.31	28.19	4.58	104.85	19.26	1.36	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	177.07	27.26	4.54	99.97	18.61	1.34	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	193.93	29.14	4.62	109.96	19.93	1.38	0.58	0.70	0.30
5^{th} percentile	95.04	18.24	1.00	53.51	11.18	0.18	0.35	0.50	0.09
95^{th} percentile	339.40	43.89	14.60	199.59	29.57	6.65	0.82	0.88	0.68
SD	76.33	8.2	4.53	46.1	5.83	2.2	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	51

TABLE 19A

Predicted PK parameters for maribavir 800 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	PREDICTED								
Mean	99.72	14.66	3.12	113.63	20.08	2.32	1.14	1.37	0.74
Median	93.26	14.23	2.40	112.62	19.90	1.62	1.21	1.40	0.68
Geometric Mean	92.65	14.09	2.29	104.85	19.26	1.36	1.13	1.37	0.59
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	99.97	18.61	1.34	1.13	1.37	0.59
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	109.96	19.93	1.38	1.13	1.37	0.60
5^{th} percentile	47.52	9.12	0.50	53.51	11.18	0.18	1.13	1.23	0.36
95^{th} percentile	169.70	21.94	7.30	199.59	29.57	6.65	1.18	1.35	0.91
SD	38.6	4.1	2.26	46.1	5.83	2.2	1.21	1.42	0.97
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of 300 mg Phenytoin QD

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses) at 1200 mg BID in the absence and presence of 300 mg phenytoin once daily from day 1 to day 12 were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}24h)}$ and C_{12h}) for the population are shown in Table 20A. Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID are presented in Table 21A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 300 mg phenytoin QD were about 2-fold higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 1200 mg BID can counteract the effect of phenytoin from an efficacy perspective (mainly driven by C_{min}). The impact of the higher exposure (AUC and C_{max}) will have to be evaluated.

TABLE 20A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of phenytoin 300 mg QD.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
	PREDICTED								
Mean	299.15	43.98	9.36	170.44	30.13	3.48	0.58	0.69	0.36
Median	279.77	42.69	7.21	168.93	29.85	2.42	0.58	0.69	0.35
Geometric Mean	277.96	42.28	6.88	157.27	28.88	2.05	0.57	0.68	0.30
90% CI around geometric mean (lower limit)	265.61	40.90	6.82	149.96	27.91	2.02	0.55	0.67	0.30
90% CI around geometric mean (upper limit)	290.89	43.71	6.94	164.94	29.89	2.08	0.58	0.70	0.30
5^{th} percentile	142.56	27.35	1.50	80.26	16.76	0.27	0.35	0.50	0.09
95^{th} percentile	509.10	65.83	21.89	299.38	44.35	9.98	0.82	0.88	0.68
SD	114.5	12.3	6.8	69.01	8.7	3.29	0.14	0.11	0.18
% CV	38	28	73	40	29	95	24	17	52

TABLE 21A

Predicted PK parameters for maribavir 1200 mg BID in the presence of phenytoin 300 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 300 mg QD Phenytoin			Ratio		
	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	99.72	14.66	3.12	170.44	30.13	3.48	1.71	2.06	1.12
Median	93.26	14.23	2.40	168.93	29.85	2.42	1.81	2.10	1.01
Geometric Mean	92.65	14.09	2.29	157.27	28.88	2.05	1.70	2.05	0.90
90% CI around geometric mean (lower limit)	88.54	13.63	2.27	149.96	27.91	2.02	1.69	2.05	0.89
90% CI around geometric mean (upper limit)	96.96	14.57	2.31	164.94	29.89	2.08	1.70	2.05	0.90
5^{th} percentile	47.52	9.12	0.50	80.26	16.76	0.27	1.69	1.84	0.54
95^{th} percentile	169.70	21.94	7.30	299.38	44.35	9.98	1.76	2.02	1.37
SD	38.2	4.1	2.26	69.01	8.7	3.29	1.81	2.12	1.46
% CV	38	28	72	40	29	95	1.05	1.04	1.30

Simulation of Plasma Concentration-Time Profiles for Maribavir 400 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 400 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0\text{-}12h)}$ and C_{12h}) for the population are shown in Table 22A.

Due to CYP3A4 induction by carbamazepine, a mean decrease of 46% in maribavir C_{12h} (efficacy marker) and a 29% and 23% decrease in AUC and C_{max} respectively, is predicted when 400 mg BID maribavir is combined with carbamazepine.

TABLE 22A

Predicted PK parameters for maribavir 400 mg BID in the presence and absence of carbamazepine.

	Maribavir 400 mg BID			Maribavir 400 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0\text{-}12\ h)}$ mg/L*h	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$ mg/L*h PREDICTED	C_{max} mg/L	$C_{12\ h}$ mg/L	$AUC_{(0\text{-}12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	67.85	11.12	1.65	0.71	0.77	0.54
Median	88.34	13.91	2.34	63.37	10.80	1.29	0.71	0.79	0.56
Geometric Mean	91.25	14.01	2.18	63.82	10.74	1.15	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	61.25	10.41	1.14	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	66.49	11.08	1.16	0.71	0.78	0.53
5^{th} percentile	48.05	8.74	0.52	36.23	7.02	0.25	0.54	0.61	0.36
95^{th} percentile	176.30	21.99	7.77	108.25	16.29	4.07	0.84	0.90	0.70
SD	40.96	4.23	2.53	24.40	2.99	1.34	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.20
3^{rd} quartile	123.18	17.26	4.22	82.72	12.92	2.34	0.77	0.09	0.61
1^{st} quartile	69.63	11.43	1.20	51.60	8.95	0.63	0.65	0.84	0.46
10^{th} percentile	55.45	9.76	0.75	40.08	7.78	0.42	0.58	0.72	0.39
90^{th} percentile	154.24	20.66	6.58	100.38	15.22	3.34	0.82	0.65	0.67

Simulation of Plasma Concentration-Time Profile Maribavir 800 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 800 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h}) for the population are shown in Table 23A. Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 24A.

Maribavir AUC and C_{max} from 800 mg BID in the presence of 400 mg carbamazepine QD were marginally higher than those from maribavir 400 mg BID alone, but C_{min} was similar to that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID can counteract the effect of carbamazepine from an efficacy perspective (mainly driven by C_{min}).

TABLE 23A

Predicted PK parameters for maribavir 800 mg BID in the presence and absence of carbamazepine.

	Maribavir 800 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	197.86	29.19	6.21	135.70	22.25	3.30	0.71	0.77	0.54
Median	176.68	27.81	4.68	126.73	21.61	2.58	0.71	0.79	0.56
Geometric Mean	182.50	28.01	4.37	127.63	21.48	2.30	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	174.10	27.08	4.35	122.50	20.82	2.29	0.69	0.76	0.52
90% CI around geometric mean (upper limit)	191.31	28.97	4.39	132.99	22.16	2.31	0.71	0.78	0.53
5th percentile	96.09	17.47	1.03	72.46	14.03	0.50	0.54	0.61	0.36
95th percentile	352.60	43.98	15.54	216.50	32.57	8.13	0.84	0.90	0.70
SD	81.92	8.47	5.05	48.79	5.97	2.67	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3rd quartile	246.37	34.52	8.44	165.43	25.84	4.67	0.77	0.84	0.61
1st quartile	139.25	22.86	2.41	103.20	17.89	1.27	0.65	0.72	0.46
10th percentile	110.89	19.53	1.50	80.17	15.56	0.85	0.58	55.45	0.39
90th percentile	308.48	41.32	13.15	200.76	30.43	6.68	0.82	154.24	0.67

TABLE 24A

Predicted PK parameters for maribavir 800 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 800 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	135.70	22.25	3.30	1.37	1.52	1.06
Median	88.34	13.91	2.34	126.73	21.61	2.58	1.43	1.55	1.10
Geometric Mean	91.25	14.01	2.18	127.63	21.48	2.30	1.40	1.53	1.06
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	122.50	20.82	2.29	1.41	1.54	1.06
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	132.99	22.16	2.31	1.39	1.53	1.05
5th percentile	48.05	8.74	0.52	72.46	14.03	0.50	1.51	1.61	0.96
95th percentile	176.30	21.99	7.77	216.50	32.57	8.13	1.23	1.48	1.05
SD	40.96	4.23	2.53	48.79	5.97	2.67	1.19	1.41	1.06
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
3rd quartile	123.18	17.26	4.22	165.43	25.84	4.67	1.34	1.50	1.11
1st quartile	69.63	11.43	1.20	103.20	17.89	1.27	1.48	1.57	1.06
10th percentile	55.45	9.76	0.75	80.17	15.56	0.85	1.45	1.59	1.13
90th percentile	154.24	20.66	6.58	200.76	30.43	6.68	1.30	1.47	1.02

Simulation of Plasma Concentration-Time Profiles of Maribavir 1200 mg BID in the Presence and Absence of Carbamazepine

Predicted plasma concentration-time profiles of maribavir during 3 days of dosing (5 doses, starting on day 15) at 1200 mg BID in the absence and presence of once daily carbamazepine QD (200 mg on day 1 and 2, and 400 mg on day 3 to day 17) were generated. For each simulation, mean concentration-time profiles for each trial with 10 subjects, with a total virtual population of 200 subjects were used. The predicted PK parameters (C_{max}, $AUC_{(0-12h)}$ and C_{12h} for the population are shown in Table 25A. Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine compared with PK parameters for maribavir 400 mg BID are presented in Table 26A.

Maribavir AUC and C_{max} from 1200 mg BID in the presence of 400 mg carbamazepine QD were significantly higher (about 2-fold) than those from maribavir 400 mg BID alone. The C_{min} was about 50% higher than that of 400 mg maribavir alone, suggesting that increasing the maribavir dose from 400 mg BID to 800 mg BID might be preferred to a 1200 mg BID maribavir dose to counteract the effect of carbamazepine from an efficacy perspective.

TABLE 25A

Predicted PK parameters for maribavir 1200 mg BID in the presence and absence of carbamazepine.

	Maribavir 1200 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	296.79	43.79	9.32	203.54	33.37	4.95	0.71	0.77	0.54
Median	265.02	41.72	7.02	190.10	32.41	3.88	0.71	0.79	0.56
Geometric Mean	273.75	42.02	6.55	191.45	32.22	3.45	0.70	0.77	0.53
90% CI around geometric mean (lower limit)	261.14	40.62	6.48	183.75	31.23	3.42	0.69	0.76	0.53
90% CI around geometric mean (upper limit)	286.97	43.46	6.62	199.48	33.24	3.49	0.71	0.78	0.53
5th percentile	144.14	26.21	1.55	108.68	21.05	0.76	0.54	0.61	0.36
95th percentile	528.90	65.97	23.31	324.74	48.86	12.20	0.84	0.90	0.70
SD	122.87	12.70	7.58	73.19	8.96	4.01	0.10	0.09	0.11
% CV	41	29	81.28	36	27	81.02	14	12	20.2
3rd quartile	369.55	51.78	12.65	248.14	38.76	7.01	0.77	0.84	0.61
1st quartile	208.88	34.29	3.61	154.80	26.84	1.90	0.65	0.72	0.46
10th percentile	166.34	29.29	2.26	120.25	23.34	1.27	0.58	55.45	0.39
90th percentile	462.72	61.98	19.73	301.14	45.65	10.02	0.82	154.24	0.67

TABLE 26A

Predicted PK parameters for maribavir 1200 mg BID in the presence of carbamazepine 400 mg QD compared with PK parameters for maribavir 400 mg BID.

	Maribavir 400 mg BID			Maribavir 1200 mg BID + 400 mg QD carbazepine			Ratio		
	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$			
	mg/L*h	mg/L	mg/L	mg/L*h PREDICTED	mg/L	mg/L	$AUC_{(0-12\ h)}$	C_{max}	$C_{12\ h}$
Mean	98.93	14.60	3.11	203.54	33.37	4.95	2.06	2.29	1.59
Median	88.34	13.91	2.34	190.10	32.41	3.88	2.15	2.33	1.66
Geometric Mean	91.25	14.01	2.18	191.45	32.22	3.45	2.10	2.30	1.58
90% CI around geometric mean (lower limit)	87.05	13.54	2.16	183.75	31.23	3.42	2.11	2.31	1.58
90% CI around geometric mean (upper limit)	95.66	14.49	2.21	199.48	33.24	3.49	2.09	2.29	1.58
5th percentile	48.05	8.74	0.52	108.68	21.05	0.76	2.26	2.41	1.46
95th percentile	176.30	21.99	7.77	324.74	48.86	12.20	1.84	2.22	1.57
SD	40.96	4.23	2.53	73.19	8.96	4.01	1.79	2.12	1.58
% CV	41	29	81.28	36	27	81.02	0.88	0.93	1.00
	123.18	17.26	4.22	248.14	38.76	7.01	2.01	2.25	1.66
	69.63	11.43	1.20	154.80	26.84	1.90	2.22	2.35	1.58
	55.45	9.76	0.75	120.25	23.34	1.27	2.17	2.39	1.69
	154.24	20.66	6.58	301.14	45.65	10.02	1.95	2.21	1.52

Discussion

Prospective use of the model to predict the likely outcomes of interaction with rifampicin (800 mg and 1200 mg BID maribavir), phenobarbital (400 mg, 800 mg and 1200 mg BID maribavir), phenytoin (400 mg, 800 mg and 1200 mg BID maribavir), and carbamazepine (400 mg, 800 mg and 1200 mg BID maribavir) indicated geometric mean (arithmetic mean) Cmax ratios of 0.54 (0.56), 0.72 (0.73), 0.68 (0.69), and 0.77 (0.77), respectively, for maribavir. Corresponding predicted geometric mean (arithmetic mean) AUC_{0-12h} ratios were 0.35 (0.37), 0.60 (0.61), 0.57 (0.58), and 0.70 (0.71), respectively. Corresponding predicted geometric mean (arithmetic mean) C_{12h} ratios were 0.04 (0.10), 0.31 (0.37), 0.30 (0.36), and 0.53 (0.54) respectively. As there was no non-linearity in the PBPK model for maribavir, the changes in exposure as a consequence of CYP3A4-mediated induction by the perpetrators were not dependent on the dose of maribavir.

Efficacy of maribavir is correlated to the trough concentrations (C_{12h}) of the drug. Significant reductions in C_{12h} were predicted when the standard dose of 400 mg BID maribavir was combined with CYP3A4 inducers such as rifampicin, phenobarbital, phenytoin, and carbamazepine. Simulations with higher doses of maribavir (i.e. 800 mg BID, 1200 mg BID and 1600 mg BID) were performed to determine whether the reduction in C_{12h}, (and hence reduction in maribavir efficacy) could be overcome by dose increase. The reduction in C_{12h} due to the rifampicin induction of CYP3A4 could not be corrected by the use of up to 1600 mg BID maribavir. Use of a dose of 800 mg BID maribavir with phenobarbital resulted in a C_{12h} that was 15% lower than that with 400 mg BID maribavir alone while 1200 mg BID resulted a 1.28-fold increase in C12h. A 1200 mg BID dose of maribavir with phenytoin was also predicted to overcome the reduction in C_{12h}, due to CYP3A4 induction. Increasing the maribavir dose to 800 or 1200 mg BID can be used to overcome the reduction in C_{12h} due to carbamazepine. Maribavir demonstrated acceptable safety/tolerability profiles with no limiting toxicity at doses up to 1200 mg BID in Phase 2 studies for the treatment of CMV infection/diseases and the predicted mean $AUC_{(0-12h)}$ and C_{max} values are 292 mg/L*h and 43.4 mg/L for 1200 mg BID maribavir. The predicted $AUC_{(0-12h)}$ and C_{max} values seen with maribavir dose increases, as discussed above, in the presence of inducers are below the exposure predicted for 1200 mg BID maribavir alone, therefore there is no safety concern with increasing maribavir dose from 400 mg to 800 mg, or 1200 mg in the presence of inducer. Overall, CYP3A4 inducers significantly reduce systemic exposure of maribavir; therefore maribavir dose increase is necessary when co-administration with CYP3A4 inducers is necessary. When co-administration with carbamazepine or phenobartital is needed, maribavir dose increase to 800 or 1200 mg BID is recommended. When co-administration with phenytoin is needed, maribavir dose increase to 1200 mg BID recommendation. Rifampicin caused significant reduction in maribavir exposure which cannot be overcome by maribavir dose increase up to 1600 mg BID, therefore coadministration with rifampicin should be prohibited or alternative antibacterial therapy of lower CYP3A4 induction potential should be considered.

General note: The dose recommendations for inducers were based on arithmetic mean ratios due to the bias in the geometric mean estimates with coadministration. However, geometric mean ratios were used for dose recommendation for inhibitors.

Example 3—Co-Administration of Maribavir with Other Drugs

Drug interaction studies were summarized based on Example 2 and other available data (e.g., in vitro and clinical data). Considerations for each coadministered drug are provided in Table 1-A, and the effects of co-administration of other drugs on the pharmacokinetics of maribavir are summarized in Tables 1-B and 1-C.

TABLE 1-A

Summary of established and other potentially significant drug interactions (observed and predicted).

Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Antiarrhythmics		
digoxin	↑digoxin	Use caution when maribavir and digoxin are co-administered. Monitor serum digoxin concentrations. The dose of digoxin may need to be reduced with co-administered with maribavir
Anticonvulsants		
Carbamazepine	↓Maribavir	A dose adjustment of maribavir to 800 mg, twice daily when co-administered with carbamazepine
Phenobarbital	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenobarbital
Phenytoin	↓Maribavir	A dose adjustment of maribavir to 1200 mg, twice daily when co-administered with phenytoin
Antimycobacterials		
Rifabutin	↓Maribavir	Co-administration of maribavir and rifamutin may decrease efficacy of maribavir
Rifampin	↓Maribavir	Co-administration of maribavir and rifampin may decrease efficacy of maribavir
Herbal Products		
St. John's wort	↓Maribavir	Co-administration of maribavir and St. John's wort may decrease efficacy of maribavir

TABLE 1-A-continued

Summary of established and other potentially significant drug interactions (observed and predicted).

Concomitant Drug Class: Drug Name	Effect on Concentration	Considerations
Immunosuppressants		
Cyclosporine	↑Cyclosporine	Frequently monitor cyclosporine levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Everolimus	↑Everolimus	Frequently monitor everolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Sirolimus	↑Sirolimus	Frequently monitor sirolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed
Tacrolimus	↑Tacrolimus	Frequently monitor tacrolimus levels throughout treatment with maribavir, especially following initiation and after discontinuation of maribavir and adjust dose as needed

TABLE 1-B

Summary of Changes in Pharmacokinetics of Maribavir in the Presence of Co-administered Drugs.

Co-administered Drug and Regimen		Maribavir Regimen	N	Geometric Mean Ratio (90% CI] of Maribavir PK with/without Co-administered Drug (No Effect = 1.00) AUC	C_{max}	$C_{tau}{}^{C}$
Anticonvulsants						
Carbamazepine[a]	400 mg once daily	800 mg twice daily/ 400 mg twice daily	200	1.40 (1.09, 1.67)	1.53 (1.22, 1.79)	1.05 (0.71, 1.40)
Phenobarbital[a]	100 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.80 (1.18, 2.35)	2.17 (1.69, 2.57)	0.94 (0.22, 1.97)
Phenytoin[a]	300 mg once daily	1,200 twice daily/ 400 mg twice daily	200	1.70 (1.06, 2.46)	2.05 (1.49, 2.63)	0.89 (0.26, 2.04)
Antimycobacterials						
Rifampin	600 mg once daily	400 mg twice daily	14	0.40 (0.36, 0.44)	0.61 (0.52, 0.72)	0.18 (0.14, 0.25)
Antifungals						
Ketoconazole	400 mg single dose	400 mg single dose	19	1.53 (1.44, 1.63)	1.10 (1.01, 1.19)	—
Aluminum hydroxide and magnesisum hydroxide antacid	20 mL[b] single dose	100 mg single dose	15	0.89 (0.83, 0.96)	0.84 (0.75, 0.94)	

[a]Based on physiologically based pharmacokinetic modelings results from 10 trials of 20 subjects each. The maribavir dosing regimen and geometric mean ratios (5^{th} percentile, 95 percentile) correspond to dose-adjusted maribavir with inducer vs. 400 mg twice daily without inducer.

[b]Containing 800 mg aluminum hydroxide and 800 mg magnesium hydroxide.

[C]tau is maribavir dosing interval: 12 hours.

TABLE 1-C

Changes of Pharmacokinetics for Co-administered Drug in the Presence of 400 mg Twice Daily Maribavir.					
			Geometric Mean Ratio [90% CI] of Maribavir PK with/without Co-maribavir (No Effect = 1.00)		
Co-administered Drug and Regimen		N	AUC	C_{max}	C_{trough}
Immunosuppressants					
Tacrolimus	stable dose, twice daily (total daily dose: 0.5-16 mg)	20	1.51 (1.39, 1.65)	1.38 (1.20, 1.57)	1.57 (1.41, 1.74)
P-gp substrate					
digoxin	0.5 mg single dose	18	1.21 (1.10, 1.32)	1.25 (1.13, 1.38)	—

While we have described a number of embodiments of this invention, it is apparent that our basic examples may be altered to provide other embodiments that utilize the compounds and methods of this invention. Therefore, it will be appreciated that the scope of this invention is to be defined by the appended claims rather than by the specific embodiments that have been represented by way of example.

The invention claimed is:

1. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom, the method comprising administering maribavir in an amount of 1200 mg orally twice daily to the patient,

wherein the patient is a transplant recipient concomitantly exposed to or receiving an anticonvulsant and an immunosuppressant, and wherein the amount of maribavir is selected to counteract the effect of the anticonvulsant from an efficacy perspective;

wherein the method further comprises monitoring immunosuppressant drug levels in blood throughout treatment with maribavir,

wherein the patient is an adult, or a child of 12 years of age or older and weighing at least 35 kg, and wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.

2. The method of claim **1**, wherein the anticonvulsant is phenytoin.

3. The method of claim **2**, wherein phenytoin is administered to the patient in an amount of 300 mg once daily.

4. The method of claim **2**, wherein the immunosuppressant is cyclosporine.

5. The method of claim **2**, wherein the immunosuppressant is everolimus.

6. The method of claim **2**, wherein the immunosuppressant is sirolimus.

7. The method of claim **2**, wherein the immunosuppressant is tacrolimus.

8. The method of claim **7**, wherein tacrolimus is administered as a stable dose, twice daily, with a total daily dose between 0.5 to 16 mg.

9. The method of claim **1**, wherein the anticonvulsant is phenobarbital.

10. The method of claim **9**, wherein phenobarbital is administered to the patient in an amount of 100 mg once daily.

11. The method of claim **9**, wherein the immunosuppressant is cyclosporine.

12. The method of claim **9**, wherein the immunosuppressant is everolimus.

13. The method of claim **9**, wherein the immunosuppressant is sirolimus.

14. The method of claim **9**, wherein the immunosuppressant is tacrolimus.

15. The method of claim **14**, wherein tacrolimus is administered to the patient as a stable dose, twice daily, with a total daily dose between 0.5 to 16 mg.

16. The method of claim **1**, wherein the monitoring comprises monitoring immunosuppressant drug levels following initiation, and after discontinuation, of treatment with maribavir in the patient.

17. A method of treating cytomegalovirus (CMV) infection in a patient suffering therefrom, the method comprising administering maribavir in an amount of 800 mg orally twice daily to the patient,

wherein the patient is a transplant recipient concomitantly exposed to or receiving an anticonvulsant and an immunosuppressant, wherein the amount of maribavir is selected to counteract the effect of the anticonvulsant from an efficacy perspective;

wherein the method further comprises monitoring immunosuppressant drug levels in blood throughout treatment with maribavir,

wherein the patient is an adult, or a child of 12 years of age or older and weighing at least 35 kg, and wherein the patient is refractory to treatment with one or more of ganciclovir, valganciclovir, cidofovir, or foscarnet.

18. The method of claim **17**, wherein the anticonvulsant is carbamazepine.

19. The method of claim **18**, wherein carbamazepine is administered to the patient in an amount of 400 mg once daily.

20. The method of claim **18**, wherein the immunosuppressant is cyclosporine.

21. The method of claim **18**, wherein the immunosuppressant is everolimus.

22. The method of claim **18**, wherein the immunosuppressant is sirolimus.

23. The method of claim **18**, wherein the immunosuppressant is tacrolimus.

24. The method of claim **23**, wherein tacrolimus is administered to the patient as a stable dose, twice daily, with a total daily dose between 0.5 to 16 mg.

25. The method of claim **17**, wherein the monitoring comprises monitoring immunosuppressant drug levels following initiation, and after discontinuation, of treatment with maribavir in the patient.

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